
FR800x Specification

Bluetooth Low Energy SOC with SIG Mesh

Version:v1.2.1

Date:2025.2



Revision History

Version	Date	Description
V1.0	2021.6.1	Draft
V1.0.1	2021.11.17	Update UART module 5.4.2 DATA and DLL Register
V1.1.0	2021.11.29	Add DMA Scatter)/ Gather 1. Add DMA Scatter/gather Function Description 2. Add SGRx 和 DSRx Register 3. Modify DMA Register Map , add SGRx and DSRx Register 4. Modify CTLRegister, support scatter/gather 5. Modify Register, including 12.4.9 RawTfr, 12.4.10 RawErr, 12.4.11 StatusTfr, 12.4.12 StatusErr, 12.4.13 MaskTfr, 12.4.14 MaskErr, 12.4.15 ClearTfr, 12.4.16 ClearErr, 12.4.17 StatusInt,12.4.19 ChEnReg Register。
V1.1.1	2021.12.1	Update SPI module 6.4.8 TXFTLR, 6.4.9 RXFTLR,6.4.10 TXFLR,6.4.11 RXFLRRegister; 7.4.5 TXFTLR, 7.4.6 RXFTLR, 7.4.7 TXFLR, 7.4.8 RXFLR Register。
V1.1.2	2021.12.6	Modify 12.3.2DMA, modify length、data width
V1.1.3	2022.1.4	Modify Chapter 5.4.6 MCR Register
V1.1.4	2022.1.26	Modify Chapter 1.2 FIFO and Chapter 18USB endpoint
V1.1.5	2022.5.25	Modify ADKey register Delete USB debugging mode Add I2C function description Add PortA/B/C/D/E_OEN Regsister Description in Chapter PMU_GPIO Add RF_PA_RXEN、RF_PA_TXEN in IOMUX
V1.1.6	2023.2.12	Modify Chapter5.4.7 LSR Register description

Version	Date	Description
V1.1.7	2023.4.21	Modify Chapter 14.4.8 and 14.4.11 Add Chapter14.9.7 Analog_Monitor_Level Delete Chapter ANA_Monitor_Status Modify Chapter 8.4.2 and 8.4.6
V1.1.8	2023.7.23	Modify Chapter 18.3.2.2 and 18.3.2.12
V1.1.9	2023.12.1	Modify Chapter 2.2 Clock Tree
V1.2.0	2024.6.3	Modify PWM period description
V1.2.1	2025.2.14	Modify Chapter 16 Register Access Type Modify Chapter 14.9 IWDG Delete Chapter 14.7 PMU_PWM3

Contents

Contents	I
Tables	XIII
Figures	XXI
1. System Overview	1
1.1. Overview	1
1.2. Feature	1
1.3. Memory Map	6
1.4. Interrupt map	7
2. System Controller	8
2.1. Overview	8
2.2. System Clock	8
2.2.1. Clock Source	8
2.2.2. System Clock Tree	1
2.3. System Reset	1
2.4. Low power consumption of the system	1
2.5. System Control Register	1
2.5.1. System Register Mapping	1
2.5.2. Clock Control (Offset 0x00)	2
2.5.3. Clock Enable(Offset 0x04)	3
2.5.4. QSPI0 Control(Offset 0x08)	5
2.5.5. Reset Control (Offset 0x0C)	7
2.5.6. APB Clock Enable (Offset 0x10)	8
2.5.7. DMA_REQ_MUX0(Offset 0x14)	10
2.5.8. DMA_REQ_MUX1(Offset 0x18)	11
2.5.9. DMA_REQ_MUX2 (Offset 0x1C)	11
2.5.10. PortA Function MUX (Offset 0x20)	12
2.5.11. PortB Function MUX(Offset 0x24)	12
2.5.12. PortC Function MUX (Offset 0x28)	13
2.5.13. PortD Function MUX (Offset 0x2C)	13
2.5.14. Portx Pull Enable (Offset 0x30)	13
2.5.15. Portx Pull Select (Offset 0x34)	14
2.5.16. PortE_CFG (Offset 0x38)	15
2.5.17. USB_OTG_CTRL (Offset 0x3C)	15
3. GPIO	17
3.1. Overview	17
3.2. Feature	17
3.3. Block Diagram	17

3.4. GPIO Register.....	18
3.4.1. GPIO Register Address Mapping.....	18
3.4.2. PortA_DATA (Offset 0x00)	18
3.4.3. PortB_DATA (Offset 0x04)	19
3.4.4. PortC_DATA (Offset 0x08).....	19
3.4.5. PortD_DATA (Offset 0x0C).....	19
3.4.6. PortE_DATA(Offset 0x10)	19
3.4.7. PortA_OEN (Offset 0x20)	20
3.4.8. PortB_OEN (Offset 0x24).....	20
3.4.9. PortC_OEN (Offset 0x28)	20
3.4.10. PortD_OEN(Offset 0x2C)	21
3.4.11. PortE_OEN(Offset 0x30).....	21
3.4.12. Port_EXTI_EN0(Offset 0x40).....	21
3.4.13. Port_EXTI_EN1(Offset 0x44).....	22
4. External Interrupt Controller (EXTI).....	23
4.1. Overview.....	23
4.2. Feature.....	23
4.3. EXTI Block Diagram.....	23
4.4. EXTI Register.....	23
4.4.1. EXTI Register Mapping	23
4.4.2. EXTI_CTL0(Offset 0x00)	25
4.4.3. EXTI_CTL1(Offset 0x04)	25
4.4.4. EXTI_STS0(Offset 0x08)	25
4.4.5. EXTI_STS1(Offset 0x0C)	26
4.4.6. EXTI_TYPE0(Offset 0x10).....	26
4.4.7. EXTI_TYPE1(Offset 0x14).....	27
4.4.8. EXTI_TYPE2(Offset 0x18).....	28
4.4.9. EXTI_CNT0 ~ EXTI_CNT35(Offset 0x20 ~ 0xAC)	28
5. UART.....	29
5.1. Overview.....	29
5.2. Feature.....	29
5.3. Function description.....	29
5.3.1. Baud Rate Calculation.....	29
5.3.2. IrDA SIR 1.0 Protocol	30
5.4. Uart Register.....	31
5.4.1. Uart Register Memory Mapping	31
5.4.2. DATA/DLL (Offset 0x00).....	32
5.4.3. IER/DLH (Offset 0x04)	33
5.4.4. FCR/IID (Offset 0x08).....	35
5.4.5. LCR (Offset 0x0C)	37
5.4.6. MCR (Offset 0x10)	38

5.4.7. LSR (Offset 0x14)	39
5.4.8. MSR (Offset 0x18)	41
5.4.9. USR(Offset 0x7C)	42
5.4.10. DLF (Offset 0xC0)	43
5.5. Work Flow	44
5.5.1. Transmission Work Flow	44
5.5.2. Reception Work Flow	45
6. Serial Peripheral Interface Master (SPI_Master)	46
6.1. Overview	46
6.2. Features	46
6.3. Function	47
6.3.1. Transmit /Receive FIFO and Interrupt	47
6.3.2. Transfer Modes	48
6.3.3. DMA Controller Interface	49
6.4. SPI Register	49
6.4.1. SPI Register Memory Mapping	49
6.4.2. CTRLR0 (Offset 0x00)	50
6.4.3. CTRLR1 (Offset 0x04)	53
6.4.4. SSIENR (Offset 0x08)	53
6.4.5. MWCR(Offset 0x0C)	54
6.4.6. SER(Offset 0x10)	55
6.4.7. BAUDR(Offset 0x14)	55
6.4.8. TXFTLR (Offset 0x18)	55
6.4.9. RXFTLR (Offset 0x1C)	56
6.4.10. TXFLR (Offset 0x20)	56
6.4.11. RXFLR (Offset 0x24)	56
6.4.12. SR (Offset 0x28)	57
6.4.13. IMR (Offset 0x2C)	58
6.4.14. ISR (Offset 0x30)	59
6.4.15. RISR (Offset 0x34)	59
6.4.16. TXOICR (Offset 0x38)	60
6.4.17. RXOICR (Offset 0x3C)	60
6.4.18. RXUICR (Offset 0x40)	61
6.4.19. ICR (Offset 0x48)	61
6.4.20. DMACR (Offset 0x4C)	61
6.4.21. DMATDLR (Offset 0x50)	62
6.4.22. DMARDLR (Offset 0x54)	62
6.4.23. DR (offset 0x60)	63
6.4.24. RX_SAMPLE_DLY (Offset 0xF0)	63
6.4.25. CTRLR2(Offset 0xF4)	63
6.5. Processing flow	65

6.5.1. SPI standard mode transmission	65
6.5.2. SPI standard mode reception	65
6.5.3. Write Operation in Dual or Quad Mode	66
6.5.4. Read Operation in Dual or Quad Mode	67
6.6. MOTOROLA SPI timing	70
7. Serial Peripheral Interface Master (SPI_Slave)	71
7.1. Overview	71
7.2. Features	71
7.3. Function	72
7.3.1. Transmit / Receive FIFO and Interrupt	72
7.3.2. Transfer Modes	73
7.3.3. DMA Interface	73
7.4. SPI Register	74
7.4.1. SPI Register Memory Mapping	74
7.4.2. CTRLR0(Offset 0x00)	75
7.4.3. SSIENR (Offset 0x08)	77
7.4.4. MWCR(Offset 0x0C)	78
7.4.5. TXFTLR(Offset 0x18)	78
7.4.6. RXFTLR(Offset 0x1C)	78
7.4.7. TXFLR (Offset 0x20)	79
7.4.8. RXFLR (Offset 0x24)	79
7.4.9. SR (Offset 0x28)	79
7.4.10. IMR (Offset 0x2C)	81
7.4.11. ISR (Offset 0x30)	81
7.4.12. RISR (Offset 0x34)	82
7.4.13. TXOICR (Offset 0x38)	83
7.4.14. RXOICR (Offset 0x3C)	83
7.4.15. RXUICR (Offset 0x40)	83
7.4.16. ICR (Offset 0x48)	84
7.4.17. DMACR (Offset 0x4C)	84
7.4.18. DMATDLR (Offset 0x50)	84
7.4.19. DMARDLR (Offset 0x54)	85
7.4.20. DR (offset 0x60)	85
7.5. Processing flow	86
7.6. MOTOROLA SPI timing	87
8. I2C	88
8.1. Overview	88
8.2. Feature	88
8.3. Function	89
8.3.1. I2C Timing Control	89
8.3.2. I2C_SDA_HOLD function	90

8.3.3. SCL、SDA Low Timeout Function.....	91
8.3.4. 7-bit Address Format、10-bit Address Format.....	92
8.3.5. Spike Suppression	93
8.4. I2C Register	94
8.4.1. I2C Register Memory Mapping.....	94
8.4.2. CONTROL (Offset 0x00).....	96
8.4.3. TAR (Offset 0x04)	98
8.4.4. SAR (Offset 0x08).....	99
8.4.5. HS_MADDR (Offset 0x0C).....	100
8.4.6. DATA_CMD (Offset 0x10)	100
8.4.7. SS_SCL_HCNT (Offset 0x14)	101
8.4.8. SS_SCL_LCNT (Offset 0x18)	101
8.4.9. FS_SCL_HCNT (Offset 0x1C)	102
8.4.10. FS_SCL_LCNT (Offset 0x20).....	102
8.4.11. HS_SCL_HCNT(Offset 0x24).....	103
8.4.12. HS_SCL_LCNT (Offset 0x28).....	104
8.4.13. INTR_STAT (Offset 0x2C)	104
8.4.14. INTR_MASK (Offset 0x30)	108
8.4.15. RAW_INTR_STAT (Offset 0x34)	109
8.4.16. RX_TL (Offset 0x38)	110
8.4.17. TX_TL (Offset 0x3C).....	110
8.4.18. CLR_INTR (Offset 0x40).....	110
8.4.19. CLR_RX_UNDER (Offset 0x44).....	110
8.4.20. CLR_RX_OVER (Offset 0x48)	111
8.4.21. CLR_TX_OVER (Offset 0x4C).....	111
8.4.22. CLR_RD_REQ (Offset 0x50).....	111
8.4.23. CLR_TX_ABRT(Offset 0x54).....	112
8.4.24. CLR_RX_DONE (Offset 0x58).....	112
8.4.25. CLR_ACTIVITY (Offset 0x5C).....	113
8.4.26. CLR_STOP_DET (Offset 0x60).....	113
8.4.27. CLR_START_DET (Offset 0x64)	113
8.4.28. ENABLE (Offset 0x6C).....	114
8.4.29. STATUS (Offset 0x70)	115
8.4.30. TXFLR (Offset 0x74)	117
8.4.31. RXFLR (Offset 0x78)	117
8.4.32. SDA_HOLD(Offset 0x7C).....	118
8.4.33. TX_ABRT_SOURCE (Offset 0x80).....	118
8.4.34. SLV_DATA_NACK_ONLY (Offset 0x84).....	123
8.4.35. DMA_CR (Offset 0x88)	123
8.4.36. DMA_TDLR (Offset 0x8C)	124
8.4.37. DMA_RDLR (Offset 0x90).....	124

8.4.38. SDA_SETUP (Offset 0x94)	124
8.4.39. FS_SPKLEN (Offset 0xA0)	125
8.4.40. HS_SPKLEN (Offset 0xA4)	125
8.4.41. CLR_RESTART_DET (Offset 0xA8)	126
8.4.42. SCL_STUCK_AT_LOW_TIMEOUT (Offset 0xAC)	126
8.4.43. SDA_STUCK_AT_LOW_TIMEOUT (Offset 0xB0)	127
8.4.44. CLR_SCL_STUCK_DET(Offset 0xB4)	127
8.5. Work Flow	127
8.5.1. Master Mode Operation	127
8.5.2. Slave Mode Operation	128
8.5.3. Timing Diagram	130
9. I2S	133
9.1. Overview	133
9.2. Feature	133
9.3. Block Diagram	134
9.4. Function Description	134
9.4.1. CLK、WS Frequency Division	134
9.4.2. WS_INV, DLY_EN, FMT Configuration	134
9.5. I2S Register	136
9.5.1. I2S Register Mapping	136
9.5.2. CONTROL (Offset 0x00)	136
9.5.3. CLK_DIV (Offset 0x04)	138
9.5.4. WS_DIV(Offset 0x08)	138
9.5.5. DATA (Offset 0x0C)	138
9.5.6. STATUS (Offset 0x10)	138
9.5.7. INTE (Offset 0x14)	139
9.5.8. FIFO_CFG (Offset 0x20)	140
9.5.9. DMA_CFG (Offset 0x24)	140
10. TRNG	141
10.1. Overview	141
10.2. Feature	141
10.3. TRNG Block Diagram	141
10.4. TRNG Register	141
10.4.1. TRNG Register Mapping	141
10.4.2. RNG_IMR (Offset 0x100)	142
10.4.3. RNG_ISR (Offset 0x104)	143
10.4.4. RNG_ICR (Offset 0x108)	143
10.4.5. TRNG_CONFIG (Offset 0x10C)	144
10.4.6. TRNG_VALID(Offset 0x110)	144
10.4.7. HER_DATA0 ~ 5(Offset 0x114 ~ 0x128)	144
10.4.8. RND_SOURCE_ENABLE (Offset 0x12C)	145

10.4.9. SAMPLE_CNT1(Offset 0x130)	145
10.4.10. TRNG_RESET (Offset 0x140)	145
10.4.11. TRNG_BUSY (Offset 0x1B8).....	146
10.4.12. RST_BITS_COUNTER (Offset 0x1BC)	146
11. PDM.....	147
11.1. Overview.....	147
11.2. Feature.....	147
11.3. PDM System Block Diagram.....	147
11.4. PDM Register	148
11.4.1. PDM Register Mapping	148
11.4.2. PDM_Control (Offset 0x00).....	148
11.4.3. VOLUME (Offset 0x04).....	149
11.4.4. VOLSTEP (Offset 0x08).....	149
11.4.5. RX_DATA(Offset 0x0C).....	150
11.4.6. FIFO_Clear(Offset 0x10)	150
11.4.7. FIFO_Status (Offset 0x14).....	151
11.4.8. FIFO_INT_EN(Offset 0x18).....	151
11.4.9. FIFO_Almost_Level (Offset 0x0x1C)	151
11.4.10. DMA_CFG(Offset 0x20).....	152
12. DMA	153
12.1. Overview.....	153
12.2. Feature.....	153
12.3. Function Description.....	153
12.3.1. DMA Priority	154
12.3.2. Transfer Length, Data Bit Width, Burst Length Relationship	154
12.3.3. Linked Lists Transfer Function.....	155
12.3.4. Scatter/Gather.....	155
12.4. DMA Register.....	157
12.4.1. DMA Register Mapping	157
12.4.2. SARx (Offset 0x00, 0x58, 0xB0)	158
12.4.3. DARx (Offset 0x08, 0x60, 0xB8)	159
12.4.4. LLPx (Offset 0x10, 0x68, 0xC0)	159
12.4.5. CTLx (Offset 0x18, 0x70, 0xC8)	159
12.4.6. CFGx(Offset 0x40, 0x98, 0xF0)	162
12.4.7. SGRx (Offset 0x48, 0xA0, 0xF8)	165
12.4.8. DSRx (Offset 0x50, 0xA8, 0x100).....	165
12.4.9. RawTfr (Offset 0x2C0).....	166
12.4.10. RawErr (Offset 0x2E0).....	166
12.4.11. StatusTfr (Offset 0x2E8)	166
12.4.12. StatusErr (Offset 0x308)	167
12.4.13. MaskTfr (Offset 0x310).....	167

12.4.14. MaskErr (Offset 0x330)	168
12.4.15. ClearTfr (Offset 0x338).....	168
12.4.16. ClearErr (Offset 0x358).....	169
12.4.17. StatusInt (Offset 0x360).....	169
12.4.18. DmaCfgReg (Offset 0x398).....	170
12.4.19. ChEnReg (Offset 0x3A0).....	170
12.5. Note	171
13. PWM.....	172
13.1. Overview.....	172
13.2. Feature.....	172
13.3. PWM Block Diagram.....	173
13.4. PWM Register	174
13.4.1. PWM Register Mapping	174
13.4.2. PWM_Output_EN(Offset 0x04).....	175
13.4.3. PWM_ENABLE (Offset 0x08)	176
13.4.4. PWM0~7_Posedge (Offset 0x10, 0x18, 0x20, 0x28, 0x30, 0x38, 0x40, 0x48).....	176
13.4.5. PWM0~7_Negedge (Offset 0x14, 0x1C, 0x24, 0x2C, 0x34, 0x3C, 0x44, 0x4C)	176
13.4.6. PWM_Output_Select (Offset 0x90).....	177
13.4.7. Capture_Prescale (Offset 0x94).....	177
13.4.8. Capture_Status (Offset 0x98).....	178
13.4.9. Capture_INT_EN (Offset 0x9C).....	178
13.4.10. Capture_CTL(Offset 0xA0)	178
13.4.11. Capture_Value0~7 (Offset 0xA4 ~ 0xC0)	179
13.4.12. PWM_Update (Offset 0xE4).....	179
13.4.13. PWM_DAC_Mode (Offset 0xE8).....	180
13.4.14. PWM_Output_Value (Offset 0xF0).....	180
13.4.15. PWM_Inverter_EN(Offset 0xF8)	180
13.4.16. PWM_CNT_EN (Offset 0xFC)	181
13.4.17. PWM0~7_Prescale (Offset 0x100, 0x108, 0x110, 0x118, 0x120, 0x128, 0x130, 0x138) ...	181
13.4.18. PWM0~7_Period(Offset 0x104, 0x10C, 0x114, 0x11C, 0x124, 0x12C, 0x134, 0x13C).....	181
13.4.19. PWM0~7_DAC_DIV(Offset 0x180, 0x184, 0x188, 0x18C, 0x190, 0x194, 0x198, 0x19C).....	182
13.5. Process.....	182
13.5.1. PWM Mode.....	182
13.5.2. PWM_DAC Mode.....	183
14. PMU.....	185
14.1. Overview.....	185
14.2. Access Control Interface	185
14.3. PMU_IWDG	186
14.3.1. PMU_IWDG Function Description	186
14.3.2. PMU_IWDG System block diagram	186
14.3.3. PMU_IWDG Register Mapping.....	186

14.3.4. IWDG_CTRL (Block0, 0x26).....	187
14.3.5. IWDG_LEN (Block0, 0x27, 0x28, 0x29).....	187
14.3.6. IWDG_TOUT_LEN (Block0, 0x2A).....	187
14.4. PMU_GPIO.....	188
14.4.1. PMU_GPIO Function Description.....	188
14.4.2. PMU_GPIO Register Mapping.....	188
14.4.3. PortA/B/C/D/E_SEL (Block0, 0x43, 0x44, 0x45, 0x46, 0x47)	189
14.4.4. PortA/B/C/D/E_PE (Block0, 0x48, 0x49, 0x4A, 0x4B, 0x4C).....	189
14.4.5. PortA/B/C/D/E_PS (Block0, 0x4D, 0x4E, 0x4F, 0x50, 0x51).....	190
14.4.6. PortA/B/C/D_Monitor_EN (Block0, 0x52, 0x53, 0x54, 0x55).....	190
14.4.7. PortE_Monitor_EN (Block0, 0x56).....	190
14.4.8. Analog_Monitor_EN (Block0, 0x57)	191
14.4.9. PortA/B/C/D_Reference_Status (Block0, 0x58, 0x59, 0x5A, 0x5B).....	191
14.4.10. PortE_Reference_Status (Block0, 0x5C).....	192
14.4.11. Analog_Reference_Status (Block0, 0x5D).....	192
14.4.12. PortA/B/C/D/E_OEN (Block0, 0x66, 0x67, 0x68, 0x69, 0x6A).....	193
14.4.13. PortA/B/C/D/E_Data(Block0, 0x6B, 0x6C, 0x6D, 0x6E, 0x6F).....	193
14.5. PMU_RTC.....	194
14.5.1. PMU_RTC Function Description.....	194
14.5.2. PMU_RTC Register Mapping.....	194
14.5.3. RTC_CTRL (Block1, 0x07)	195
14.5.4. RTC_CNT (Block1, 0x08, 0x09, 0x0A, 0x0B).....	195
14.5.5. ALARM_A (Block1, 0x0C, 0x0D, 0x0E, 0x0F).....	196
14.5.6. ALARM_B (Block1, 0x10, 0x11, 0x12, 0x13).....	196
14.6. PMU_KeyScan.....	196
14.6.1. PMU_KeyScan Fuction Description.....	196
14.6.2. PMU_KeyScan Register Mapping.....	197
14.6.3. KeyScan_CTL (Block1, 0x18).....	197
14.6.4. Scan_Interval (Block1, 0x19, 0x1A).....	198
14.6.5. KeyScan_Config (Block1, 0x1B)	198
14.6.6. ROW_SEL (Block1, 0x1C).....	199
14.6.7. COL_SEL (Block1, 0x1D, 0x1E, 0x1F)	199
14.6.8. KEY_VAL (Block1, 0x20 ~ 0x33)	199
14.7. PMU_PWM.....	200
14.7.1. PMU_PWM function description.....	200
14.7.2. PMU_PWM IO allocation.....	200
14.7.3. PMU_PWM Register Mapping.....	200
14.7.4. PWM0 ~ 2_CTRL (Block1, 0x34, 0x35, 0x36)	201
14.7.5. PWM0_PRC (Block1, 0x38, 0x39).....	202
14.7.6. PWM0_HRC (Block1, 0x3A, 0x3B)	202
14.7.7. PWM1_PRC (Block1, 0x3C, 0x3D).....	202

14.7.8. PWM1_HRC (Block1, 0x3E, 0x3F).....	202
14.7.9. PWM2_PRC (Block1, 0x40, 0x41).....	202
14.7.10. PWM2_HRC (Block1, 0x42, 0x43).....	202
14.8. PMU_IOMux.....	203
14.8.1. Function Description.....	203
14.8.2. PMU_IOMux Register Mapping.....	203
14.8.3. PortA_IOMux (Block1, 0x6A, 0x6B).....	203
14.8.4. PortB_IOMux(Block1, 0x6C, 0x6D).....	204
14.8.5. PortC_IOMux (Block1, 0x6E, 0x6F).....	204
14.8.6. PortD_IOMux (Block1, 0x70, 0x71).....	205
14.8.7. PortE_IOMux (Block1, 0x72).....	205
14.9. PMU_INT_CTRL.....	205
14.9.1. PMU_INT_CTRL Function Description.....	205
14.9.2. PMU_INT_CTRL Register Mapping.....	205
14.9.3. BAT_Full_Filter (Block1, 0x58).....	206
14.9.4. ACOK_Filter (Block1, 0x59).....	206
14.9.5. LVD_Filter (Block1, 0x5A).....	207
14.9.6. OTD_Filter (Block1, 0x5B).....	207
14.9.7. Analog_Monitor_Level (Block1, 0x5C).....	207
14.9.8. INT_EN0 (Block1, 0x5E).....	208
14.9.9. INT_EN1 (Block1, 0x5F).....	209
14.9.10. INT_CLR0(Block1, 0x60).....	210
14.9.11. INT_CLR1 (Block1, 0x61).....	210
14.9.12. INT_STS0 (Block1, 0x62).....	211
14.9.13. INT_STS1 (Block1, 0x63).....	211
14.9.14. PortA/B/C/D_Monitor_Status (Block1, 0x64, 0x65, 0x66, 0x67).....	211
14.9.15. PortE_Monitor_Status (Block1, 0x68).....	212
15. LCD.....	213
15.1. Overview.....	213
15.2. Feature.....	213
15.3. LCD Block Diagram.....	213
15.4. LCD Register.....	214
15.4.1. LCD Register Mapping.....	214
15.4.2. LCD_TXFF_PUSH (Offset 0x00).....	214
15.4.3. LCD_CFG (Offset 0x04).....	215
15.4.4. LCD_CFGWR_STATUS (Offset 0x08).....	215
15.4.5. LCD_CFGRD_STATUS (Offset 0x0C).....	215
15.4.6. LCD_APB_RD (Offset 0x10).....	216
15.4.7. LCD_TXFF_AEMP_LV (Offset 0x14).....	216
15.4.8. LCD_DAT_WR_LEN (Offset 0x18).....	216
15.4.9. LCD_DAT_WR_CFG (Offset 0x1C).....	217

15.4.10. LCD_DMA (Offset 0x20)	218
15.4.11. LCD_CSX (Offset 0x24).....	218
15.4.12. LCD_CRM(Offset 0x28)	219
15.4.13. LCD_TXFF_CLR (Offset 0x2C)	219
15.4.14. LCD_INTF_CFG (Offset 0x30).....	220
15.4.15. LCD_INT (Offset 0x34)	220
15.4.16. LCD_TE_VSYNC_CFG (Offset 0x38).....	221
15.4.17. LCD_TE_HSYNC_CFG (Offset 0x3C)	222
15.5. Process.....	222
15.5.1. Write command or parameter	222
15.5.2. Write Data.....	223
15.5.3. Read Data.....	223
16. Timer.....	225
16.1. Overview.....	225
16.2. Feature.....	225
16.3. System Block Diagram	225
16.4. Timer Register	226
16.4.1. Timer Register Mapping.....	226
16.4.2. LOAD_VALUE (Offset 0x00)	226
16.4.3. COUNT_VALUE (Offset 0x04).....	226
16.4.4. CONTROL (Offset 0x08).....	226
16.4.5. INT_CLR (Offset 0x0C).....	227
16.5. Process.....	227
17. ADC	228
17.1. Overview.....	228
17.2. Feature.....	228
17.3. Block Diagram.....	228
17.4. Function Description.....	229
17.5. ADC Register.....	229
17.5.1. ADC Register Mapping	229
17.5.2. CONTROL (Offset 0x00).....	230
17.5.3. DATA0 ~ DATA7 (Offset 0x04 ~ 0x20)	232
17.5.4. DATA (Offset 0x24).....	232
17.5.5. FIFO_Status (Offset 0x28).....	232
17.5.6. INT (Offset 0x2C).....	233
17.5.7. INTC (Offset 0x30).....	234
17.5.8. ADC_ANA_CTRL0 (Offset 0x34).....	234
17.5.9. ADC_ANA_CTRL1 (Offset 0x38).....	235
17.5.10. ADC_ANA_CTRL2 (Offset 0x3C).....	236
18. USB	237
18.1.1. Overview	237

18.1.2. Feature.....	237
18.2. USB Block Diagram	238
18.3. USB Register	238
18.3.1. USB Register Mapping.....	238
18.3.2. USB Common Register (Offset 0x00~0x0F).....	242
18.3.3. USB endpoint 0 control status Register (Offset 0x10~0x1F).....	249
18.3.4. USB endpoints 1 to 5 control status Register (Offset 0x10~0x1F).....	253
18.3.5. USB Endpoint FIFO Register (Offset 0x20~0x34)	264
18.4. Process.....	264
18.4.1. Endpoint index.....	264
18.4.2. Endpoint FIFO Configuration.....	264
18.4.3. IN TRANSACTION HANDLING AS A PERIPHERAL	265
18.4.4. OUT TRANSACTION HANDLING AS A PERIPHERAL	266
18.4.5. Device Mode Additional Actions.....	266
18.4.6. Device Mode Suspend	267
18.4.7. Device Mode Start of frame.....	267
18.4.8. In transaction handling as a Host.....	268
18.4.9. Out Transaction handling as a Host	269
18.4.10. Host Mode Transaction Scheduling.....	269
18.4.11. Host Mode Reset.....	270
18.4.12. Host Mode Suspend	270
Contact Information	271
19. Appendix I	272
20. Appendix II.....	277

Tables

table 5-1 DLH Register.....	35
table 7-1 SPI Register Mapping.....	74
table 8-1 SDA_HOLD Register.....	118
table 8-2 TX_ABRT_SOURCE Register.....	118
table 8-3 SLV_DATA_NACK_ONLY Register	123
table 8-4 DMA_CR Register	123
table 8-5 DMA_TDLR Register	124
table 8-6 DMA_RDLR Register	124
table 8-7 SDA_SETUP Register	124
table 8-8 FS_SPKLEN Register.....	125
table 8-9 HS_SPKLEN Register.....	125
table 8-10 CLR_RESTART_DET Register	126
table 8-11 SCL_STUCK_AT_LOW_TIMEOUT Register.....	126
table 8-12 SDA_STUCK_AT_LOW_TIMEOUT Register.....	127
table 8-13 CLR_SCL_STUCK_DET Register	127
table 9-1 I2S Register Mapping.....	136
table 9-2 CONTROL Register.....	136
table 9-3 CLK_DIV Register	138
table 9-4 WS_DIV Register	138
table 9-5 DATA Register.....	138
table 9-6 STATUS Register	138
table 9-7 INTE Register	139
table 9-8 FIFO_CFG Register.....	140
table 9-9 DMA_CFG Register.....	140
table 10-1 TRNG Register Mapping	142
table 10-2 RNG_IMR Register	142
table 10-3 RNG_ISR Register.....	143

table 10-4 RNG_ICR Register	143
table 10-5 TRNG_CONFIG Register.....	144
table 10-6 TRNG_VALID Register	144
table 10-7 HER_DATA0~5 Register	144
table 10-8 RND_SOURCE_ENABLE Register	145
table 10-9 SAMPLE_CNT1 Register	145
table 10-10 TRNG_RESET Register	145
table 10-11 TRNG_BUSY Register	146
table 10-12 RST_BITS_COUNTER Register	146
table 11-1 PDM Register Mapping	148
table 11-2 PDM_Control	148
table 11-3 VOLUME Register	149
table 11-4 VOLSTEP Register	149
table 11-5 RX_DATA Register	150
table 11-6 FIFO_Clear Register	150
table 11-7 FIFO_Status Register	151
table 11-8 FIFO_INT_EN Register	151
table 11-9 FIFO_Almost_Level Register	151
table 11-10 DMA_CFG Register.....	152
table 12-1 DMA Register Mapping.....	158
table 12-2 SARx Register.....	158
table 12-3 DARx Register.....	159
table 12-4 LLPx Register	159
table 12-5 CTLx Register	159
table 12-6 CFGx Register.....	162
table 12-7 RawTfr Register	166
table 12-8 RawErr Register	166
table 12-9 StatusTfr Register.....	166
table 12-10 StatusErr Register	167

table 12-11 MaskTfr Register.....	167
table 12-12 MaskErr Register.....	168
table 12-13 ClearTfr Register.....	168
table 12-14 ClearErr Register.....	169
table 12-15 StatusInt Register.....	169
table 12-16 DmaCfgRegRegister.....	170
table 12-17 ChEnReg Register.....	170
table 13-1 PWM Register Mapping.....	174
table 13-2 PWM_Output_EN Register.....	175
table 13-3 PWM_ENABLE Register.....	176
table 13-4 PWM0~7_Posedge Register.....	176
table 13-5 PWM0~7_Negedge Register.....	176
table 13-6 PWM_Output_Select Register.....	177
table 13-7 Capture_Prescale Register.....	177
table 13-8 Capture_Status Register.....	178
table 13-9 Capture_INT_EN Register.....	178
table 13-10 Capture_CTL Register.....	178
table 13-11 Capture_Value0~7 Register.....	179
table 13-12 PWM_Update Register.....	179
table 13-13 PWM_DAC_Mode Register.....	180
table 13-14 PWM_Output_Value Register.....	180
table 13-15 PWM_Inverter_EN Register.....	180
table 13-16 PWM_CNT_EN Register.....	181
table 13-17 PWM0~7_Prescale Register.....	181
table 13-18 PWM0~7_Period Register.....	181
table 13-19 PWM0~7_DAC_DIV Register.....	182
table 14-1 PMU_IWDG Register Mapping.....	187
table 14-2 IWDG_CTRL Register.....	187
table 14-3 IWDG_LEN Register.....	187

table 14-4 IWDG_TOUT_LEN Register	187
table 14-5 PMU_GPIO Register Mapping	188
table 14-6 PortA/B/C/D/E_SEL Register	189
table 14-7 PortA/B/C/D/E_PE Register.....	189
table 14-8 PortA/B/C/D/E_PSRegister	190
table 14-9 PortA/B/C/D Monitor EN Register	190
table 14-10 PortE Monitor EN Register.....	190
table 14-11 Analog Monitor_EN Register.....	191
table 14-12 PortA/B/C/D Reference Status Register	191
table 14-13 PortE Reference Status Register.....	192
table 14-14 Analog Reference Status Register	192
table 14-15 PortA/B/C/D/E_OEN Register.....	193
table 14-16 PortA/B/C/D/E_Data Register	193
table 14-17 PMU_RTC Register Mapping	194
table 14-18 RTC_CTRL Register.....	195
table 14-19 RTC_CNT Register	195
table 14-20 ALARM_A Register.....	196
table 14-21 ALARM_B Register	196
table 14-22 PMU_KeyScan Register Mapping	197
table 14-23 KeyScan_CTL Register	197
table 14-24 Scan_Interval Register.....	198
table 14-25 KeyScan_Config Register.....	198
table 14-26 ROW_SEL Register	199
table 14-27 COL_SEL Register	199
table 14-28 KEY_VAL Register.....	199
table 14-29 PMU_PWM Register Mapping	201
table 14-30 PWM0~3_CTRL Register	201
table 14-31 PWM0~2_PRC Register.....	202
table 14-32 PWM0~2_HRC Register.....	202

table 14-33 PMU_IOMux Register Mapping.....	203
table 14-34 PortA IOMux Register.....	203
table 14-35 PortB IOMux Register	204
table 14-36 PortC IOMux Register.....	204
table 14-37 PortD IOMux Register.....	205
table 14-38 PortE IOMux Register	205
table 14-39 PMU_INT_CTRL Register Mapping.....	206
table 14-40 BAT_Full_Filter Register.....	206
table 14-41 ACOK_Filter Register.....	206
table 14-42 LVD_Filter Register	207
table 14-43 OTD_Filter Register.....	207
table 14-44 INT_EN0 Register	208
table 14-45 INT_EN1 Register	209
table 14-46 INT_CLR0 Register.....	210
table 14-47 INT_CLR1 Register.....	210
table 14-48 INT_STS0 Register	211
table 14-49 INT_STS1 Register	211
table 14-50 PortA/B/C/D_Monitor_Status Register.....	211
table 14-51 PortE_Monitor_Status Register.....	212
table 15-1 LCD Register Mapping	214
table 15-2 LCD_TXFF_PUSH Register	214
table 15-3 LCD_CFG Register	215
table 15-4 LCD_CFGWR_STATUS Register	215
table 15-5 LCD_CFGRD_STATUS Register	215
table 15-6 LCD_APB_RD Register.....	216
table 15-7 LCD_TXFF_AEMP_LV Register	216
table 15-8 LCD_DAT_WR_LEN Register	216
table 15-9 LCD_DAT_WR_CFG Register	217
table 15-10 LCD_DMA Register.....	218

table 15-11 LCD_CSX Register	218
table 15-12 LCD_CRM Register	219
table 15-13 LCD_TXFF_CLR Register	219
table 15-14 LCD_INTF_CFG Register	220
table 15-15 LCD_INT Register	220
table 15-16 LCD_TE_VSYNC_CFG Register	221
table 15-17 LCD_TE_HSYNC_CFG Register	222
table 16-1 Timer Register Mapping	226
table 16-2 LOAD_VALUE Register	226
table 16-3 COUNT_VALUE Register	226
table 16-4 CONTROL Register	226
table 16-5 INT_CLR Register	227
table 17-1 ADC Register Mapping	229
table 17-2 CONTROL Register	230
table 17-3 DATA0~DATA7 Register	232
table 17-4 DATA Register	232
table 17-5 FIFO_Status Register	232
table 17-6 INT Register	233
table 17-7 INTC Register	234
table 17-8 ADC_ANA_CTRL0 Register	234
table 17-9 ADC_ANA_CTRL1 Register	235
table 17-10 ADC_ANA_CTRL2 Register	236
table 18-1 IntrTx1 Register	243
table 18-2 IntrTx2 Register	244
table 18-3 IntrRx1 Register	244
table 18-4 IntrRx2 Register	244
table 18-5 IntrUSB Register	245
table 18-6 IntrTx1E Register	245
table 18-7 IntrTx2E Register	246

table 18-8 IntrRx1E Register	246
table 18-9 IntrRx2E Register	246
table 18-10 IntrUSB E Register	246
table 18-11 Frame1 Register.....	247
table 18-12 Frame2 Register.....	247
table 18-13 Index Register.....	248
table 18-14 DevCtl Register	248
table 18-15 Device Mode CSR0 Register	249
table 18-16 Host Mode CSR0 Register	250
table 18-17 CSR2 Register	251
table 18-18 Count0 Register.....	252
table 18-19 NAKLimit0 Register	252
table 18-20 TxMaxP Register	253
table 18-21 Device Mode TxCSR1 Register	253
table 18-22 Host Mode TxCSR1 Register.....	254
table 18-23 TxCSR2 Register.....	255
table 18-24 RxMaxP Register.....	256
table 18-25 Device Mode RxCSR1 Register	257
table 18-26 Host Mode RxCSR1 Register.....	258
table 18-27 RxCSR2 Register.....	259
table 18-28 RxCount1 Register	260
table 18-29 RxCount2 Register	260
table 18-30 TxType Register	260
table 18-31 TxInterval Register.....	261
table 18-32 RxType Register	261
table 18-33 RxInterval Register	262
table 18-34 TxFIFO1、TxFIFO2 Register	262
table 18-35 RxFIFO1、RxFIFO2 Register.....	263
table 18-36 USB Endpoint FIFO Register.....	264

table 19-1 PortA MUX..... 272

table 19-2 PortB MUX..... 273

table 19-3 PortC MUX 274

table 19-4 PortD MUX..... 275

table 19-5 PortE MUX..... 276

table 20-1 PMU PortA MUX..... 277

table 20-2 PMU PortB MUX..... 277

table 20-3 PMU PortC MUX 277

table 20-4 PMU PortD MUX..... 278

table 20-5 PMU PortE MUX..... 278

Figures

Figure 3-1 GPIO Block Diagram	17
Figure 4-1 EXTI Block Diagram	23
Figure 5-1 IrDA Block Diagram	31
Figure 5-2 IrDA Timing Diagram	31
Figure 6-1 SPI Master Wiring Schematics	46
Figure 6-2 Instruction and Address in Standard Mode	66
Figure 6-3 Instruction in Standard and Address in Enhanced SPI Format	67
Figure 6-4 Instruction and Address Transmitted in Enhanced SPI Format	67
Figure 6-5 Instruction only transfer in enhanced SPI Format	67
Figure 6-6 Instruction and Address in Standard Mode	68
Figure 6-7 Instruction in Standard and Address in Enhanced SPI Format	68
Figure 6-8 Instruction and Address Transmitted in Enhanced SPI Format	69
Figure 6-9 No instruction, no address, only data transmission	69
Figure 6-10 MOTOROLA SPI Timing Diagram	70
Figure 7-1 SPI Slave Wiring Schematics	71
Figure 7-2 MOTOROLA SPI timing Diagram	87
Figure 13-1 PWM_DAC timing diagram	183
Figure 13-2 Input Capture timing diagram	184
Figure 14-1 PMU_IWDG system diagram	186

1. System Overview

1.1. Overview

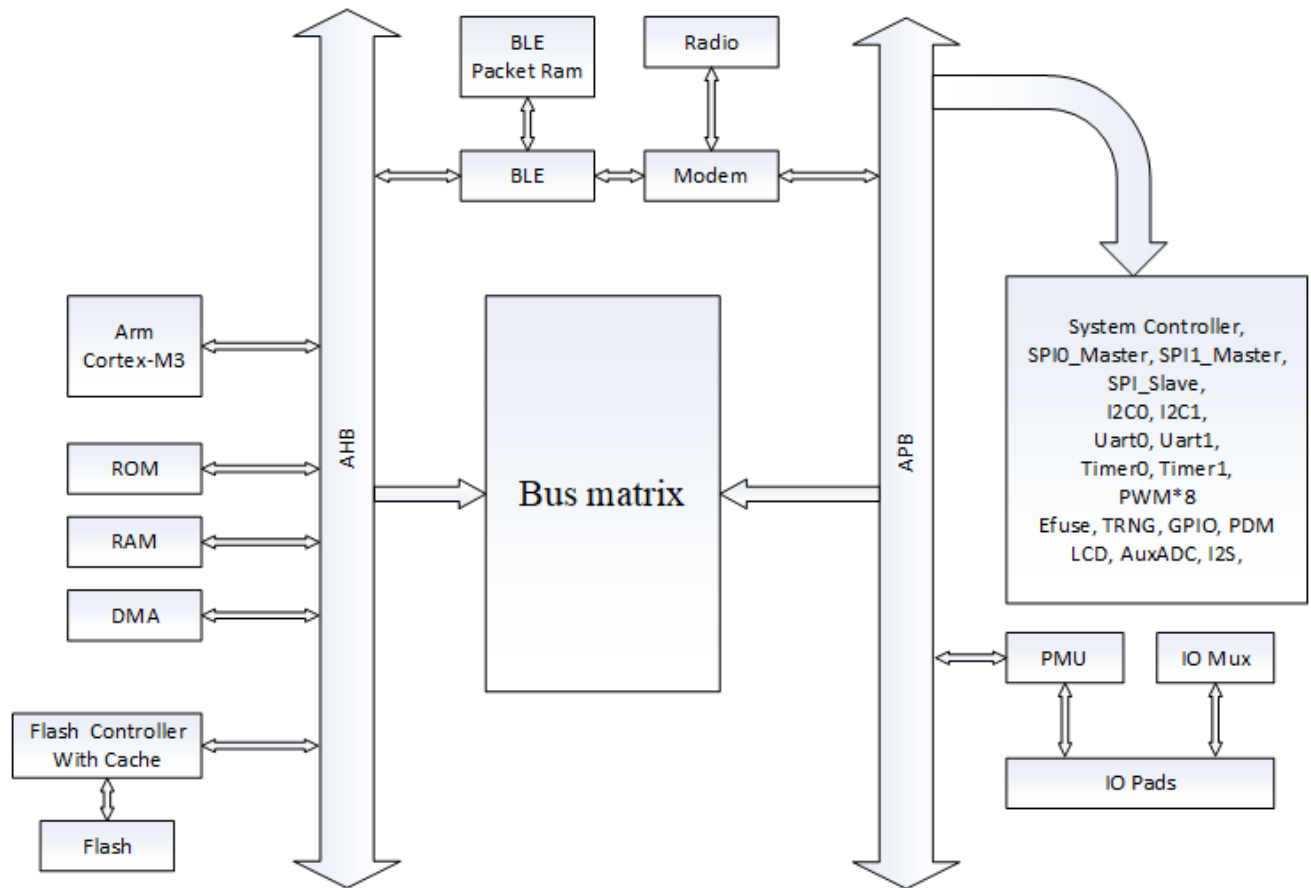


Figure 1-1 system block diagram

1.2. Feature

- RF Transceiver & Receiver
 - Bluetooth Radio integrated balun
 - External PA Support
 - TX Power -20dBm~10dBm
 - RX Sensitivity -96dBm@1Mbps
 - RX Sensitivity -93dBm@2Mbps
 - RX Sensitivity -101dBm@125Kbps

- RX Sensitivity -98dBm@500Kbps
- PMU
 - DC-DC Buck Inside
 - IO LDO
 - CORE/Analog LDO
 - Charger
 - Battery QS
 - Rcosc 32K
 - Power mode switch
 - BLE sniff mode clock & counter
- Baseband & Modem
 - Bluetooth5.0 BLE 1M/2M/125K/500K compatible
 - RSSI
 - RF rx & tx block dynamic controller
- CPU
 - Cortex-M3 for baseband & peripheral driver
 - 48MHz、96MHz Configurable frequency
- Memory
 - 128KB ROM
 - RAM
 - ◆ 8KB(BLE Packet RAM)
 - ◆ 32KB(System RAM0)+16KB(System RAM1)+8KB(System RAM2)
 - ◆ 4KB(Cache)
 - ◆ All the above memory block could be retention individual
- Logic Interface
 - ◆ Timer*2
 - ✧ System cpu/bus clock as reference clock
 - ✧ 32-bits counter with reload supports
 - ◆ Efuse 128bits

- ◆ SPIM*2
 - Master clock supports 96MHz(sclk up to 48MHz)
 - RX/TX FIFO configure 32 x 16
 - X2/X4 enhanced SPI mode
 - DMA hardware handshake
 - Transfer data SPIM0->SPIM1 bypass bus-matrix
- ◆ Flash2LCD Engine
- ◆ 8080 Parallel interfaces with FIFO 32*32 & DMA handshake
- ◆ SPIS
 - RX/TX FIFO configure -> 16 x 16
 - Only standard SPI access mode
 - DMA hardware handshake
- ◆ QSPI
 - STIG for normal SPI access with x1/x2/x4
 - DAC mode
 - Cacheable for XIP
- ◆ QSPI0
 - STIG for normal SPI access with x1/x2/x4
 - DAC mode
 - Programmable psram page boundary & refresh period
- ◆ I2C0
 - ✧ RX/TX FIFO depth 32x
 - ✧ Standard mode & High speed mode
 - ✧ DMA hardware handshake
- ◆ I2C1
 - ✧ RX/TX FIFO depth 8x
 - ✧ Only Standard mode
 - ✧ DMA hardware handshake
- ◆ UART0

- ✧ RX/TX FIFO depth 32x
- ✧ 9 bits mode
- ✧ Flow control
- ✧ IrDA SIR1.0
- ✧ RS485 interface support
- ✧ DMA hardware handshake
- ◆ UART1
 - ✧ RX/TX FIFO depth 16x
 - ✧ DMA hardware handshake
- ◆ GPIO*34(PA/PA/PC/PD/PE)
 - ✧ PA~PD x8, PEx2
 - ✧ External Interrupt supports
- ◆ PDM
 - ✧ Mono PDM with 16-bit word
 - ✧ RX FIFO depth 32x
 - ✧ 1M、1.5M、2M、3M clock supported
 - ✧ Configurable 62.5 or 125 down sample rate conversion
- ◆ TRNG
 - ✧ Produce 5K bits/second of entropy when core running at 96MHz
 - ✧ Built-in hardware tests for auto correlation and CRNGT
 - ✧ Random numbers are accessible to the applications in 6 of 32-bit read-only registers
- ◆ Internal FRSPIM
 - ✧ PMU access serial interface
 - ✧ Burst support @max length 4bytes
 - ✧ CSNA/CSNB for PMU PK/PD
- ◆ AUXADC(SARADC)
 - ✧ Max sample data rate @1Mbps
 - ✧ Loop mode for X8 input

- ✧ Fixed mode for FIFO
- ✧ Sample trigger comply with PWM pulse.

◆ PWMX8

- ✧ Full configurable for period & Duty cycle
- ✧ Timer based trigger to AUXADC

◆ GPIO Pads

- ✧ User usable x34
- ✧ Support Interrupts source x34
- ✧ Support SARADC*8
- ✧ UART/SPI/I2C/PDM/GPIO Mux output
- ✧ Pull up/down floating configurable

1.3. Memory Map

table 1-1 memory map

Start Address	Description	Bus
0x1000_0000 ~ 0x10FF_FFFF	Flash supports up to 16M bytes. (Different packages may have different storage sizes)	AHB
0x1100_0000 ~ 0x1100_DFFF	RAM. The 8KB at the bottom of the system RAM can be allocated to BLE Packet RAM.	
0x4000_4000 ~ 0x4000_5FFF	BLE Packet RAM. Package cache memory, with an exclusive 8KB. The 8KB from the system can be expanded as BLE Packet RAM.	
0x2200_0000 ~ 0x22FF_FFFF	PSRAM/Flash Extend. Expand RAM or Flash through QSPI1. After expansion, it can be directly accessed through the address.	
0x2002_0000	DMA	
0x2004_0000	USB	
0x5000_0000	System Controller	APB
0x5001_0000	Timer	
0x5002_0000	PMU	
0x5002_8000	Efuse	
0x5003_0000	SPI0_Master	
0x5003_4000	SPI1_Master	
0x5003_8000	SPI_Slave	
0x5004_0000	I2C0	
0x5004_8000	I2C1	
0x5005_0000	Uart0	
0x5005_8000	Uart1	
0x5006_0000	GPIO	
0x5007_0000	I2S	
0x5008_0000	PWM	
0x5009_0000	AuxADC	
0x500A_0000	PDM	
0x500C_0000	TRNG	
0x500D_0000	LCD	
0x500F_0000	QSPI0	

1.4. Interrupt map

table 1-2 Interrupt map

Interrupt Number	Interrupt Source
0	BLE_Int
1	DMA_Int
2	Timer0_Int
3	Timer1_Int
4	I2C0_Int
5	I2C1_Int
6	SPI0_Master_Int
7	SPI1_Master_Int
8	SPI_Slave_Int
9	Uart0_Int
10	Uart1_Int
11	PDM_Int
12	TRAN_Int
13	ADC_Int
14	GPIO_Int
15	PMU_Int
16	FC_Int
17	
18	PWM_Int
19	USB_Int
20	
21	LCD_Int
22	QSPI0_Int
23	I2S_Int

2. System Controller

2.1. Overview

System control unit. It provides management of the system clock, core clock, and peripheral clocks; provides reset control for each peripheral; and provides I/O port control

2.2. System Clock

2.2.1. Clock Source

table 2-1 Clock Source

Clock Name	Clock Frequency	Description
CLK_24M	24MHz	The clock generated by the external high-speed crystal
CLK_48M	48MHz	The clock generated by doubling the frequency of the external high-speed crystal internally.
CLK_96M	96MHz	The clock generated by quadrupling the frequency of the external high-speed crystal internally
CLK_16M	16MHz	The clock generated by first doubling the frequency of the external high-speed crystal internally and then dividing the resulting frequency by 3
CLK_32M	32MHz	The clock generated by first quadrupling the frequency of the external high-speed crystal internally and then dividing the resulting frequency by 3.
LPRC	64KHz	Internal low-speed RC clock.

2.2.2. System Clock Tree

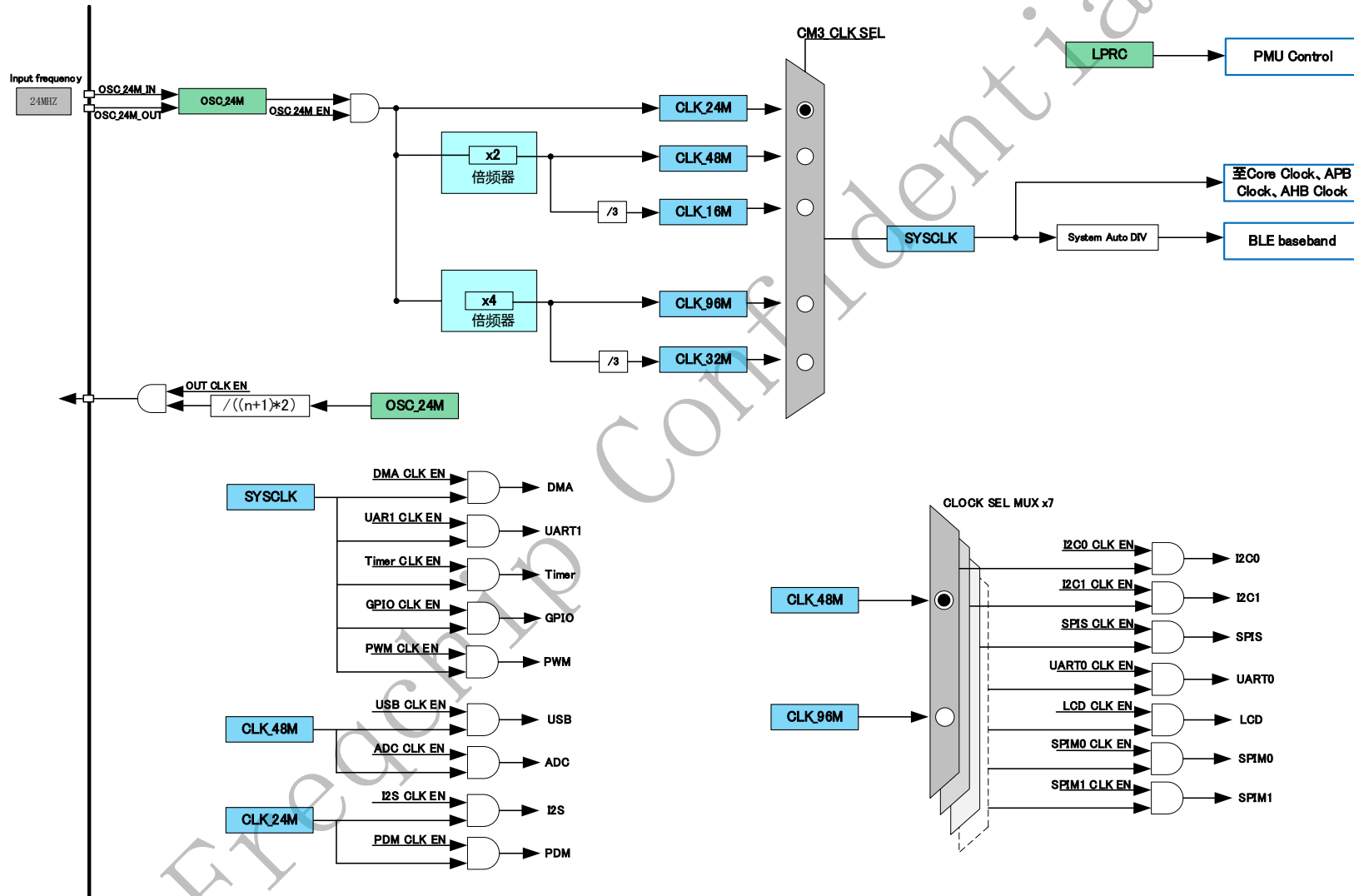


Figure 2-1 System clock tree

2.3. System Reset

table 2-2 System Reset Source

Reset source	Description
Chip Power on Reset(POR)	CPU core reset, all peripherals reset, PMU reset
External IO Reset	
IWDG Reset	
Software Debugging Reset	CPU core reset, all peripherals reset, and PMU does not reset

2.4. Low power consumption of the system

table 2-3 Low Power Consumption Mode

Mode	Mode Description	Entry Conditions	Exit conditions
SLEEP	CPU and peripheral interface module are powered down, and some memory modules are in retention mode; Chip IO supports external wake-up according to the selected configuration	The system application scheduling is idle, so it is no longer necessary to send or receive peripheral data, and no Bluetooth timeslot is sent or received	Wake up when the sleep timer reaches the count value; External interrupt triggers wake-up
SHUT-DOWN	CPU, peripheral interface module and all memories are in power down mode	No need for system operation and quick wake-up response	External interrupt wake-up or timer wake-up to trigger system startup

2.5. System Control Register

2.5.1. System Register Mapping

System Register Base Address: 0x5000_0000

table 2-4 System Register Mapping

Start Address	Register	Description
0x5000_0000	Clock Control	Clock Control
0x5000_0004	Clock Enable	Clock Enable
0x5000_0008	QSPI0 Control	MISC Control
0x5000_000C	Reset Control	Reset Control
0x5000_0010	APB Clock Enable	APB Bus Peripheral Clock Enable
0x5000_0014	DMA_REQ_MUX0	DMA Request Configuration 0
0x5000_0018	DMA_REQ_MUX1	DMA Request Configuration 1
0x5000_001C	DMA_REQ_MUX2	DMA Request Configuration 2
0x5000_0020	PortA Function MUX	PortA Function MUX
0x5000_0024	PortB Function MUX	PortB Function MUX
0x5000_0028	PortC Function MUX	PortC Function MUX
0x5000_002C	PortD Function MUX	PortD Function MUX
0x5000_0030	Portx Pull Enable	Portx Pull Up/Down Enable
0x5000_0034	Portx Pull Select	Portx Pull Up/Down Select
0x5000_0038	PortE_CFG	PortE_CFG
0x5000_003C	USB_OTG_CTRL	USB PHY Configurate

2.5.2. Clock Control (Offset 0x00)

table 2-5 Clock Control

Bit	Name	Access	Default	Description
32:24	Reserved	R/W	8'h08	Reserved
23:16	OUT_CLK_DIV	R/W	8'h0C	Output clock divider, only support even divider OUT_CLK = CLK_OSC/2*(OUT_CLK_DIV+1)
15:12	Reserved	R/W	4'h0	Reserved
11	Reserved	R/W	1'b0	Reserved
10	LCD_CLK_SEL	R/W	1'b0	LCD para block master clock select 0: 48MHZ 1: 96MHZ
9	UART0_CLK_SEL	R/W	1'b0	UART0(RICH_UART) serial clock select 0: 48MHZ 1: 96MHZ
8	SPI_Slave_CLK_SEL	R/W	1'b0	SPI Slave clock select

Bit	Name	Access	Default	Description
				0: 48MHZ 1: 96MHZ
7	SPI1_Master_CLK_SEL	R/W	1'b0	SPI1 Master serial clock select 0: 48MHZ 1: 96MHZ
6	SPI0_Master_CLK_SEL	R/W	1'b0	SPI0 Master serial clock select 0: 48MHZ 1: 96MHZ
5	I2C1_CLK_SEL	R/W	1'b0	I2C1 serial clock select 0: 48MHZ 1: 96MHZ
4	I2C0_CLK_SEL	R/W	1'b0	I2C0(RICH_I2C) serial clock select 0: 48MHZ 1: 96MHZ
3	Reserved	R/W	1'b0	Reserved
2:0	CM3_CLK_SEL	R/W	3'b001	CM3 & AHB/APB bus clock configure: 3'b000: 8MHz(No bt running supports) 3'b001: 16MHz 3'b010: 24MHz 3'b011: 32MHz 3'b100: 48MHz 3'b101: 96MHz

2.5.3. Clock Enable(Offset 0x04)

table 2-6 Clock Enable

Bit	Name	Access	Default	Description
31	USB12M_CLK_EN	R/W	1'b0	USB12M clock enable 0: Disable 1: Enable
30	USB48M_CLK_EN	R/W	1'b0	USB48M clock enable 0: Disable 1: Enable
29	USB_HCLK_EN	R/W	1'b0	USB clock enable 0: Disable 1: Enable
28	Reserved	R/W	1'b0	Reserved
27	Reserved	R/W	1'b0	Reserved
26	MDM_RXV1_CLK_EN	R/W	1'b0	Modem Rx VER1 clock enable 0: Disable 1: Enable

Bit	Name	Access	Default	Description
25	MDM_RXV0_CLK_EN	R/W	1'b0	Modem Rx VER0 clock enable 0: Disable 1: Enable
24	RF_CLK_EN	R/W	1'b0	FR clock always on enable 0: Disable 1: Enable
23	MDM_CLK_EN	R/W	1'b0	Modem clock always on enable 0: Disable 1: Enable
22	PMU_Interface_CLK_EN	R/W	1'b1	PMU clock enable 0: Disable 1: Enable
21	APB_CLK_EN	R/W	1'b1	APB bus clock enable 0: Disable 1: Enable
20	I2S_CLK_EN	R/W	1'b0	I2S clock enable 0: Disable 1: Enable
19	DMA_CLK_EN	R/W	1'b0	DMA clock enable 0: Disable 1: Enable
18	AHB_CLK_EN	R/W	1'b1	AHB bus clock enable 0: Disable 1: Enable
17	RAM_CLK_EN	R/W	1'b1	RAM block clock enable 0: Disable 1: Enable
16	ROM_CLK_EN	R/W	1'b1	ROM block clock enable 0: Disable 1: Enable
15	TRNG	R/W	1'b0	TRAN master clock enable 0: Disable 1: Enable
14	SPI_Slave_CLK_EN	R/W	1'b0	SPI Slave clock enable 0: Disable 1: Enable
13	SPI1_Master_CLK_EN	R/W	1'b0	SPI1 Master clock enable 0: Disable 1: Enable
12	SPI0_Master_CLK_EN	R/W	1'b0	SPI0 Master clock enable 0: Disable 1: Enable
11	I2C1_CLK_EN	R/W	1'b0	I2C1 clock enable 0: Disable

Bit	Name	Access	Default	Description
				1: Enable
10	I2C0_CLK_EN	R/W	1'b0	I2C0 clock enable 0: Disable 1: Enable
9	OUT_CLK_EN	R/W	1'b0	Output clock enable 0: Disable 1: Enable
8	PDM_CLK_EN	R/W	1'b0	PDM Master clock enable 0: Disable 1: Enable
7	UART0_CLK_EN	R/W	1'b1	Uart0 Master clock enable 0: Disable 1: Enable
6	AUXADC_CLK_EN	R/W	1'b0	AUXADC Master clock enable 0: Disable 1: Enable
5	GPIO_CLK_EN	R/W	1'b0	GPIO Master clock enable 0: Disable 1: Enable
4	EFUSE_Interface_CLK_EN	R/W	1'b0	EFUSE Master clock enable 0: Disable 1: Enable
3	BLE_EM_CLK_EN	R/W	1'b1	BLE packet memory clock enable 0: Disable 1: Enable
2	BLE_CLK_EN	R/W	1'b1	BLE hclk enable 0: Disable 1: Enable
1	LCD_CLK_EN	R/W	1'b0	LCDpara interface block clock enable 0: Disable 1: Enable
0	CM3_CLK_EN	R/W	1'b1	CM3 clock enable 0: Disable 1: Enable

Note: Clock enables all bits of the register. Write '0': Disable, write '1': Enable.

2.5.4. QSPI0 Control (Offset 0x08)

table 2-7 QSPI0 Control Register

Bit	Name	Access	Default	Description
31:28	QSPI0_SCLK_DLY_ADJ	R/W	4'h0	QSPI0 sclk delay adjust(TBD)

Bit	Name	Access	Default	Description
				(The Portx port pull-up and pull-down enable may cause the online signal to fail to reach the optimal state due to the influence of the user's peripheral hardware, which may lead to data sampling errors. At this time, adjust this value to achieve an accurate sampling value)
27:24	Reserved	R		The value configured by the manufacturer, which cannot be changed by the user
23	QSPI0_IO_CS_OEN	R/W	1'b1	QSPI0 CSN output enable @PORT_IO 0: output enable 1: output disable
22	QSPI0_IO_SCLK_OEN	R/W	1'b1	QSPI0 SCLK output enable @PORT_IO 0: output enable 1: output disable
21:20	Reserved	R		The value configured by the manufacturer, which cannot be changed by the user
19	QSPI0_HGCLK_EN	R/W	1'b0	QSPI0 hclk enable 0: Clock enable in non sleep state of CPU 1: always enable
18	Reserved	R		The value configured by the manufacturer, which cannot be changed by the user
17	QSPI0_REF_GLCK_EN	R/W	1'b0	QSPI0 reference clock enable 0: disable 1: enable
16:12	Reserved	R		The value configured by the manufacturer, which cannot be changed by the user
11:10	QSPI0_REF_CLK_SEL	R/W	2'b00	QSPI0 reference clock select: 00: 48MHz 01: 96MHz 10: 128MHz 11: 192MHz
9:8	Reserved	R		The value configured by the manufacturer, which cannot be changed by the user
7	QSPI0_REAMP	R/W	1'b0	QSPI0 remap enable 0: QSPI0 mapped to PortC 1: disable
6:0	Reserved	R		The value configured by the manufacturer, which cannot be changed by the user

2.5.5. Reset Control (Offset 0x0C)

table 2-8 Reset Control

Bit	Name	Access	Default	Description
31:27	Reserved	R/W	5'b0	Reserved
26	I2S_RST	R/W	1'b0	I2S soft reset, write 1 reset and then hardware clear to 0
25	QSPI0_REF_RST	R/W	1'b0	Qspi0 reference soft reset, write 1 reset and then hardware clear to 0
24	LCD_PARA_RST	R/W	1'b0	LCD para soft reset, write 1 reset and then hardware clear to 0
23	Reserved	R/W	1'b0	Reserved
22	Reserved	R/W	1'b0	Reserved
21	TRNG_RST	R/W	1'b0	TRNG soft reset, write 1 reset and then hardware clear to 0
20	PDM_RST	R/W	1'b0	PDM soft reset, write 1 reset and then hardware clear to 0
19	AUXADC_RST	R/W	1'b0	Aux-adc soft reset, write 1 reset and then hardware clear to 0
18	PWM_RST	R/W	1'b0	PWM reset, write 1 reset and then hardware clear to 0
17	UART1_RST	R/W	1'b0	Uart1 reset, write 1 reset and then hardware clear to 0
16	UART0_RST	R/W	1'b0	Uart0 reset, write 1 reset and then hardware clear to 0
15	GPIO_RST	R/W	1'b0	GPIO reset, write 1 reset and then hardware clear to 0
14	I2C1_RST	R/W	1'b0	I2C1 reset, write 1 reset and then hardware clear to 0
13	I2C0_RST	R/W	1'b0	I2C0 reset, write 1 reset and then hardware clear to 0
12	SPI_SLAVE_RST	R/W	1'b0	SPI Slave reset, write 1 reset and then hardware clear to 0
11	SPI1_Master_RST	R/W	1'b0	SPI1_Master reset, write 1 reset and then hardware clear to 0
10	SPI0_Master_RST	R/W	1'b0	SPI0_Master reset, write 1 reset and then hardware clear to 0

Bit	Name	Access	Default	Description
9	EFUSE_Interface_RST	R/W	1'b0	EFUSE reset, write 1 reset and then hardware clear to 0
8	PMU_Interface_RST	R/W	1'b0	PMU reset, write 1 reset and then hardware clear to 0s
7:0	Reserved	R/W	1'b0	Reserved

2.5.6. APB Clock Enable (Offset 0x10)

table 2-9 APB Clock Enable

Bit	Name	Access	Default	Description
31	LCD_APB_CLKEN	R/W	1'b0	LCD parallel block APB clock enable 0: disable 1: enable
30	Reserved	R/W	1'b0	Reserved
29	PDM_APB_CLKEN	R/W	1'b0	PDM block APB clock enable 0: disable 1: enable
28	AuxADC_APB_CLKEN	R/W	1'b0	AuxADC block APB clock enable 0: disable 1: enable
27	PWM_APB_CLKEN	R/W	1'b0	PWM block APB clock enable 0: disable 1: enable
26	GPIO_APB_CLKEN	R/W	1'b0	GPIO block APB clock enable 0: disable 1: enable
25	UART1_APB_CLKEN	R/W	1'b0	Uart1 block APB clock enable 0: disable 1: enable
24	UART0_APB_CLKEN	R/W	1'b1	Uart0 block APB clock enable 0: disable 1: enable
23	I2C1_APB_CLKEN	R/W	1'b0	I2C1 block APB clock enable

Bit	Name	Access	Default	Description
				0: disable 1: enable
22	I2C0_APB_CLKEN	R/W	1'b0	I2C0 block APB clock enable 0: disable 1: enable
21	SPI_Slave_APB_CLKEN	R/W	1'b0	SPI Slave block APB clock enable 0: disable 1: enable
20	SPI1_Master_APB_CLKEN	R/W	1'b0	SPI1 Master block APB clock enable 0: disable 1: enable
19	SPI0_Master_APB_CLKEN	R/W	1'b0	SPI0 Master block APB clock enable 0: disable 1: enable
18	EFUSE_Interface_APB_CLKEN	R/W	1'b0	Efuse block APB clock enable 0: disable 1: enable
17	PMU_Interface_APB_CLKEN	R/W	1'b1	PMU block APB clock enable 0: disable 1: enable
16	TIMER_APB_CLKEN	R/W	1'b0	Timer APB clock enable 0: disable 1: enable
15:8	GPIO_CLK_DIV	R/W	8'h08	GPIO master clock divider GPIO Clock = $APB_bus_clk / 2 * (GPIO_CLK_DIV + 1)$ The clock after frequency division is used to detect GPIO input signals, which can detect pin signals more quickly. When GPIO input is interrupted, the real-time performance can be further improved.
7:5	Reserved	R/W	3'b0	Reserved
4	I2S_APB_CLKEN	R/W	1'b0	I2S block APB clock enable 0: disable

Bit	Name	Access	Default	Description
				1: enable
3	Reserved	R/W	1' b0	Reserved
2	Reserved	R/W	1' b0	Reserved
1	MEMORY_SEL	R/W	1'b0	8KB Shared memory address map 0: mapped to system memory 1: mapped to Bluetooth packet memory
0	Reserved	R/W	1' b0	Reserved

2.5.7. DMA_REQ_MUX0(Offset 0x14)

table 2-10 DMA_REQ_MUX0

Bit	Name	Access	Default	Description
31:28	SPI1_Master_TX_DMA_REQ	R/W	4'b0	SPI1_Master_Tx DMA Request Request Numer: 0x0 ~ 0xF
27:24	SPI1_Master_RX_DMA_REQ	R/W	4'b0	SPI1_Master_Rx DMA Request Request Numer: 0x0 ~ 0xF
23:20	SPI0_Master_TX_DMA_REQ	R/W	4'b0	SPI0_Master_Tx DMA Request Request Numer: 0x0 ~ 0xF
19:16	SPI0_Master_RX_DMA_REQ	R/W	4'b0	SPI0_Master_Rx DMA Request Request Numer: 0x0 ~ 0xF
15:12	UART1_TX_DMA_REQ	R/W	4'b0	Uart1_Tx DMA Request Request Numer: 0x0 ~ 0xF
11:8	UART1_RX_DMA_REQ	R/W	4'b0	Uart1_Rx DMA Request Request Numer: 0x0 ~ 0xF
7:4	UART0_TX_DMA_REQ	R/W	4'b0	Uart0_Tx DMA Request Request Numer: 0x0 ~ 0xF
3:0	UART0_RX_DMA_REQ	R/W	4'b0	Uart0_Rx DMA Request Request Numer: 0x0 ~ 0xF

2.5.8. DMA_REQ_MUX1(Offset 0x18)

table 2-11 DMA_REQ_MUX1

Bit	Name	Access	Default	Description
31:28	PDM_Rx_DMA_REQ	R/W	4'b0	PDM Digital Microphone Input DMA Request Request Numer: 0x0 ~ 0xF
27:24	AuxADC_RX_DMA_REQ	R/W	4'b0	Aux ADC RX DMA Request Request Numer: 0x0 ~ 0xF
23:20	SPI_Slave_TX_DMA_REQ	R/W	4'b0	SPI1_Slave_Tx DMA Request Request Numer: 0x0 ~ 0xF
19:16	SPI_Slave_RX_DMA_REQ	R/W	4'b0	SPI1_Slave_Rx DMA Request Request Numer: 0x0 ~ 0xF
15:12	I2C1_TX_DMA_REQ	R/W	4'b0	I2C1_Tx DMA Request Request Numer: 0x0 ~ 0xF
11:8	I2C1_RX_DMA_REQ	R/W	4'b0	I2C1_Rx DMA Request Request Numer: 0x0 ~ 0xF
7:4	I2C0_TX_DMA_REQ	R/W	4'b0	I2C0_Tx DMA Request Request Numer: 0x0 ~ 0xF
3:0	I2C0_RX_DMA_REQ	R/W	4'b0	I2C0_Rx DMA Request Request Numer: 0x0 ~ 0xF

2.5.9. DMA_REQ_MUX2 (Offset 0x1C)

table 2-12 DMA_REQ_MUX2

Bit	Name	Access	Default	Description
31:16	Reserved	R/W	4'b0	Reserved
15:12	I2S_TX_DMA_REQ	R/W	4'b0	I2S_TX DMA Request Request Numer: 0x0 ~ 0xF
11:8	I2S_RX_DMA_REQ	R/W	4'b0	I2S_RX DMA Request Request Numer: 0x0 ~ 0xF
7:4	SPI1_Master_CMD_DMA_REQ	R/W	4'b0	SPI1_Master_CMD DMA Request Request Numer: 0x0 ~ 0xF
3:0	SPI0_Master_CMD_DMA_REQ	R/W	4'b0	SPI0_Master_CMD DMA Request

Bit	Name	Access	Default	Description
				Request Numer: 0x0 ~ 0xF

2.5.10. PortA Function MUX (Offset 0x20)

table 2-13 PortA Function MUX

Bit	Name	Access	Default	Description
31:28	PA7_MUX	R/W	4'h0	PA7 MUX. value: 0x0 ~ 0xF
27:24	PA6_MUX	R/W	4'h0	PA6 MUX. value: 0x0 ~ 0xF
23:20	PA5_MUX	R/W	4'h0	PA5 MUX. value: 0x0 ~ 0xF
19:16	PA4_MUX	R/W	4'h0	PA4 MUX. value: 0x0 ~ 0xF
15:12	PA3_MUX	R/W	4'h0	PA3 MUX. value: 0x0 ~ 0xF
11:8	PA2_MUX	R/W	4'h0	PA2 MUX. value: 0x0 ~ 0xF
7:4	PA1_MUX	R/W	4'h0	PA1 MUX. value: 0x0 ~ 0xF
3:0	PA0_MUX	R/W	4'h0	PA0 MUX. value: 0x0 ~ 0xF

For more information, please refer to [Appendix I](#).

2.5.11. PortB Function MUX (Offset 0x24)

table 2-14 PortB Function MUX

Bit	Name	Access	Default	Description
31:28	PB7_MUX	R/W	4'h0	PB7 MUX. value: 0x0 ~ 0xF
27:24	PB6_MUX	R/W	4'h0	PB6 MUX. value: 0x0 ~ 0xF
23:20	PB5_MUX	R/W	4'h0	PB5 MUX. value: 0x0 ~ 0xF
19:16	PB4_MUX	R/W	4'h0	PB4 MUX. value: 0x0 ~ 0xF
15:12	PB3_MUX	R/W	4'h0	PB3 MUX. value: 0x0 ~ 0xF
11:8	PB2_MUX	R/W	4'h0	PB2 MUX. value: 0x0 ~ 0xF
7:4	PB1_MUX	R/W	4'h0	PB1 MUX. value: 0x0 ~ 0xF
3:0	PB0_MUX	R/W	4'h0	PB0 MUX. value: 0x0 ~ 0xF

For more information, please refer to [Appendix I](#).

2.5.12. PortC Function MUX (Offset 0x28)

table 2-15 PortC Function MUX

Bit	Name	Access	Default	Description
31:28	PC7_MUX	R/W	4'h8	Default as CM3_SWDIO
27:24	PC6_MUX	R/W	4'h8	Default as CM3_SWCLK
23:20	PC5_MUX	R/W	4'h0	PC5 MUX. value: 0x0 ~ 0xF
19:16	PC4_MUX	R/W	4'h0	PC4 MUX. value: 0x0 ~ 0xF
15:12	PC3_MUX	R/W	4'h0	PC3 MUX. value: 0x0 ~ 0xF
11:8	PC2_MUX	R/W	4'h0	PC2 MUX. value: 0x0 ~ 0xF
7:4	PC1_MUX	R/W	4'h0	PC1 MUX. value: 0x0 ~ 0xF
3:0	PC0_MUX	R/W	4'h0	PC0 MUX. value: 0x0 ~ 0xF

For more information, please refer to [Appendix I](#).

2.5.13. PortD Function MUX (Offset 0x2C)

table 2-16 PortD Function MUX

Bit	Name	Access	Default	Description
31:28	PD7_MUX	R/W	4'h0	PD7 MUX. value: 0x0 ~ 0xF
27:24	PD6_MUX	R/W	4'h0	PD6 MUX. value: 0x0 ~ 0xF
23:20	PD5_MUX	R/W	4'h0	PD5 MUX. value: 0x0 ~ 0xF
19:16	PD4_MUX	R/W	4'h0	PD4 MUX. value: 0x0 ~ 0xF
15:12	PD3_MUX	R/W	4'h0	PD3 MUX. value: 0x0 ~ 0xF
11:8	PD2_MUX	R/W	4'h0	PD2 MUX. value: 0x0 ~ 0xF
7:4	PD1_MUX	R/W	4'h0	PD1 MUX. value: 0x0 ~ 0xF
3:0	PD0_MUX	R/W	4'h0	PD0 MUX. value: 0x0 ~ 0xF

For more information, please refer to [Appendix I](#).

2.5.14. Portx Pull Enable (Offset 0x30)

table 2-17 Portx Pull Enable

Bit	Name	Access	Default	Description
31:24	PD_PE	R/W	8'hFF	Port D pull enable. 1 bit corresponds to 1 pin. 0: disable

Bit	Name	Access	Default	Description
				1: enable
23:16	PC_PE	R/W	8'hFF	Port C pull enable. 1 bit corresponds to 1 pin. 0: disable 1: enable
15:8	PB_PE	R/W	8'hFF	Port B pull enable. 1 bit corresponds to 1 pin. 0: disable 1: enable
7:0	PA_PE	R/W	8'hFF	Port A pull enable. 1 bit corresponds to 1 pin. 0: disable 1: enable

2.5.15. Portx Pull Select (Offset 0x34)

table 2-18 Portx Pull Select Register

Bit	Name	Access	Default	Description
31:24	PD_PS	R/W	8'h00	Port D pull select. 1 bit corresponds to 1 pin. 0: pull down 1: pull up
23:16	PC_PS	R/W	8'h00	Port C pull select. 1 bit corresponds to 1 pin. 0: pull down 1: pull up
15:8	PB_PS	R/W	8'h00	Port B pull select. 1 bit corresponds to 1 pin. 0: pull down 1: pull up
7:0	PA_PS	R/W	8'h00	Port A pull select. 1 bit corresponds to 1 pin. 0: pull down 1: pull up

2.5.16. PortE_CFG (Offset 0x38)

table 2-19 PortE_CFG Register

Bit	Name	Access	Default	Description
31:28	PE1_MUX	R/W	4'h00	PE1 function Mux. value: 0x0 ~ 0xF
27:24	PE0_MUX	R/W	4'h00	PE0 function Mux. value: 0x0 ~ 0xF
23:22	PE_PS	R/W	2'b00	Port E pull select. 1 bit corresponds to 1 pin. 0: pull down 1: pull up
21:20	PE_PE	R/W	2'b00	Port E pull enable. 1 bit corresponds to 1 pin. 0: disable 1: enable
19:14	Reserved	R	6'h00	Reserved
13:8	QSPI_PS	R/W	6'h3F	QSPI pull select
7:6	Reserved	R	2'h00	Reserved
5:0	QSPI_PE	R/W	6'h3F	QSPI pull enable

2.5.17. USB_OTG_CTRL (Offset 0x3C)

table 2-20 USB_OTG_CTRL Register

Bit	Name	Access	Default	Description
31:21	Reserved	R/W	11'h0	Reserved
20	USB_UART_SEL	R/W	1'b0	USB PHY select 0: uart0 1: uart1
19	USB_IO_PE1	R/W	1'b1	USB_DM IO pull enable 0: disable 1: enable
18	USB_IO_PE0	R/W	1'b1	USB_DP IO pull enable 0: disable 1: enable
17	USB_IO_PS1	R/W	1'b0	USB_DM IO pull select 0: pull down 1: pull up

Bit	Name	Access	Default	Description
16	USB_IO_PS0	R/W	1'b1	USB_DPO pull select 0: pull down 1: pull up
15	USB_STDBY_REG	R/W	1'b0	USB standby configure value 0: standby disable 1: standby enable
14	USB_STDBY_SEL	R/W	1'b0	USB standby select 0: from USB OTG Controller 1: from static register(software, this register bit[15])
13	USB_UART_EN	R/W	1'b0	USB PHY as Uart rx/tx 0: USB PAD not as UART PAD 1: USB PAD as UART PAD
12	USB_PHY_SEL	R/W	1'b0	USB PHY select 0: normal PHY 1: GPIO pads
11	OTG_CID_I	R/W	1'b0	OTG_CID_I behave from software
10	OTG_VBUSLO_I	R/W	1'b0	Analog VBUSLO behave from software
9	OTG_VBUS_SES_I	R/W	1'b0	VBUS_SES behave from software
8	OTG_VBUS_VAL_I	R/W	1'b0	Analog VBUS_VAL behave from software
7:0	USB_PHY_DAP_CFG	R/W	8'h09	USB PHY adapter configure

3. GPIO

3.1. Overview

It integrates the input and output functions of general digital IO, and also supports external interrupt detection processing (supports hardware anti shake) in the input mode.

GPIO has five groups of controllers, PortA, PortB, PortC, PortD, and PortE. Each GPIO controller is completely independent and does not share any resources with each other.

3.2. Feature

- Five groups of controllers
- Support input and output modes and internal anti shake

3.3. Block Diagram

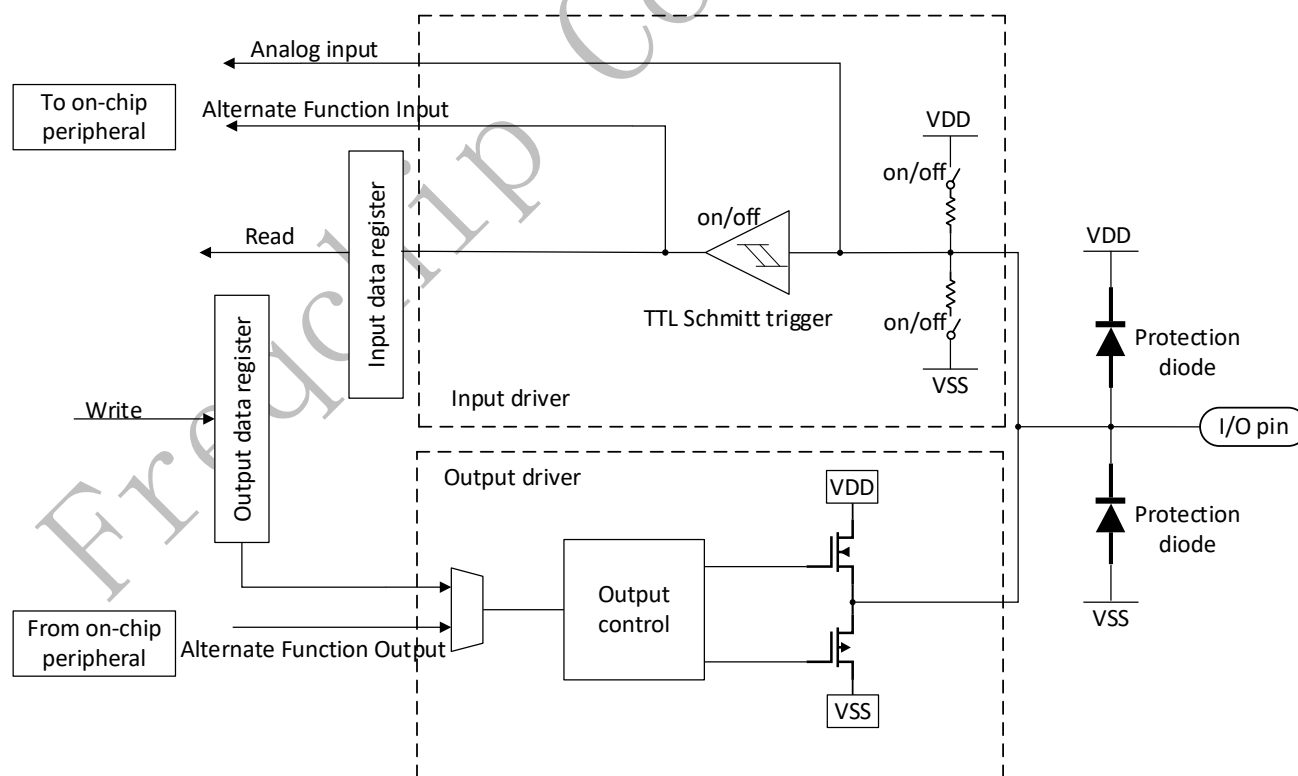


Figure 3-1 GPIO Block Diagram

3.4. GPIO Register

3.4.1. GPIO Register Address Mapping

GPIO Base Address: 0x5006_0000

table 3-1 GPIO Register Address Mapping

Offset	Register	Description
0x00	PortA_DATA	GPIO Port A Data Register
0x04	PortB_DATA	GPIO Port B Data Register
0x08	PortC_DATA	GPIO Port C Data Register
0x0C	PortD_DATA	GPIO Port D Data Register
0x10	PortE_DATA	GPIO Port E Data Register
0x20	PortA_OEN	GPIO Port A Output Enable Register
0x24	PortB_OEN	GPIO Port B Output Enable Register
0x28	PortC_OEN	GPIO Port C Output Enable Register
0x2C	PortD_OEN	GPIO Port D Output Enable Register
0x30	PortE_OEN	GPIO Port E Output Enable Register
0x40	Port_EXTI_EN0	PortA~PortD EXTI Interrupt enable
0x44	Port_EXTI_EN1	PortE EXTI Interrupt enable

3.4.2. PortA_DATA (Offset 0x00)

table 3-2 PortA DATA Register

Bit	Name	Access	Default	Description
7:0	PA_DATA	R/W	8'h00	Port A data register 1 bit corresponds to 1 pin. Write '1' output high, write '0' output low, read as current pin state

3.4.3. PortB_DATA (Offset 0x04)

table 3-3 PortB DATA Register

Bit	Name	Access	Default	Description
7:0	PB_DATA	R/W	8'h00	Port B data register 1 bit corresponds to 1 pin. Write '1' output high, write '0' output low, read as current pin state

3.4.4. PortC_DATA (Offset 0x08)

table 3-4 PortC DATA Register

Bit	Name	Access	Default	Description
7:0	PC_DATA	R/W	8'h00	Port C data register 1 bit corresponds to 1 pin. Write '1' output high, write '0' output low, read as current pin state

3.4.5. PortD_DATA (Offset 0x0C)

table 3-5 PortB DATA Register

Bit	Name	Access	Default	Description
7:0	PD_DATA	R/W	8'h00	Port D data register 1 bit corresponds to 1 pin. Write '1' output high, write '0' output low, read as current pin state

3.4.6. PortE_DATA(Offset 0x10)

table 3-6 PortE DATA Register

Bit	Name	Access	Default	Description
7:0	PE_DATA	R/W	8'h00	Port E data register

Bit	Name	Access	Default	Description
				1 bit corresponds to 1 pin. Write '1' output high, write '0' output low, read as current pin state

3.4.7. PortA_OEN (Offset 0x20)

table 3-7 PortA OEN Register

Bit	Name	Access	Default	Description
7:0	PA_OEN	R/W	8'h00	PortA Output enable register 1 bit corresponds to 1 pin. 1: disable 0: enable

3.4.8. PortB_OEN (Offset 0x24)

table 3-8 PortB OEN Register

Bit	Name	Access	Default	Description
7:0	PB_OEN	R/W	8'h00	PortB Output enable register 1 bit corresponds to 1 pin. 1: disable 0: enable

3.4.9. PortC_OEN (Offset 0x28)

table 3-9 PortC OEN Register

Bit	Name	Access	Default	Description
7:0	PC_OEN	R/W	8'h00	PortC Output enable register 1 bit corresponds to 1 pin. 1: disable 0: enable

3.4.10. PortD_OEN(Offset 0x2C)

table 3-10 PortD OEN Register

Bit	Name	Access	Default	Description
7:0	PD_OEN	R/W	8'h00	PortD Output enable register 1 bit corresponds to 1 pin. 1: disable 0: enable

3.4.11. PortE_OEN (Offset 0x30)

table 3-11 PortE OEN Register

Bit	Name	Access	Default	Description
7:0	PE_OEN	R/W	8'h00	PortE Output enable register 1 bit corresponds to 1 pin. 1: Disable 0: Enable

3.4.12. Port_EXTI_EN0(Offset 0x40)

table 3-12 Port_EXTI_EN0 Register

Bit	Name	Access	Default	Description
31:24	PortD_EXTI_EN	R/W	8'h00	PortD EXTINT enable 1 bit corresponds to 1 pin. 0: Disable。 1: Enable。
23:16	PortC_EXTI_EN	R/W	8'h00	PortC EXTINT enable 1 bit corresponds to 1 pin. 0: Disable。 1: Enable。
15:8	PortB_EXTI_EN	R/W	8'h00	PortB EXTINT enable 1 bit corresponds to 1 pin. 0: Disable。

Bit	Name	Access	Default	Description
				1: Enable。
7:0	PortA_EXTI_EN	R/W	8'h00	PortA EXTINT enable 1 bit corresponds to 1 pin. 0: Disable。 1: Enable。

3.4.13. Port_EXTI_EN1(Offset 0x44)

table 3-13 Port_EXTI_EN1 Register

Bit	Name	Access	Default	Description
31:4	Reserved	R/W	28'h00	Reserved
3:0	PortE_EXTI_EN	R/W	4'h0	PortE EXTINT enable 1 bit corresponds to 1 pin. 0: Disable。 1: Enable。

4. External Interrupt Controller (EXTI)

4.1. Overview

EXTI supports 36 independent detection circuits and can generate interrupt requests to the processor. Each external interrupt controller is completely independent and does not share any resources with each other.

EXTI provides four trigger types: high level trigger, low level trigger, posedge trigger and negedge trigger. Each detection circuit can be configured separately.

4.2. Feature

- Up to 36 independent external interrupt controls
- Level trigger hardware anti shake

4.3. EXTI Block Diagram

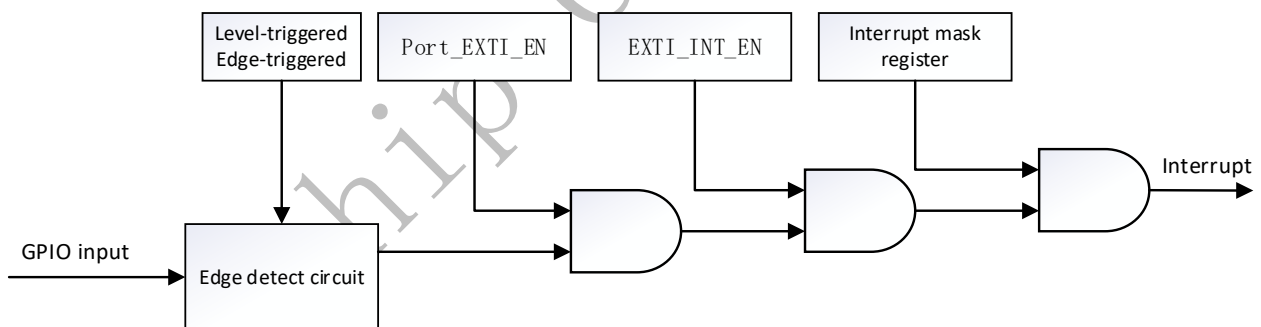


Figure 4-1 EXTI Block Diagram

4.4. EXTI Register

4.4.1. EXTI Register Mapping

EXTI Base Address: 0x5006_8000

table 4-1 EXTI Register Mapping

Offset	Register	Description
0x00	EXTI_CTL0	EXTI source enable0
0x04	EXTI_CTL1	EXTI source enable1
0x08	EXTI_STS0	EXTI status0
0x0C	EXTI_STS1	EXTI status1
0x10	EXTI_TYPE0	EXTI type0 set the external interrupt source PortX[n] trigger type: 00: low level trigger 01: high level trigger 10: negedge trigger 11: posedge trigger
0x14	EXTI_TYPE1	EXTI type1 set the external interrupt source PortX[n] trigger type: 00: low level trigger 01: high level trigger 10: negedge trigger 11: posedge trigger
0x18	EXTI_TYPE2	EXTI type2 set the external interrupt source PortX[n] trigger type: 00: low level trigger 01: high level trigger 10: negedge trigger 11: posedge trigger
0x20 ~ 0x3C	EXTI_CNT0 ~ EXTI_CNT7	Anti shake count value of EXTI0~ EXTI7
0x40 ~ 0x5C	EXTI_CNT8 ~ EXTI_CNT15	Anti shake count value of EXTI8~ EXTI15
0x60 ~ 0x7C	EXTI_CNT16 ~ EXTI_CNT23	Anti shake count value of EXTI16~ EXTI23
0x80 ~ 0x9C	EXTI_CNT24 ~ EXTI_CNT31	Anti shake count value of EXTI24~ EXTI31
0xA0 ~ 0xAC	EXTI_CNT32 ~ EXTI_CNT35	Anti shake count value of EXTI32~ EXTI35

4.4.2. EXTI_CTL0(Offset 0x00)

table 4-2 EXTI_CTL0 Register

Bit	Name	Access	Default	Description
31:0	EXTI_INT0_EN (EXTI request 0 ~ 31)	R/W	32'h00	EXTI(0~31): External interrupt source enable EXTI_INT0_EN[7:0]: PA7 ~ PA0。 EXTI_INT0_EN[15:8]: PB7 ~ PB0。 EXTI_INT0_EN[23:16]: PC7 ~ PC0。 EXTI_INT0_EN[31:24]: PD7 ~ PD0。 1 bit corresponds to 1 request 0: Disable 1: Enable

4.4.3. EXTI_CTL1(Offset 0x04)

table 4-3 EXTI_CTL1 Register

Bit	Name	Access	Default	Description
31:4	Reserved	R/W	28'h00	Reserved
3:0	EXTI_INT1_EN (EXTI request 32 ~ 35)	R/W	4'h0	EXTI(32~35): External interrupt source enable EXTI_INT1_EN[3:0]: PE3 ~ PE0。 1 bit corresponds to 1 request 0: Disable 1: Enable

4.4.4. EXTI_STS0(Offset 0x08)

table 4-4 EXTI_STS0 Register

Bit	Name	Access	Default	Description
31:0	EXTI_STS0	R/W	32'h00	EXTI(0~31) External interrupt status EXTI_STS0 [7:0]: PA7 ~ PA0。 EXTI_STS0 [15:8]: PB7 ~ PB0。 EXTI_STS0 [23:16]: PC7 ~ PC0。

Bit	Name	Access	Default	Description
				EXTI_STS0 [31:24] : PD7 ~ PD0。 1 bit corresponds to 1 request Write Operation: Write 1 to clear the set status Read Operation: Read the current interrupt status

4.4.5. EXTI_STS1(Offset 0x0C)

table 4-5 EXTI_STS1 Register

Bit	Name	Access	Default	Description
31:4	Reserved	R/W	24'h00	Reserved
3:0	EXTI_STS1	R/W	4'h0	EXTI(32~35)External interrupt status EXTI_STS1 [3:0]: PE3 ~ PE0 1 bit corresponds to 1 request Write Operation: Write 1 to clear the set status Read Operation: Read the current interrupt status

4.4.6. EXTI_TYPE0(Offset 0x10)

table 4-6 EXTI_TYPE0 Register

Bit	Name	Access	Default	Description
31:16	EXTI_PortB_TYPE	R/W	16'h00	PortB external interrupt source trigger type 2 bit corresponds to 1 pin EXTI_PortB_TYPE[1:0]: PB0 EXTI_PortB_TYPE[3:2]: PB1 Type: 00: low level trigger 01: high level trigger 10: negedge trigger 11: posedge trigger
15:0	EXTI_PortA_TYPE	R/W	16'h00	PortA external interrupt source trigger type 2 bit corresponds to 1 pin

Bit	Name	Access	Default	Description
				EXTI_PortA_TYPE[1:0]: PA0 EXTI_PortA_TYPE[3:2]: PA1 Type: 00: low level trigger 01: high level trigger 10: negedge trigger 11: posedge trigger

4.4.7. EXTI_TYPE1(Offset 0x14)

table 4-7 EXTI_TYPE1 Register

Bit	Name	Access	Default	Description
31:16	EXTI_PortD_TYPE	R/W	16'h00	PortD external interrupt source trigger type 2 bit corresponds to 1 pin EXTI_PortD_TYPE[1:0]: PD0 EXTI_PortD_TYPE[3:2]: PD1 Type: 00: low level trigger 01: high level trigger 10: negedge trigger 11: posedge trigger
15:0	EXTI_PortC_TYPE	R/W	16'h00	PortC external interrupt source trigger type 2 bit corresponds to 1 pin EXTI_PortC_TYPE[1:0]: PC0 EXTI_PortC_TYPE[3:2]: PC1 Type: 00: low level trigger 01: high level trigger 10: negedge trigger 11: posedge trigger

4.4.8. EXTI_TYPE2(Offset 0x18)

table 4-8 EXTI_TYPE2 Register

Bit	Name	Access	Default	Description
31:8	Reserved	R/W	24'h00	Reserved
7:0	EXTI_PortE_TYPE	R/W	8'h0	PortE external interrupt source trigger type 2 bit corresponds to 1 pin EXTI_PortE_TYPE[1:0]: PE0 EXTI_PortE_TYPE[3:2]: PE1 Type: 00: low level trigger 01: high level trigger 10: negedge trigger 11: posedge trigger

4.4.9. EXTI_CNT0 ~ EXTI_CNT35(Offset 0x20 ~ 0xAC)

table 4-9 EXTI_CNT0~EXTI_CNT35 Register

Bit	Name	Access	Default	Description
31:16	Reserved	R/W	16'h00	Reserved
15:8	PRE_VAL	R/W	8'h0	EXTI(0 ~ 35) Precaler Value $CLK = APB_CLK / (1 + PRE_VAL)$ 。
7:4	Reserved	R/W	4'h00	Reserved
3:0	DES_CNT	R/W	4'h0	Counter Value: If the external interrupt source 0 last longer than the value set in this register, the corresponding External interrupt status would set.(only used for level trigger mode)

5. UART

5.1. Overview

UART (Universal Asynchronous Receiver/Transmitter) provides a flexible method for full duplex data exchange with other peripherals that support the standard NRZ asynchronous serial data format.

5.2. Feature

- Uart0 FIFO depth 32, width 8
- Uart1 FIFO depth 16, width 8
- Uart0 supports IrDA SIR 1.0
- Programmable fractional baud rate support

5.3. Function description

5.3.1. Baud Rate Calculation

The programmable fractional baud clock divider allows for the programmability of both an integer divisor as well as fractional component. The average frequency of the baud clock from the fractional baud rate divisor is dependent upon both the integer divisor and the fractional component, thereby providing a finer resolution to the average frequency of the baud clock.

48MHz or 96MHz can be configured to Uart0, Uart1 only uses APB_CLK.

The baud rate generator divisor value, DIVISOR (composed of DLH & DLL registers).

DLL is set to the low 8 bits, and the DLH is set to the high 8 bits.

$$\text{Baud Rate Divisor} = \frac{\text{Serial Clock Frequency}}{16 \times \text{Required Baud Rate}} = \text{BRD}_I + \text{BRD}_F$$

BRD_I: Integer part divisor(DLH register、DLL register)

BRD_F: Fractional part of the divisor(DLF register)

5.3.2. IrDA SIR 1.0 Protocol

The Infrared Data Association (IrDA) 1.0 Serial Infrared (SIR) mode supports bi-directional data communications with remote devices using infrared radiation as the transmission medium. IrDA 1.0 SIR mode specifies a maximum baud rate of 115.2Kbaud.

The IrDA SIR physical layer specifies the use of Return to Zero Invert (RZI), which uses an infrared optical pulse to represent logic '0'. The SIR transmission encoder modulates the NRZ (non return to zero) bit stream output from the UART. The output pulse stream is transmitted to an external output driver and infrared LED. Only 115.2Kbps is supported. In normal mode, the pulse width is specified as 3/16 of a bit period.

The SIR receiving decoder demodulates the zeroing bit stream from the infrared receiver, and outputs the received NRZ serial bit stream to the UART. In the idle state, the decoder input is usually high. The polarity of the transmission encoder output is opposite to that of the decoder input. When the decoder input is low, a start bit is detected.

- SIR sending logic sends '0' as high pulse and '1' as low level
- SIR receiving logic interprets the high level status as '1' and the low pulse as '0'
- Transmission encoder output has opposite polarity to the decoder input. When idle, SIR output is in low state
- In IrDA mode, one stop must be configured

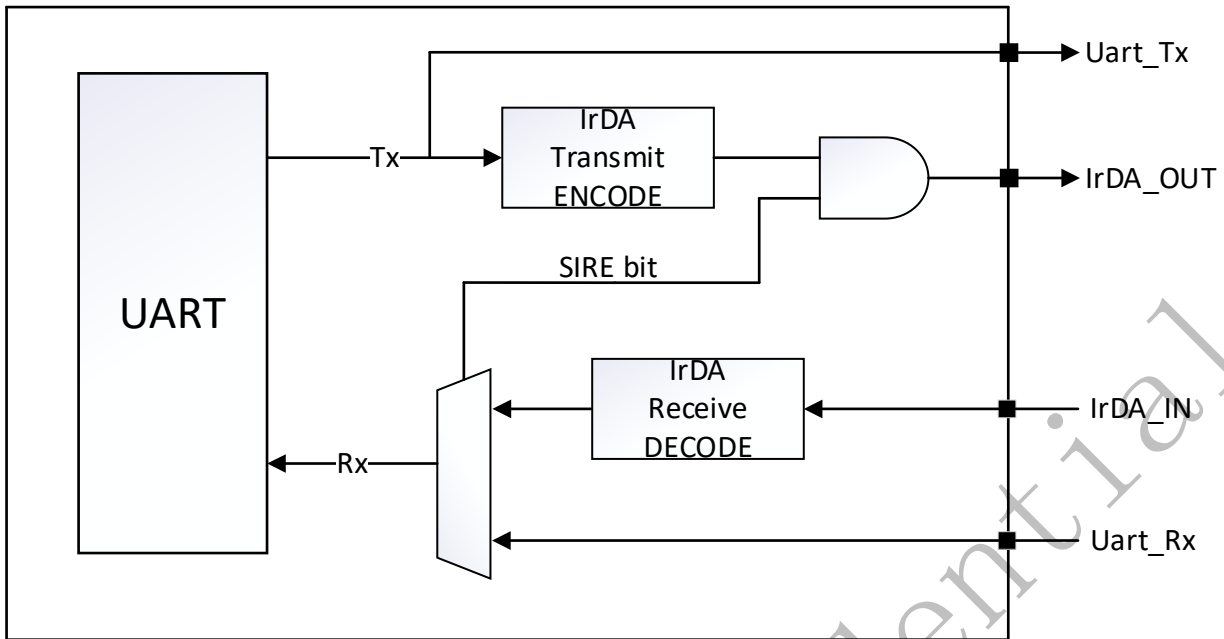


Figure 5-1 IrDA Block Diagram

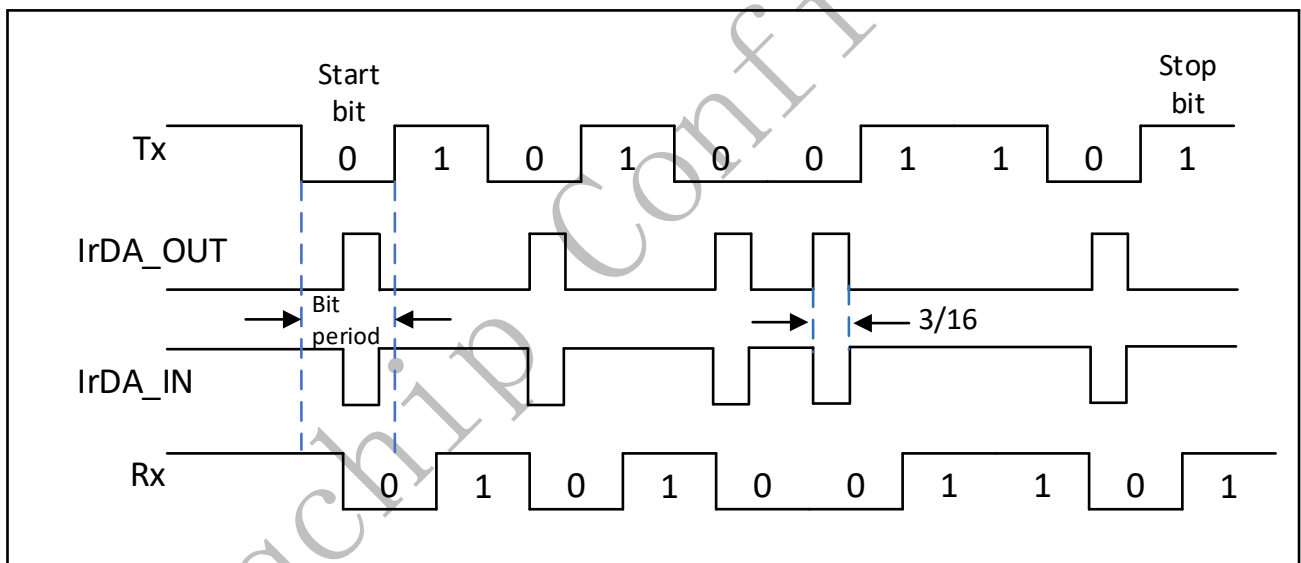


Figure 5-2 IrDA Timing Diagram

5.4. Uart Register

5.4.1. Uart Register Memory Mapping

Uart0 Register Base Addr: 0x5005_0000

Uart1 Register Base Addr: 0x5005_8000

table 5-1 Uart Register Memory Mapping

Offset	Name	Description
0x00	DATA/DLL	DATA Register/Divisor Latch(Low)
0x04	IER/DLH	Interrupt Enable Register/Divisor Latch High
0x08	FCR/IID	FIFO Control Register/Interrupt ID
0x0C	LCR	Line Control Register
0x10	MCR	Modem Control Register
0x14	LSR	Line Status Register
0x18	MSR	Modem Status Register
0x7C	USR	UART Status Register
0xC0	DLF	Divisor Latch Fraction Register

5.4.2. DATA/DLL (Offset 0x00)

table 5-2 DATA/DLL Register

Bit	Name	Access	Default	Description
31:9	Reserved	R/W	0x0	Reserved
8:0	DATA	R/W	9'h00	Data to be received or sent Read Operation: If in non-FIFO mode (FIFO_MODE == NONE) or FIFOs are disabled, the data in the Register must be read before the next data arrives, otherwise it will be overwritten, resulting in an over-run error. If in FIFO mode and FIFOs are enabled, this register accesses the head of the receive FIFO. If the receive FIFO is full and this register is not read before the next data character arrives, then the data already in the FIFO will be preserved but any incoming data will be lost and an over-run error occurs. Write Operation: If in non-FIFO mode or FIFO's are disabled, Any additional writes to the this register will cause the data to be overwritten.

Bit	Name	Access	Default	Description
				If in FIFO mode and FIFO's are enabled, x number of characters of data may be written to the register before the FIFO is full. The number x is determined by the value of FIFO Depth that is set during configuration. Any attempt to write data when the FIFO is full results in the write data being lost.

table 5-3 DLL Register

Bit	Name	Access	Default	Description
31:8	Reserved	R/W	0x0	Reserved
7:0	DLL	R/W	8'h00	Divisor Latch(LOW) This register makes up the lower 8-bits of a 16-bit, read/ write, Divisor Latch register that contains the baud rate divisor for the UART. Note: This register can only be set when DLAB of LCR register is set to '1', otherwise it will be mapped to DATA register

Note: When the DLAB of LCR register is set to '1', the DLL register is accessed; otherwise, the DATA register is accessed.

5.4.3. IER/DLH (Offset 0x04)

table 5-4 IER Register

Bit	Name	Access	Default	Description
31:8	Reserved	R/W	0x0	Reserved
7	PTIME	R/W	1'b0	Programmable THRE Interrupt Mode Enable. Writeable only when THRE_MODE_USER == Enabled, always readable. This is used to enable/disable the generation of THRE Interrupt. Reset Value: 0x0 Values: 0x0 (DISABLED): Disable Programmable THRE Interrupt Mode 0x1 (ENABLED): Enable Programmable THRE Interrupt Mode

Bit	Name	Access	Default	Description
				Note: This interrupt should be used with the ETBEI bit
6:4	Reserved	R/W	0x0	Reserved
3	EDSSI	R/W	1'b0	Enable Modem Status Interrupt. This is used to enable/ disable the generation of Modem Status Interrupt. This is the fourth highest priority interrupt. Reset Value: 0x0 Values: 0x0 (DISABLED): Disable Modem Status Interrupt 0x1 (ENABLED): Enable Modem Status Interrupt
2	ELSI	R/W	1'b0	Enable Receiver Line Status Interrupt. This is used to enable/disable the generation of Receiver Line Status Interrupt. This is the highest priority interrupt. Overrun/parity/framing in LSR status register Errors and break, all these signals will trigger interrupts 0x0 (DISABLED): Disable Receiver Line Status Interrupt 0x1 (ENABLED): Enable Receiver Line Status Interrupt
1	ETBEI	R/W	1'b0	The transmit register is empty (FIFO is not used). Transmit threshold interrupt (when FIFO is used, the data in TxFIFO is equal to or less than the threshold set by TE in FCR register, and PTIME=1 needs to be set for this mode). This is the third highest priority interrupt. 0x0 (DISABLED): Disable Transmit empty interrupt 0x1 (ENABLED): Enable Transmit empty interrupt Note: Refer to TET of FCR register for threshold setting
0	ERBFI	R/W	1'b0	Enable Received Data Available Interrupt. This is used to enable/disable the generation of Received Data Available Interrupt and the Character Timeout Interrupt (if in FIFO mode

Bit	Name	Access	Default	Description
				and FIFO's enabled). These are the second highest priority interrupts. Reset Value: 0x0 Values: 0x0 (DISABLED): Disable Receive data Interrupt 0x1 (ENABLED): Enable Receive data Interrupt Note: Refer to RT of FCR register for threshold setting

table 5-1 DLH Register

Bit	Name	Access	Default	Description
31:9	Reserved	R/W	0x0	Reserved
8:0	DLH	R/W	8'h00	Divisor Latch High Register. Note: This register can only be set when DLAB of LCR register is set to '1', otherwise it will be mapped to IER register

Note: When DLAB of LCR register is set to '1', the description content of this register can be accessed; otherwise, the content of IER register is accessed.

5.4.4. FCR/IID (Offset 0x08)

The following Register description will be used for the 'write' operation.

table 5-5 FCR Register

Bit	Name	Access	Default	Description
31:8	Reserved	W	0x0	Reserved
7:6	RT	W	2'h0	RCVR Trigger (or RT). This is used to select the trigger level in the receiver FIFO at which the Received Data Available Interrupt will be generated. In auto flow control mode, it is used to determine when the rts_n signal will be de-asserted only when RTC_FCT is disabled. It also determines when the dma_rx_req_n signal will be asserted when in certain modes of operation. For details on DMA support, refer to 'DMA Support' section of data book. Values:

Bit	Name	Access	Default	Description
				0x0 (FIFO_CHAR_1): 1 character in FIFO 0x1 (FIFO_QUARTER_FULL): FIFO 1/4 full 0x2 (FIFO_HALF_FULL): FIFO 1/2 full 0x3 (FIFO_FULL_2): FIFO 2 less than full
5:4	TET	W	2'h0	TX Empty Trigger (or TET). This is used to select the empty threshold level at which the THRE Interrupts will be generated when the mode is active. It also determines when the dma_tx_req_n signal will be asserted when in certain modes of operation. Values: 0x0 (FIFO_EMPTY): FIFO Empty 0x1 (FIFO_CHAR_2): 2 characters in FIFO 0x2 (FIFO_QUARTER_FULL): FIFO 1/4 full 0x3 (FIFO_HALF_FULL): FIFO 1/2 full
3	DMAM	W	1'b0	DMA Mode (or DMAM). 0: DMA mode selection when FIFO is disabled 1: DMA mode selection when FIFO is enabled
2	XFIFOR	W	1'b0	XMIT FIFO Reset 0x1 (RESET): Transmit FIFO reset
1	RFIFOR	W	1'b0	RCVR FIFO Reset 0x1 (RESET): Receive FIFO reset
0	FIFOE	W	1'b0	FIFO enable 0: FIFO Disable 1: FIFO Enable

The following Register description will be used during the 'read' operation

table 5-6 IID Register

Bit	Name	Access	Default	Description
31:4	Reserved	R	0x0	Reserved
3:0	IID	R	4'h1	Interrupt ID (or IID). This indicates the highest priority pending interrupt which can

Bit	Name	Access	Default	Description
				be one of the following types specified in Values. 0x0 (MODEM_STATUS): modem status 0x1 (NO_INTERRUPT_PENDING): no interrupt pending 0x2 (THR_EMPTY): THR empty 0x4 (RECEIVED_DATA_AVAILABLE): received data available 0x6 (RECEIVER_LINE_STATUS): receiver line status (read LSR clear) 0xc (CHARACTER_TIMEOUT): character timeout (read RxFIFO clear)

5.4.5. LCR (Offset 0x0C)

table 5-7 LCR Register

Bit	Name	Access	Default	Description
31:8	Reserved	R/W	0x0	Reserved
7	DLAB	R/W	1'b0	Divisor Latch Access Bit. 0x0 (DISABLED): Divisor Latch register is writable only when UART Not BUSY 0x1 (ENABLED): Divisor Latch register is always readable and writable Note: After the initial baud rate is set, this bit must be cleared to access other registers
6	BC	R/W	1'b0	Break Control Bit. This is used to cause a break condition to be transmitted to the receiving device. If set to one the serial output is forced to the spacing (logic 0) state. 0x0 (DISABLED): Serial output is released for data transmission 0x1 (ENABLED): Serial output is forced to spacing state
5	SP	R/W	1'b0	Stick Parity.

Bit	Name	Access	Default	Description
				SP = 1, EPS = 1, PEN = 1, the parity bit is transmitted and checked as logic 0 SP = 1, EPS = 0, PEN = 1, parity bit is transmitted and checked as a logic 1. 0x0 (DISABLED): Stick parity disabled 0x1 (ENABLED): Stick parity enabled
4	EPS	R/W	1'b0	Even Parity Select 0x0 (ODD_PARITY): an odd parity is transmitted or checked 0x1 (EVEN_PARITY): an even parity is transmitted or checked
3	PEN	R/W	1'b0	Parity Enable 0x0 (DISABLED): disable parity 0x1 (ENABLED): enable parity
2	STOP	R/W	1'b0	Number of stop bits 0x0 (STOP_1BIT): 1 stop bit 0x1 (STOP_1_5BIT_OR_2BIT): 1.5 stop bits when DLS (LCR[1:0]) is zero, else 2 stop bit
1:0	DLS	R/W	2'h0	Data Length Select (0x0 (CHAR_5BITS): 5 data bits per character 0x1 (CHAR_6BITS): 6 data bits per character 0x2 (CHAR_7BITS): 7 data bits per character 0x3 (CHAR_8BITS): 8 data bits per character

5.4.6. MCR (Offset 0x10)

table 5-8 MCR Register

Bit	Name	Access	Default	Description
31:7	Reserved	R/W	0x0	Reserved
6	SIRE	R/W	1'b0	SIR Mode Enable 0x0 (DISABLED): IrDA SIR Mode disabled

Bit	Name	Access	Default	Description
				0x1 (ENABLED): IrDA SIR Mode enable (only Uart0)
5	AFCE	R/W	1'b0	Auto Flow Control Enable 0x0 (DISABLED): Auto Flow Control Mode disabled 0x1 (ENABLED): Auto Flow Control Mode enabled (only Uart0)
4:2	Reserved	R/W	0x0	Reserved
1	RTS	R/W	1'b0	Request to send This is used to directly control the Request to Send (rts_n) output. The Request To Send (rts_n) output is used to inform the modem or data set that the UART is ready to exchange data. 0x0 (INACTIVE): Request to Send rts_n de-asserted (logic 1) 0x1 (ACTIVE): Request to Send rts_n asserted (logic 0) Note: When automatic flow control is used, if it is set to 0, RTS is invalid.
0	Reserved	R/W	0x0	Reserved

5.4.7. LSR (Offset 0x14)

table 5-9 LSR Register

Bit	Name	Access	Default	Description
31:8	Reserved	R/W	0x0	Reserved
7	RFE	R	1'b0	Receiver FIFO Error bit This is used to indicate if there is at least one parity error, framing error, or break indication in the FIFO. This bit is cleared when the LSR is read and when the character with the error is at the top of the receiver

Bit	Name	Access	Default	Description
				FIFO and there are no subsequent errors in the FIFO, this bit will be cleared. 0x0 (NO_RX_FIFO_ERROR): No error in RX FIFO 0x1 (RX_FIFO_ERROR): Error in RX FIFO
6	TEMT	R	1'b1	Transmitter Empty bit If in FIFO mode, this bit is set whenever the Transmitter Shift Register and the FIFO are both empty. If in the non-FIFO mode or FIFO's are disabled, this bit is set whenever the Transmitter Holding Register and the Transmitter Shift Register are both empty. 0: 0x0 (DISABLED): Transmitter not empty 1: 0x1 (ENABLED): Transmitter empty
5	THRE	R	1'b1	If PTIME in the IER register is equal to 1 and FIFOE in the FCR register is equal to 1, when the value is equal to or less than the transmit threshold set by TET in the FCR register, it indicates a transmit threshold interrupt
4	BI	R	1'b0	Break error. When receiving data, a continuous low-level signal is detected and a Break signal is generated. Read the LSR register to clear the BI bit
3	FE	R	1'b0	Framing Error bit. This is used to indicate the occurrence of a framing error in the receiver. A framing error occurs when the receiver does not detect a valid STOP bit in the received data. Values: 0x0 (NO_FRAMING_ERROR): no framing error 0x1 (FRAMING_ERROR): framing error
2	PE	R	1'b0	Parity Error bit. This is used to indicate the occurrence of a parity error in the receiver if the Parity Enable (PEN) bit (LCR[3]) is set.

Bit	Name	Access	Default	Description
				Values: 0x0 (NO_PARITY_ERROR): no parity error 0x1 (PARITY_ERROR): parity error
1	OE	R	1'b0	Overrun error bit. This is used to indicate the occurrence of an overrun error. This occurs if a new data character was received before the previous data was read. Values: 0x0 (NO_OVER_RUN_ERROR): no overrun error 0x1 (OVER_RUN_ERROR): overrun error Read the LSR register to clear the OE bit.
0	DR	R	1'b0	Data Ready bit. This is used to indicate that the receiver contains at least one character in the RBR or the receiver FIFO. This bit is cleared when the RBR is read in the non-FIFO mode, or when the receiver FIFO is empty, in the FIFO mode.

5.4.8. MSR (Offset 0x18)

table 5-10 MSR Register

Bit	Name	Access	Default	Description
31:5	Reserved	R/W	0x0	Reserved
4	CTS	R	1'b0	Clear to Send. This is used to indicate the current state of the modem control line cts_n. 0: CTS Inactive 1: CTS Active
3:0	Reserved	R/W	0x0	Reserved

5.4.9. USR(Offset 0x7C)

table 5-11 USR Register

Bit	Name	Access	Default	Description
31:5	Reserved	R/W	0x0	Reserved
4	RFF	R	1'b0	Receive FIFO Full This is used to indicate that the receive FIFO is completely full. That is: This bit is cleared when the RX FIFO is no longer full. Values: 0x0 (NOT_FULL): Receive FIFO not full 0x1 (FULL): Receive FIFO full
3	RFNE	R	1'b0	Receive FIFO Not Empty This is used to indicate that the receive FIFO contains one or more entries. This bit is cleared when the RX FIFO is empty. Values: 0: Receive FIFO is empty 1: Receive FIFO is not empty
2	TFE	R	1'b1	Transmit FIFO Empty This is used to indicate that the transmit FIFO is completely empty. This bit is cleared when the TX FIFO is no longer empty 0: Transmit FIFO is not empty 1: Transmit FIFO is empty
1	TFNF	R	1'b1	Transmit FIFO Not Full. This is used to indicate that the transmit FIFO is not full 0: Transmit FIFO is full 1: Transmit FIFO is not full
0	Reserved	R/W	0x0	Reserved

5.4.10. DLF (Offset 0xC0)

table 5-12 DLF Register

Bit	Name	Access	Default	Description
31:4	Reserved	R/W	0x0	Reserved
3:0	DLF	R/W	4'h0	Fractional part of divisor. 0x0: $0/16 = 0.0000$ 0x1: $1/16 = 0.0625$ 0x2: $2/16 = 0.125$ 0x3: $3/16 = 0.1875$ 0x4: $4/16 = 0.25$ 0x5: $5/16 = 0.3125$ 0x6: $6/16 = 0.375$ 0x7: $7/16 = 0.4375$ 0x8: $8/16 = 0.5$ 0x9: $9/16 = 0.5625$ 0xA: $10/16 = 0.625$ 0xB: $11/16 = 0.6875$ 0xC: $12/16 = 0.75$ 0xD: $13/16 = 0.8125$ 0xE: $14/16 = 0.875$ 0xF: $15/16 = 0.9375$ For more information ,please refer to Baud rate calculation

5.5. Work Flow

5.5.1. Transmission Work Flow

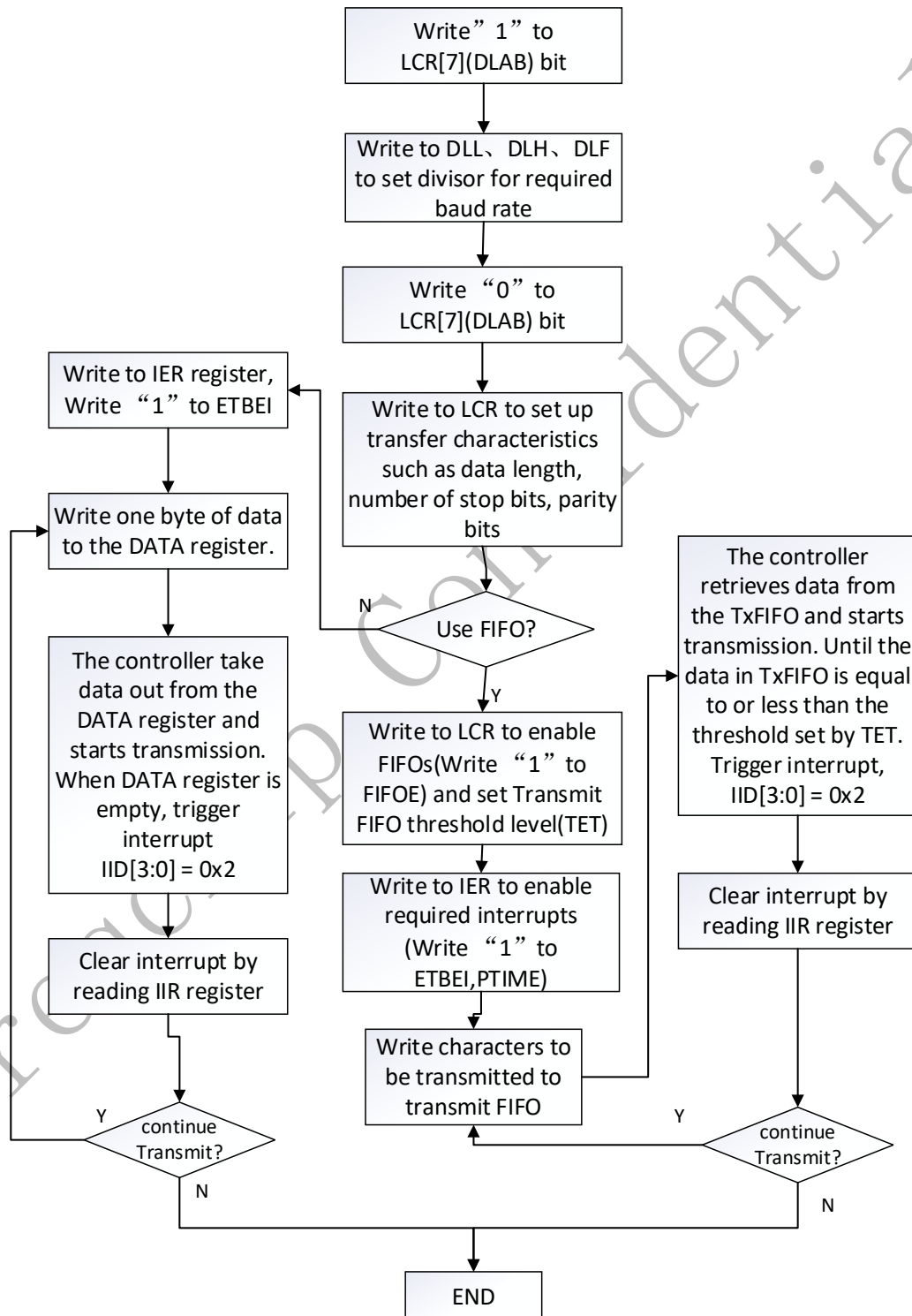


Figure 5-3 Flow for Transmission Programming Example

5.5.2. Reception Work Flow

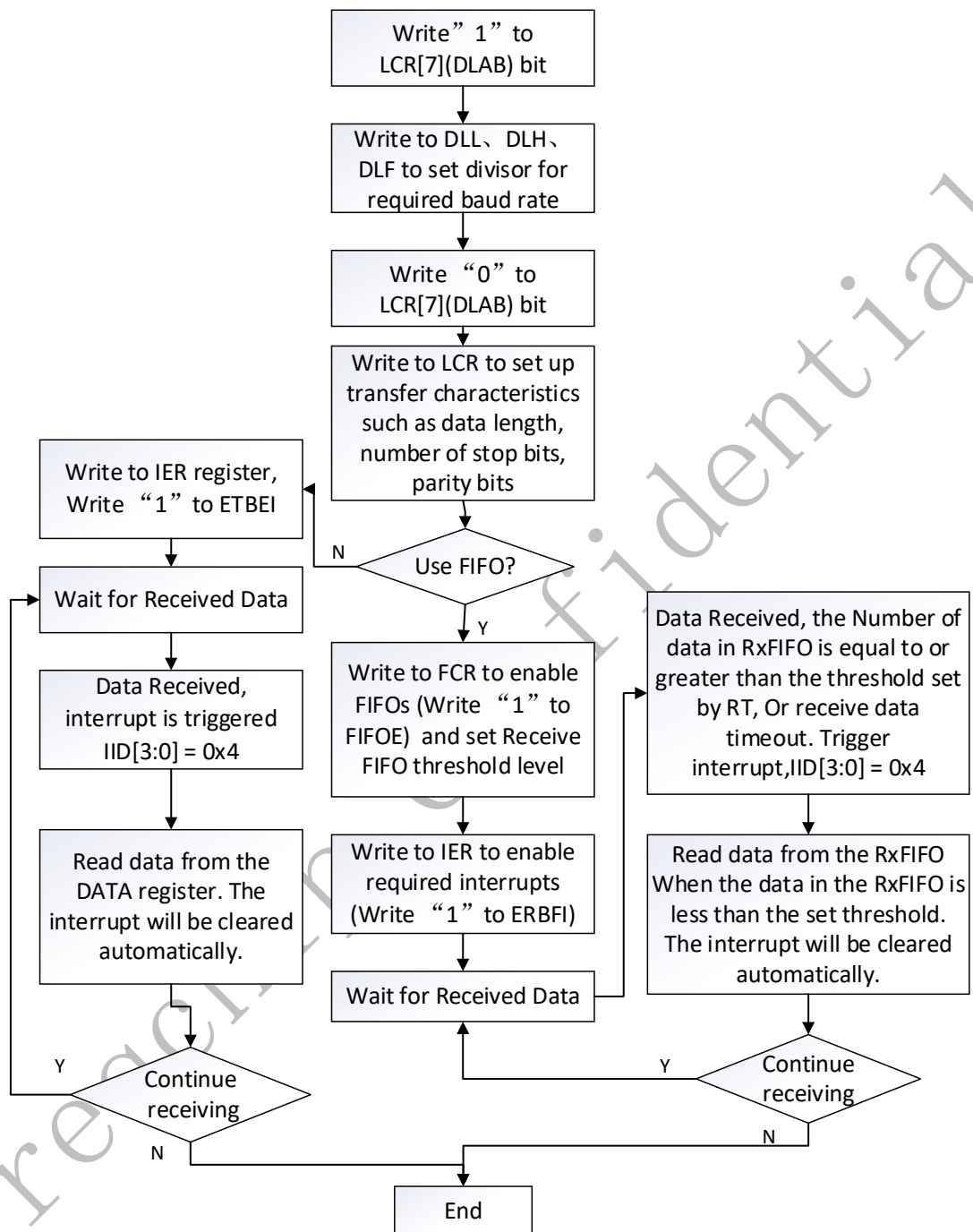


Figure 5-4 Flow for Reception Programming Example

6. Serial Peripheral Interface Master (SPI_Master)

6.1. Overview

Serial Peripheral Interface Master (SPIM) provides an industry standard 4-wire SPI interface for master devices. It supports Motorola, TI, National Semiconductor Microwire interfaces and allows communication with any SPI slave device. In addition to the standard 8-bit word length, the SPI master device also supports the configuration of 4-16 bit word length for communication with non-standard SPI word length.

SPIM signals include standard serial clock (SCLK), slave selection (CS), data transmission (TXD), data reception (RXD), and data output enable (SSOE).

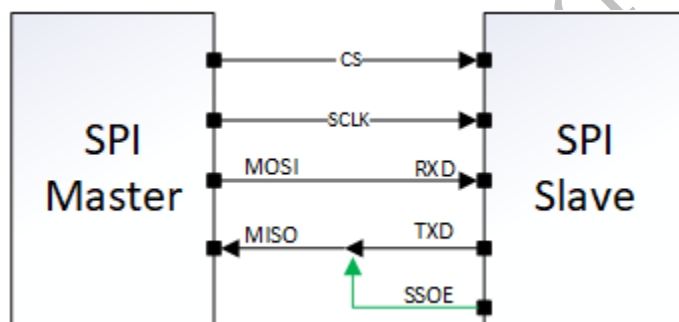


Figure 6-1 SPI Master Wiring Schematics

6.2. Features

- Supports APB data bus widths of 4 to 16 bits
- Up to 48Mbps
- Support DMA
- Configurable and programmable Standard/Dual/Quad mode
- FIFO depth 32, width 16

6.3. Function

6.3.1. Transmit /Receive FIFO and Interrupt

The FIFO buffers can be configured in depth 32 and width 16 bits. Each data entry in the FIFO buffers contains a single data frame. It is impossible to store multiple data frames in a single FIFO location. for example, you may not store two 8-bit data frames in a single FIFO location. If an 8-bit data frame is required, the upper 8-bits of the FIFO entry are ignored or unused when the serial shifter transmits the data.

The transmit FIFO is loaded to the data register (DR). Data are popped (removed) from the transmit FIFO by the shift control logic into the transmit shift register. The transmit FIFO generates a FIFO empty interrupt request when the number of entries in the FIFO is less than or equal to the FIFO threshold value. The threshold value, set through the programmable register TXFTLR, determines the level of FIFO entries at which an interrupt is generated. The threshold value allows you to provide early indication to the processor that the transmit FIFO is nearly empty. A transmit FIFO overflow interrupt is generated if you attempt to write data into an already full transmit FIFO.

Data are popped from the receive FIFO by APB read commands to the data register (DR). The receive FIFO is loaded from the receive shift register by the shift control logic. The receive FIFO generates a FIFO-full interrupt request when the number of entries in the FIFO is greater than or equal to the FIFO threshold value plus 1. The threshold value, set through programmable register RXFTLR, determines the level of FIFO entries at which an interrupt is generated. The threshold value allows you to provide early indication to the processor that the receive FIFO is nearly full. A receive FIFO overrun interrupt is generated when the receive shift logic attempts to load data into a completely full receive FIFO. However, this newly received data are lost. A receive FIFO underflow interrupt is generated if you attempt to read from an empty receive FIFO. This alerts the processor that the read data are invalid.

6.3.2. Transfer Modes

The transfer mode (TMOD) is set by writing to control register 0 (CTRLR0). TMOD is ignored for Microwire transfers.

When $TMOD = 0$, both transmit and receive logic are valid. The data transfer occurs as normal according to the selected frame format (serial protocol). Transmit data are popped from the transmit FIFO and sent through the txd line to the target device, which replies with data on the rxd line. The receive data from the target device is moved from the receive shift register into the receive FIFO at the end of each data frame.

When $TMOD = 1$, the receive data are invalid and should not be stored in the receive FIFO. The data transfer occurs as normal, according to the selected frame format (serial protocol). Transmit data are popped from the transmit FIFO and sent through the txd line to the target device, which replies with data on the rxd line. At the end of the data frame, the receive shift register does not load its newly received data into the receive FIFO. The data in the receive shift register is overwritten by the next transfer. You should mask interrupts originating from the receive logic when this mode is entered.

When $TMOD = 2$ the transmit data are invalid. When configured as a slave, the transmit FIFO is never popped in Receive Only mode. The txd output remains at a constant logic level during the transmission. The data transfer occurs as normal according to the selected frame format (serial protocol). The receive data from the target device is moved from the receive shift register into the receive FIFO at the end of each data frame. You should mask interrupts originating from the transmit logic when this mode is entered.

When $TMOD = 3$, the transmit data is used to transmit an opcode and/or an address to the EEPROM device. Typically this takes three data frames (8-bit opcode followed by 8-bit upper address and 8-bit lower address). During the transmission of the opcode and address, no data is captured by the receive logic (as long as the DW_apb_ssi master is transmitting data on its txd line, data on the rxd line is ignored). The DW_apb_ssi master continues to transmit data until the transmit FIFO is empty. Therefore, you should ONLY have enough data frames in the transmit FIFO to supply the opcode and address to the

EEPROM. If more data frames are in the transmit FIFO than are needed, then read data is lost.

When the transmit FIFO becomes empty (all control information has been sent), data on the receive line (rxd) is valid and is stored in the receive FIFO; the txd output is held at a constant logic level. The serial transfer continues until the number of data frames received by the DW_apb_ssi master matches the value of the NDF field in the CTRLR1 register + 1.

6.3.3. DMA Controller Interface

SPIS uses two DMA channels, one for the transmit data and one for the receive data. SPIS has these DMA registers:

- DMACR – Control register to enable DMA operation.
- DMATDLR – Register to set the transmit the FIFO level at which a DMA request is made.
- DMARDLR – Register to set the receive FIFO level at which a DMA request is made.
DMA request generated when the level is greater than threshold

6.4. SPI Register

6.4.1. SPI Register Memory Mapping

SPIM0 Base Addr: 0x5003_0000

SPIM1 Base Addr: 0x5003_4000

table 6-1 SPI Register Mapping

Offset	Register	Description
0x00	CTRLR0	Control Register 0
0x04	CTRLR1	Control Register 1
0x08	SSIENR	SSI Enable Register
0x0C	MWCR	Microwire Control Register
0x10	SER	Slave Enable Register
0x14	BAUDR	Baud Rate Select

Offset	Register	Description
0x18	TXFTLR	Transmit FIFO Threshold Level
0x1C	RXFTLR	Receive FIFO Threshold Level
0x20	TXFLR	Transmit FIFO Level Register
0x24	RXFLR	Receive FIFO Level Register
0x28	SR	Status Register
0x2C	IMR	Interrupt Mask Register
0x30	ISR	Interrupt Status Register
0x34	RISR	Raw Interrupt Status Register
0x38	TXOICR	Transmit FIFO Overflow Interrupt Clear Register
0x3C	RXOICR	Receive FIFO Overflow Interrupt Clear Register
0x40	RXUICR	Receive FIFO Underflow Interrupt Clear Register
0x48	ICR	Interrupt Clear Register
0x4C	DMACR	DMA Control Register
0x50	DMATDLR	: DMA Transmit Data Level
0x54	DMARDLR	DMA Receive Data Level
0x60	DR	Data Register
0xF0	RX_SAMPLE_DLY	RX Sample Delay Register
0xF4	CTRLR2	Control Register 2(only valid in Dual/Quad mode)

6.4.2. CTRLR0 (Offset 0x00)

table 6-2 CTRLR0 Register

Bit	Name	Access	Default	Description
31:25	Reserved	R	0x0	Reserved
24	SSTE	R/W	1'b0	Slave Select Toggle Enable. When operating in SPI mode with clock phase (SCPH) set to 0, this register controls the behavior of the slave select line between data frames. If this register field is set to 1 the slave select line will

Bit	Name	Access	Default	Description
				toggle between consecutive data frames, with the serial clock (sclk) being held to its default value while slave select is high; if this register field is set to 0 the slave select will stay low and sclk will run continuously for the duration of the transfer.
23	Reserved	R	0x0	Reserved
22:21	SPI_FRF	R/W	2'b00	SPI Frame Format: 00: Standard 01: Dual 10: Quad 11: Reserved
20:16	Reserved	R	0x0	Reserved
15:12	CFS	R/W	4'h0	Control Frame Size. Selects the length of the control word for the Microwire frame format. 0x0(SIZE_01_BIT): 1-bit Control Word 0x1(SIZE_02_BIT): 2-bit Control Word 0x2(SIZE_03_BIT) : 3-bit Control Word 0x3 (SIZE_04_BIT): 4-bit Control Word 0x4 (SIZE_05_BIT): 5-bit Control Word 0x5 (SIZE_06_BIT): 6-bit Control Word 0x6 (SIZE_07_BIT): 7-bit Control Word 0x7 (SIZE_08_BIT): 8-bit Control Word 0x8 (SIZE_09_BIT): 9-bit Control Word 0x9 (SIZE_10_BIT): 10-bit Control Word 0xa (SIZE_11_BIT): 11-bit Control Word 0xb (SIZE_12_BIT): 12-bit Control Word 0xc (SIZE_13_BIT): 13-bit Control Word 0xd (SIZE_14_BIT): 14-bit Control Word 0xe (SIZE_15_BIT): 15-bit Control Word 0xf (SIZE_16_BIT): 16-bit Control Word
11	SRL	R/W	1'b0	Shift Register Loop. Used for testing purposes only. When internally active, connects the transmit shift register output to the receive shift register input.

Bit	Name	Access	Default	Description
				0x1 (TESTING_MODE): Test mode: Tx & Rx shift reg connected 0x0 (NORMAL_MODE): Normal mode operation
10	Reserved	R	0x0	Reserved
9:8	TMOD	R/W	2'b00	Transfer Mode. 00: Transmit & Receive 01: Transmit Only(not valid in Standard mode) 10: Receive Only(not valid in Standard mode)s 11: Flash Read Note: When Dual or Quad is selected, TMOD can only be configured to send or receive.
7	SCPOL	R/W	1'b0	Serial Clock Polarity. Valid when the frame format (FRF) is set to Motorola SPI. Used to select the polarity of the inactive serial clock, which is held inactive when the SSI master is not actively transferring data on the serial bus. 0: Inactive state of serial clock is low 1: Inactive state of serial clock is high
6	SCPH	R/W	1'b0	Serial Clock Phase. Valid when the frame format (FRF) is set to Motorola SPI. 0: Serial clock toggles in middle of first data bit 1: Serial clock toggles at start of first data bit
5:4	FRF	R/W	2'b00	Frame Format 0: MOTOROLA SPI: Motorola SPI Frame Format 1: TEXAS SSP: Texas Instruments SSP Frame Format 2: NS_MICROWIRE: National Microwire Frame Format 3: RESERVED
3:0	DFS	R/W	4'h7	Data Frame Size. 0x3 (FRAME_04BITS): 4-bit serial data transfer 0x4 (FRAME_05BITS): 5-bit serial data transfer 0x5 (FRAME_06BITS): 6-bit serial data transfer

Bit	Name	Access	Default	Description
				0x6 (FRAME_07BITS): 7-bit serial data transfer 0x7 (FRAME_08BITS): 8-bit serial data transfer 0x8 (FRAME_09BITS): 9-bit serial data transfer 0x9 (FRAME_10BITS): 10-bit serial data transfer 0xa (FRAME_11BITS): 11-bit serial data transfer 0xb (FRAME_12BITS): 12-bit serial data transfer 0xc (FRAME_13BITS): 13-bit serial data transfer 0xd (FRAME_14BITS): 14-bit serial data transfer 0xe (FRAME_15BITS): 15-bit serial data transfer 0xf (FRAME_16BITS): 16-bit serial data transfer

6.4.3. CTRLR1 (Offset 0x04)

table 6-3 CTRLR1 Register

Bit	Name	Access	Default	Description
31:16	Reserved	R	0x0	Reserved
15:0	NDF	R/W	16'h00	Number of Data Frames. When TMOD = 10 or TMOD = 11 , this register field sets the number of data frames to be continuously received. Continues to receive serial data until the number of data frames received is equal to this register value plus 1, which enables you to receive up to 64 KB of data in a continuous transfer.

6.4.4. SSIENR (Offset 0x08)

table 6-4 SSIENR Register

Bit	Name	Access	Default	Description
31:1	Reserved	R	0x0	Reserved
0	SSI_EN	R/W	1'b0	SSI Enable. Enables and disables all DW_apb_ssi operations. When disabled, all serial transfers are halted immediately. Transmit and receive FIFO buffers are cleared

Bit	Name	Access	Default	Description
				when the device is disabled. 1: Enables Serial Transfer 0: Disables Serial Transfer

6.4.5. MWCR(Offset 0x0C)

table 6-5 MWCR Register

Bit	Name	Access	Default	Description
31:3	Reserved	R	0x0	Reserved
2	MHS	R/W	1'b0	Microwire Handshaking. Used to enable and disable the busy/ready handshaking interface for the Microwire protocol. When enabled, the DW_apb_ssi checks for a ready status from the target slave, after the transfer of the last data/control bit, before clearing the BUSY status in the SR register. 0: Handshaking interface is disabled 1: Handshaking interface is enabled
1	MDD	R/W	1'b0	Microwire Control. Defines the direction of the data word when the Microwire serial protocol is used. 0: SSI receives data 1: SSI transmits data
0	MWMOD	R/W	1'b0	Microwire Transfer Mode. es whether the Microwire transfer is sequential or nonsequential. 0: Non-Sequential Microwire Transfer 1: Sequential Microwire Transfer

6.4.6. SER(Offset 0x10)

table 6-6 SER Register

Bit	Name	Access	Default	Description
31:1	Reserved	R	0x0	Reserved
0	SER	R/W	1'b0	Slave Select Enable Flag Each bit in this register corresponds to a slave select line. 0: No slave selected 1: Slave is selected

6.4.7. BAUDR(Offset 0x14)

table 6-7 BAUDR Register

Bit	Name	Access	Default	Description
31:16	Reserved	R	0x0	Reserved
15:0	SCKDV	R/W	16'h00	SSI Clock Divider. The frequency of the sclk_out is derived from the following equation: $SPI_CLK_OUT = APB_SPI_CLK / SCKDV$ where SCKDV is any even value between 2 and 65534 注: APB_SPI_CLK can use SPI0/1_Master_CLK_SEL of Clock Control Register to choose 48MHz or 96MHz.

6.4.8. TXFTLR (Offset 0x18)

table 6-8 TXFTLR Register

Bit	Name	Access	Default	Description
31:5	Reserved	R	0x0	Reserved
4:0	TFT	R/W	5'h0	Transmit FIFO Threshold.

Bit	Name	Access	Default	Description
				Controls the level of entries (or below) at which the transmit FIFO controller triggers an interrupt.

6.4.9. RXFTLR (Offset 0x1C)

table 6-9 RXFTLR Register

Bit	Name	Access	Default	Description
31:5	Reserved	R	0x0	Reserved
4:0	RFT	R/W	5'h0	Receive FIFO Threshold. Controls the level of entries (or above) at which the receive FIFO controller triggers an interrupt.

6.4.10. TXFLR (Offset 0x20)

table 6-10 TXFLR Register

Bit	Name	Access	Default	Description
31:6	Reserved	R	0x0	Reserved
5:0	TXTFL	R	6'h0	Transmit FIFO Level. Contains the number of valid data entries in the transmit FIFO.

6.4.11. RXFLR (Offset 0x24)

table 6-11 RXFLR Register

Bit	Name	Access	Default	Description
31:6	Reserved	R	0x0	Reserved
5:0	RXTFL	R	6'h0	Receive FIFO Level.

Bit	Name	Access	Default	Description
				Contains the number of valid data entries in the receive FIFO.

6.4.12. SR (Offset 0x28)

table 6-12 SR Register

Bit	Name	Access	Default	Description
31:5	Reserved	R	0x0	Reserved
4	RFF	R	1'b0	Receive FIFO Full. When the receive FIFO is completely full, this bit is set. When the receive FIFO contains one or more empty location, this bit is cleared. 0: Receive FIFO is not full 1: Receive FIFO is full
3	RFNE	R	1'b0	Receive FIFO Not Empty. Set when the receive FIFO contains one or more entries and is cleared when the receive FIFO is empty. 0: Receive FIFO is empty 1: Receive FIFO is not empty
2	TFE	R	1'b1	Transmit FIFO Empty. When the transmit FIFO is completely empty, this bit is set. When the transmit FIFO contains one or more valid entries, this bit is cleared. 0: Transmit FIFO is not empty 1: Transmit FIFO is empty
1	TFNF	R	1'b1	Transmit FIFO Not Full. Set when the transmit FIFO contains one or more empty locations, and is cleared when the FIFO is full. 0: Transmit FIFO is full 1: Transmit FIFO is not Full

Bit	Name	Access	Default	Description
0	BUSY	R	1'b0	SSI Busy Flag. When set, indicates that a serial transfer is in progress; when cleared indicates that the SSI is idle or disabled. 0: SPI is idle or disabled 1: SPI is actively transferring data

6.4.13. IMR (Offset 0x2C)

table 6-13 IMR Register

Bit	Name	Access	Default	Description
31:5	Reserved	R	0x0	Reserved
4	RXFIM	R/W	1'b1	Receive FIFO Full Interrupt Mask 0x0: interrupt is masked 0x1: interrupt is not masked
3	RFOIM	R/W	1'b1	Receive FIFO Overflow Interrupt Mask
2	RXUIM	R/W	1'b1	Receive FIFO Underflow Interrupt Mask Values: 0x0: interrupt is masked 0x1: interrupt is not masked
1	TXOIM	R/W	1'b1	Transmit FIFO Overflow Interrupt Mask Values: 0x0 (MASKED): interrupt is masked 0x1 (UNMASKED): interrupt is not masked
0	TXEIM	R/W	1'b1	Transmit FIFO Empty Interrupt Mask 0: interrupt is masked 1: interrupt is not masked

6.4.14. ISR (Offset 0x30)

table 6-14 ISR Register

Bit	Name	Access	Default	Description
31:5	Reserved	R	0x0	Reserved
4	RXFIS	R	1'b0	Receive FIFO Full Interrupt Status (after IMR IMR masking) 0x0: interrupt is not active after masking 0x1: interrupt is full after masking
3	RXOIS	R	1'b0	Receive FIFO Overflow Interrupt Status 0x0: interrupt is not active after masking 0x1: interrupt is active after masking
2	RXUIS	R	1'b0	Receive FIFO Underflow Interrupt Status 0x0: interrupt is not active after masking 0x1: interrupt is active after masking
1	TXOIS	R	1'b0	Transmit FIFO Overflow Interrupt Status 0x0: interrupt is not active after masking 0x1: interrupt is active after masking
0	TXEIS	R	1'b0	Transmit FIFO Empty Interrupt Status 0: interrupt is not active after masking 1: interrupt is active after masking

6.4.15. RISR (Offset 0x34)

table 6-15 RISR Register

Bit	Name	Access	Default	Description
31:5	Reserved	R	0x0	Reserved
4	RXFIR	R	1'b0	Receive FIFO Full Raw Interrupt Status 0x0: interrupt is not active prior to masking 0x1: interrupt is active prior to masking
3	RXOIR	R	1'b0	Receive FIFO Overflow Raw Interrupt Status 0x1: interrupt is not active prior to masking 0x0: interrupt is active prior masking

Bit	Name	Access	Default	Description
2	RXUIR	R	1'b0	Receive FIFO Underflow Raw Interrupt Status 0x0: interrupt is not active prior to masking 0x1: interrupt is active prior to masking
1	TXOIR	R	1'b0	Transmit FIFO Overflow Raw Interrupt Status 0x0: interrupt is not active prior to masking 0x1: interrupt is active prior masking
0	TXEIR	R	1'b0	Transmit FIFO Empty Raw Interrupt Status 0x0: interrupt is not active prior to masking 0x1: interrupt is active prior masking

6.4.16. TXOICR (Offset 0x38)

table 6-16 TXOICR Register

Bit	Name	Access	Default	Description
31:1	Reserved	R	0x0	Reserved
0	TXOICR	R	1'b0	Clear Transmit FIFO Overflow Interrupt. This register reflects the status of the interrupt. A read from this register clears the ssi_txo_intr interrupt; writing has no effect.

6.4.17. RXOICR (Offset 0x3C)

table 6-17 RXOICR Register

Bit	Name	Access	Default	Description
31:1	Reserved	R	0x0	Reserved
0	RXOICR	R	1'b0	Clear Receive FIFO Overflow Interrupt. This register reflects the status of the interrupt. A read from this register clears the ssi_rxo_intr interrupt; writing has no effect.

6.4.18. RXUICR (Offset 0x40)

table 6-18 RXUICR Register

Bit	Name	Access	Default	Description
31:1	Reserved	R	0x0	Reserved
0	RXUICR	R	1'b0	Clear Receive FIFO Underflow Interrupt. This register reflects the status of the interrupt. A read from this register clears the ssi_rxu_intr interrupt; writing has no effect.

6.4.19. ICR (Offset 0x48)

table 6-19 ICR Register

Bit	Name	Access	Default	Description
31:1	Reserved	R	0x0	Reserved
0	ICR	R	1'b0	Clear Interrupts. This register is set if any of the interrupts below are active. A read clears the ssi_txo_intr, ssi_rxu_intr, ssi_rxo_intr, and the ssi_mst_intr interrupts. Writing to this register has no effect.

6.4.20. DMACR (Offset 0x4C)

table 6-20 DMACR Register

Bit	Name	Access	Default	Description
31:2	Reserved	R	0x0	Reserved
1	TDMAE	R/W	1'b0	Transmit DMA Enable. This bit enables/disables the transmit FIFO DMA channel. 0x0: Transmit DMA disabled 0x1: Transmit DMA enabled
0	RDMAE	R/W	1'b0	Receive DMA Enable. This bit

Bit	Name	Access	Default	Description
				enables/disables the receive FIFO DMA channel 0x0: Receive DMA disabled 0x1: Receive DMA enabled

6.4.21. DMATDLR (Offset 0x50)

table 6-21 DMATDLR Register

Bit	Name	Access	Default	Description
31:4	Reserved	R	0x0	Reserved
4:0	DMATDL	R/W	5'h0	Transmit Data Level. This bit field controls the level at which a DMA request is made by the transmit logic. It is equal to the watermark level; that is, the dma_tx_req signal is generated when the number of valid data entries in the transmit FIFO is equal to or below this field value, and TDMAE = 1.

6.4.22. DMARDLR (Offset 0x54)

table 6-22 DMARDLR Register

Bit	Name	Access	Default	Description
31:4	Reserved	R	0x0	Reserved
4:0	DMARDL	R/W	5'h0	Receive Data Level. This bit field controls the level at which a DMA request is made by the receive logic. The watermark level = DMARDL+1; that is, dma_rx_req is generated when the number of valid data entries in the receive FIFO is equal to or above this field value + 1, and RDMAE=1.

6.4.23. DR (offset 0x60)

table 6-23 DR Register

Bit	Name	Access	Default	Description
31:16	Reserved	R	0x0	Reserved
15:0	DR	R/W	16'h00	Data register. When reading this register, the value is taken from the receive FIFO. When writing this register, it is written to the transmit FIFO.

6.4.24. RX_SAMPLE_DLY (Offset 0xF0)

table 6-24 RX_SAMPLE_DLY Register

Bit	Name	Access	Default	Description
31:16	Reserved	R	0x0	Reserved
7:0	RSD	R/W	8'h00	Rxd Sample Delay. This register is used to delay the sample of the rxd input port.

6.4.25. CTRLR2(Offset 0xF4)

table 6-25 CTRLR2 Register

Bit	Name	Access	Default	Description
31:16	Reserved	R	0x0	Reserved
15:11	WAIT_CYCLES	R/W	5'h0	Wait cycles Number of wait cycles in Dual/Quad mode between control frames transmit and data reception.
10	Reserved	R	0x0	Reserved
9:8	INST_L	R/W	2'h0	Instruction Length Dual/Quad mode instruction length in bits. 0x0: 0-bit (No Instruction) 0x1: 4-bit Instruction 0x2: 8-bit Instruction 0x3: 16-bit Instruction
7:6	Reserved	R	0x0	Reserved

Bit	Name	Access	Default	Description
5:2	ADDR_L	R/W	4'h0	Address Length. This bit defines Length of Address to be transmitted. Only after this much bits are programmed in to the FIFO the transfer can begin 0x0: 0bit Address Width 0x1: 4bit Address Width 0x2: 8bit Address Width 0x3: 12bit Address Width 0x4: 16bit Address Width 0x5: 20bit Address Width 0x6: 24bit Address Width 0x7: 28bit Address Width 0x8: 32bit Address Width 0x9: 36bit Address Width 0xA: 40bit Address Width 0xB: 44bit Address Width 0xC: 48bit Address Width 0xD: 52bit Address Width 0xE: 56bit Address Width 0xF: 60bit Address Width
1:0	TRANS_TYPE	R/W	2'b00	Address and instruction transfer format 00: Instruction and Address will be sent in Standard SPI Mode. 01: Instruction will be sent in Standard SPI Mode and Address will be sent in the mode specified by CTRLR0.SPI_FRF. 10: Both Instruction and Address will be sent in the mode specified by SPI_FRF. 11: Reserved Note: SPI_FRF determines whether to use two or four wires in Enhanced SPI format

Note: CTRLR2 Register is only valid in Dual or Quad modes.

6.5. Processing flow

6.5.1. SPI standard mode transmission

1. Configure SPI control register CTRLR0.
FRF - Select which serial protocol transfers the data.
DFS – Data Frame Size.
SPI_FRF – select data frame format for Transmitting/Receiving the data,
Standord/Dual/Quad mode
SCPL – Select the polarity of the inactive serial clock.
SCPH- The serial clock phase selects the relationship of the serial clock with the slave select signal.
2. Configure BAUDR register to set clock frequency division.
3. Configure SER register to set SSI_EN bit. Enable SSI.
4. Configure DR register, Write data to FIFO.
5. SPI transfer data
6. When the FIFO data is not full or empty, repeat step 4 until the data transmission is completed.
7. Configure SER register, clear SSI_EN, release CSs

6.5.2. SPI standard mode reception

1. Configure SPI control register CTRLR0.
FRF - Select which serial protocol transfers the data.
DFS – Data Frame Size.
SPI_FRF – select data frame format for Transmitting/Receiving the data,
Standord/Dual/Quad mode
SCPL – Select the polarity of the inactive serial clock.

SCPH- The serial clock phase selects the relationship of the serial clock with the slave select signal.

2. Configure BAUDR register to set clock frequency division.
3. Configure SER register to set SSI_EN bit. Enable SSI.
4. Write data to DR register (used to generate the Clock). Shouldn't exceed FIFO depth.
5. When TxFIFO is not empty, read data in TxFIFO.

6.5.3. Write Operation in Dual or Quad Mode

Dual(SPI_FRF = 1), Quad (SPI_FRF = 2) write operations can be divided into three parts:

- Instruction phase
- Address phase
- Data phase

Compared with the standard mode, related registers below need to be configured:

- CTRLR0.SPI_FRF: Specifies the format in which the transmission happens for the frame.
- CTRLR0.DFS: Specifies data length.
- CTRLR2.INST_L: Specifies length of an instruction.
- CTRLR2.ADDR_L: Specifies address length.
- CTRLR2.TRANS_TYPE: Specifies Address and Instruction transfer format.

An instruction takes one FIFO location and address can take more than one FIFO locations. Both the instruction and address write in TxFIFO, and when the written data matches the configured length, transmission starts.

Dual mode is used as below:

Case 1: Instruction and Address in Standard Mode. CTRLR2.TRANS_TYPE = 0.

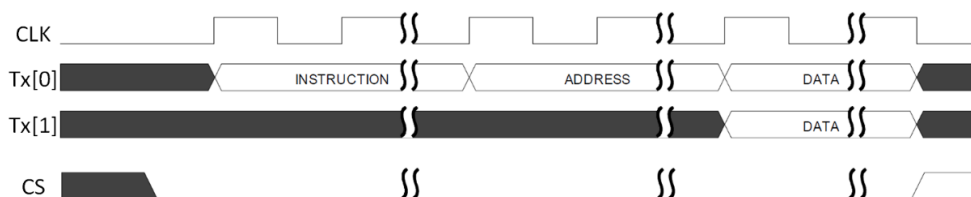


Figure 6-2 Instruction and Address in Standard Mode

Case 2: Instruction in Standard and Address in Enhanced SPI Format.

CTRLR2.TRANS_TYPE = 1。

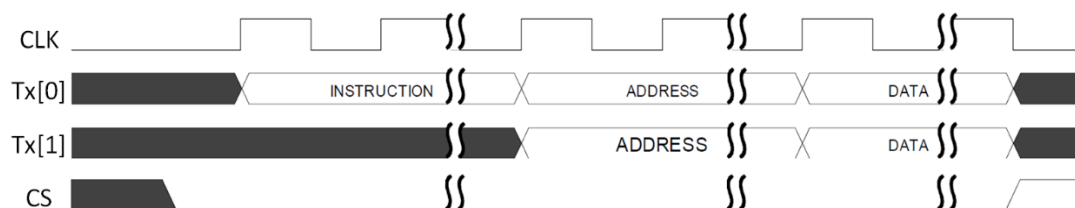


Figure 6-3 Instruction in Standard and Address in Enhanced SPI Format

Case 3: Instruction and Address Transmitted in Enhanced SPI Format.

CTRLR2.TRANS_TYPE = 2。

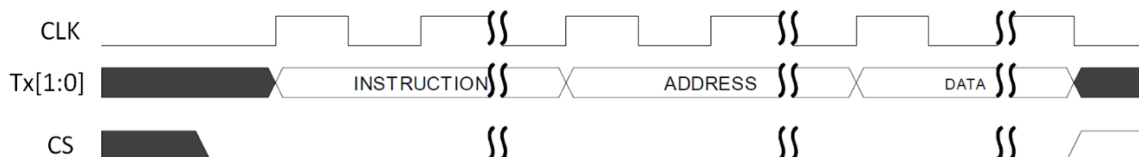


Figure 6-4 Instruction and Address Transmitted in Enhanced SPI Format

Case 4: Instruction only transfer in enhanced SPI Format. CTRLR2.TRANS_TYPE = 2。

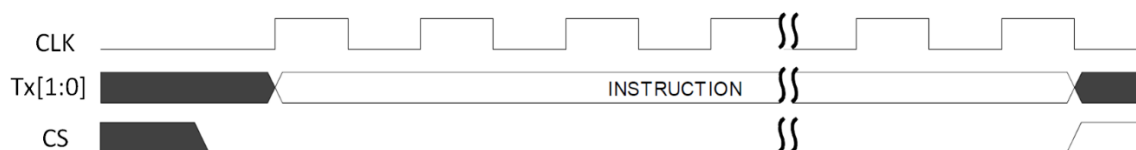


Figure 6-5 Instruction only transfer in enhanced SPI Format

6.5.4. Read Operation in Dual or Quad Mode

A Dual or Quad SPI read operation can be divided into four phases:

- Instruction phase
- Address phase
- Wait cycles
- Data phase

Compared with the standard mode, related registers below need to be configured:

- CTRLR0.SPI_FRF: Select which serial protocol transfers the data, Standord/Dual/Quad mode

- CTRLR0.DFS: Data Frame Size.
- CTRLR2.INST_L: Instruction Length
- CTRLR2.ADDR_L: Address Length
- CTRLR2.TRANS_TYPE: Address and instruction transfer format
- CTRLR2.WAIT_CYCLES: Wait cycles

An instruction takes one FIFO location and address can take more than one FIFO locations. Both the instruction and address write in TxFIFO, and when the written data matches the configured length, transmission starts.

Dual mode is used as below:

Case 1: Instruction and Address in Standard Mode. CTRLR2.TRANS_TYPE = 0.



Figure 6-6 Instruction and Address in Standard Mode

Case 2: Instruction in Standard and Address in Enhanced SPI Format.

CTRLR2.TRANS_TYPE = 1.

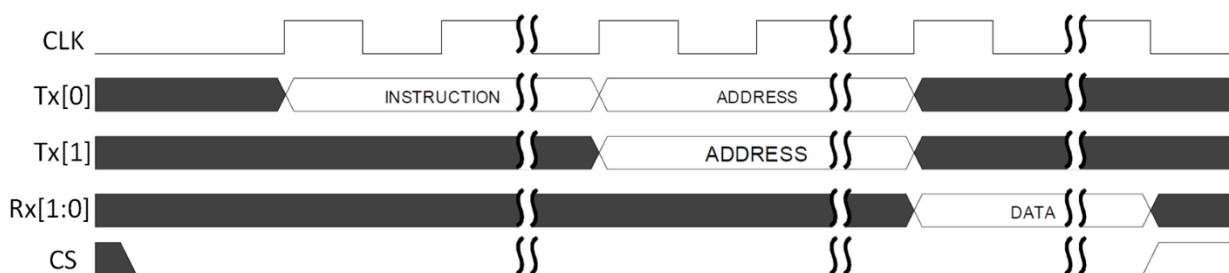


Figure 6-7 Instruction in Standard and Address in Enhanced SPI Format

Case 3: Instruction and Address Transmitted in Enhanced SPI Format.

CTRLR2.TRANS_TYPE = 2.

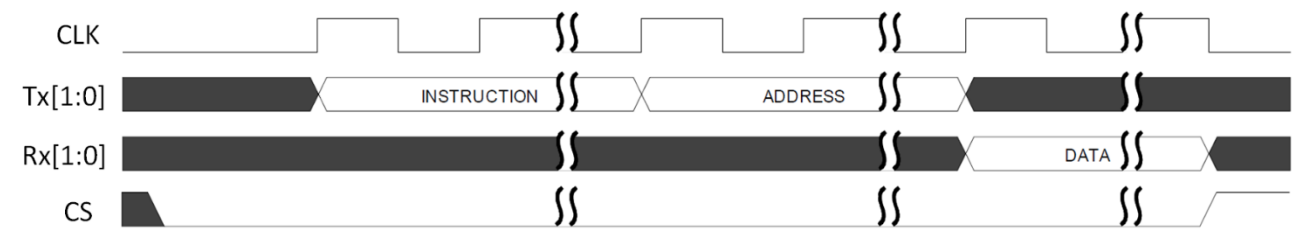


Figure 6-8 Instruction and Address Transmitted in Enhanced SPI Format

Case 4: No instruction, no address, only data transmission.

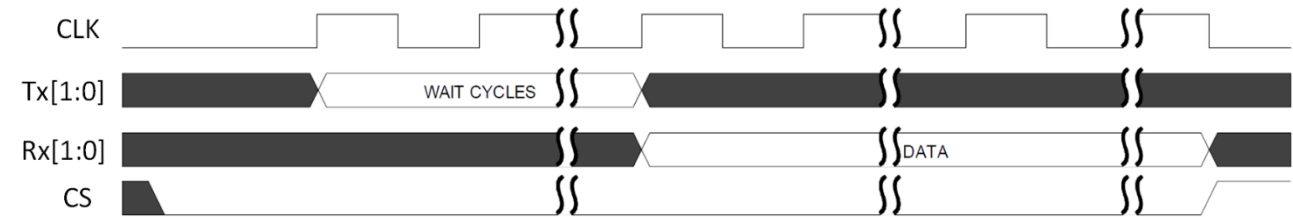


Figure 6-9 No instruction, no address, only data transmission

6.6. MOTOROLA SPI timing

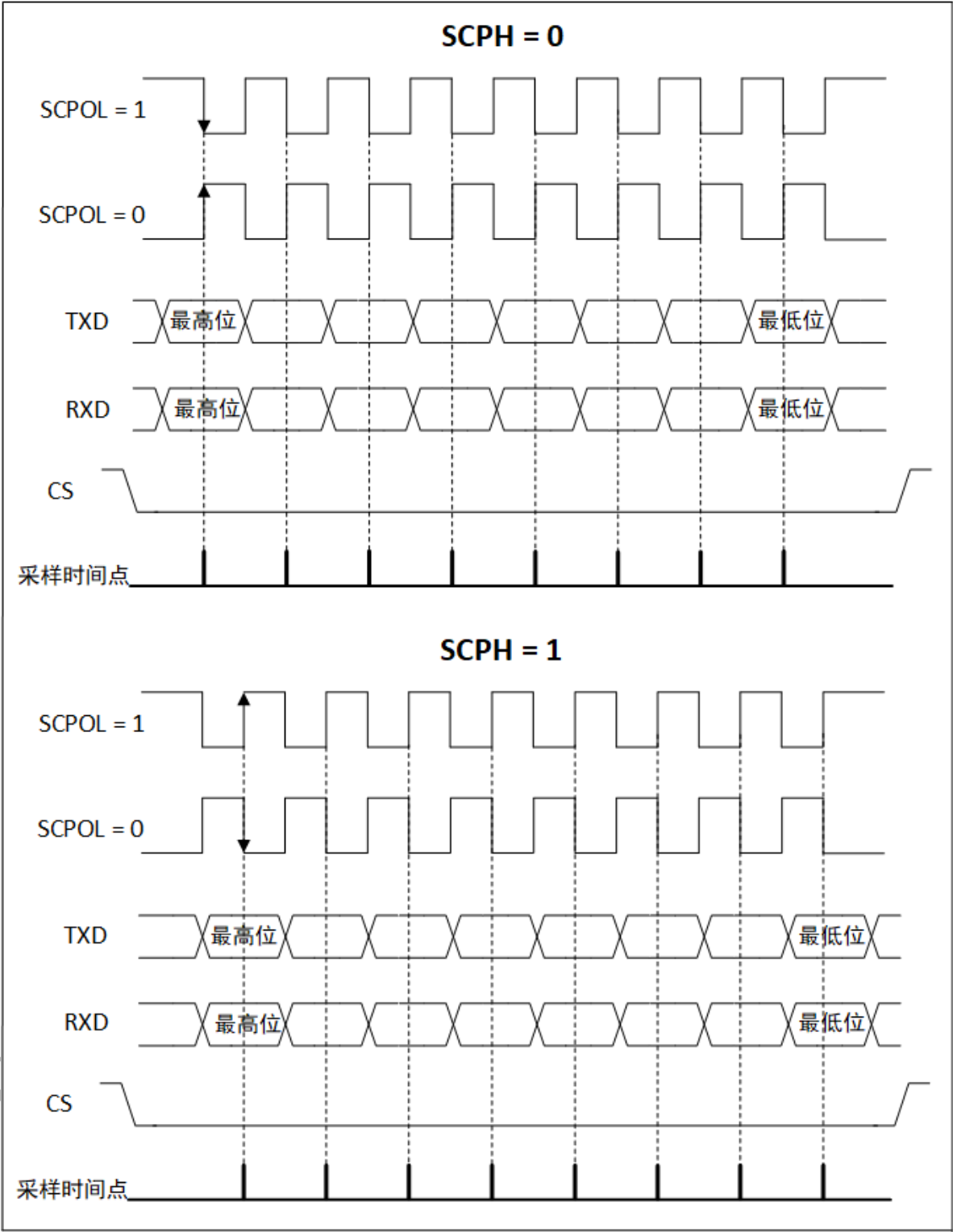


Figure 6-10 MOTOROLA SPI Timing Digram

7. Serial Peripheral Interface Master (SPI_Slave)

7.1. Overview

Serial Peripheral Interface Master (SPIM) provides an industry standard 4-wire SPI interface for master devices. It supports Motorola, TI, National Semiconductor Microwire interfaces and allows communication with any SPI slave device. In addition to the standard 8-bit word length, the SPI master device also supports the configuration of 4-16 bit word length for communication with non-standard SPI word length.

SPIM signals include standard serial clock (SCLK), slave selection (CS), data transmission (TXD), data reception (RXD), and data output enable (SSOE).

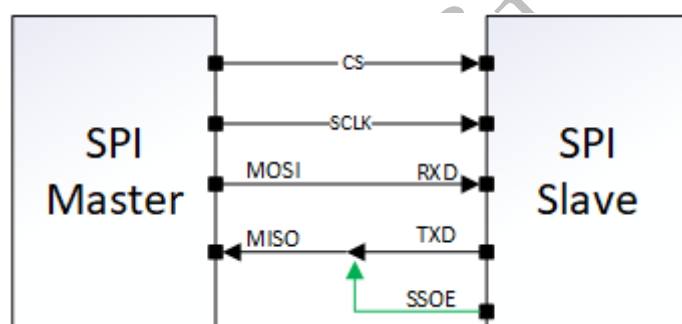


Figure 7-1 SPI Slave Wiring Schematics

7.2. Features

- Supports APB data bus widths of 4 to 16 bits
- Up to 48Mbps
- Support DMA
- Only Standard SPI mode(Dual/Quad Mode not supported)
- FIFO depth 16, width 16

7.3. Function

7.3.1. Transmit / Receive FIFO and Interrupt

The FIFO buffers can be configured in depth 16 and width 16 bits. Each data entry in the FIFO buffers contains a single data frame. It is impossible to store multiple data frames in a single FIFO location; for example, you may not store two 8-bit data frames in a single FIFO location. If an 8-bit data frame is required, the upper 8-bits of the FIFO entry are ignored or unused when the serial shifter transmits the data.

The transmit FIFO is loaded to the data register (DR). Data are popped (removed) from the transmit FIFO by the shift control logic into the transmit shift register. The transmit FIFO generates a FIFO empty interrupt request when the number of entries in the FIFO is less than or equal to the FIFO threshold value. The threshold value, set through the programmable register TXFTLR, determines the level of FIFO entries at which an interrupt is generated. The threshold value allows you to provide early indication to the processor that the transmit FIFO is nearly empty. A transmit FIFO overflow interrupt is generated if you attempt to write data into an already full transmit FIFO.

Data are popped from the receive FIFO by APB read commands to the data register (DR). The receive FIFO is loaded from the receive shift register by the shift control logic. The receive FIFO generates a FIFO-full interrupt request when the number of entries in the FIFO is greater than or equal to the FIFO threshold value plus 1. The threshold value, set through programmable register RXFTLR, determines the level of FIFO entries at which an interrupt is generated. The threshold value allows you to provide early indication to the processor that the receive FIFO is nearly full. A receive FIFO overrun interrupt is generated when the receive shift logic attempts to load data into a completely full receive FIFO. However, this newly received data are lost. A receive FIFO underflow interrupt is generated if you attempt to read from an empty receive FIFO. This alerts the processor that the read data are invalid.

7.3.2. Transfer Modes

The transfer mode (TMOD) is set by writing to control register 0 (CTRLR0). TMOD is ignored for Microwire transfers.

When $TMOD = 0$, both transmit and receive logic are valid. The data transfer occurs as normal according to the selected frame format (serial protocol). Transmit data are popped from the transmit FIFO and sent through the txd line to the target device, which replies with data on the rxd line. The receive data from the target device is moved from the receive shift register into the receive FIFO at the end of each data frame.

When $TMOD = 1$, the receive data are invalid and should not be stored in the receive FIFO. The data transfer occurs as normal, according to the selected frame format (serial protocol). Transmit data are popped from the transmit FIFO and sent through the txd line to the target device, which replies with data on the rxd line. At the end of the data frame, the receive shift register does not load its newly received data into the receive FIFO. The data in the receive shift register is overwritten by the next transfer. You should mask interrupts originating from the receive logic when this mode is entered.

When $TMOD = 2$ the transmit data are invalid. When configured as a slave, the transmit FIFO is never popped in Receive Only mode. The txd output remains at a constant logic level during the transmission. The data transfer occurs as normal according to the selected frame format (serial protocol). The receive data from the target device is moved from the receive shift register into the receive FIFO at the end of each data frame. You should mask interrupts originating from the transmit logic when this mode is entered.

7.3.3. DMA Interface

SPIS uses two DMA channels, one for the transmit data and one for the receive data. SPIS has these DMA registers:

- DMACR – Control register to enable DMA operation.

- DMATDLR – Register to set the transmit the FIFO level at which a DMA request is made.
- DMARDLR – Register to set the receive FIFO level at which a DMA request is made.

7.4. SPI Register

7.4.1. SPI Register Memory Mapping

SPIS Register BaseAddr: 0x5003_8000

table 7-1 SPI Register Mapping

Offset	Register	Description
0x00	CTRLR0	Control Register 0
0x08	SSIENR	SSI Enable Register
0x0C	MWCR	Microwire Control Register
0x18	TXFTLR	Transmit FIFO Threshold Level
0x1C	RXFTLR	Receive FIFO Threshold Level
0x20	TXFLR	Transmit FIFO Level Register
0x24	RXFLR	Receive FIFO Level Register
0x28	SR	Status Register
0x2C	IMR	Interrupt Mask Register
0x30	ISR	Interrupt Status Register
0x34	RISR	Raw Interrupt Status Register
0x38	TXOICR	Transmit FIFO Overflow Interrupt Clear Register
0x3C	RXOICR	Receive FIFO Overflow Interrupt Clear Register
0x40	RXUICR	Receive FIFO Underflow Interrupt Clear Register
0x48	ICR	Interrupt Clear Register
0x4C	DMACR	DMA Control Register
0x50	DMATDLR	DMA Transmit Data Level
0x54	DMARDLR	DMA Receive Data Level
0x60	DR	Data Register

7.4.2. CTRLR0(Offset 0x00)

table 7-1 CTRLR0 Register

Bit	Name	Access	Default	Description
31:25	Reserved	R	0x0	Reserved
24	SSTE	R/W	1'b0	Slave Select Toggle Enable. When operating in SPI mode with clock phase (SCPH) set to 0, this register controls the behavior of the slave select line between data frames. If this register field is set to 1 the slave select line will toggle between consecutive data frames, with the serial clock (sclk) being held to its default value while slave select is high; if this register field is set to 0 the slave select will stay low and sclk will run continuously for the duration of the transfer.
23:16	Reserved	R	0x0	Reserved
15:12	CFS	R/W	4'h0	Control Frame Size. Selects the length of the control word for the Microwire frame format. 0x0(SIZE_01_BIT): 1-bit Control Word 0x1(SIZE_02_BIT): 2-bit Control Word 0x2(SIZE_03_BIT) : 3-bit Control Word 0x3 (SIZE_04_BIT): 4-bit Control Word 0x4 (SIZE_05_BIT): 5-bit Control Word 0x5 (SIZE_06_BIT): 6-bit Control Word 0x6 (SIZE_07_BIT): 7-bit Control Word 0x7 (SIZE_08_BIT): 8-bit Control Word 0x8 (SIZE_09_BIT): 9-bit Control Word 0x9 (SIZE_10_BIT): 10-bit Control Word 0xa (SIZE_11_BIT): 11-bit Control Word 0xb (SIZE_12_BIT): 12-bit Control Word 0xc (SIZE_13_BIT): 13-bit Control Word 0xd (SIZE_14_BIT): 14-bit Control Word 0xe (SIZE_15_BIT): 15-bit Control Word 0xf (SIZE_16_BIT): 16-bit Control Word
11	SRL	R/W	1'b0	Shift Register Loop. Used for testing purposes

Bit	Name	Access	Default	Description
				only. When internally active, connects the transmit shift register output to the receive shift register input. 0x1 (TESTING_MODE): Test mode: Tx & Rx shift reg connected 0x0 (NORMAL_MODE): Normal mode operation
10	SLV_OE	R/W	1'b0	Slave output enable. 1: Disable 0: Enable
9:8	TMOD	R/W	2'b00	Transfer Mode. 00: Transmit & Receive 01: Transmit Only(not valid in Standard mode) 10: Receive Only(not valid in Standard mode)s 11: Flash Read Note: When Dual or Quad is selected, TMOD can only be configured to send or receive.
7	SCPOL	R/W	1'b0	Serial Clock Polarity. Valid when the frame format (FRF) is set to Motorola SPI. Used to select the polarity of the inactive serial clock, which is held inactive when the SSI master is not actively transferring data on the serial bus. 0: Inactive state of serial clock is low 1: Inactive state of serial clock is high
6	SCPH	R/W	1'b0	Serial Clock Phase. Valid when the frame format (FRF) is set to Motorola SPI. 0: Serial clock toggles in middle of first data bit 1: Serial clock toggles at start of first data bit
5:4	FRF	R/W	2'b00	Frame Format 0: MOTOROLA SPI: Motorola SPI Frame Format 1: TEXAS SSP: Texas Instruments SSP Frame Format 2: NS_MICROWIRE: National Microwire

Bit	Name	Access	Default	Description
				Frame Format 3: RESERVED
3:0	DFS	R/W	4'h7	Data Frame Size. 0x3 (FRAME_04BITS): 4-bit serial data transfer 0x4 (FRAME_05BITS): 5-bit serial data transfer 0x5 (FRAME_06BITS): 6-bit serial data transfer 0x6 (FRAME_07BITS): 7-bit serial data transfer 0x7 (FRAME_08BITS): 8-bit serial data transfer 0x8 (FRAME_09BITS): 9-bit serial data transfer 0x9 (FRAME_10BITS): 10-bit serial data transfer 0xa (FRAME_11BITS): 11-bit serial data transfer 0xb (FRAME_12BITS): 12-bit serial data transfer 0xc (FRAME_13BITS): 13-bit serial data transfer 0xd (FRAME_14BITS): 14-bit serial data transfer 0xe (FRAME_15BITS): 15-bit serial data transfer 0xf (FRAME_16BITS): 16-bit serial data transfer

7.4.3. SSIENR (Offset 0x08)

table 7-2 SSIENR Register

Bit	Name	Access	Default	Description
31:1	Reserved	R	0x0	Reserved
0	SSI_EN	R/W	1'b0	SSI Enable. Enables and disables all DW_apb_ssi operations. When disabled, all serial transfers are halted immediately. Transmit and receive FIFO buffers are cleared when the device is disabled. 1: Enables Serial Transfer 0: Disables Serial Transfer

7.4.4. MWCR(Offset 0x0C)

table 7-3 MWCR Register

Bit	Name	Access	Default	Description
31:2	Reserved	R	0x0	Reserved
1	MDD	R/W	1'b0	Microwire Control. Defines the direction of the data word when the Microwire serial protocol is used. 0: SSI receives data 1: SSI transmits data
0	MWMOD	R/W	1'b0	Microwire Transfer Mode. es whether the Microwire transfer is sequential or nonsequential. 0: Non-Sequential Microwire Transfer 1: Sequential Microwire Transfer

7.4.5. TXFTLR(Offset 0x18)

table 7-4 TXFTLR Register

Bit	Name	Access	Default	Description
31:4	Reserved	R	0x0	Reserved
3:0	TFT	R/W	4'h0	Transmit FIFO Threshold. Controls the level of entries (or below) at which the transmit FIFO controller triggers an interrupt.

7.4.6. RXFTLR(Offset 0x1C)

table 7-5 RXFTLR Register

Bit	Name	Access	Default	Description
31:4	Reserved	R	0x0	Reserved
3:0	RFT	R/W	4'h0	Receive FIFO Threshold.

Bit	Name	Access	Default	Description
				Controls the level of entries (or above) at which the receive FIFO controller triggers an interrupt.

7.4.7. TXFLR (Offset 0x20)

table 7-6 TXFLR Register

Bit	Name	Access	Default	Description
31:5	Reserved	R	0x0	Reserved
4:0	TXTFL	R	5'h0	Transmit FIFO Level. Contains the number of valid data entries in the transmit FIFO.

7.4.8. RXFLR (Offset 0x24)

table 7-7 RXFLR Register

Bit	Name	Access	Default	Description
31:5	Reserved	R	0x0	Reserved
4:0	RXTFL	R	5'h0	Receive FIFO Level. Contains the number of valid data entries in the receive FIFO.

7.4.9. SR (Offset 0x28)

table 7-8 SR Register

Bit	Name	Access	Default	Description
31:6	Reserved	R	0x0	Reserved
5	TXE	R	1'b0	Transmission Error. Set if the transmit FIFO is empty when a transfer is started. 0: No Error

Bit	Name	Access	Default	Description
				1: Transmission Error
4	RFF	R	1'b0	Receive FIFO Full. When the receive FIFO is completely full, this bit is set. When the receive FIFO contains one or more empty location, this bit is cleared. 0: Receive FIFO is not full 1: Receive FIFO is full
3	RFNE	R	1'b0	Receive FIFO Not Empty. Set when the receive FIFO contains one or more entries and is cleared when the receive FIFO is empty. 0: Receive FIFO is empty 1: Receive FIFO is not empty
2	TFE	R	1'b1	Transmit FIFO Empty. When the transmit FIFO is completely empty, this bit is set. When the transmit FIFO contains one or more valid entries, this bit is cleared. 0: Transmit FIFO is not empty 1: Transmit FIFO is empty
1	TFNF	R	1'b1	Transmit FIFO Not Full. Set when the transmit FIFO contains one or more empty locations, and is cleared when the FIFO is full. 0: Transmit FIFO is full 1: Transmit FIFO is not Full
0	BUSY	R	1'b0	SSI Busy Flag. When set, indicates that a serial transfer is in progress; when cleared indicates that the SSI is idle or disabled. 0: SPI is idle or disabled 1: SPI is actively transferring data

7.4.10. IMR (Offset 0x2C)

table 7-9 IMR Register

Bit	Name	Access	Default	Description
31:5	Reserved	R	0x0	Reserved
4	RXFIM	R/W	1'b1	Receive FIFO Full Interrupt Mask 0x0: interrupt is masked 0x1: interrupt is not masked
3	RFOIM	R/W	1'b1	Receive FIFO Overflow Interrupt Mask 0x0: interrupt is masked 0x1: interrupt is not masked
2	RXUIM	R/W	1'b1	Receive FIFO Underflow Interrupt Mask Values: 0x0: interrupt is masked 0x1: interrupt is not masked
1	TXOIM	R/W	1'b1	Transmit FIFO Overflow Interrupt Mask Values: 0x0 (MASKED): interrupt is masked 0x1 (UNMASKED): interrupt is not masked
0	TXEIM	R/W	1'b1	Transmit FIFO Empty Interrupt Mask 0: interrupt is masked 1: interrupt is not masked

7.4.11. ISR (Offset 0x30)

table 7-10 ISR Register

Bit	Name	Access	Default	Description
31:5	Reserved	R	0x0	Reserved
4	RXFIS	R	1'b0	Receive FIFO Full Interrupt Status (after IMR masking) 0x0: interrupt is not active after masking 0x1: interrupt is full after masking
3	RXOIS	R	1'b0	Receive FIFO Overflow Interrupt Status 0x0: interrupt is not active after masking

Bit	Name	Access	Default	Description
				0x1: interrupt is active after masking
2	RXUIS	R	1'b0	Receive FIFO Underflow Interrupt Status 0x0: interrupt is not active after masking 0x1: interrupt is active after masking
1	TXOIS	R	1'b0	Transmit FIFO Overflow Interrupt Status 0x0: interrupt is not active after masking 0x1: interrupt is active after masking
0	TXEIS	R	1'b0	Transmit FIFO Empty Interrupt Status 0: interrupt is not active after masking 1: interrupt is active after masking

7.4.12. RISR (Offset 0x34)

table 7-11 RISR Register

Bit	Name	Access	Default	Description
31:5	Reserved	R	0x0	Reserved
4	RXFIR	R	1'b0	Receive FIFO Full Raw Interrupt Status 0x0: interrupt is not active prior to masking 0x1: interrupt is active prior to masking
3	RXOIR	R	1'b0	Receive FIFO Overflow Raw Interrupt Status 0x1: interrupt is not active prior to masking 0x0: interrupt is active prior masking
2	RXUIR	R	1'b0	Receive FIFO Underflow Raw Interrupt Status 0x0: interrupt is not active prior to masking 0x1: interrupt is active prior to masking
1	TXOIR	R	1'b0	Transmit FIFO Overflow Raw Interrupt Status 0x0: interrupt is not active prior to masking 0x1: interrupt is active prior masking
0	TXEIR	R	1'b0	Transmit FIFO Empty Raw Interrupt Status 0x0: interrupt is not active prior to masking 0x1: interrupt is active prior masking

7.4.13. TXOICR (Offset 0x38)

table 7-12 TXOICR Register

Bit	Name	Access	Default	Description
31:1	Reserved	R	0x0	Reserved
0	TXOICR	R	1'b0	Clear Transmit FIFO Overflow Interrupt. This register reflects the status of the interrupt. A read from this register clears the ssi_txo_intr interrupt; writing has no effect.

7.4.14. RXOICR (Offset 0x3C)

table 7-13 RXOICR Register

Bit	Name	Access	Default	Description
31:1	Reserved	R	0x0	Reserved
0	RXOICR	R	1'b0	Clear Receive FIFO Overflow Interrupt. This register reflects the status of the interrupt. A read from this register clears the ssi_rxo_intr interrupt; writing has no effect.

7.4.15. RXUICR (Offset 0x40)

table 7-14 RXUICR Register

Bit	Name	Access	Default	Description
31:1	Reserved	R	0x0	Reserved
0	RXUICR	R	1'b0	Clear Receive FIFO Underflow Interrupt. This register reflects the status of the interrupt. A read from this register clears the ssi_rxu_intr interrupt; writing has no effect.

7.4.16. ICR (Offset 0x48)

table 7-15 ICR Register

Bit	Name	Access	Default	Description
31:1	Reserved	R	0x0	Reserved
0	ICR	R	1'b0	Clear Interrupts. This register is set if any of the interrupts below are active. A read clears the ssi_txo_intr, ssi_rxu_intr, ssi_rxo_intr, and the ssi_mst_intr interrupts. Writing to this register has no effect.

7.4.17. DMACR (Offset 0x4C)

table 7-16 DMACR Register

Bit	Name	Access	Default	Description
31:2	Reserved	R	0x0	Reserved
1	TDMAE	R/W	1'b0	Transmit DMA Enable. This bit enables/disables the transmit FIFO DMA channel. 0x0: Transmit DMA disabled 0x1: Transmit DMA enabled
0	RDMAE	R/W	1'b0	Receive DMA Enable. This bit enables/disables the receive FIFO DMA channel 0x0: Receive DMA disabled 0x1: Receive DMA enabled

7.4.18. DMATDLR (Offset 0x50)

table 7-17 DMATDLR Register

Bit	Name	Access	Default	Description
31:4	Reserved	R	0x0	Reserved
3:0	DMATDL	R/W	4'h0	Transmit Data Level. This bit field controls

Bit	Name	Access	Default	Description
				the level at which a DMA request is made by the transmit logic. It is equal to the watermark level; that is, the dma_tx_req signal is generated when the number of valid data entries in the transmit FIFO is equal to or below this field value, and TDMAE = 1.

7.4.19. DMARDLR (Offset 0x54)

table 7-18 DMARDLR Register

Bit	Name	Access	Default	Description
31:4	Reserved	R	0x0	Reserved
3:0	DMARDL	R/W	4'h0	Receive Data Level. This bit field controls the level at which a DMA request is made by the receive logic. The watermark level = DMARDL+1; that is, dma_rx_req is generated when the number of valid data entries in the receive FIFO is equal to or above this field value + 1, and RDMAE=1.

7.4.20. DR (offset 0x60)

table 7-19 DR Register

Bit	Name	Access	Default	Description
31:16	Reserved	R	0x0	Reserved
15:0	DR	R/W	16'h00	Data register. When reading this register, the value is taken from the receive FIFO. When writing this register, it is written to the transmit FIFO.

7.5. Processing flow

SPIS data transmission is controlled by SCLK generated by the master device. When the CS signal is not pulled down, SPIS remains in the idle state, and the TXD line needs to be kept at a high level to avoid interfering with the data transmission of other SPI slave devices. SSOE signal line can control the output buffer of TXD and choose whether to output TXD signal.

SPI slave Software configuration process of continuous data transmission:

1. Configure CTRLR0 (the SCPH and SCPOL must be consistent with the master device), set FIFO interrupt thresholds (TXFTLR and RXFTLR), and enable related interrupts (IMR). Note: Configure when SSI_EN=0.
2. Enable SPIS (SSI_EN= 1)
3. If the transfer mode is the transmit and receive mode (TMOD=0) or the transmit only mode (TMOD=1), write the data into the transmission FIFO (write DR); If the transmit mode is the receive only mode (TMOD=2), it is not necessary to fill in any data to the transmit FIFO.
4. When transmitting, you can confirm whether the transmission is completed by querying the BUSY bit. If a transmit FIFO empty interrupt is generated, write the transmit FIFO (write DR). If a receive FIFO full interrupt is generated, read data from the receive FIFO (read DR).
5. When the master device pulls up the CS signal or all FIFO data are sent, the BUSY bit will be reset.

7.6. MOTOROLA SPI timing

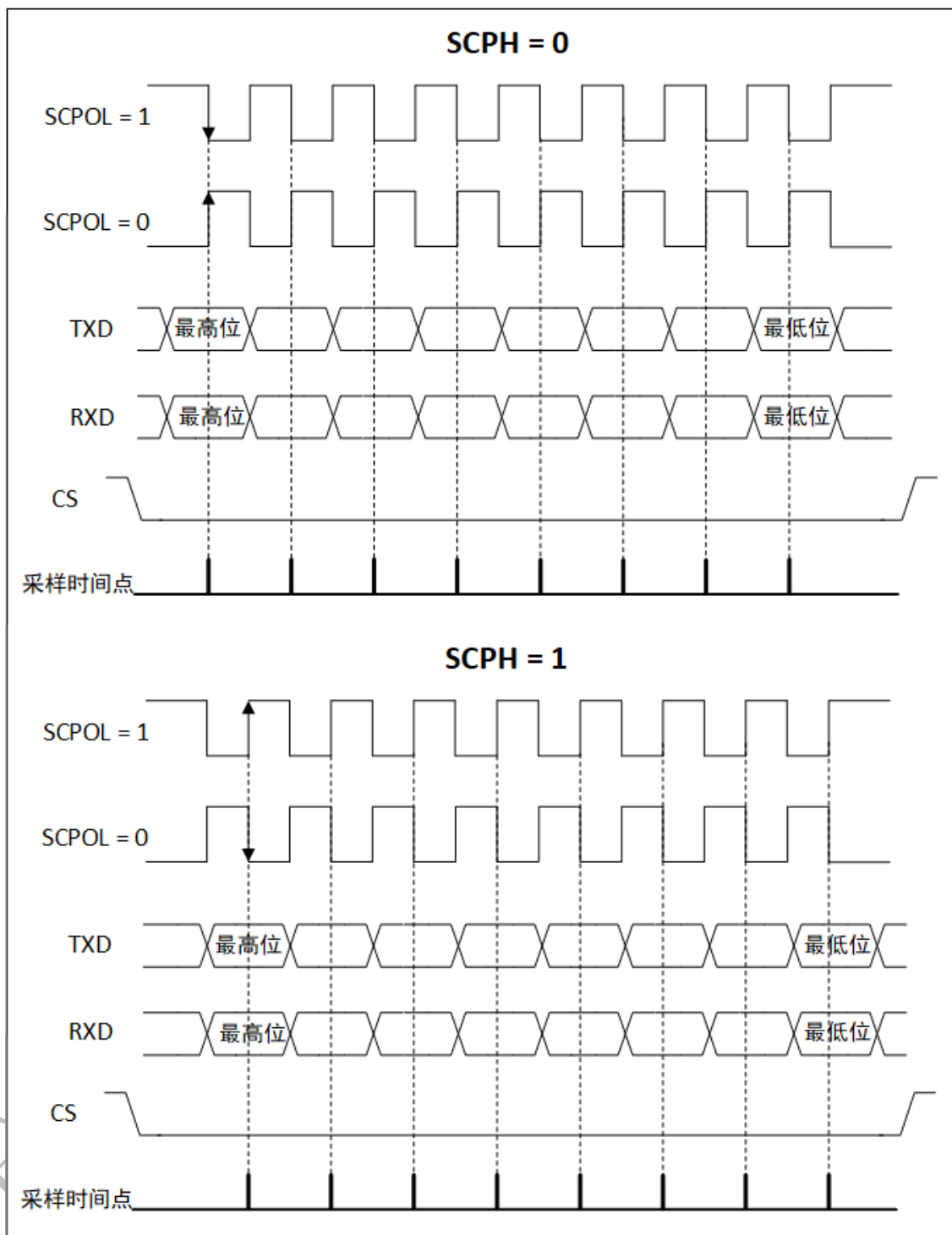


Figure 7-2 MOTOROLA SPI timing Diagram

8. I2C

8.1. Overview

The I2C bus interface connects the microcontroller and the serial I2C bus. It allows multiple masters and controls all I2C bus specific timing, protocol and arbitration.

8.2. Feature

- Two-wire I2 C serial interface – consists of a serial data line (SDA) and a serial clock (SCL)
- Three speeds:
 - ❑ Standard mode
 - ❑ Fast mode or fast mode plus
 - ❑ High-speed mode
- Tx FIFO depth 32-bit, width 8-bit
- Rx FIFO depth 32-bit, width 8-bit
- Master OR slave I2C operation
- 7- or 10-bit addressing
- SCL, SDA low-level timeout detection
- Spike Suppression

8.3. Function

8.3.1. I2C Timing Control

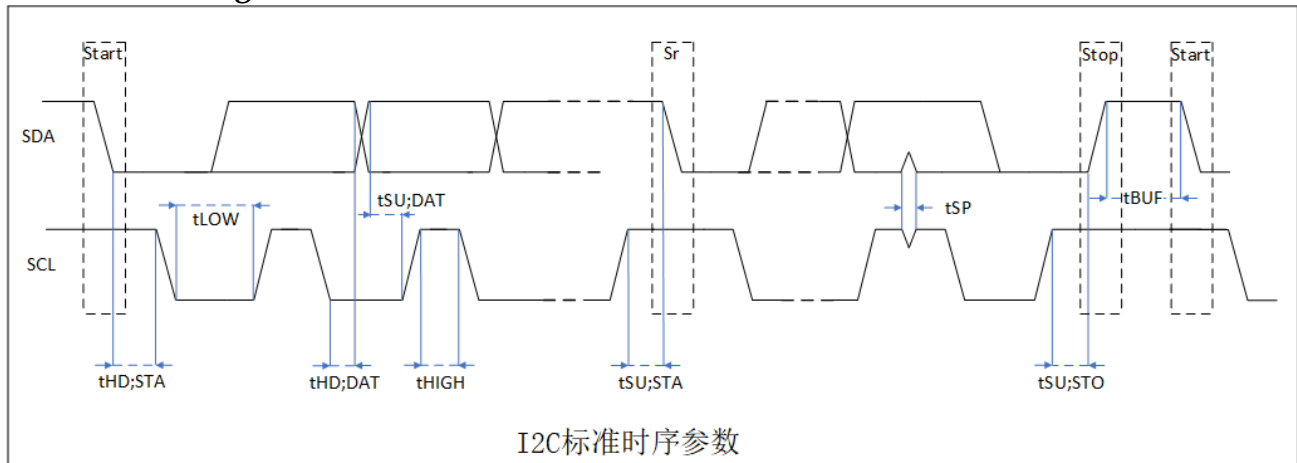


Figure 8-1 I2C Timing Diagram

table 8-1 The Relationship Between I2C Timing Parameters and Register

Timing Parameter	Symbol	Standard Speed	Fast Speed	High Speed
LOW period of the SCL clock	tLOW	SS_SCL_LCNT	FS_SCL_LCNT	HS_SCL_LCNT
HIGH period of the SCL clock	tHIGH	SS_SCL_HCNT	FS_SCL_HCNT	HS_SCL_HCNT
Setup time for a repeated START condition	tSU;STA	SS_SCL_LCNT	FS_SCL_HCNT	HS_SCL_LCNT
Hold time (repeated) START condition	tHD;STA	SS_SCL_HCNT	FS_SCL_HCNT	HS_SCL_LCNT
Setup time for STOP condition	tSU;STO	SS_SCL_HCNT	FS_SCL_HCNT	HS_SCL_LCNT
Bus free time between a STOP and a START condition	tBUF	SS_SCL_LCNT	FS_SCL_LCNT	
Spike length	tSP	FS_SPKLEN	FS_SPKLEN	HS_SPKLEN
Data hold time	tHD;DAT	SDA_HOLD	SDA_HOLD	SDA_HOLD
Data setup time	tSU;DAT	SDA_SETUP	SDA_SETUP	SDA_SETUP

I2C provides a relatively flexible method for timing modification. The relevant registers are shown in Table 8 - 1, and the adjusted timing can be referred to in Figure 8 - 1. The adjustment units of the above - mentioned registers are all based on APB_I2C_CLK.

- Taking Standard Speed as an example, in one clock cycle:

The high - level duration of the clock signal = (SS_SCL_HCNT + FS_SPKLEN + 6) *

APB_I2C_CLK

The low - level duration of the clock signal = $(SS_SCL_LCNT + 1) * APB_I2C_CLK$

- Taking Fast Speed as an example, in one clock cycle:

The high - level duration of the clock signal = $(FS_SCL_HCNT + FS_SPKLEN + 3) * APB_I2C_CLK$

The low - level duration of the clock signal = $(FS_SCL_LCNT + 1) * APB_I2C_CLK$

Note: Fast Speed is selected by default.

8.3.2. I2C_SDA_HOLD function

I2C contains a software programmable register ([SDA_HOLD](#)) to enable dynamic adjustment of the SDA hold-time. The bits [15:0] are used to control the hold time of SDA during transmit in both slave and master mode. The bits [23:16] are used to extend the SDA transition (if any) whenever SCL is HIGH in the receiver.

- The IC_SDA_TX_HOLD register can be used to alter the timing of the generated SDA (ic_data_oe) signal by the DW_apb_i2c.

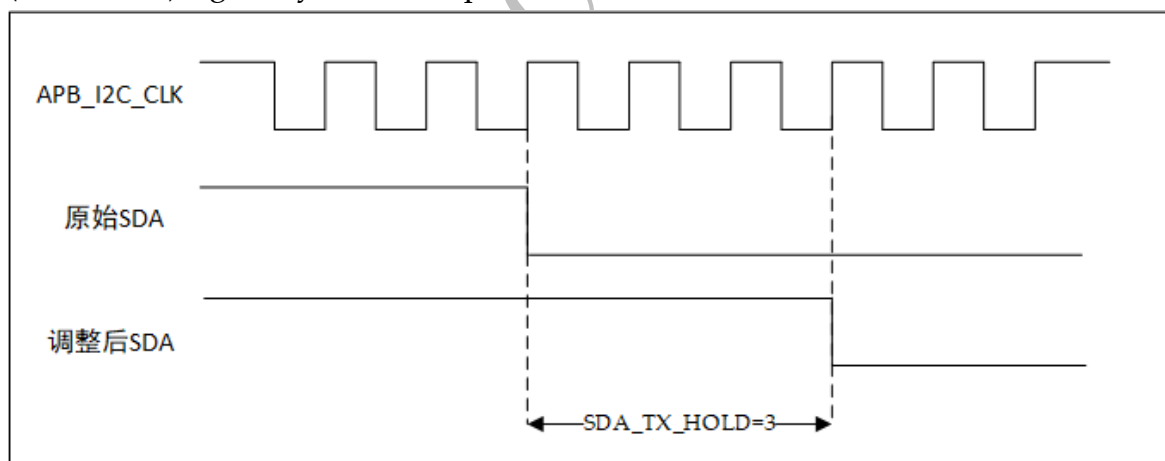


Figure 8-2 SDA_TX_HOLD Diagram

- IC_SDA_RX_HOLD can be used to alter the internal hold time which DW_apb_i2c applies to the incoming SDA line

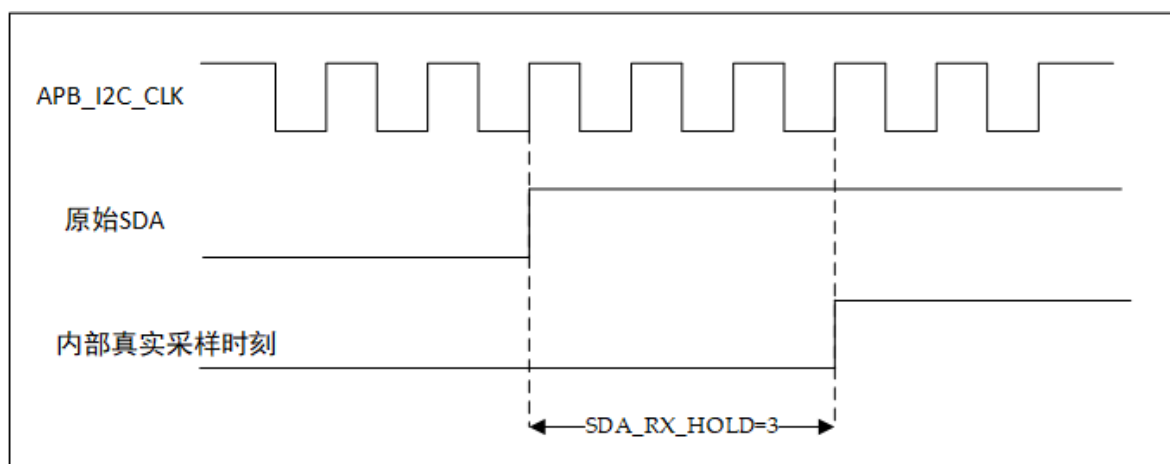


Figure 8-3 SDA_RX_HOLD Diagram

8.3.3. SCL、SDA Low Timeout Function

■ SCL Low Timeout Detect

A malfunctioning device holds the SDA line low indefinitely. This would prevent the master from issuing a STOP condition and ending a transaction. The host can perform SCL pull down timeout detection by configuring related registers.

Note: This function is only supported by I2C0.

Enable BUS_DETEC_FEATURE_CTRL bit in CONTROL register, enable detect function. Users can use the [SCL STUCK AT LOW TIMEOUT](#) Register to program timeout value to detect the SCL low timeout. If SCL is stuck at low, SCL_STUCK_AT_LOW bit in [RAW_INTR_STAT](#) register is set, and can generate interrupt. At this time, I2C can be reset through the system register, or detect the cause being pulled down.

■ SDA Low Timeout Detect

A malfunctioning device holds the SDA line low indefinitely. This would prevent the master from issuing a STOP condition and ending a transaction. If SDA is still low $t_{\text{TIMEOUT, MAX}}$ after SCL has gone high at the end of a transaction the master should hold SMBCLK low for at least $t_{\text{TIMEOUT, MAX}}$ in an attempt to reset the SMBus interface of all of the devices on the bus. The DW_apb_i2c enables the Bus clear feature in SMBus mode and the user can use the SDA_STUCK_TIMEOUT Register to program the SDA timeout value to detect the SMBDAT low timeout. If SDA line is stuck at low, the

SDA_STUCK_TIMEOUT abort is generated and software can enable the SMBUS_CLK_RESET register bit of ENABLE register to hold the SCL low for SCL_STUCK_TIMEOUT which in turn resets the SMBus interface of all devices on the bus. Note: This function is only supported by I2C0.

SDA recovery mechanism:

The host sends 9 I2C clock cycles to try to recover the bus. If the SDA is not recovered, it will continue to send up to 9 SCL clocks again to try to recover the bus. If the SDA line is recovered within 9 clock pulses, the host will send STOP to release the bus.

If it has not been recovered, you can query the SDA of the STATUS register_ STUCK_NOT_RECOVERED bit to check whether SDA is recovered successfully. If it is not successful, reset I2C through the system register, or detect the reason why the slave continues to pull down.

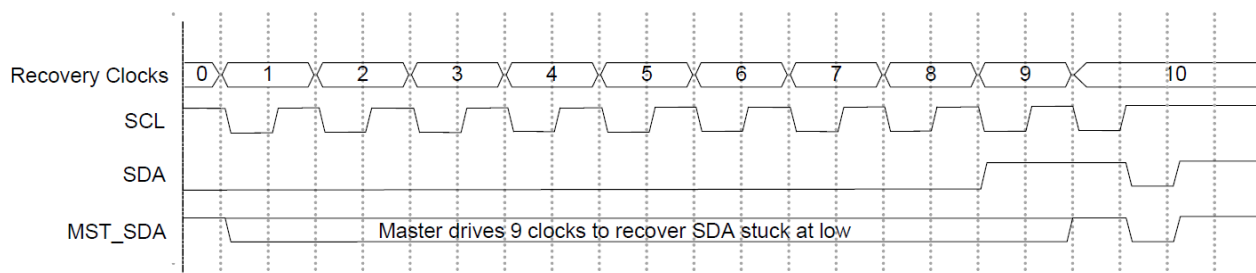


Figure 8-4 SDA Recovery with 9 SCL Clocks

8.3.4. 7-bit Address Format, 10-bit Address Format

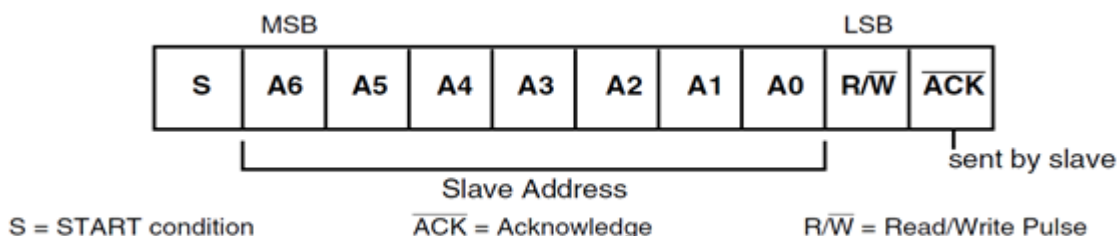


Figure 8-5 7bit Address Format

During the 7-bit address format, the first seven bits (bits 7:1) of the first byte set the slave address and the LSB bit (bit 0) is the R/W bit as shown in Figure 8-5. When bit 0 (R/W)

is set to 0, the master writes to the slave. When bit 0 (R/W) is set to 1, the master reads from the slave.

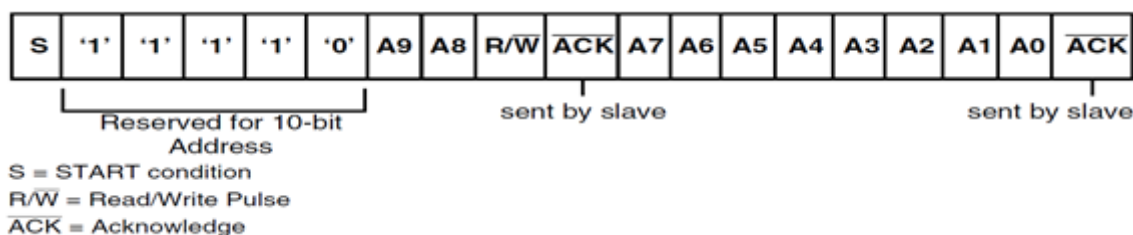


Figure 8-6 10-bit Address Format

During 10-bit addressing, two bytes are transferred to set the 10-bit address. The transfer of the first byte contains the following bit definition. The first five bits (bits 7:3) notify the slaves that this is a 10-bit transfer followed by the next two bits (bits 2:1), which set the slaves address bits 9:8, and the LSB bit (bit 0) is the R/W bit. The second byte transferred sets bits 7:0 of the slave address.

8.3.5. Spike Suppression

Spike Suppression, which acts on the tSP part of the I2C timing, can be found in the [I2C Timing Control](#) section for details. The Register related to this feature is [FS SPKLEN](#), [HS SPKLEN](#).

This logic is based on counters that monitor the input signals (SCL and SDA), checking if they remain stable for a predetermined amount of ic_clk cycles before they are sampled internally. There is one separate counter for each signal (SCL and SDA). The number of APB_I2C_CLK cycles can be programmed by the user.

Each counter is started whenever its input signal changes its value. Depending on the behavior of the input signal, one of the following scenarios occurs:

- The input signal remains unchanged until the counter reaches its count limit value. When this happens, the internal version of the signal is updated with the input value, and the counter is reset and stopped. The counter is not restarted until a new change on the input signal is detected.

- The input signal changes again before the counter reaches its count limit value. When this happens, the counter is reset and stopped, but the internal version of the signal is not updated. The counter remains stopped until a new change on the input signal is detected.

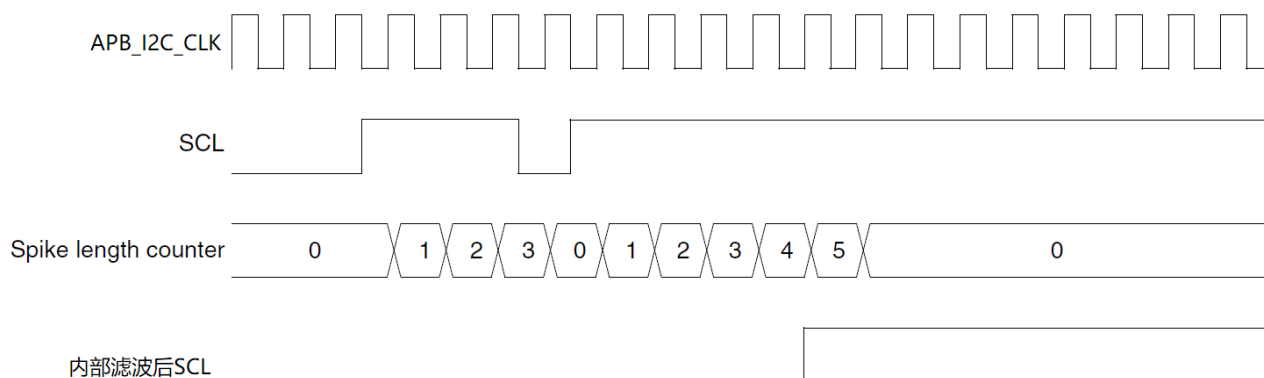


Figure 8-7 Spike Suppression Example

As shown in the figure below, the FS_SPKLEN or HS_SPKLEN is set to 5, see the spike Suppression working process. The first time a SCL change is detected, start counting. If the count is less than 5 times, the SCL changes. The internal SCL is not sampled and remains in the original state. If the SCL changes again, count again. If the count is full for 5 times, it is considered that the SCL is valid for sampling, and the internal SCL changes.

8.4. I2C Register

8.4.1. I2C Register Memory Mapping

Note: Although there are many registers in Section I2C, they are not complex, as shown in the register mapping table. The green background is the registers related to timing control, and the blue background is the registers related to status, interrupt status, and status clearing. Related to other register control functions.

I2C0 Register Base Addr: 0x5004_0000

I2C1 Register Base Addr: 0x5004_8000

table 8-2 I2C Register Mapping

Offset	Register	Description
0x00	CONTROL	I2C Control Register.
0x04	TAR	I2C Target Address Register
0x08	SAR	I2C Slave Address Register
0x0C	HS_MADDR	I2C High Speed Master Mode Code Address Register
0x10	DATA_CMD	I2C Rx/Tx Data Buffer and Command Register
0x14	SS_SCL_HCNT	Standard Speed I2C Clock SCL High Count Register
0x18	SS_SCL_LCNT	Standard Speed I2C Clock SCL Low Count Register
0x1C	FS_SCL_HCNT	Fast Mode or Fast Mode Plus I2C Clock SCL High Count Register (Only I2C0 supports)
0x20	FS_SCL_LCNT	Fast Mode or Fast Mode Plus I2C Clock SCL Low Count Register (Only I2C0 supports)
0x24	HS_SCL_HCNT	High Speed I2C Clock SCL High Count Register (Only I2C0 supports)
0x28	HS_SCL_LCNT	High Speed I2C Clock SCL Low Count Register (Only I2C0 supports)
0x2C	INTR_STAT	I2C Interrupt Status Register
0x30	INTR_MASK	I2C Interrupt Mask Register.
0x34	RAW_INTR_STAT	I2C Raw Interrupt Status Register
0x38	RX_TL	I2C Receive FIFO Threshold Register
0x3C	TX_TL	I2C Transmit FIFO Threshold Register
0x40	CLR_INTR	Clear Combined and Individual Interrupt Register
0x44	CLR_RX_UNDER	Clear RX_UNDER Interrupt Register
0x48	CLR_RX_OVER	Clear RX_OVER Interrupt Register

Offset	Register	Description
0x4C	CLR_TX_OVER	Clear TX_OVER Interrupt Register
0x50	CLR_RD_REQ	Clear RD_REQ Interrupt Register
0x54	CLR_TX_ABRT	Clear TX_ABRT Interrupt Register
0x58	CLR_RX_DONE	Clear RX_DONE Interrupt Register
0x5C	CLR_ACTIVITY	Clear ACTIVITY Interrupt Register
0x60	CLR_STOP_DET	Clear STOP_DET Interrupt Register
0x64	CLR_START_DET	Clear START_DET Interrupt Register
0x6C	ENABLE	I2C Enable Register
0x70	STATUS	I2C Status Register
0x74	TXFLR	I2C Transmit FIFO Level Register
0x78	RXFLR	I2C Receive FIFO Level Register
0x7C	SDA_HOLD	I2C SDA Hold Time Length Register
0x80	TX_ABRT_SOURCE	I2C Transmit Abort Source Register
0x84	SLV_DATA_NACK_ONLY	Generate Slave Data NACK Register
0x88	DMA_CR	DMA Control Register
0x8C	DMA_TDLR	DMA Transmit Data Level Register
0x90	DMA_RDLR	I2C Receive Data Level Register
0x94	SDA_SETUP	I2C SDA Setup Register
0xA0	FS_SPKLEN	I2C SS, FS or FM+ spike suppression limit
0xA4	HS_SPKLEN	I2C HS spike suppression limit register
0xA8	CLR_RESTART_DET	Clear RESTART_DET Interrupt Register
0xAC	SCL_STUCK_AT_LOW_TIMEOUT	I2C SCL Stuck at Low Timeout
0xB0	SDA_STUCK_AT_LOW_TIMEOUT	I2C SDA Stuck at Low Timeout
0xB4	CLR_SCL_STUCK_DET	Clear SCL Stuck at Low Detect Interrupt Register

8.4.2. CONTROL (Offset 0x00)

table 8-3 CONTROL Register

Bit	Name	Access	Default	Description
31:12	Reserved	R	0x0	Reserved
11	BUS_DETEC_FEATURE_CTRL	R/W	1'b0	In Master mode:

Bit	Name	Access	Default	Description
				0: Bus Detect Feature is Disabled. 1: Bus Detect Feature is Enabled. Detect whether SCL or SDA is Stuck low Note: Only Supports I2C0, Only in Master mode
10	STOP_DET_IF_MASTER_ACTIVE	R/W	1'b0	In Master mode: 0: issues the STOP_DET irrespective of whether master is active or not. 1: issues the STOP_DET interrupt only when master is active.
9	RX_FIFO_FULL_HLD_CTRL	R/W	1'b0	This bit controls whether DW_apb_i2c should hold the bus when the Rx FIFO is physically full to its RX_BUFFER_DEPTH 0: Overflow when RX_FIFO is full 1: Hold bus when RX_FIFO is full
8	TX_EMPTY_CTRL	R/W	1'b0	This bit controls the generation of the TX_EMPTY interrupt 0: Default behaviour of TX_EMPTY interrupt 1: Controlled generation of TX_EMPTY interrupt
7	STOP_DET_IFADDRESSED	R/W	1'b0	Slave mode: 0: The slave can always generate a STOP_DET interrupt. 1: The slave can generate a STOP_DET interrupt only after the address is matched.
6	SLAVE_DISABLE	R/W	1'b1	This bit controls whether I2C has its slave disabled 0: Slave mode is enabled 1: Slave mode is disabled

Bit	Name	Access	Default	Description
				Note: MASTER_MODE should be set to 0 when DISABLE=0.
5	RESTART_EN	R/W	1'b1	Determines whether RESTART conditions may be sent when acting as a master.. 0: Master restart disabled 1: Master restart enabled
4	Reserved	R/W	1'b0	Reserved
3	10BITADDR_SLAVE	R/W	1'b1	When acting as a slave, this bit controls whether the DW_apb_i2c responds to 7- or 10-bit addresses. 0: Slave 7Bit addressing 1: Slave 10Bit addressing
2:1	SPEED	R/W	2'b11	Speed mode 01: Standard(100kbit/s) 10: Fast(400 kbit/s <= Speed<=1000 kbit/s) 11: High Speed(3.4Mbit/s)
0	MASTER_MODE	R/W	1'b1	This bit controls whether the DW_apb_i2c master is enabled. 0: Master mode is disabled 1: Master mode is enabled NOTE: Software should ensure that if this bit is written with '1' then bit 6 should also be written with a '1'.

8.4.3. TAR (Offset 0x04)

table 8-4 TAR Register

Bit	Name	Access	Default	Description
31:14	Reserved	R	0x0	Reserved
13	DEVICE_ID	R/W	1'b0	If bit 11 (SPECIAL) is set to 1, then this

Bit	Name	Access	Default	Description
				bit indicates whether a Device-ID of a particular slave mentioned in TAR[9:0] is to be performed by the DW_apb_i2c Master. 0: Disables programming of DEVICE-ID transmission 1: Enables programming of DEVICE-ID transmission
12	10BITADDR_MASTER	R/W	1'b1	This bit controls whether the DW_apb_i2c starts its transfers in 7- or 10-bit addressing mode when acting as a master. 0: Address 7Bit transmission format 1: Address 10Bit transmission format
11	SPECIAL	R/W	1'b0	This bit indicates whether software performs a Device-ID or General Call or START BYTE command. 0: : Disables programming of GENERAL_CALL or START_BYTE transmission 1: Enables programming of GENERAL_CALL or START_BYTE transmission
10	Reserved	R	0x0	Reserved
9:0	TAR	R/W	10'h55	This is the target address for any master transaction.

8.4.4. SAR (Offset 0x08)

table 8-5 SAR Register

Bit	Name	Access	Default	Description
31:10	Reserved	R	0x0	Reserved
9:0	SAR	R/W	10'h00	SAR holds the slave address when the

Bit	Name	Access	Default	Description
				I2C is operating as a slave. For 7-bit addressing, only IC_SAR[6:0] is used.

8.4.5. HS_MADDR (Offset 0x0C)

table 8-6 HS_MADDR Register

Bit	Name	Access	Default	Description
31:3	Reserved	R	0x0	Reserved
2:0	HS_MAR	R/W	3'b000	This bit field holds the value of the I2C HS mode master code. HS-mode master codes are reserved 8-bit codes (00001xxx) that are not used for slave addressing or other purposes.

8.4.6. DATA_CMD (Offset 0x10)

table 8-7 DATA_CMD Register

Bit	Name	Access	Default	Description
31:12	Reserved	R	0x0	Reserved
11	FIRST_DATA_BYTE	R	1'b0	Indicates the first data byte received after the address phase for receive transfer in Master receiver or Slave receiver mode. 0: The first byte after receiving the address. 1: The first byte after receiving the address has not been received.
10	RESTART	W	1'b0	This bit controls whether a RESTART is issued before the byte is sent or received. 0: Do not Issue RESTART before this

Bit	Name	Access	Default	Description
				command 1: Issue RESTART before this command
9	STOP	W	1'b0	This bit controls whether a STOP is issued after the byte is sent or received. 0: Do not Issue STOP after this command 1: Issue STOP after this command
8	CMD	W	1'b0	This bit controls whether a read or a write is performed. 0: Master Write Command 1: Master Read Command
7:0	DAT	R/W	8'h00	This register contains the data to be transmitted or received on the I2C bus.

8.4.7. SS_SCL_HCNT (Offset 0x14)

table 8-8 SS_SCL_HCNT Register

Bit	Name	Access	Default	Description
31:16	Reserved	R	0x0	Reserved
15:0	SS_SCL_HCNT	R/W	16'h190	This register sets the SCL clock high-period count for standard speed. The minimum valid value is 6. Note: This register can be written only when the I2C interface is disabled which corresponds to the ENABLE[0] register being set to 0. Writes at other times have no effect.

Please refer to the [I2C Timing Control](#) chapter for details.

8.4.8. SS_SCL_LCNT (Offset 0x18)

table 8-9 SS_SCL_LCNT Register

Bit	Name	Access	Default	Description
31:16	Reserved	R	0x0	Reserved

Bit	Name	Access	Default	Description
15:0	SS_SCL_LCNT	R/W	16'h1D6	This register must be set before any I2C bus transaction can take place to ensure proper I/O timing. This register sets the SCL clock low period count for standard speed. The minimum valid value is 8. Note: This register can be written only when the I2C interface is disabled which corresponds to the ENABLE[0] register being set to 0. Writes at other times have no effect.

Please refer to the [I2C Timing Control](#) chapter for details.

8.4.9. FS_SCL_HCNT (Offset 0x1C)

table 8-10 FS_SCL_HCNT Register

Bit	Name	Access	Default	Description
31:16	Reserved	R	0x0	Reserved
15:0	FS_SCL_HCNT	R/W	16'h3C	This register must be set before any I2C bus transaction can take place to ensure proper I/O timing. This register sets the SCL clock high-period count for fast mode or fast mode plus. The minimum valid value is 6. Note: This register can be written only when the I2C interface is disabled, which corresponds to the IC_ENABLE[0] register being set to 0. Writes at other times have no effect.

Please refer to the [I2C Timing Control](#) chapter for details.

8.4.10. FS_SCL_LCNT (Offset 0x20)

table 8-11 FS_SCL_LCNT Register

Bit	Name	Access	Default	Description
31:16	Reserved	R	0x0	Reserved

Bit	Name	Access	Default	Description
15:0	FS_SCL_LCNT	R/W	16'h82	This register must be set before any I2C bus transaction can take place to ensure proper I/O timing. This register sets the SCL clock low period count for fast speed. The minimum valid value is 8. Note: This register can be written only when the I2C interface is disabled, which corresponds to the IC_ENABLE[0] register being set to 0. Writes at other times have no effect.

Please refer to the [I2C Timing Control](#) chapter for details.

8.4.11. HS_SCL_HCNT(Offset 0x24)

table 8-12 HS_SCL_HCNT Register

Bit	Name	Access	Default	Description
31:16	Reserved	R	0x0	Reserved
15:0	HS_SCL_HCNT	R/W	16'h6	This register must be set before any I2C bus transaction can take place to ensure proper I/O timing. This register sets the SCL clock high period count for high speed. The minimum valid value is 6. Note: This register can be written only when the I2C interface is disabled, which corresponds to the ENABLE[0] register being set to 0. Writes at other times have no effect.

Please refer to the [I2C Timing Control](#) chapter for details.

8.4.12. HS_SCL_LCNT (Offset 0x28)

table 8-13 HS_SCL_LCNT Register

Bit	Name	Access	Default	Description
31:16	Reserved	R	0x0	Reserved
15:0	HS_SCL_LCNT	R/W	16'h10	This register must be set before any I2C bus transaction can take place to ensure proper I/O timing. This register sets the SCL clock low period count for high speed. The minimum valid value is 8. Note: This register can be written only when the I2C interface is disabled, which corresponds to the IC_ENABLE[0] register being set to 0. Writes at other times have no effect.

Please refer to the [I2C Timing Control](#) chapter for details.

8.4.13. INTR_STAT (Offset 0x2C)

table 8-14 INTR_STAT Register

Bit	Name	Access	Default	Description
31:15	Reserved	R	0x0	Reserved
14	SCL_STUCK_AT_LOW	R	1'b0	Indicates whether the SCL Line is stuck at low for the IC_SCL_STUCK_LOW_TIMEOUT number of ic_clk periods. 0: Status inactive 1: Status active (Only I2C0 supports)
13	MASTER_ON_HOLD	R	1'b0	Indicates whether master is holding the bus and TX FIFO is empty. 0: MASTER_ON_HOLD interrupt is inactive 1: MASTER_ON_HOLD interrupt is active

Bit	Name	Access	Default	Description
12	RESTART_DET	R	1'b0	Indicates whether a RESTART condition has occurred on the I2C interface when DW_apb_i2c is operating in Slave mode and the slave is being addressed. 0: RESTART_DET interrupt is active 1: RESTART_DET interrupt is inactive
11	Reserved	R	1'b0	Reserved
10	START_DET	R	1'b0	Indicates whether a START or RESTART condition has occurred on the I2C interface regardless of whether DW_apb_i2c is operating in slave or master mode. 0: START_DET interrupt is inactive 1: START_DET interrupt is active
9	STOP_DET	R	1'b0	Indicates whether a STOP condition has occurred on the I2C interface regardless of whether DW_apb_i2c is operating in slave or master mode. Slave Mode: If STOP_DET_IFADDRESSED = 1, The In Slave Mode: STOP_DET interrupt is generated only when the transmitted address matches the slave address (SAR). If STOP_DET_IFADDRESSED = 0, the STOP_DET interrupt is issued irrespective of whether it is being addressed. In Master Mode: if STOP_DET_IF_MASTER_ACTIVE = 1, the STOP_DET interrupt will be issued only if Master is active. If STOP_DET_IF_MASTER_ACTIVE = 0, the STOP_DET interrupt will be issued irrespective of whether master is active or

Bit	Name	Access	Default	Description
				not. 0: STOP_DET interrupt is inactive 1: STOP_DET interrupt is active
8	ACTIVITY	R	1'b0	This bit captures activity and stays set until it is cleared. 0: Status inactive 1: Status active ----- There are three ways to clear it: 1、Reading the CLR_ACTIVITY register 2、Reading the IC_CLR_INTR register 3、I2C Disable Note: Once this bit is set, it stays set unless one of the three methods is used to clear it.
7	RX_DONE		1'b0	When the I2C is acting as a slave-transmitter, this bit is set to 1 if the master does not acknowledge a transmitted byte. This occurs on the last byte of the transmission, indicating that the transmission is done. 0: RX_DONE interrupt is inactive 1: RX_DONE interrupt is active
6	TX_ABRT	R	1'b0	This bit indicates if I2C, as an I2C transmitter, is unable to complete the intended actions on the contents of the transmit FIFO. the IC_TX_ABRT_SOURCE register indicates the reason why the transmit abort takes places. 0: TX_ABRT interrupt is inactive 1: TX_ABRT interrupt is active
5	RD_REQ	R	1'b0	his bit is set to 1 when I2C is acting as a slave and another I2C master is attempting to read data from I2C. The I2C holds the I2C bus in a

Bit	Name	Access	Default	Description
				wait state (SCL=0) until this interrupt is serviced, which means that the slave has been addressed by a remote master that is asking for data to be transferred. The processor must respond to this interrupt and then write the requested data to the DATA_CMD register. This bit is set to 0 just after the processor reads the CLR_RD_REQ register. 0: Status inactive 1: Status active
4	TX_EMPTY	R	1'b0	When TX_EMPTY_CTRL = 0: This bit is set to 1 when the transmit buffer is at or below the threshold value set in the TX_TL register. When TX_EMPTY_CTRL = 1: This bit is set to 1 when the transmit buffer is at or below the threshold value set in the TX_TL register and the transmission of the address/data from the internal shift register for the most recently popped command is completed. 0: TX_EMPTY interrupt is inactive 1: TX_EMPTY interrupt is active Note: It is automatically cleared by hardware when the buffer level goes above the threshold.
3	TX_OVER	R	1'b0	Set during transmit if the transmit buffer is filled to TX_BUFFER_DEPTH and the processor attempts to issue another I2C command by writing to the DATA_CMD register. 0: TX_OVER interrupt is inactive 1: TX_OVER interrupt is active
2	RX_FULL	R	1'b0	Set when the receive buffer reaches or goes

Bit	Name	Access	Default	Description
				above the RX_TL threshold in the RX_TL register. Note: It is automatically cleared by hardware when buffer level goes below the threshold. 0: RX_FULL interrupt is inactive 1: RX_FULL interrupt is active
1	RX_OVER	R	1'b0	Set if the receive buffer is completely filled to RX_BUFFER_DEPTH and an additional byte is received from an external I2C device. 0: Status inactive 1: Status active Note: if RX_FIFO_FULL_HLD_CTRL = 1, when Rx FIFO is full, SCL keep its level, No more data will be received. If RX_FIFO_FULL_HLD_CTRL = 0, when Rx FIFO is full, data will continue to be received but all received data will be lost.
0	RX_UNDER	R	1'b0	Set if the processor attempts to read the receive buffer when it is empty by reading from the DATA_CMD register.

8.4.14. INTR_MASK (Offset 0x30)

table 8-15 INTR_MASK Register

Bit	Name	Access	Default	Description
31:16	Reserved	R/W	0x0	Reserved
14	SCL_STUCK_AT_LOW	R/W	1'b1	This bit masks the corresponding interrupt bit in INTR_STAT register. 0: interrupt is masked 1: interrupt is unmasked Note: after mask, corresponding bit in the STAT register will be masked, not be set.
13	MASTER_ON_HOLD	R/W	1'b0	
12	RESTART_DET	R/W	1'b0	
11	Reserved	R/W	1'b0	
10	START_DET	R/W	1'b0	
9	STOP_DET	R/W	1'b0	
8	ACTIVITY	R/W	1'b0	

Bit	Name	Access	Default	Description
7	RX_DONE	R/W	1'b1	
6	TX_ABRT	R/W	1'b1	
5	RD_REQ	R/W	1'b1	
4	TX_EMPTY	R/W	1'b1	
3	TX_OVER	R/W	1'b1	
2	RX_FULL	R/W	1'b1	
1	RX_OVER	R/W	1'b1	
0	RX_UNDER	R/W	1'b1	

8.4.15. RAW_INTR_STAT (Offset 0x34)

table 8-16 RAW_INTR_STAT Register

Bit	Name	Access	Default	Description
31:16	Reserved	R	0x0	Reserved
14	SCL_STUCK_AT_LOW	R	1'b0	I2C Raw Interrupt Status Register 0: Interrupt is inactive. 1: interrupt is active Note: Unlike the IC_INTR_STAT register, these bits are not masked so they always show the true status of the I2C.
13	MASTER_ON_HOLD	R	1'b0	
12	RESTART_DET	R	1'b0	
11	Reserved	R	1'b0	
10	START_DET	R	1'b0	
9	STOP_DET	R	1'b0	
8	ACTIVITY	R	1'b0	
7	RX_DONE	R	1'b0	
6	TX_ABRT	R	1'b0	
5	RD_REQ	R	1'b0	
4	TX_EMPTY	R	1'b0	
3	TX_OVER	R	1'b0	
2	RX_FULL	R	1'b0	
1	RX_OVER	R	1'b0	
0	RX_UNDER	R	1'b0	

8.4.16. RX_TL (Offset 0x38)

table 8-17 RX_TL Register

Bit	Name	Access	Default	Description
31:8	Reserved	R	0x0	Reserved
7:0	RX_TL	R/W	8'h00	Receive FIFO Threshold Level.

8.4.17. TX_TL (Offset 0x3C)

table 8-18 TX_TL Register

Bit	Name	Access	Default	Description
31:8	Reserved	R	0x0	Reserved
7:0	TX_TL	R/W	8'h00	Transmit FIFO Threshold Level.

8.4.18. CLR_INTR (Offset 0x40)

table 8-19 CLR_INTR Register

Bit	Name	Access	Default	Description
31:1	Reserved	R	0x0	Reserved
0	CLR_INTR	R	1'b0	Read this register to clear the combined interrupt, all individual interrupts, and the IC_TX_ABRT_SOURCE register. This bit does not clear hardware clearable interrupts but software clearable interrupts.

8.4.19. CLR_RX_UNDER (Offset 0x44)

table 8-20 CLR_RX_UNDER Register

Bit	Name	Access	Default	Description
-----	------	--------	---------	-------------

31:1	Reserved	R	0x0	Reserved
0	CLR_RX_UNDER	R	1'b0	Read this register to clear the RX_UNDER interrupt (bit 0) of the RAW_INTR_STAT register.

8.4.20. CLR_RX_OVER (Offset 0x48)

table 8-21 CLR_RX_OVER Register

Bit	Name	Access	Default	Description
31:1	Reserved	R	0x0	Reserved
0	CLR_RX_OVER	R	1'b0	Read this register to clear the RX_OVER interrupt (bit 1) of the RAW_INTR_STAT register.

8.4.21. CLR_TX_OVER (Offset 0x4C)

table 8-22 CLR_TX_OVER Register

Bit	Name	Access	Default	Description
31:1	Reserved	R	0x0	Reserved
0	CLR_TX_OVER	R	1'b0	Read this register to clear the TX_OVER interrupt (bit 3) of the RAW_INTR_STAT register.

8.4.22. CLR_RD_REQ (Offset 0x50)

table 8-23 CLR_RD_REQ Register

Bit	Name	Access	Default	Description
31:1	Reserved	R	0x0	Reserved
0	CLR_RD_REQ	R	1'b0	Read this register to clear the RD_REQ interrupt (bit 5) of the RAW_INTR_STAT register.

8.4.23. CLR_TX_ABRT(Offset 0x54)

table 8-24 CLR_TX_ABRT Register

Bit	Name	Access	Default	Description
31:1	Reserved	R	0x0	Reserved
0	CLR_TX_ABRT	R	1'b0	Read this register to clear the TX_ABRT interrupt (bit 6) of the RAW_INTR_STAT register, and the TX_ABRT_SOURCE register.

8.4.24. CLR_RX_DONE (Offset 0x58)

table 8-25 CLR_RX_DONE Register

Bit	Name	Access	Default	Description
31:1	Reserved	R	0x0	Reserved
0	CLR_RX_DONE	R	1'b0	Read this register to clear the RX_DONE interrupt (bit 7) of the RAW_INTR_STAT register.

8.4.25. CLR_ACTIVITY (Offset 0x5C)

table 8-26 CLR_ACTIVITY Register

Bit	Name	Access	Default	Description
31:1	Reserved	R	0x0	Reserved
0	CLR_ACTIVITY	R	1'b0	Reading this register clears the ACTIVITY interrupt if the I2C is not active anymore. If the I2C module is still active on the bus, the ACTIVITY interrupt bit continues to be set. It is automatically cleared by hardware if the module is disabled and if there is no further activity on the bus.

8.4.26. CLR_STOP_DET (Offset 0x60)

table 8-27 CLR_STOP_DET Register

Bit	Name	Access	Default	Description
31:1	Reserved	R	0x0	Reserved
0	CLR_STOP_DET	R	1'b0	Read this register to clear the STOP_DET interrupt (bit 9) of the RAW_INTR_STAT register.

8.4.27. CLR_START_DET (Offset 0x64)

table 8-28 CLR_START_DET Register

Bit	Name	Access	Default	Description
31:1	Reserved	R	0x0	Reserved
0	CLR_START_DET	R	1'b0	Read this register to clear the START_DET interrupt (bit 10) of the IC_RAW_INTR_STAT register.

8.4.28. ENABLE (Offset 0x6C)

table 8-29 ENABLE Register

Bit	Name	Access	Default	Description
31:4	Reserved	R	0x0	Reserved
3	SDA_STUCK_RECOVERY	R/W	1'b0	If SDA is stuck at low indicated through the TX_ABORT interrupt (TX_ABORT_SOURCE[17]), then this bit is used as a control knob to initiate the SDA Recovery Mechanism. 0: Master disabled the SDA stuck at low recovery mechanism. 1: Master initiates the SDA stuck at low recovery mechanism. (Only I2C0 supports)
2	TX_CMD_BLOCK	R/W	1'b0	In Master mode: 0: The transmission of data starts on I2C bus automatically, as soon as the first data is available in the Tx FIFO. 1: Blocks the transmission of data on I2C bus even if Tx FIFO has data to transmit. Note: To block the execution of Master commands, set the TX_CMD_BLOCK bit only when Tx FIFO is empty
1	ABORT	R/W	1'b0	The software can abort the I2C transfer in master mode by setting this bit. In response to an ABORT, the controller issues a STOP and flushes the Tx FIFO after completing the current transfer, then sets the TX_ABORT interrupt after the abort operation. The ABORT bit is cleared automatically after the abort operation. 0: ABORT operation not in progress 1: ABORT operation in progress Note: The software can set this bit only when ENABLE is already set; otherwise,

Bit	Name	Access	Default	Description
				the controller ignores any write to ABORT bit.
0	ENABLE	R/W	1'b0	Controls whether I2C is enabled. 0: I2C Disable 1: I2C Enable

8.4.29. STATUS (Offset 0x70)

table 8-30 STATUS Register

Bit	Name	Access	Default	Description
31:12	Reserved	R	0x0	Reserved
11	SDA_STUCK_NOT_RECOVERED	R	1'b0	This bit indicates that SDA stuck at low is not recovered after the recovery mechanism. In Slave mode, this register bit is not applicable. 0: SDA Stuck at low is not recovered after recovery mechanism. 1: SDA Stuck at low is recovered after recovery mechanism. (Only I2C0 supports)
10	SLV_HOLD_RX_FIFO_FULL	R	1'b0	This bit indicates the BUS Hold in Slave mode due to Rx FIFO is Full. 0: Slave is not holding the bus or Bus hold is not due to Rx FIFO is full 1: Slave holds the bus due to Rx FIFO is full
9	SLV_HOLD_TX_FIFO_EMPTY	R	1'b0	This bit indicates the BUS Hold in Slave mode for the Read request when the Tx FIFO is empty. 0: Slave is not holding the bus or Bus hold is not due to Tx FIFO is empty

Bit	Name	Access	Default	Description
				1: Slave holds the bus due to Tx FIFO is empty
8	MST_HOLD_RX_FIFO_FULL	R	1'b0	This bit indicates the BUS Hold in Master mode due to Rx FIFO is Full. 0: Master is not holding the bus or Bus hold is not due to Rx FIFO is full 1: Master holds the bus due to Rx FIFO is full
7	MST_HOLD_TX_FIFO_EMPTY	R	1'b0	This bit indicates the BUS hold when the master holds the bus because of the Tx FIFO being empty. 0: Master is not holding the bus or Bus hold is not due to Tx FIFO is empty 1: Master holds the bus due to Tx FIFO is empty
6	SLV_ACTIVITY	R	1'b0	Slave FSM Activity Status. When the Slave Finite State Machine (FSM) is not in the IDLE state, this bit is set. 0: Slave is idle 1: Slave not idle
5	MST_ACTIVITY	R	1'b0	Master FSM Activity Status. When the Master Finite State Machine (FSM) is not in the IDLE state, this bit is set. 0: Master is idle 1: Master not idle
4	RFF	R	1'b0	Receive FIFO Completely Full. When the receive FIFO is completely full, this bit is set. 0: Rx FIFO not full 1: Rx FIFO is full
3	RFNE	R	1'b0	Receive FIFO Not Empty.

Bit	Name	Access	Default	Description
				0: Rx FIFO is empty 1: Rx FIFO not empty
2	TFE	R	1'b0	Transmit FIFO Completely Empty. 0: Tx FIFO not empty 1: Tx FIFO is empty
1	TFNF	R	1'b0	Transmit FIFO Not Full. 0: Tx FIFO is full 1: Tx FIFO not full
0	ACTIVITY	R	1'b0	I2C Activity Status. 0: I2C is idle 1: I2C is active

8.4.30. TXFLR (Offset 0x74)

table 8-31 TXFLR Register

Bit	Name	Access	Default	Description
31:6	Reserved	R	0x0	Reserved
5:0	TXFLR	R	6'h0	Transmit FIFO Level.

8.4.31. RXFLR (Offset 0x78)

table 8-32 RXFLR Register

Bit	Name	Access	Default	Description
31:6	Reserved	R	0x0	Reserved
5:0	RXFLR	R	6'h0	Receive FIFO Level.

8.4.32. SDA_HOLD(Offset 0x7C)

table 8-1 SDA_HOLD Register

Bit	Name	Access	Default	Description
31:24	Reserved	R	0x0	Reserved
23:16	SDA_RX_HOLD	R/W	8'h0	Sets the required SDA hold time.
15:0	SDA_TX_HOLD	R/W	16'h1	Sets the required SDA hold time.

Please refer to the [I2C SDA_HOLD func](#) chapter for details.

8.4.33. TX_ABRT_SOURCE (Offset 0x80)

table 8-2 TX_ABRT_SOURCE Register

Bit	Name	Access	Default	Description
31:23	TX_FLUSH_CNT	R	9'h0	This field indicates the number of Tx FIFO Data Commands which are flushed due to TX_ABRT interrupt.
22:21	Reserved	R	2'b00	Reserved
20	ABRT_DEVICE_WRITE	R	1'b0	In Master mode. 0: Not terminated 1: During the host's transmission process, the CMD bit of DATA_CMD was modified, resulting in a change in direction and an error, and the transmission was aborted.
19	ABRT_DEVICE_SLVADDR_NOACK	R	1'b0	This is a master-mode-only bit. Master is initiating the DEVICE_ID transfer and the slave address sent was not acknowledged by any slave. 0: This abort is not generated 1: This abort is generated because of NOACK for Slave

Bit	Name	Access	Default	Description
				address (Only I2C0 supports)
18	Reserved	R	1'b0	Reserved
17	ABRT_SDA_STUCK_AT_LOW	R	1'b0	This is a master-mode-only bit. Master detects the SDA Stuck at low for the SDA_STUCK_AT_LOW_TIMEOUT. 0: This abort is not generated 1: This abort is generated because of Sda stuck at low for SDA_STUCK_AT_LOW_TIMEOUT value
16	ABRT_USER_ABRT	R	1'b0	This is a master-mode-only bit. Master has detected the transfer abort. 0: Transfer abort detected by master- scenario not present 1: Transfer abort detected by master
15	ABRT_SLVRD_INTX	R	1'b0	0: Not aborted 1: When slave transfer, write 1 to CMD bit of DATA_CMD register
14	ABRT_SLV_ARBLOST	R	1'b0	This field indicates that a Slave has lost the bus while transmitting data to a remote master. 0: Slave lost arbitration to remote master- scenario not present 1: Slave lost arbitration to remote master
13	ABRT_SLVFLUSH_TXFIFO	R	1'b0	This field specifies that the Slave

Bit	Name	Access	Default	Description
				has received a read command and some data exists in the TX FIFO, so the slave issues a TX_ABRT interrupt to flush old data in TX FIFO. 0: Slave flushes existing data in TX-FIFO upon getting read command-scenario not present 1: : Slave flushes existing data in TX-FIFO upon getting read command
12	ARB_LOST	R	1'b0	This field specifies that the Master has lost arbitration, or if TX_ABRT_SOURCE[14] is also set, then the slave transmitter has lost arbitration 0: Master or Slave-Transmitter lost arbitration- scenario not present 1: Master or Slave Transmitter lost arbitration
11	ABRT_MASTER_DIS	R	1'b0	This field indicates that the User tries to initiate a Master operation with the Master mode disabled. 0: User initiating master operation when MASTER disabled- scenario not present 1: User initiating master operation when MASTER disabled
10	ABRT_10B_RD_NORSTRT	R	1'b0	This field indicates that the restart is disabled and the master sends a read command in 10-bit addressing mode. 0: Master not trying to read in

Bit	Name	Access	Default	Description
				10Bit addressing mode when RESTART disabled 1: Master trying to read in 10Bit addressing mode when RESTART disabled
9	Reserved	R	1'b0	Reserved
8	ABRT_HS_NORSTRT	R	1'b0	This field indicates that the restart is disabled and the user is trying to use the master to transfer data in High-Speed mode. 0: User trying to switch Master to HS mode when RESTART disabled- scenario not present 1: User trying to switch Master to HS mode when RESTART disabled
7	Reserved	R	1'b0	Reserved
6	ABRT_SBYTE_ACKDET	R	1'b0	This field indicates that the Master is in High-Speed mode and the High Speed Master code was acknowledged (wrong behavior). 0: HS Master code ACKed in HS Mode- scenario not present 1: HS Master code ACKed in HS Mode
5:4	Reserved	R	2'b00	Reserved
3	ABRT_TXDATA_NOACK	R	1'b0	This field indicates the master-mode only bit. When the master receives an acknowledgement for the address, but when it sends data byte(s) following the address, it did not receive an

Bit	Name	Access	Default	Description
				acknowledge from the remote slave(s). 0: Transmitted data non-ACKed by addressed slave-scenario not present 1: Transmitted data not ACKed by addressed slave
2	ABRT_10ADDR2_NOACK	R	1'b0	This field indicates that the Master is in 10-bit address mode and that the second address byte of the 10-bit address was not acknowledged by any slave. 0: : This abort is not generated 1: Byte 2 of 10Bit Address not ACKed by any slave
1	ABRT_10ADDR1_NOACK	R	1'b0	This field indicates that the Master is in 10-bit address mode and the first 10-bit address byte was not acknowledged by any slave. 0: This abort is not generated 1: Byte 1 of 10Bit Address not ACKed by any slave
0	ABRT_7ADDR_NOACK	R	1'b0	This field indicates that the Master is in 7-bit addressing mode and the address sent was not acknowledged by any slave. 0: This abort is not generated 1: This abort is generated because of NOACK for 7-bit address

8.4.34. SLV_DATA_NACK_ONLY (Offset 0x84)

table 8-3 SLV_DATA_NACK_ONLY Register

Bit	Name	Access	Default	Description
31:1	Reserved	R	0x0	Reserved
0	NACK	R	1'b0	Generate NACK. This NACK generation only occurs when I2C is a slave-receiver. 0: Slave receiver generates NACK normally 1: Slave receiver generates NACK upon data reception only

8.4.35. DMA_CR (Offset 0x88)

table 8-4 DMA_CR Register

Bit	Name	Access	Default	Description
31:2	Reserved	R	0x0	Reserved
1	TDMAE	R/W	1'b0	Transmit DMA Enable. This bit enables/disables the transmit FIFO DMA channel. 0: Transmit FIFO DMA channel enabled 1: transmit FIFO DMA channel disabled
0	RDMAE	R/W	1'b0	Receive DMA Enable. This bit enables/disables the receive FIFO DMA channel. 0: Receive FIFO DMA channel disabled 1: Receive FIFO DMA channel enabled

8.4.36. DMA_TDLR (Offset 0x8C)

table 8-5 DMA_TDLR Register

Bit	Name	Access	Default	Description
31:6	Reserved	R	0x0	Reserved
5:0	DMATDL	R/W	6'h00	Transmit Data Level. This bit field controls the level at which a DMA request is made by the transmit logic. the dma_tx_req signal is generated When the number of valid data entries in the transmit FIFO is equal to or below this field value, and TDMAE = 1.

8.4.37. DMA_RDLR (Offset 0x90)

table 8-6 DMA_RDLR Register

Bit	Name	Access	Default	Description
31:6	Reserved	R	0x0	Reserved
5:0	DMARDL	R/W	6'h00	Receive Data Level. This bit field controls the level at which a DMA request is made by the receive logic. dma_rx_req is generated when the number of valid data entries in the receive FIFO is equal to or more than this field value + 1.

8.4.38. SDA_SETUP (Offset 0x94)

table 8-7 SDA_SETUP Register

Bit	Name	Access	Default	Description
31:8	Reserved	R	0x0	Reserved
7:0	SDA_SETUP	R/W	8'h64	The minimum value is 8, unit is APB_I2C_CLK.s

Bit	Name	Access	Default	Description
				Please refer to tSU:DAT of I2C timing in the I2C Timing Control chapter for details. Note: It is configured when I2C is not enabled (ENABLE=0), otherwise it is invalid.

8.4.39. FS_SPKLEN (Offset 0xA0)

table 8-8 FS_SPKLEN Register

Bit	Name	Access	Default	Description
31:8	Reserved	R	0x0	Reserved
7:0	FS_SPKLEN	R/W	8'h5	This register sets the duration, measured in ic_clk cycles, of the longest spike in the SCL or SDA lines that will be filtered out by the spike suppression logic. The minimum value is 1. Please refer to tSP of I2C timing in the I2C Timing Control chapter for details. Note: This register can be written only when the I2C interface is disabled which corresponds to the ENABLE[0] register being set to 0. Writes at other times have no effect.

8.4.40. HS_SPKLEN (Offset 0xA4)

table 8-9 HS_SPKLEN Register

Bit	Name	Access	Default	Description
31:8	Reserved	R	0x0	Reserved
7:0	HS_SPKLEN	R/W	8'h1	This register sets the duration, measured in ic_clk cycles, of the longest spike in the SCL or SDA lines that will be filtered out by the

Bit	Name	Access	Default	Description
				spike suppression logic. The minimum value is 1. Please refer to tSP of I2C timing in the I2C Timing Control chapter for details. Note: This register can be written only when the I2C interface is disabled which corresponds to the ENABLE[0] register being set to 0. Writes at other times have no effect.

8.4.41. CLR_RESTART_DET (Offset 0xA8)

table 8-10 CLR_RESTART_DET Register

Bit	Name		Access	Default	Description
31:1	Reserved		R	0x0	Reserved
0	CLR_RESTART_DET		R	1'b0	Read this register to clear the RESTART_DET interrupt of RAW_INTR_STAT register.

8.4.42. SCL_STUCK_AT_LOW_TIMEOUT (Offset 0xAC)

table 8-11 SCL_STUCK_AT_LOW_TIMEOUT Register

Bit	Name	Access	Default	Description
31:0	SCL_STUCK_LOW_TIMEOUT	R/W	0xFFFFFFFF	generate the interrupt to indicate SCL stuck at low (SCL_STUCK_AT_LOW) if it detects the SCL stuck at low for the SCL_STUCK_LOW_TIMEOUT in units of clk period.

8.4.43. SDA_STUCK_AT_LOW_TIMEOUT (Offset 0xB0)

table 8-12 SDA_STUCK_AT_LOW_TIMEOUT Register

Bit	Name	Access	Default	Description
31:0	SDA_STUCK_LOW_TIMEOUT	R/W	0xFFFFFFFF	I2C initiates the recovery of SDA line through enabling the SDA_STUCK_RECOVERY_EN (ENABLE[3]) register bit, if it detects the SDA stuck at low for the SDA_STUCK_LOW_TIMEOUT in units of clk period.

8.4.44. CLR_SCL_STUCK_DET(Offset 0xB4)

table 8-13 CLR_SCL_STUCK_DET Register

Bit	Name	Access	Default	Description
31:1	Reserved	R	0x0	Reserved
0	CLR_SCL_STUCK_DET	R	1'b0	Read this register to clear the SCL_STUCT_AT_LOW interrupt of the RAW_INTR_STAT register.

8.5. Work Flow

8.5.1. Master Mode Operation

1. Disable I2C by writing 0 to bit0 of ENABLE register.
2. Write to the CONTROL register to set SPEED mode, 10BITADDR_MASTER, 7-bit or 10-bit addressing.
3. Write to the TAR register the address of the I2C device to be addressed (bits 9:0). Set 10BITADDR_MASTER, 7-bit or 10-bit addressing.

4. Only applicable for high-speed mode transfers. Write to the HS_MADDR register the desired master code for the I2C.
5. Enable I2C by writing a 1 to the bit0 of ENABLE register.
6. Now write transfer direction and data to be sent to the DATA_CMD register. If the DATA_CMD register is written before the I2C is enabled, the data and commands are lost as the buffers are kept cleared when I2C is disabled. This step generates the START condition and the address byte on the I2C. Once I2C is enabled and there is data in the TX FIFO, I2C starts reading the data.

Note:

- When the master performs the write operation, it writes the data to be transmitted to DAT field of DATA_CMD register. The CMD, RESTART, and STOP statuses need to be written at the same time. Keep CMD '0' unchanged during transmission; When a STOP or RESTART condition is required, write '1' to STOP or RESTART at the same time when writing the corresponding byte.
(Note: When data is transmitted, write the whole DATA_CMD register instead of just writing DAT field)
- When the master performs the read operation, if writes to DATA_CMD register, the I2C controller will automatically ignore the DAT part and only retain the CMD, RESTART, and STOP (all called as command). The number of commands to be written to Tx FIFO depends on the number of data to be received.
- (Note: Each byte is received by writing a command to DATA_CMD, and then the controller automatically stores it in Tx FIFO. The number of commands to be written depends on the number of data to be received.)
Note: If the interrupt is not enabled, you can query RAW_INTR_STAT register.

8.5.2. Slave Mode Operation

To use the I2C as a slave, perform the following steps:

1. Disable the I2C by writing a '0' to bit0 of the ENABLE register.

2. Write to the SAR register (bits 9:0) to set the slave address. This is the address to which the I2C responds.
3. Write to the CONTROL register to specify which type of addressing is supported (7- or 10-bit by setting bit 3). Enable the DW_apb_i2c in slave-only mode by writing a '0' into bit 6 (SLAVE_DISABLE) and a '0' to bit 0 (MASTER_MODE).
4. Enable the I2C by writing a '1' in bit 0 of the ENABLE register.

Note: If the interrupt is not enabled, you can query RAW_INTR_STAT register.

8.5.2.1. Slave-Transmitter Operation

1. The other I2C master device initiates an I2C transfer with an address that matches the slave address in the SAR register of the I2C.
2. The I2C acknowledges the sent address and recognizes the direction of the transfer to indicate that it is acting as a slave-transmitter. The I2C asserts the RD_REQ interrupt (bit 5 of the RAW_INTR_STAT register) and holds the SCL line low. It is in a wait state until software responds.
3. If there is any data remaining in the Tx FIFO before receiving the read request, then the I2C asserts a TX_ABRT interrupt (bit 6 of the RAW_INTR_STAT register) to flush the old data from the TX FIFO.
4. Software writes to the DATA_CMD register with the data to be written (by writing a '0' in bit 8).
5. Software must clear the RD_REQ and TX_ABRT interrupts (bits 5 and 6, respectively) of the RAW_INTR_STAT register before proceeding.
6. The I2C releases the SCL and transmits the byte.

Note: If the interrupt is not enabled, you can query RAW_INTR_STAT register.

8.5.2.2. Slave-Receiver Operation

1. The other I2C master device initiates an I2C transfer with an address that matches the I2C's slave address in the SAR register.

2. The I2C acknowledges the sent address and recognizes the direction of the transfer to indicate that the I2C is acting as a slave-receiver. I2C receives the transmitted byte and places it in the receive buffer.
3. I2C asserts the RX_FULL interrupt (RAW_INTR_STAT[2] register).
4. Software may read the byte from the DATA_CMD register (bits 7:0).

Note: If the interrupt is not enabled, you can query RAW_INTR_STAT register.

8.5.3. Timing Diagram

Start、Stop、Restart are controlled by [DATA_CMD](#) Register.

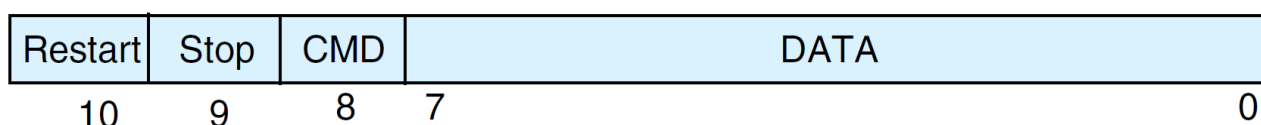


Figure 8-8 DATA_CMD Register Diagram

- DATA: Read/Write field; data retrieved from slave is read from this field; data to be sent to slave is written to this field.
- CMD: Write-only field; this bit determines whether transfer to be carried out is Read (CMD=1) or Write (CMD=0)
- Stop: Write-only field; this bit determines whether STOP is generated after data byte is sent or received
- Restart: Write-only field; this bit determines whether RESTART (or STOP followed by START in case of restart capability is not enabled) is generated before data byte is sent or received.

1. Master Transmitter -Tx FIFO Empties/STOP Generation

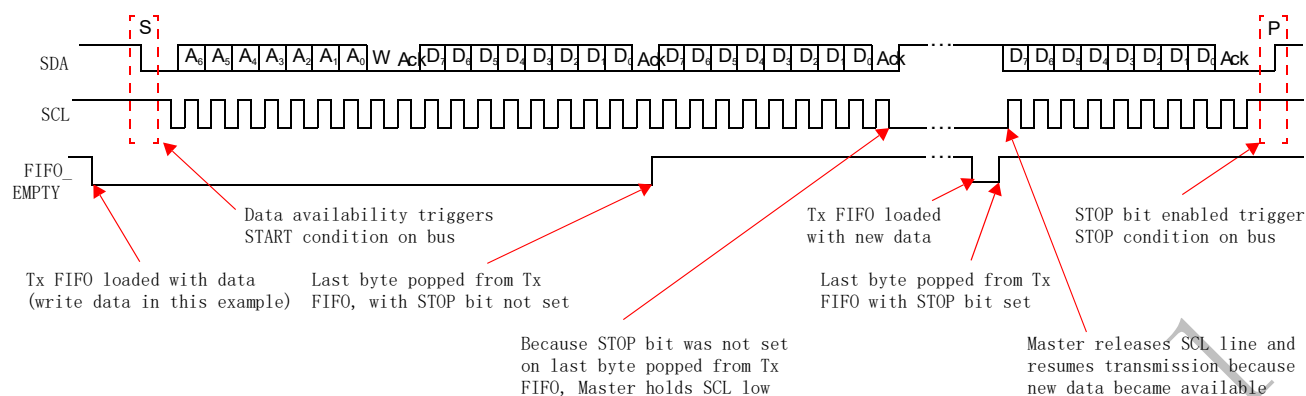


Figure 8-9 Master Transmitter-Tx FIFO Empties/STOP Generation

2. Master Receiver – Tx FIFO Empties/STOP Generation

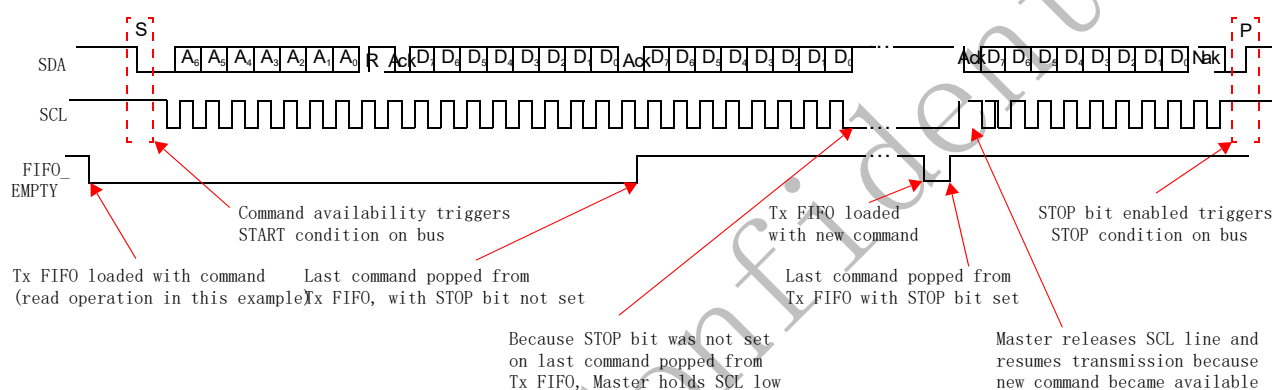


Figure 8-10 Master Receiver - Tx FIFO Empties/STOP Generation.

Figure 8-10 and Figure 8-11 illustrate configurations where the user can control the generation of RESTART conditions on the I2C bus. If bit 10 (Restart) of the DATA_CMD register is set and the restart capability is enabled (RESTART_EN=1), a RESTART is generated before the data byte is written to or read from the slave. If the restart capability is not enabled a STOP followed by a START is generated in place of the RESTART.

3. Master Transmitter — Restart Bit of IC_DATA_CMD Is Set

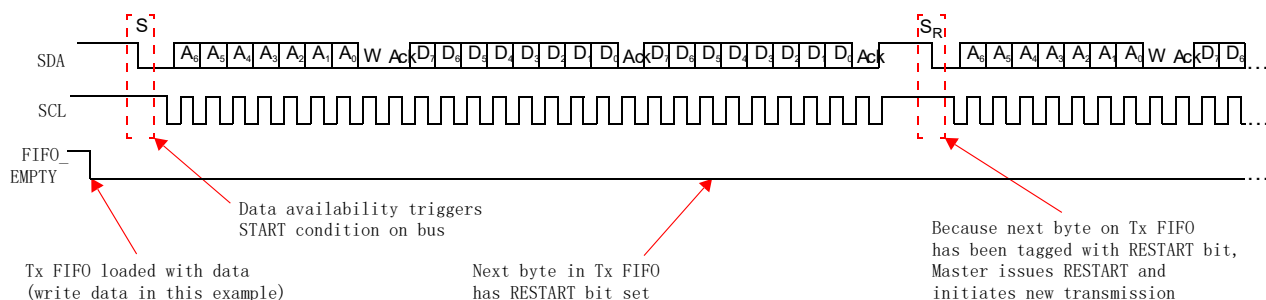


Figure 8-11 Master Transmitter-Restart Bit of IC_DATA_CMD Is Set

4. Master Receiver – Restart Bit of DATA CMD Is Set

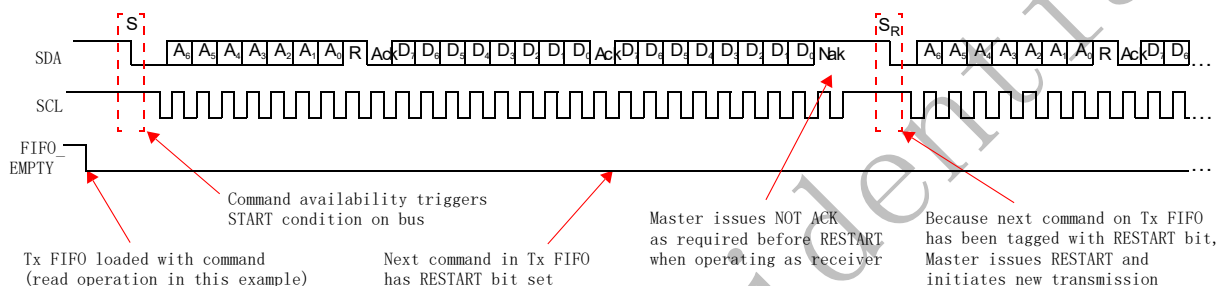


Figure 8-12 Master Receiver - Restart Bit of DATA CMD Is Set

5. Master Transmitter — Stop Bit of IC DATA CMD Set/Tx FIFO Not Empty

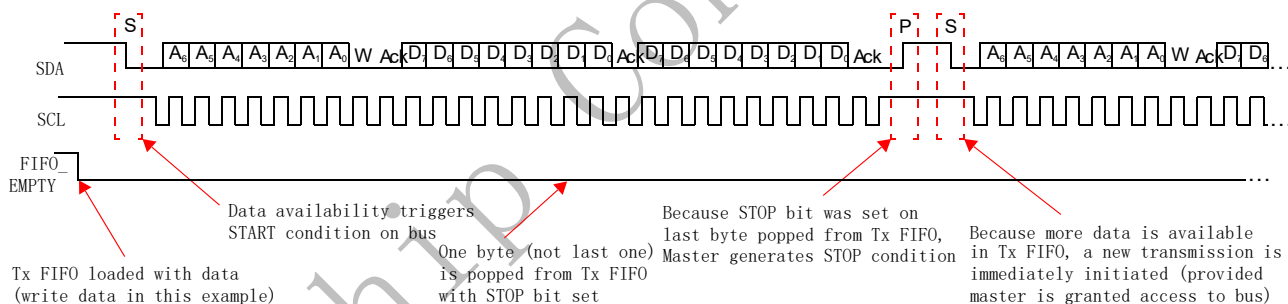


Figure 8-13 Master Transmitter — Stop Bit of IC_DATA_CMD Set/Tx FIFO Not Empty

6. Master Receiver — Stop Bit of IC_DATA_CMD Set/Tx FIFO Not Empty

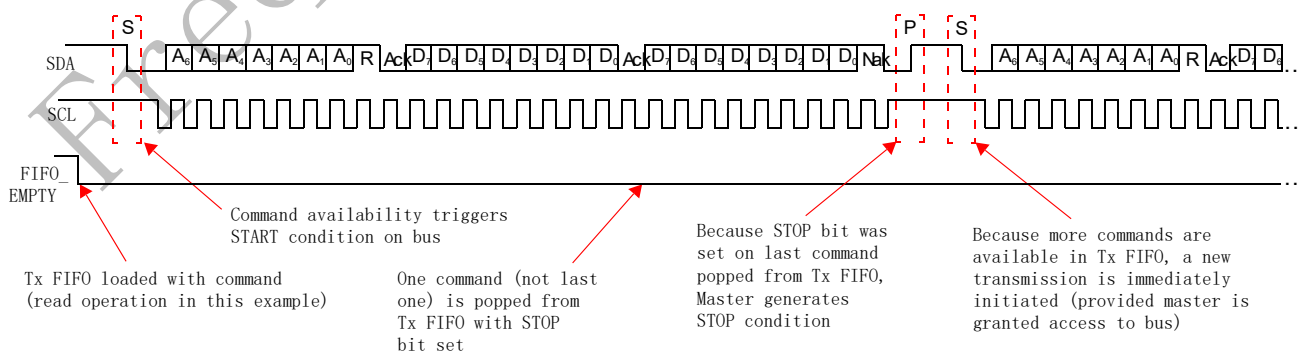


Figure 8-14 Master Receiver — Stop Bit of IC DATA_CMD Set/Tx FIFO Not Empty

9. I2S

9.1. Overview

Inter -IC- Sound(I2S) bus, Integrated Interchip Sound, a serial interface developed for digital audio to enable easy connectivity and ensure successful designs.

9.2. Feature

- MSB(left-aligned), PCM, Philips formats
- Only 16 bit effective data transmission with master mode
- Tx FIFO and Rx FIFO have 32 depth and 32bitwidth FIFO respectively
- Full-duplex data TX and RX

9.3. Block Diagram

9.4. Function Description

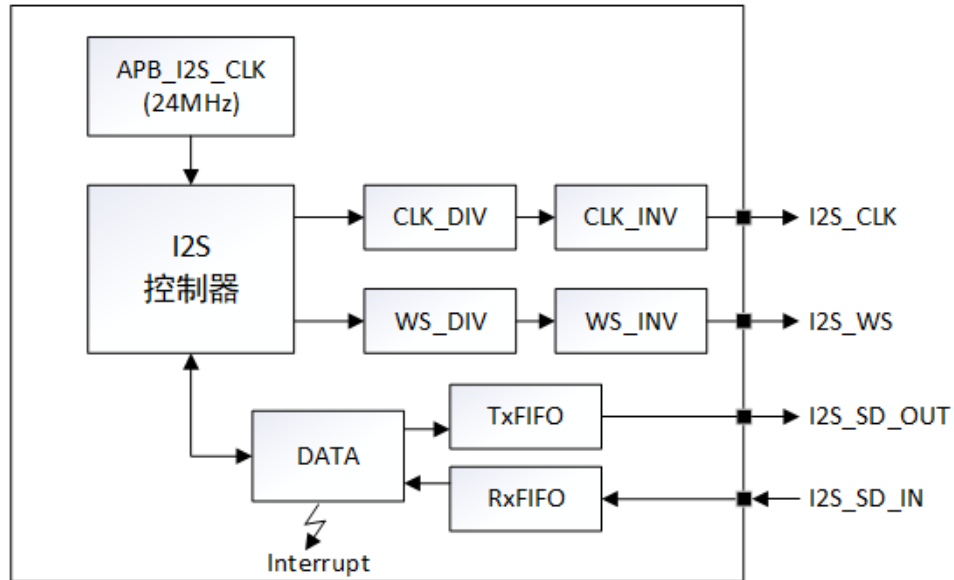


Figure 9-1 Block Diagram

9.4.1. CLK、WS Frequency Division

Clock signal output frequency division, and audio channel signal output frequency division.

CLK frequency division calculation formula:
$$CLK = \frac{APB_I2S_CLK}{2 \times (CLK_DIV + 1)}$$

WS frequency division calculation formula:
$$WS = \frac{CLK}{(WS_DIV + 1)}$$

Note: APB_I2S_CLK fixed at 24MHz.

9.4.2. WS_INV, DLY_EN, FMT Configuration

WS_INV = 0, DLY_EN = 0, FMT = 0

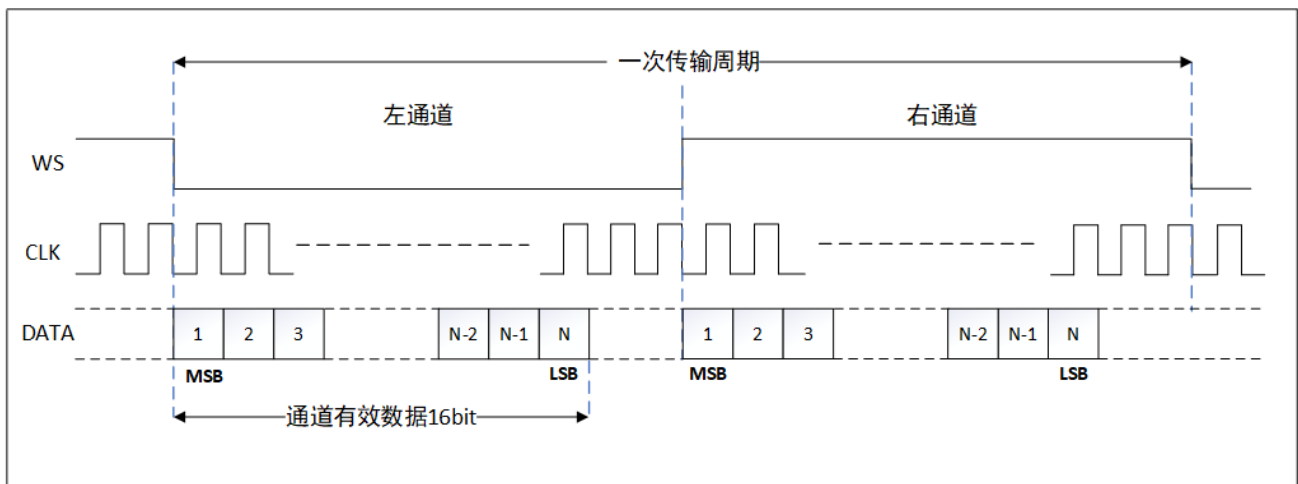


Figure 9-2 Timing Diagram (Left-aligned)

WS_INV = 1, DLY_EN = 0, FMT = 0

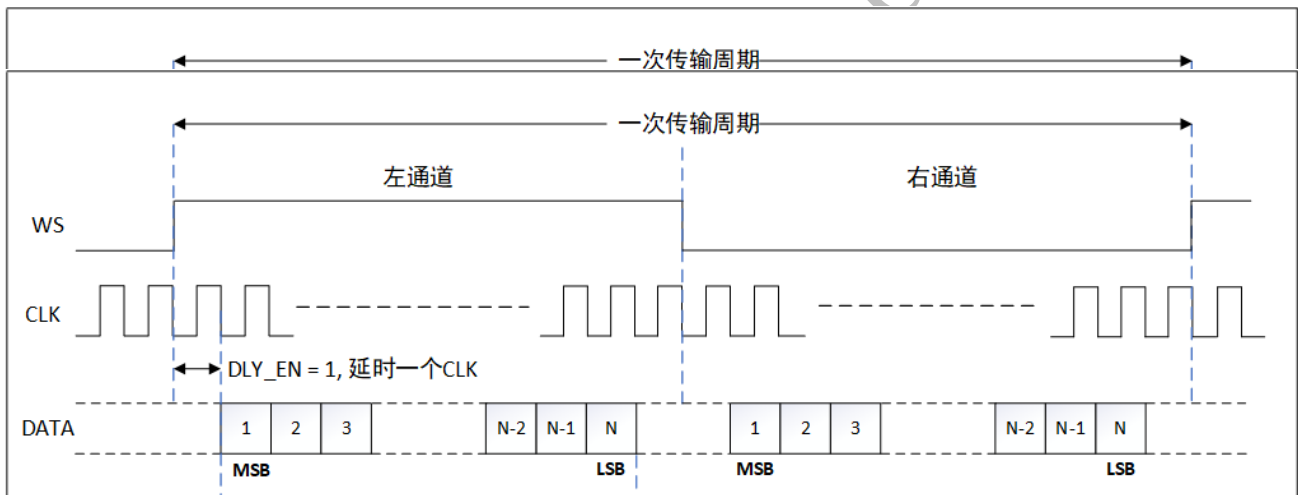


Figure 9-3 Timing Diagram (Left-aligned, WS inverted)

Figure 9-4 Timing Diagram (Left-aligned, WS inverted, DLY_EN = 1)

WS_INV = 1, DLY_EN = 1, FMT = 0

WS_INV = 0, DLY_EN = 0, FMT = 1

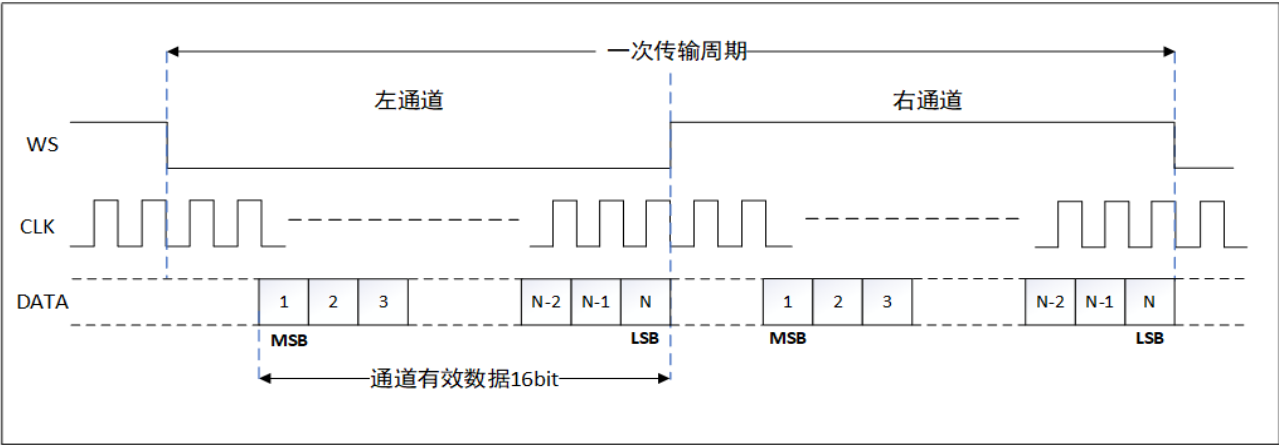


Figure 9-5 Timing Diagram (Right-aligned)

9.5. I2S Register

9.5.1. I2S Register Mapping

I2S Base Addr: 0x5007_0000

table 9-1 I2S Register Mapping

Offset	Name	Description
0x00	CONTROL	I2S Control Register
0x04	CLK_DIV	I2S Clock Division Register
0x08	WS_DIV	I2S WS Division Register
0x0C	DATA	I2S Data Register
0x10	STATUS	I2S Status Register
0x14	INTE	Interrupt Enable Register
0x20	FIFO_CFG	FIFO Configuration Register
0x24	DMA_CFG	DMA Configuration Register

9.5.2. CONTROL (Offset 0x00)

table 9-2 CONTROL Register

Bit	Name	Access	Default	Description
31:11	Reserved	R/W	0x0	Reserved

Bit	Name	Access	Default	Description
10	HLSEL	R/W	1'b0	data storage 0: When receiving data, the DATA register is stored as follows: right stereo data is placed in the upper four bytes, and left stereo data is placed in the lower four bytes. When receiving data, the DATA register is stored as follows: left stereo data is placed in the upper four bytes, and right stereo data is placed in the lower four bytes.
9	INT_EN	R/W	1'b0	I2S Interrupt Enable
8	TX_INT_EN	R/W	1'b0	Transmit Interrupt Enable
7	RX_INT_EN	R/W	1'b0	Receive Interrupt Enable
6:5	Reserved	R/W	0x0	Reserved
4	CLK_INV	R/W	1'b0	0: No effect 1: CLK signal reverse
3	WS_INV	R/W	1'b0	0: No effect 1: WS signal reverse
2	DLY_EN	R/W	1'b0	Delay Enable 0: No effect 1: Delay 1 CLK when transmitting data frame
1	FMT	R/W	1'b0	Format Selection 0: PCM(Short frame) 1: left-aligned
0	I2S_EN	R/W	1'b0	I2S Enable

Note: When FMT selects left alignment, if DLY_EN=0, it is the MSB (left aligned) format.

If DLY_EN=1, it is Philips format. When the FMT selects PCM, the channel width must ≥ 32 bit.

9.5.3. CLK_DIV (Offset 0x04)

table 9-3 CLK_DIV Register

Bit	Name	Access	Default	Description
31:16	Reserved	R/W	0x0	Reserved
15:0	CLK_DIV	R/W	16'h00	CLK division Clock = APB Clock / (CLK_DIV + 1) * 2

9.5.4. WS_DIV(Offset 0x08)

table 9-4 WS_DIV Register

Bit	Name	Access	Default	Description
31:16	Reserved	R/W	0x0	Reserved
15:0	WS_DIV	R/W	16'h00	WS division WS = Clock/ (WS_DIV +1)

9.5.5. DATA (Offset 0x0C)

table 9-5 DATA Register

Bit	Name	Access	Default	Description
31:0	DATA	R/W	32'h00	Write operation: write data to TxFIFO. Read operation: read data from Rx FIFO

9.5.6. STATUS (Offset 0x10)

table 9-6 STATUS Register

Bit	Name	Access	Default	Description
31:6	Reserved	R/W	0x0	Reserved
5	TX_FIFO_Empty	R	1'b0	TxFIFO Empty
4	TX_FIFO_HalfEmpty	R	1'b0	TxFIFO Half Empty
3	TX_FIFO_Full	R	1'b0	TxFIFO Full

Bit	Name	Access	Default	Description
2	RX_FIFO_Empty	R	1'b0	RxFIFO Empty
1	RX_FIFO_HalfFull	R	1'b0	RxFIFO Half Full
0	RX_FIFO_Full	R	1'b0	RxFIFO Full

9.5.7. INTE (Offset 0x14)

table 9-7 INTE Register

Bit	Name	Access	Default	Description
31:6	Reserved	R/W	0x0	Reserved
5	TX_FIFO_Empty_INT	R	1'b0	TxFIFO Empty Interrupt Enable 0: Disable 1: Enable
4	TX_FIFO_HalfEmpty_INT	R	1'b0	TxFIFO Half Empty Interrupt Enable 0: Disable 1: Enable
3	TX_FIFO_Full_INT	R	1'b0	TxFIFO Full Interrupt Enable 0: Disable 1: Enable
2	RX_FIFO_Empty_INT	R	1'b0	RxFIFO Empty Interrupt Enable 0: Disable 1: Enable
1	RX_FIFO_HalfFull_INT	R	1'b0	RxFIFO Half Full Interrupt Enable 0: Disable 1: Enable
0	RX_FIFO_Full_INT	R	1'b0	RxFIFO Full Interrupt Enable 0: Disable 1: Enable

9.5.8. FIFO_CFG (Offset 0x20)

table 9-8 FIFO_CFG Register

Bit	Name	Access	Default	Description
31:24	RxFIFO_CFG	R/W	8'h32	RxFIFO full threshold If number of data in RxFIFO >= RxFIFO_CFG, set RX_FIFO_Full
23:8	Reserved	R/W	0x0	Reserved
7:0	TxFIFO_CFG	R/W	8'h0E	TxFIFO empty threshold. If number of data in TxFIFO <= TxFIFO_CFG, set TX_FIFO_Empty

9.5.9. DMA_CFG (Offset 0x24)

table 9-9 DMA_CFG Register

Bit	Name	Access	Default	Description
31:12	Reserved	R/W	0x0	Reserved
11:7	DMA_TX_Level	R/W	5'h10	DMA Tx threshold If number of data in Tx FIFO is equal to or less than DMA_TX_Level, DMA Tx request is generated.
6:2	DMA_RX_Level	R/W	5'h10	DMA Rx threshold If number of data in Rx FIFO is equal to or more than DMA_TX_Level, DMA Rx request is generated.
1	DMA_RX_EN	R/W	1'b0	DMA Rx Enable 0: Disable 1: Enable
0	DMA_TX_EN	R/W	1'b0	DMA Tx Enable 0: Disable 1: Enable

10. TRNG

10.1. Overview

True Random Number Generator (TRNG) allows the generation and collection of true random bitstreams from digital logic.

10.2. Feature

- 192bit True Random Number Generator

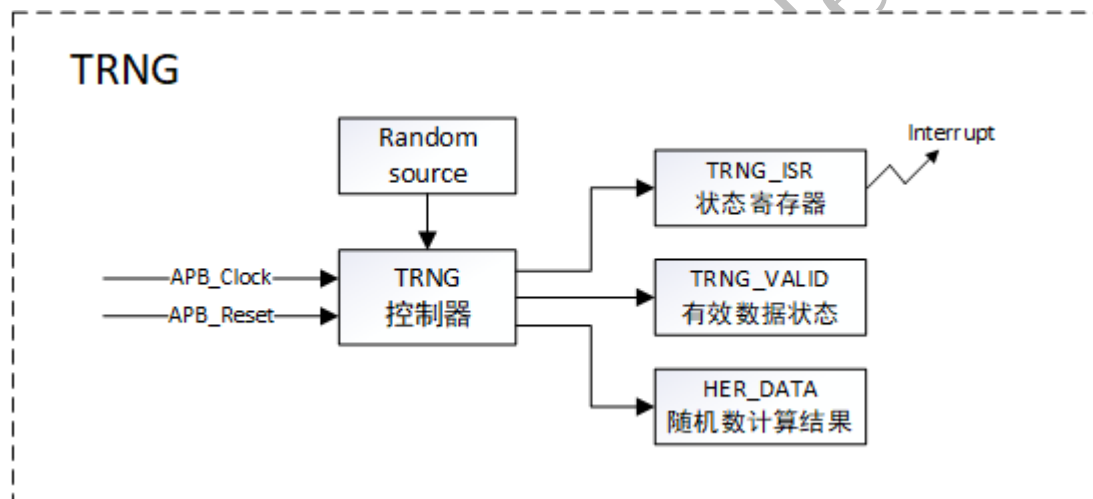


Figure 10-1 TRNG Block Diagram

10.3. TRNG Block Diagram

10.4. TRNG Register

10.4.1. TRNG Register Mapping

TRNG Base Addr: 0x500C_0000

table 10-1 TRNG Register Mapping

Offset	Name	Description
0x100	RNG_IMR	Interrupt Mask register
0x104	RNG_ISR	Interrupt Status Register
0x108	RNG_ICR	Interrupt Clear Register
0x10C	TRNG_CONFIG	TRNG Config Register
0x110	TRNG_VALID	VALID Status Register
0x114	HER_DATA0	Random Number Result Register
0x118	HER_DATA1	
0x11C	HER_DATA2	
0x120	HER_DATA3	
0x124	HER_DATA4	
0x128	HER_DATA5	
0x12C	RND_SOURCE_ENABLE	Random Number Source Enable Register
0x130	SAMPLE_CNT1	Sample Count Register
0x140	TRNG_RESET	TRNG Reset Register
0x1B8	TRNG_BUSY	TRNG Busy Status Register
0x1BC	RST_BITS_COUNTER	Counter Reset Register

10.4.2. RNG_IMR (Offset 0x100)

table 10-2 RNG_IMR Register

Bit	Name	Access	Default	Description
31:4	Reserved	R/W	28'h00	Reserved
3	VN_ERR_INT_MASK	R/W	1'b1	Von Neumann Errors Interrupt Write 1 to mask
2	CRNGT_ERR_INT_MASK	R/W	1'b1	CRNGT Errors Interrupt Write 1 to mask
1	AUTOCORR_ERR_INT_MASK	R/W	1'b1	Autocorrelation Errors Interrupt Write 1 to mask
0	EHR_VALID_INT_MASK	R/W	1'b1	TRNG has collected 192bit Interrupt Write 1 to mask

10.4.3. RNG_ISR (Offset 0x104)

table 10-3 RNG_ISR Register

Bit	Name	Access	Default	Description
31:4	Reserved	R/W	28'h00	Reserved
3	VN_ERR	R	1'b0	Von Neumann Error status flag If the 32 consecutive collection bits are the same, that is, 32 0s or 32 1s, a Von Neumann error will occur.
2	CRNGT_ERR	R	1'b0	CRNGT errors Error status flag The continuous random number generation test fails. If two consecutive blocks of 16 acquisition bits are equal, the fault occurs.
1	AUTOCORR_ERR	R	1'b0	Autocorrelation errors Error status flag Indicates four consecutive autocorrelation test failures. Note: After setting, TRNG will stop working until the next reset.
0	EHR_VALID	R	1'b0	TRNG has collected 192bit status flags. When '1' is set, HER_DATA [0,1,2,3,4,5] register is ready to be read.

10.4.4. RNG_ICR (Offset 0x108)

table 10-4 RNG_ICR Register

Bit	Name	Access	Default	Description
31:4	Reserved	R/W	28'h00	Reserved
3	VN_ERR	W	1'b0	Write '1' to clear the Von Neumann errors status flag.
2	CRNGT_ERR	W	1'b0	Write '1' to clear the CRNGT errors status flag.
1	AUTOCORR_ERR	W	1'b0	The software cannot operate this bit. TRNG is automatically cleared after reset.

Bit	Name	Access	Default	Description
0	EHR_VALID	W	1'b0	After reading HER_DATA [0~5] register, write '1' to clear the status flag.

10.4.5. TRNG_CONFIG (Offset 0x10C)

table 10-5 TRNG_CONFIG Register

Bit	Name	Access	Default	Description
31:2	Reserved	R/W	30'h00	Reserved
1:0	ENG_SRC_SEL	R/W	2'b0	Configure the inverter chain length in the ring oscillator. 00: Select the shortest inverter chain length 01: Select a shorter inverter chain length 10: Select a longer inverter chain length 11: Select the longest inverter chain length

10.4.6. TRNG_VALID(Offset 0x110)

table 10-6 TRNG_VALID Register

Bit	Name	Access	Default	Description
31:1	Reserved	R/W	31'h00	Reserved
0	EHR_VALID	R	1'b0	When set to "1", it indicates that bit collection in TRNG is completed and can read from HER_DATA [0~5] register.

10.4.7. HER_DATA0 ~ 5(Offset 0x114 ~ 0x128)

table 10-7 HER_DATA0~5 Register

Bit	Name	Access	Default	Description
31:0	HER_DATA5(0x124)	R	32'h00	random number [191:160]

Bit	Name	Access	Default	Description
31:0	HER_DATA4(0x124)	R	32'h00	random number [159:128]
31:0	HER_DATA3(0x120)	R	32'h00	random number [127:96]
31:0	HER_DATA2(0x11C)	R	32'h00	random number [95:64]
31:0	HER_DATA1(0x118)	R	32'h00	random number [63:32]
31:0	HER_DATA0(0x114)	R	32'h00	random number [31:0]

10.4.8. RND_SOURCE_ENABLE (Offset 0x12C)

table 10-8 RND_SOURCE_ENABLE Register

Bit	Name	Access	Default	Description
31:1	Reserved	R/W	31'h00	Reserved
0	RND_SRC_EN	R/W	1'b0	Random number source enable 0: Disable 1: Enable

10.4.9. SAMPLE_CNT1(Offset 0x130)

table 10-9 SAMPLE_CNT1 Register

Bit	Name	Access	Default	Description
31:0	SAMPLE_CNT1	R/W	32'hFFFF	Controls the frequency of a single bit output of the TRNG sampling ring oscillator

10.4.10. TRNG_RESET (Offset 0x140)

table 10-10 TRNG_RESET Register

Bit	Name	Access	Default	Description
31:1	Reserved	R/W	31'h00	Reserved
0	TRNG_RESET	R	1'b0	Writing '1' reset internal TRNG.

10.4.11. TRNG_BUSY (Offset 0x1B8)

table 10-11 TRNG_BUSY Register

Bit	Name	Access	Default	Description
31:1	Reserved	R/W	31'h00	Reserved
0	TRNG_BUSY	R	1'b0	Reflect the status of TRNG busy signal

10.4.12. RST_BITS_COUNTER (Offset 0x1BC)

table 10-12 RST_BITS_COUNTER Register

Bit	Name	Access	Default	Description
31:1	Reserved	R/W	31'h00	Reserved
0	RST_BITS_COUNTER	W	1'b0	Writing any value to this bit resets the bit counter and TRNG VALID register. Note: Only valid when RND_SRC_EN==0

11. PDM

11.1. Overview

Pulse Density Modulation (PDM), Pulse density modulation signal can be input from external audio (such as digital microphone) to support single channel data input.

11.2. Feature

- FIFO with depth 32 bits and width 16 bits
- Hardware high-pass filter

11.3. PDM System Block Diagram

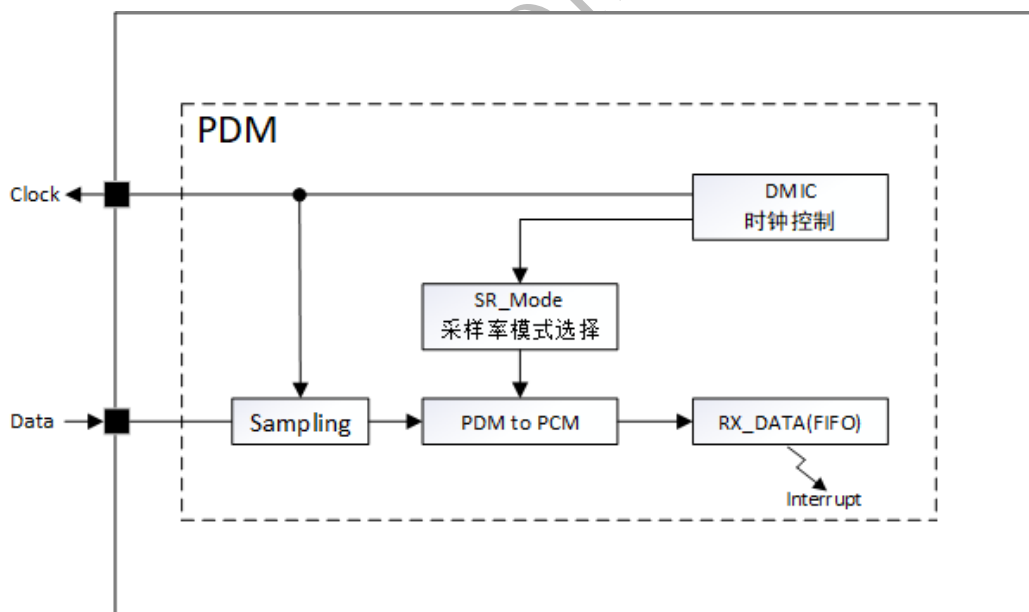


Figure 11-1 PDM System Block Diagram

11.4. PDM Register

11.4.1. PDM Register Mapping

PDM Base Addr: 0x500A_0000

table 11-1 PDM Register Mapping

Offset	Name	Description
0x00	PDM_Control	PDM Control Register
0x04	VOLUME	Volume value
0x08	VOLSTEP	Volume Configuration
0x0C	RX_DATA	Receive Data Register
0x10	FIFO_Clear	FIFO Clear Register
0x14	FIFO_Status	FIFO Status Register
0x18	FIFO_INT_EN	FIFO Interrupt Enable Register
0x1C	FIFO_Almost_Level	FIFO Almost Threshold Level Setting Register
0x20	DMA_CFG	DMA Control Register

11.4.2. PDM_Control (Offset 0x00)

table 11-2 PDM_Control

Bit	Name	Access	Default	Description
31:8	Reserved	R	0x0	Reserved
7:6	CLK_CFG	R/W	2'b00	DMIC Clock configuration 00: 1M 01: 1.5M 10: 2M 11: 3M
5	FIFO_Enable	R/W	1'b0	FIFO Enable 0: Disable FIFO 1: Enable FIFO
4	ZERO_DET	R/W	1'b1	Whether the volume adjustment is at zero. 0: Adjust the volume at any time

Bit	Name	Access	Default	Description
				1: Adjust the volume when zero crossing
3	HPF_EN	R/W	1'b1	High-Pass Filter Enable 0: High-Pass Filter Disable 1: High-Pass Filter Enable
2	SR_Mode	R/W	1'b0	Sample Rate Mode Selection 0: 62.5 1: 125 PCM rate = DMIC_Clock / SR_Mode For example: Set DMIC to 1M, SR_MODE is 1. Then the sampling rate is 1M/125=8KHz
1	EDGE_Select	R/W	1'b1	Sampling Edge Selection 0: Falling Edge Sampling 1: Rising Edge Sampling
0	PDM_EN	R/W	1'b0	PDM Enable 0: Disable 1: Enable

11.4.3. VOLUME (Offset 0x04)

table 11-3 VOLUME Register

Bit	Name	Access	Default	Description
31:8	Reserved	R	0x0	Reserved
11:0	Volume	R/W	12'h172	Volume Value Volume Value = Volume_Gain / 256

11.4.4. VOLSTEP (Offset 0x08)

table 11-4 VOLSTEP Register

Bit	Name	Access	Default	Description
31:17	Reserved	R	0x0	Reserved
16	VOL_DIRECT	R/W	1'b0	Adjust the volume directly. Directly reach the target value when adjusting

Bit	Name	Access	Default	Description
				the volume
15:10	Reserved	R	0x0	Reserved
9:0	VOL_STEP	R/W	10'h01	Adjust the volume step by step. When adjusting the volume, gradually increase or decrease the volume to reach the target value. Each increment or decrement value is (VOL_STEP)

11.4.5. RX_DATA(Offset 0x0C)

table 11-5 RX_DATA Register

Bit	Name	Access	Default	Description
31:16	Reserved	R	0x0	Reserved
15:0	RxDATA	R	1'b0	Receive Data

11.4.6. FIFO_Clear(Offset 0x10)

table 11-6 FIFO_Clear Register

Bit	Name	Access	Default	Description
31:2	Reserved	R	0x0	Reserved
1	Read_FIFO_Pointer_CLR	R/W	1'b0	Reset Read FIFO Pointer 1: reset Read FIFO pointer 0: release
0	Write_FIFO_Pointer_CLR	R/W	1'b0	Reset Write FIFO Pointer 1: reset Write FIFO pointer 0: release

11.4.7. FIFO_Status (Offset 0x14)

table 11-7 FIFO_Status Register

Bit	Name	Access	Default	Description
31:3	Reserved	R	0x0	Reserved
2	FIFO_Empty	R/W	1'b0	FIFO Empty
1	FIFO_Almost_Full	R/W	1'b0	FIFO Almost Full
0	FIFO_Full	R/W	1'b0	FIFO Full

11.4.8. FIFO_INT_EN(Offset 0x18)

table 11-8 FIFO_INT_EN Register

Bit	Name	Access	Default	Description
31:3	Reserved	R	0x0	Reserved
2	FIFO_Empty_INTEN	R/W	1'b0	FIFO Empty Interrupt Enable 0: Disable 1: Enable
1	FIFO_Almost_Full_INTEN	R/W	1'b0	FIFO Almost Full Interrupt Enable 0: Disable 1: Enable
0	FIFO_Full_INTEN	R/W	1'b0	FIFO Full Interrupt Enable 0: Disable 1: Enable

11.4.9. FIFO_Almost_Level (Offset 0x0x1C)

table 11-9 FIFO_Almost_Level Register

Bit	Name	Access	Default	Description
31:5	Reserved	R	0x0	Reserved
4:0	FIFO_Almost_Level	R/W	5'b10	FIFO almost threshold level setting When data in FIFO reaches the set value, FIFO_Almost_Full bit in FIFO_Stats register is set.

11.4.10. DMA_CFG(Offset 0x20)

table 11-10 DMA_CFG Register

Bit	Name	Access	Default	Description
31:6	Reserved	R	0x0	Reserved
5	DMA_EN	R/W	1'b0	DMA Enable 0: Disable 1: Enable
4:0	DMA_Rx_Level	R/W	5'h1E	DMA Rx Threshold Settings DMA Rx request is generated when FIFO data is equal to or greater than the set value

12. DMA

12.1. Overview

Direct memory access (DMA) uses hardware to transfer data, which is used to provide high-speed data transmission between memory and memory and between peripherals and memory. No CPU intervention is required, which saves CPU resources. Each channel is independent of each other, specifically managing access requests of its own channel.

12.2. Feature

- 3 independent configurable channels (requests)
- Support for memory-to-memory DMA transfer
- Support for peripheral-to-memory, memory-to-peripheral DMA transfer
- Block chaining using linked lists
- Configurable source transfer width and destination width (byte, half word, word)
- hardware priority (priority can be configured. Under the same priority, the lower the channel number, the higher the priority)
- Channel 0 supports 64 Byte FIFO depth
- Channel 1 and channel 2 support 32 Byte FIFO depth

12.3. Function Description

DMA controller and CPU core share system data bus, and perform direct memory data transfer. When the CPU and DMA access the same destination (RAM or peripherals) at the same time, the DMA request will suspend the CPU from accessing the system bus for several cycles, and the bus arbiter will execute circular scheduling to ensure the normal execution of the CPU after obtaining access resources.

12.3.1. DMA Priority

Programmable channel priority, can be configured by CFGx.CH_PROOR, 7 is the highest priority and 0 is the lowest priority. For channels with the same priority, the lower the channel number, the higher the priority.

Note: When the DMA controller receives multiple requests at the same time, the arbiter will decide which peripheral request to respond to according to the hardware priority.

If the DMA is transmitting data for a low priority channel and a high priority channel request is generated, the DMA will complete the data transmission of the low priority channel first, and then process the high priority channel request.

12.3.2. Transfer Length, Data Bit Width, Burst Length Relationship

In CTLx. register, BLOCK_TS is block transfer size, and the unit is source transfer width (SRC_TR_WIDTH), SRC_TR_WIDTH is source transfer width, and DST_TR_WIDTH is destination transfer width. SRC_MSIZ is source burst transaction length and DEST_MSIZ is destination burst transaction length.

For example:

BLOCK_TS = 5.

SRC_TR_WIDTH = 0x2.(4 byte)

DST_TR_WIDTH = 0x2.(4 byte)

SRC_MSIZ = 0x1.(Number of data items to be transferred is 4)

DEST_MSIZ = 0x1.(Number of data items to be transferred is 4)

The total number of bytes to be transferred is

$5(\text{BLOCK_TS}) \times 4(\text{SRC_TR_WIDTH}) = 20\text{byte}$.

After receiving a transaction request, data items to be transferred $4 * 4 \text{ byte} = 16 \text{ byte}$ will be read into the FIFO. Then move from FIFO to the destination address.

DMA receives another transaction request, and moves the remaining 4bytes($20 - 16 = 4$) of data to the destination address.

Note: SRC_TR_WIDTH * SRC_ The MSIZE data cannot exceed the DMA channel FIFO depth. If it is a transfer between memory and peripherals, SRC_TR_WIDTH * SRC_MSIZ The amount of data that needs to carry cannot exceed the depth of the peripheral FIFO.

12.3.3. Linked Lists Transfer Function

Multi-Block Transfer can use linked lists. Each linked list contains 1. destination address, 2. source address, 3. linked list base address of the next transaction (if this transaction is the last one, the value is NULL), and 4. control register.

After the first linked list is transferred, the *Next value of the first linked list will be queried. If it is not NULL, the next linked list will be transferred.....

12.3.4. Scatter/Gather

Direct memory access (DMA) is a technology that transfers data blocks between memories , memories and peripherals without the participation of CPU. DMA not only reduces the work of the system processing unit, but also transfers data at a much higher rate than the read and write rates of the processor.

The traditional block DMA can only transfer data of one physically continuous block at a time, and an interrupt is initiated after the transaction is completed. While Scatter /Gather provides data transfer from one discontinuous memory to another through a series of smaller continuous data blocks. Multiple physically discontinuous blocks can be transmitted at one time, and only one interrupt is initiated after the transmission is completed. The advantage of this is intuitive, greatly reducing the number of interruptions, and improving the efficiency of data transmission.

12.3.4.1. Scatter

Scatter is relevant to a destination transfer. The destination address is incremented or decremented by a programmed amount – the scatter increment – when a scatter boundary is reached. 错误!未找到引用源。 shows an example destination scatter transfer. The destination address is incremented or decremented by the value stored in the destination scatter increment(DSRx.DSI) field, multiplied by the number of bytes in a single transfer to the destination (decoded value of CTLx.DST_TR_WIDTH)/8 -When a scatter boundary is reached. The number of destination transfers between successive scatter boundaries is programmed into the Destination Scatter Count(DSC)field of the DSRx register.

Scatter is enabled by writing a 1 to the CTLx.DST_SCATTER_EN field. The CTLx.DINC field determines if the address is incremented, decremented, or remains fixed when a scatter boundary is reached. If the CTLx.DINC field indicates a fixed-address control throughout a DMA transfer, then the CTLx.DST_SCATTER_EN field is ignored, and the scatter feature is automatically disabled.

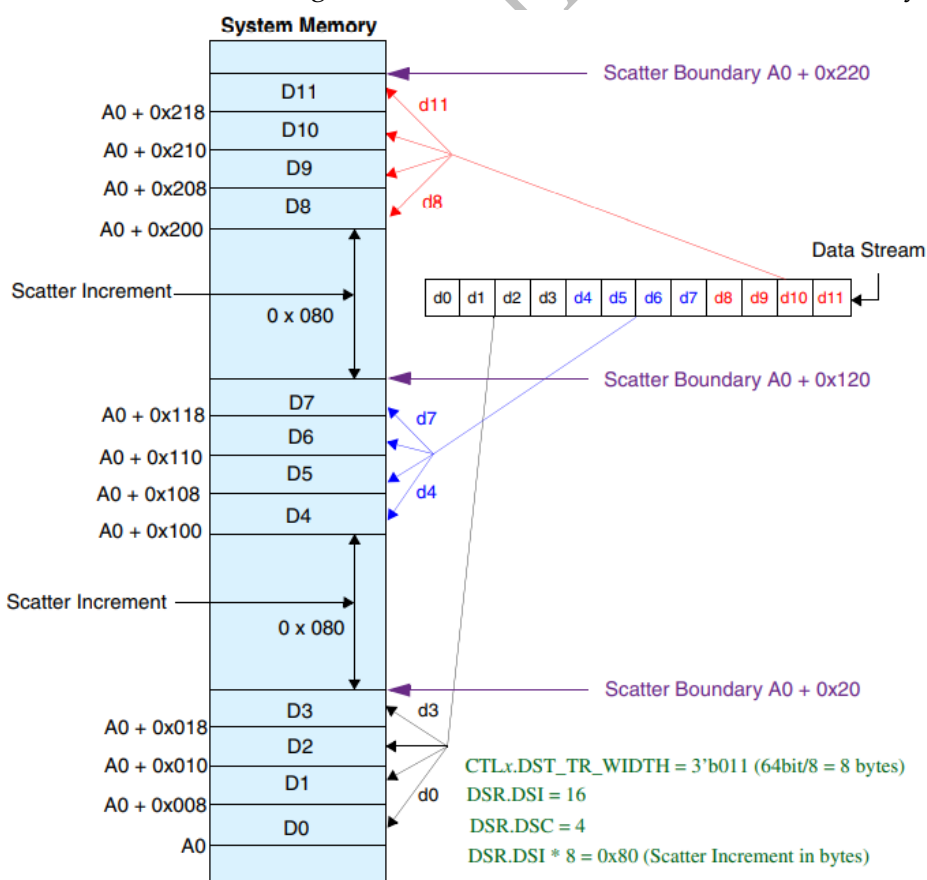


Figure 12-1 Example of Destination Scatter Transfer

Gather is relevant to a source transfer. The source address is incremented or decremented by a programmed amount when a gather boundary is reached. The number of source transfers between successive gather boundaries is programmed into the Source Gather Count (SGRx.SGC) field. The source address is incremented or decremented by the value stored in the source gather increment (SGRx.SGI) field, multiplied by the number of bytes in a single AHB transfer from the source – (decoded value of CTLx.SRC_TR_WIDTH)/8 – when a gather boundary is reached. The following figure shows an example of Source Gather.

Gather is enabled by writing a 1 to the CTLx.SRC_GATHER_EN field. The CTLx.SINC field determines if the address is incremented, decremented, or remains fixed when a gather boundary is reached. If the CTLx.SINC field indicates a fixed-address control throughout a DMA transfer, then the CTLx.SRC_GATHER_EN field is ignored, and the gather feature is automatically disabled.

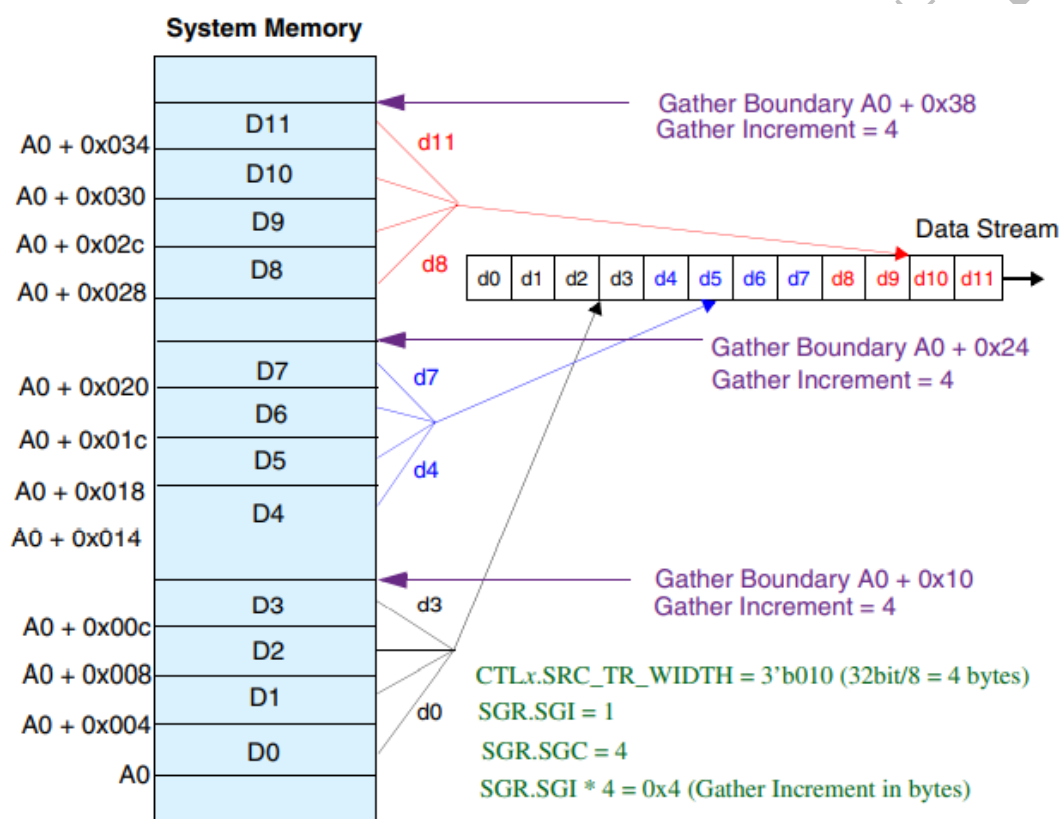


Figure 12-2 Example of Source Gather

12.4. DMA Register

12.4.1. DMA Register Mapping

DMA Base Addr: 0x2002_0000

table 12-1 DMA Register Mapping

Offset	Name	Description
0x00, 0x58, 0xB0	SARx	Channel 0, 1, 2 Source Address Register
0x08, 0x60, 0xB8	DARx	Channel 0, 1, 2 Destination Address Register
0x10, 0x68, 0xC0	LLPx	Channel 0, 1, 2 Linked List Pointer Register
0x18, 0x70, 0xC8	CTLx	Channel 0, 1, 2 Control Register
0x40, 0x98, 0xF0	CFGx	Channel 0, 1, 2 Config Register
0x48, 0xA0, 0xF8	SGRx	Channel 0, 1, 2 Source Gather Register
0x50, 0xA8, 0x100	DSRx	Channel 0, 1, 2 Destination Scatter Register
0x2C0	RawTfr	Raw Status for IntTfr Interrupt
0x2E0	RawErr	Raw Status for IntErr Interrupt
0x2E8	StatusTfr	Status for IntTfr Interrupt
0x308	StatusErr	Status for IntErr Interrupt
0x310	MaskTfr	Mask for IntTfr Interrupt
0x330	MaskErr	Mask for IntErr Interrupt
0x338	ClearTfr	Clear for IntTfr Interrupt
0x358	ClearErr	Clear for IntErr Interrupt
0x360	StatusInt	Status for each Interrupt type
0x398	DmaCfgReg	DMA Configuration Register
0x3A0	ChEnReg	DMA Channel Enable Register

12.4.2. SARx (Offset 0x00, 0x58, 0xB0)

table 12-2 SARx Register

Bit	Name	Access	Default	Description
63:32	Reserved	R/W	0x0	Reserved
31:0	SAR	R/W	32'h00	Current Source Address of DMA transfer.

12.4.3. DARx (Offset 0x08, 0x60, 0xB8)

table 12-3 DARx Register

Bit	Name	Access	Default	Description
63:32	Reserved	R/W	0x0	Reserved
31:0	DAR	R/W	32'h00	Current Source Address of DMA transfer.

12.4.4. LLPx (Offset 0x10, 0x68, 0xC0)

table 12-4 LLPx Register

Bit	Name	Access	Default	Description
63:32	Reserved	R/W	0x0	Reserved
31:2	LOC	R/W	30'h00	Starting Address In Memory of next LLI if block chaining is enabled.
1:0	LMS	R/W	2'b00	List Master Select. 0: AHB Master1 1: AHB Master2 2: AHB Master3 3: AHB Master4

12.4.5. CTLx (Offset 0x18, 0x70, 0xC8)

table 12-5 CTLx Register

Bit	Name	Access	Default	Description
63:44	Reserved	R/W	0x0	Reserved
43:32	BLOCK_TS	R/W	12'h000	Block Transfer Size.
31:29	Reserved	R/W	0x0	Reserved
28	LLP_SRC_EN	R/W	1'b0	Block chaining is enabled on the source side only if the LLP_SRC_EN field is high and LLPx.LOC is non-zero.
27	LLP_DST_EN	R/W	1'b0	Block chaining is enabled on the destination side only if LLP_DST_EN field is high and LLPx.LOC is non-zero.

Bit	Name	Access	Default	Description
26:25	SMS	R/W	2'b00	Source Master Select. 0: AHB Master1 1: AHB Master2 2: AHB Master3 3: AHB Master4
24:23	DMS	R/W	2'b00	Destination Master Select. 0: AHB Master1 1: AHB Master2 2: AHB Master3 3: AHB Master4
22:20	TT_FC	R/W	3'b000	Transfer Type and Flow Control. 0: Transfer type is Memory to Memory and Flow Controller is DMA. 1: Transfer type is Memory to Peripheral and Flow Controller is DMA. 2: Transfer type is Peripheral to Memory and Flow Controller is DMA. 3: Transfer type is Peripheral to Peripheral and Flow Controller is DMA.
19	Reserved	R/W	0x0	Reserved
18	DST_SCATTER_EN	R/W	0x0	Destination scatter enable. 0x0: Destination scatter is disabled. 0x1: Destination scatter is enable.
17	SRC_GATHER_EN	R/W	0x0	Source gather enable. 0x0: Source gather disable. 0x1: Source gather enable.
16:14	SRC_MSIZE	R/W	3'b000	Source Burst Transaction Length. Number of data items, each of width CTLx.SRC_TR_WIDTH, to be read from the source every time a burst transferred request is made from either the corresponding hardware or software handshaking interface. 0x0: Number of data items to be transferred is 1 0x1: Number of data items to be transferred is 4

Bit	Name	Access	Default	Description
				0x2: Number of data items to be transferred is 8 0x3: Number of data items to be transferred is 16 0x4: Number of data items to be transferred is 32 0x5: Number of data items to be transferred is 64 Note: the configuration of SRC_MSIZ should be same with the configuration of DEST_MSIZ.
13:11	DEST_MSIZ	R/W	3'b000	Destination Burst Transaction Length. Number of data items, each of width CTLx.DST_TR_WIDTH, to be written to the destination every time a destination burst transaction request is made from either the corresponding hardware or software handshaking interface. 0x0: Number of data items to be transferred is 1 0x1: Number of data items to be transferred is 4 0x2: Number of data items to be transferred is 8 0x3: Number of data items to be transferred is 16 0x4: Number of data items to be transferred is 32 0x5: Number of data items to be transferred is 64
10:9	SINC	R/W	2'b0	Source Address Increment. Indicates whether to increment or decrement the Source address on every Source transfer. 0x0: Increments the source address 0x1: Decrements the source address 0x2: No change in the source address
8:7	DINC	R/W	2'b0	Destination Address Increment.

Bit	Name	Access	Default	Description
				Indicates whether to increment or decrement the destination address on every destination transfer. 0x0: Increments the destination address 0x1: Decrements the destination address 0x2: No change in the destination address
6:4	SRC_TR_WIDTH	R/W	3'b000	Source Transfer Width. 0x0: 1 byte 0x1: 2 byte 0x2: 4 byte 0x3: 8 byte 0x4: 16 byte 0x5: 32 byte
3:1	DST_TR_WIDTH	R/W	3'b000	Destination Transfer Width. 0x0: 1 byte 0x1: 2 byte 0x2: 4 byte 0x3: 8 byte 0x4: 16 byte 0x5: 32 byte
0	INT_EN	R/W	1'b0	Interrupt Enable Bit. 0x0: Interrupt is disabled 0x1: Interrupt is enabled.

12.4.6. CFGx(Offset 0x40, 0x98, 0xF0)

table 12-6 CFGx Register

Bit	Name	Access	Default	Description
63:47	Reserved	R/W	0x0	Reserved
46:43	DEST_PER	R/W	4'h0	Destination hardware interface. Assigns a hardware handshaking interface (0: DMAH_NUM_HS_INT -1) to the destination of channel if the CFGx.HS_SEL_DST field is 0; otherwise, this

Bit	Name	Access	Default	Description
				field is ignored. The channel can then communicate with the destination peripheral connected to that interface through the assigned hardware handshaking interface. NOTE1: For correct DW_ahb_dmac operation, only one peripheral (source or destination) should be assigned to the same handshaking interface.
42:39	SRC_PER	R/W	4'h0	Source Hardware Interface. Assigns a hardware handshaking interface (0: DMAH_NUM_HS_INT -1) to the source of channel if the CFGx.HS_SEL_SRC field is 0; otherwise, this field is ignored. The channel can then communicate with the source peripheral connected to that interface through the assigned hardware handshaking interface. NOTE1: For correct DW_ahb_dmac operation, only one peripheral (source or destination) should be assigned to the same handshaking interface.
38:34	Reserved	R/W	0x0	Reserved
33	FIFO_MODE	R/W	1'b0	FIFO Mode Select. Determines how much space or data needs to be available in the FIFO before a burst transaction request is serviced. 0x0: FIFO_MODE_0, Space/data available for single AHBB transfer of the specified transfer width 0x1: FIFO_MODE_1, Data available is greater than or equal to half the FIFO depth for destination transfers and space available is greater than half the FIFO depth for source transfers. The exceptions are at the end of a burst transaction request or at the end of a block transfer.
32:20	Reserved	R/W	0x0	Reserved
19	SRC_HS_POL	R/W	1'b0	Source Handshaking Interface Polarity. 0x0: Source Handshaking Interface Polarity

Bit	Name	Access	Default	Description
				is Active high. 0x1: Source Handshaking Interface Polarity is Active low. Note: when the source address is a peripheral, this bit needs to be set.
18	DST_HS_POL	R/W	1'b0	Destination Handshaking Interface Polarity. 0x0: Destination Handshaking Interface Polarity is Active high. 0x1: Destination Handshaking Interface Polarity is Active low. Note: when the destination address is a peripheral, this bit needs to be set.
17:12	Reserved	R/W	0x0	Reserved
11	HS_SEL_SRC	R/W	1'b0	Source Software or Hardware Handshaking Select. If the source peripheral is memory, then this bit is ignored. 0x0: Hardware handshaking interface. Software initiated transaction requests are ignored. 0x1: Software handshaking interface. Hardware initiated transaction requests are ignored.
10	HS_SEL_DST	R/W	1'b0	Destination Software or Hardware Handshaking Select. 0x0: Hardware handshaking interface. Software initiated transaction requests are ignored. 0x1: Software handshaking interface. Hardware initiated transaction requests are ignored.
9	FIFO_EMPTY	R	1'b0	Channel FIFO status. 0x0: Channel FIFO is not empty 0x1: Channel FIFO is empty
8	CH_SUSP	R/W	1'b0	Channel Suspend. 0x0: DMA transfer from the source is not suspended

Bit	Name	Access	Default	Description
				0x1: Suspend DMA transfer from the source
7:5	CH_PROOR	R/W	3'b000	Channel Priority. A priority of 7 is the highest priority, and 0 is the lowest. For channels with the same priority, the lower the channel number, the higher the priority. 0x0: Channel priority is 0(the lowest priority) 0x1: Channel priority is 1 0x2: Channel priority is 2 0x3: Channel priority is 3 0x4: Channel priority is 4 0x5: Channel priority is 5 0x6: Channel priority is 6 0x7: Channel priority is 7(the highest priority)
4:0	Reserved	R/W	0x0	Reserved

12.4.7. SGRx (Offset 0x48, 0xA0, 0xF8)

Bit	Name	Access	Default	Description
31:20	SGC	R/W	0x0	Source Gather Count. Source contiguous tranfer count between successive gather boundaries.
19:0	SGI	R/W	3'b000	Source Gather Interval.

12.4.8. DSRx (Offset 0x50, 0xA8, 0x100)

Bit	Name	Access	Default	Description
31:20	DSC	R/W	0x0	Destination Scatter Counter. Destination contiguous transfer count between successive scatter boundaries.

Bit	Name	Access	Default	Description
19:0	RAW	R/W	3'b000	Destination Scatter Interval.

12.4.9. RawTfr (Offset 0x2C0)

table 12-7 RawTfr Register

Bit	Name	Access	Default	Description
31:3	Reserved	R/W	0x0	Reserved
2:0	RAW	R/W	3'b000	Raw Status for IntTfr Interrupt Each bit corresponds to one channel, and bit0 corresponds to channel 0. 0x0: Inactive Raw Interrupt Status 0x1: active Raw Interrupt Status

12.4.10. RawErr (Offset 0x2E0)

table 12-8 RawErr Register

Bit	Name	Access	Default	Description
31:3	Reserved	R/W	0x0	Reserved
2:0	Raw	R	3'b000	Raw Status for IntErr Interrupt Each bit corresponds to one channel, and bit0 corresponds to channel 0. 0x0: Inactive Raw Interrupt Status 0x1: Active Raw Interrupt Status

12.4.11. StatusTfr (Offset 0x2E8)

table 12-9 StatusTfr Register

Bit	Name	Access	Default	Description
31:3	Reserved	R/W	0x0	Reserved
2:0	Status	R	3'b000	Status for IntTfr Interrupt

Bit	Name	Access	Default	Description
				Each bit corresponds to one channel, and bit0 corresponds to channel 0. 0x0: Inactive Interrupt Status 0x1: Active Interrupt Status

12.4.12. StatusErr (Offset 0x308)

table 12-10 StatusErr Register

Bit	Name	Access	Default	Description
31:3	Reserved	R/W	0x0	Reserved
2:0	Status	R	3'b000	Status for IntErr Interrupt Each bit corresponds to one channel, and bit0 corresponds to channel 0. 0x0: Inactive Interrupt Status 0x1: Active Interrupt Status

12.4.13. MaskTfr (Offset 0x310)

table 12-11 MaskTfr Register

Bit	Name	Access	Default	Description
31:11	Reserved	R/W	0x0	Reserved
10:8	INT_MASK_WE	W	3'b000	Interrupt Mask Write Enable Each bit corresponds to one channel, and bit8 corresponds to channel 0. When writing the MaskTfr register, the channel write enable needs to be written together with the corresponding channel number. 0x0: Interrupt mask write disable 0x1: Interrupt mask write enable
7:3	Reserved	R/W	0x0	Reserved
2:0	INT_MASK	R/W	3'b000	Mask for IntTfr Interrupt 0x0: Mask the interrupts

Bit	Name	Access	Default	Description
				0x1: Unmask the interrupts

12.4.14. MaskErr (Offset 0x330)

table 12-12 MaskErr Register

Bit	Name	Access	Default	Description
31:11	Reserved	R/W	0x0	Reserved
10:8	INT_MASK_WE	W	3'b000	Interrupt Mask Write Enable Interrupt Mask Write Enable Each bit corresponds to one channel, and bit8 corresponds to channel 0. When writing the MaskTfr register, the channel write enable needs to be written together with the corresponding channel number. 0x0: Interrupt mask write disable 0x1: Interrupt mask write enable
7:3	Reserved	R/W	0x0	Reserved
2:0	INT_MASK	R/W	3'b000	Mask for IntErr Interrupt Each bit corresponds to one channel, and bit0 corresponds to channel 0 0x0: Mask the interrupts 0x1: Unmask the interrupts

12.4.15. ClearTfr (Offset 0x338)

table 12-13 ClearTfr Register

Bit	Name	Access	Default	Description
31:3	Reserved	W	0x0	Reserved
2:0	CLEAR	W	3'b000	Clear for IntTfr Interrupt Each bit corresponds to one channel, and bit0 corresponds to channel 0

Bit	Name	Access	Default	Description
				0x0: No effect 0x1: Clears interrupts

12.4.16. ClearErr (Offset 0x358)

table 12-14 ClearErr Register

Bit	Name	Access	Default	Description
31:3	Reserved	W	0x0	Reserved
2:0	CLEAR	W	3'b000	Clear for IntErr Interrupt Each bit corresponds to one channel, and bit0 corresponds to channel 0 0x0: No effect 0x1: Clears interrupts

12.4.17. StatusInt (Offset 0x360)

table 12-15 StatusInt Register

Bit	Name	Access	Default	Description
31:5	Reserved	R	0x0	Reserved
4	ERR	R	0x0	OR of the contents of StatusErr 0x0: OR of the contents of StatusErr register is 0 0x1: OR of the contents of StatusErr register is 1
3	DSTT	R	0x0	OR of the contents of StatusDstTran 0x0: OR of the contents of StatusDstTran register is 0 0x1: OR of the contents of StatusDstTran register is 1
2	SRCT	R	0x0	OR of the contents of StatusSrcTran 0x0: OR of the contents of StatusSrcTran register is 0 0x1: OR of the contents of StatusSrcTran register is 1
1	BLOCK	R	0x0	OR of the contents of StatusBlock

Bit	Name	Access	Default	Description
				0x0: OR of the contents of StatusBlock register is 0 0x1: OR of the contents of StatusBlock register is 1
0	TFR	R	0x0	OR of the contents of StatusTfr 0x0: OR of the contents of StatusTfr register is 0 0x1: OR of the contents of StatusTfr register is 1

12.4.18. DmaCfgReg (Offset 0x398)

table 12-16 DmaCfgRegRegister

Bit	Name	Access	Default	Description
31:1	Reserved	R	0x0	Reserved
0	DAM_EN	R/W	1'b0	DMA Enable bit. 0x0: DMA Disabled. 0x1: DMA Enabled.

12.4.19. ChEnReg (Offset 0x3A0)

table 12-17 ChEnReg Register

Bit	Name	Access	Default	Description
31:11	Reserved	R/W	0x0	Reserved
10:8	CH_EN_WE	W	3'b000	Channel enable register Each bit corresponds to one channel, and bit 8 corresponds to channel 0. When writing the ChEnReg register, need to write the channel enable together with the corresponding channel number. 0x0: Disabled 0x1: Enabled.
7:3	Reserved	R/W	0x0	Reserved
2:0	CH_EN	R/W	3'b000	Channel Enable. The ChEnReg.CH_EN bit is automatically cleared by hardware to disable the channel after the last AMBA

Bit	Name	Access	Default	Description
				transfer of the DMA transfer to the destination has completed. Software can therefore poll this bit to determine when this channel is free for a new DMA transfer. Each bit corresponds to one channel, and bit0 corresponds to channel 0. Transfer starts after the channel is enabled. 0x0: Disable the channel 0x1: Enable the channel

12.5. Note

1. After the channel is enabled, no write operation can be performed on the channel register. If the channel register must be written, it can only be written after channel disabled.
2. When using the transfer between the memory and the peripheral, need to set the request ID. CFGx.DEST_PER and CFGx.SRC_PER should be configured too. The value should be the same as the ID in the system register [DMA_REQ_MUX0](#)、[DMA_REQ_MUX1](#)、[DMA_REQ_MUX2](#).

13. PWM

13.1. Overview

CorePWM is a general purpose, multi-channel pulse width modulator (PWM) module for motor control, tone generation, battery charging, heating elements, and more.

In General Purpose PWM mode, duty cycle updates can be performed asynchronously or synchronously, based on parameter selection. In synchronous mode, all channels are updated at the beginning of the PWM period, which is useful for motor control and can be used to keep a constant dead band space between channel waveforms. Asynchronous mode is relevant to applications such as LED control, where synchronous updates are not required. Asynchronous mode lowers the area size, reducing shadow register requirements.

In addition to the general purpose PWM modes, there is a "Low Ripple DAC" mode that creates a minimum period pulse train whose High/Low average is that of the chosen duty cycle. When used with a low-pass filter (such as a simple RC circuit), a DAC can be created with far better bandwidth and ripple performance than a standard PWM algorithm can achieve. This type of DAC is ideally suited for fine tuning of power supply output levels.

13.2. Feature

- 8-channel PWM output or 8-channel input capture
- Two PWM output modes are optional (PWM mode, PWM_DAC mode)

- Each PWM channel has independent frequency division, duty cycle and cycle configuration
- PWM inverter output optional
- Input capture interrupt response
- 8-channel PWM output or 8-channel input capture
- Two PWM output modes are optional (PWM mode, PWM_DAC mode)
- Each PWM channel has independent configuration of frequency division, duty cycle and period
- Optional inverted PWM output
- Input capture interrupt response

13.3. PWM Block Diagram

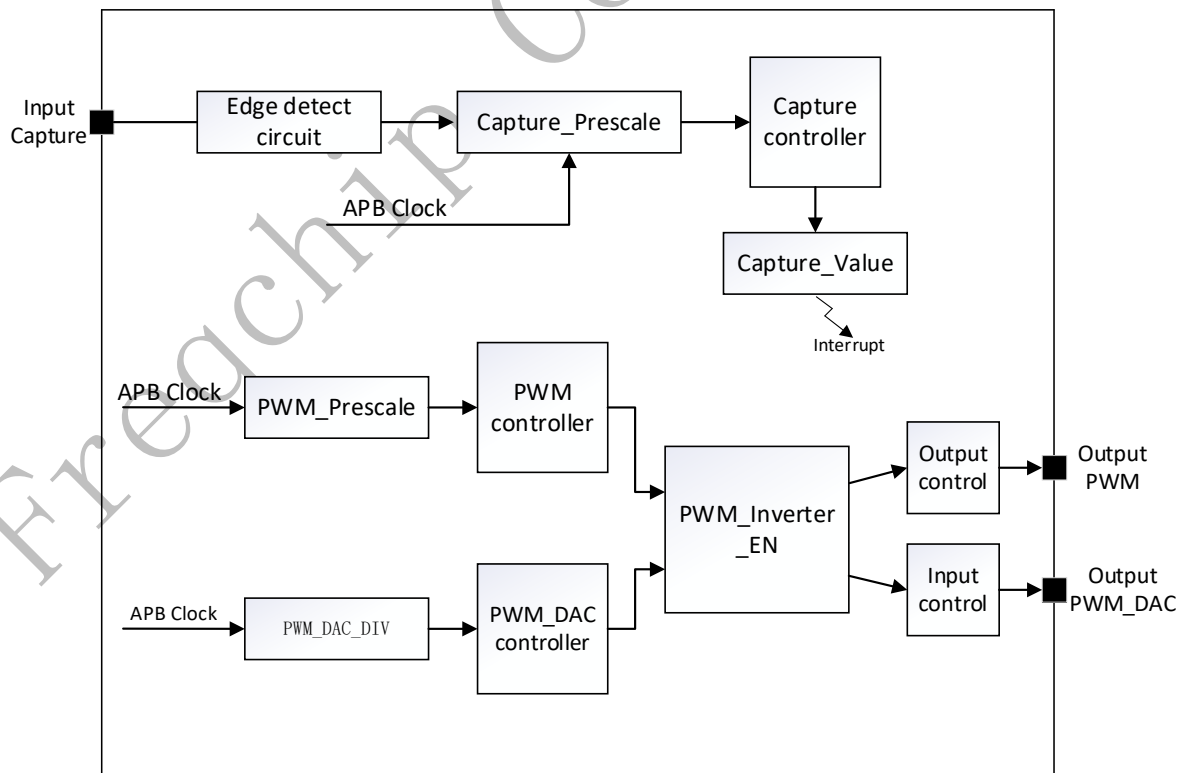


Figure 13-1 PWM Block Diagram

13.4. PWM Register

13.4.1. PWM Register Mapping

PWM Base Addr: 0x5008_0000

table 13-1 PWM Register Mapping

Offset	Name	Description
0x04	PWM_Output_EN	PWM Output Enable
0x08	PWM_ENABLE	PWM Enable
0x10	PWM0_Posedge	PWM Channel 0 ~ 7 Posedge Register PWM Channel 0 ~ 7 Negedge Register
0x14	PWM0_Negedge	
0x18	PWM1_Posedge	
0x1C	PWM1_Negedge	
0x20	PWM2_Posedge	
0x24	PWM2_Negedge	
0x28	PWM3_Posedge	
0x2C	PWM3_Negedge	
0x30	PWM4_Posedge	
0x34	PWM4_Negedge	
0x38	PWM5_Posedge	
0x3C	PWM5_Negedge	
0x40	PWM6_Posedge	
0x44	PWM6_Negedge	
0x48	PWM7_Posedge	
0x4C	PWM7_Negedge	
0x90	PWM_Output_Select	PWM Output Select Register
0x94	Capture_Prescale	Input Capture Prescale Register
0x98	Capture_Status	Input Capture Status Register
0x9C	Capture_INT_Mask	Input Capture INT Mask Register
0xA0	Capture_CTL	Input Capture Control Register
0xA4	Capture_Value0	Channel 0 ~ 7 Input Capture Value
0xA8	Capture_Value1	
0xAC	Capture_Value2	
0xB0	Capture_Value3	
0xB4	Capture_Value4	
0xB8	Capture_Value5	
0xBC	Capture_Value6	
0xC0	Capture_Value7	
0xE4	PWM_Update	PWM Output Synchronous Update
0xE8	PWM_DAC_Mode	DAC Mode Selection
0xF0	PWM_Output_Value	PWM Output Value

Offset	Name	Description
0xF8	PWM_Inverter_EN	PWM Inverter Output Enable
0xFC	PWM_CNT_EN	PWM Counter, Divider Enable Register
0x100	PWM0_Prescale	PWM Channel 0 ~ 7 Prescale Register PWM Channel 0 ~ 7 Period Register
0x104	PWM0_Period	
0x108	PWM1_Prescale	
0x10C	PWM1_Period	
0x110	PWM2_Prescale	
0x114	PWM2_Period	
0x118	PWM3_Prescale	
0x11C	PWM3_Period	
0x120	PWM4_Prescale	
0x124	PWM4_Period	
0x128	PWM5_Prescale	
0x12C	PWM5_Period	
0x130	PWM6_Prescale	
0x134	PWM6_Period	
0x138	PWM7_Prescale	
0x13C	PWM7_Period	
0x180	PWM0_DAC_DIV	PWM Channel 0 ~ 7 Frequency Division Register (Valid only in DAC mode)
0x184	PWM1_DAC_DIV	
0x188	PWM2_DAC_DIV	
0x18C	PWM3_DAC_DIV	
0x190	PWM4_DAC_DIV	
0x194	PWM5_DAC_DIV	
0x198	PWM6_DAC_DIV	
0x19C	PWM7_DAC_DIV	

13.4.2. PWM_Output_EN(Offset 0x04)

table 13-2 PWM_Output_EN Register

Bit	Name	Access	Default	Description
31:8	Reserved	R/W	24'h00	Reserved
7:0	PWM_Output_EN	R/W	8'hFF	PWM Output Enable 1 bit corresponds to 1 channel 0: Enable 1: Disable

13.4.3. PWM_ENABLE (Offset 0x08)

table 13-3 PWM_ENABLE Register

Bit	Name	Access	Default	Description
31:8	Reserved	R/W	24'h00	Reserved
7:0	PWM_Enable	R/W	8'h00	PWM Channel Enable 1 bit corresponds to 1 channel 0: Disable 1: Enable

13.4.4. PWM0~7_Posedge (Offset 0x10, 0x18, 0x20, 0x28, 0x30, 0x38, 0x40, 0x48)

table 13-4 PWM0~7_Posedge Register

Bit	Name	Access	Default	Description
31:0	PWM_Posedge	R/W	32'h00	PWM Mode: Sets the positive edge of the output with respect to the PERIOD

Note: The register descriptions of PWM0_Posedge to PWM7_Posedge are the same.

13.4.5. PWM0~7_Negedge (Offset 0x14, 0x1C, 0x24, 0x2C, 0x34, 0x3C, 0x44, 0x4C)

table 13-5 PWM0~7_Negedge Register

Bit	Name	Access	Default	Description
31:0	PWM_Negedge	R/W	32'h00	PWM Mode: Sets the negative edge of the output with respect to the PERIOD resolution. PWM_DAC Mode: Percentage of high level time respect to the PERIOD. 0%~100%(0 ~ 0xFFFFFFFF)。

Note: The register descriptions of PWM0_Negedge ~ PWM7_Negedge are the same.

13.4.6. PWM_Output_Select (Offset 0x90)

table 13-6 PWM_Output_Select Register

Bit	Name	Access	Default	Description
31:8	Reserved	R/W	24'h00	Reserved
7:0	PWM_Output_Select	R/W	8'h00	PWM Output Control Selection 1 bit corresponds to 1 channel. 0: use PWM Posedge and PWM Negedge to control output waveform 1: use PWM Output Value control output waveform

13.4.7. Capture_Prescale (Offset 0x94)

table 13-7 Capture_Prescale Register

Bit	Name	Access	Default	Description
31:4	Reserved	R/W	28'h00	Reserved
3:0	Capture_Prescale	R/W	4'h0	Clock prescale setting. Determines effective clockrate for the counter based on APB_Clock 0000 = divide by 1 (default) 0001 = divide by 2 0010 = divide by 4 0011 = divide by 8 0100 = divide by 16 0101 = divide by 32 0110 = divide by 64 0111 = divide by 128 1000 = divide by 256 1001 = divide by 512 1010 = divide by 1024 1011 = divide by 2048 Others = divide by 2048

13.4.8. Capture_Status (Offset 0x98)

table 13-8 Capture_Status Register

Bit	Name	Access	Default	Description
31:8	Reserved	R/W	24'h00	Reserved
7:0	Capture_Status	R/W	8'h00	Input capture status. 1 bit corresponds to 1 channel. Write operation; Write the '1' clear status bit. Read operation: whether the current channel has updated the capture value

13.4.9. Capture_INT_EN (Offset 0x9C)

table 13-9 Capture_INT_EN Register

Bit	Name	Access	Default	Description
31:8	Reserved	R/W	24'h00	Reserved
7:0	Capture_INT_EN	R/W	8'h00	Input capture interrupt enable 1 bit corresponds to 1 channel. 0: Disable 1: Enable

13.4.10. Capture_CTL(Offset 0xA0)

table 13-10 Capture_CTL Register

Bit	Name	Access	Default	Description
31:16	Reserved	R/W	16'h00	Reserved
15:8	Capture_EN	R/W	8'h00	Input capture enable 1 bit corresponds to 1 channel 0: Disable 1: Enable
7:0	Capture_Mode	R/W	8'h00	Input capture mode selection. 1 bit corresponds to 1 channel

Bit	Name	Access	Default	Description
				0: The captured signals are continuously stored in the data registers of each channel. 1: When the corresponding channel bit of Capture_Status is cleared for a capture, the capture is stopped after the corresponding channel bit of Capture_Status is set.

13.4.11. Capture_Value0~7 (Offset 0xA4 ~ 0xC0)

table 13-11 Capture_Value0~7 Register

Bit	Name	Access	Default	Description
31:0	Capture_Value	R	32'h00	Stores the number of timer ticks between two successive positive (or negative) edges. The edge to be used is configurable. If the number of timer ticks exceeds the maximum register value, the value of 0 shall be stored instead.

13.4.12. PWM_Update (Offset 0xE4)

• table 13-12 PWM_Update Register

Bit	Name	Access	Default	Description
31:24	Reserved	R/W	8'h00	Reserved
23:16	PWM_Status	R	8'h00	PWM instructions 1 bit corresponds to 1 channel 0: The corresponding channel stops working 1: The corresponding channel is working
15:8	Reserved	R/W	8'h00	Reserved
7:0	PWM_Update	R/W	8'h00	PWM configuration updates 1 bit corresponds to 1 channel Write '0': the Period, Prescale, Posedge , Negedge register of the corresponding channel are not updated. Write '1': the Period, Prescale, Posedge ,

Bit	Name	Access	Default	Description
				Negedge register of the corresponding channel are updated synchronously.

13.4.13. PWM_DAC_Mode (Offset 0xE8)

table 13-13 PWM_DAC_Mode Register

Bit	Name	Access	Default	Description
31:8	Reserved	R/W	24'h00	Reserved
7:0	DAC_Mode_EN	R/W	8'h00	Output normal PWM mode or DAC mode 1 bit corresponds to 1 channel 0: General purpose PWM mode 1: Low ripple DAC mode

13.4.14. PWM_Output_Value (Offset 0xF0)

table 13-14 PWM_Output_Value Register

Bit	Name	Access	Default	Description
31:8	Reserved	R/W	24'h00	Reserved
7:0	PWM_Output_Value	R/W	8'h00	PWM output control (valid when PWM_Output_Select is 1) 1 bit corresponds to 1 channel 0: output high level of corresponding channel 1: output low level of corresponding channel

13.4.15. PWM_Inverter_EN(Offset 0xF8)

table 13-15 PWM_Inverter_EN Register

Bit	Name	Access	Default	Description
31:8	Reserved	R/W	24'h00	Reserved
7:0	PWM_Inverter_EN	R/W	8'h00	PWM Inverter output enable 1 bit corresponds to 1 channel

Bit	Name	Access	Default	Description
				0: PWM normal output 1: PWM inverted output

13.4.16. PWM_CNT_EN (Offset 0xFC)

table 13-16 PWM_CNT_EN Register

Bit	Name	Access	Default	Description
31:8	Reserved	R/W	24'h00	Reserved
7:0	PWM_CNT_EN	R/W	8'hFF	PWM cycle and prescaler counter enable 1 bit corresponds to 1 channel 0: Disable 1: Enable

13.4.17. PWM0~7_Prescale (Offset 0x100, 0x108, 0x110, 0x118, 0x120, 0x128, 0x130, 0x138)

table 13-17 PWM0~7_Prescale Register

Bit	Name	Access	Default	Description
31:0	PWM_Prescale	R/W	32'h08	PWM corresponding channel prescale. $PWMx_Clock = APB_Clock / PWM_Prescale + 1$

Note: The register description of PWM0_Prescale ~ PWM7_Prescale are the same.

13.4.18. PWM0~7_Period(Offset 0x104, 0x10C, 0x114, 0x11C, 0x124, 0x12C, 0x134, 0x13C)

table 13-18 PWM0~7_Period Register

Bit	Name	Access	Default	Description
31:0	PWM_Period	R/W	32'h08	PWM period of corresponding channel $PWMx_Clock = APB_Clock / (PWM_Prescale + 1)$ $PWMx_Clock_Period = 1 / PWMx_Clock$ $Period = PWMx_Clock_Period * (PWM_Period + 1)$

Note: The register description of PWM0_Period ~ PWM7_Period are the same.

13.4.19. PWM0~7_DAC_DIV(Offset 0x180, 0x184, 0x188, 0x18C, 0x190, 0x194, 0x198, 0x19C)

table 13-19 PWM0~7_DAC_DIV Register

Bit	Name	Access	Default	Description
31:16	Reserved	R/W	16'h00	Reserved
15:0	PWM_DIV	R/W	16'h08	Frequency division of corresponding channel in PWM_DAC mode $PWMx_Pulse = APB_Clock / 2 * (PWM_DIV + 1)$ Note: This register is only valid in PWM_DAC mode.

13.5. Process

13.5.1. PWM Mode

Use normal PWM mode, mainly configure Prescale, Period, PWM_Posedge, PWM_Negedge register.

The relationship is as shown in the figure:

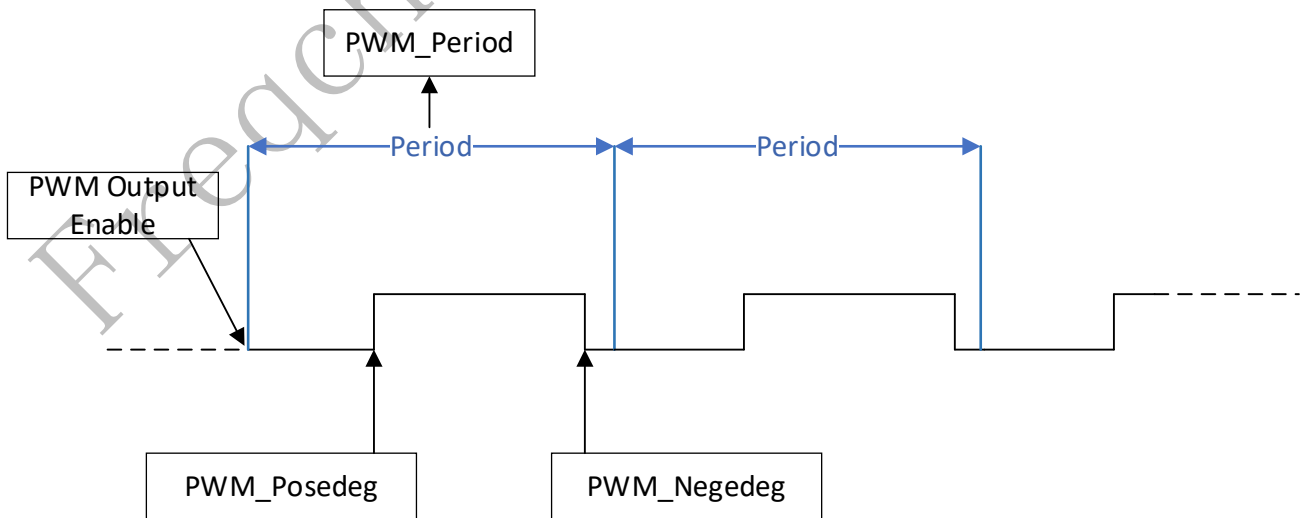


Figure 13-2 PWM Timing Diagram

Synchronous update: There is a problem of time sequence when configuring prescale, period, PWM_Posedge, PWM_Negedge register, which leads to problems in motor and other applications. PWM registers can be referenced for synchronous update. After all the registers to be set, PWM can be synchronous updated by setting PWM_Update.

Asynchronous update: In some simple applications, synchronous update is not required. At this time, corresponding registers can be modified directly to modify PWM frequency or duty cycle.

13.5.2. PWM_DAC Mode

A very small cycle pulse width can be created in PWM_DAC mode.

To use DAC mode, first configure [PWM DAC Mode](#) registers and select PWM_DAC mode. Then configure the [PWM DAC DIV](#) register to set the clock frequency division to obtain the high-level pulse width PWMx_Pulse. Write [PWM Negedge](#) register to set proportion of high level in the whole cycle (0%~100%).

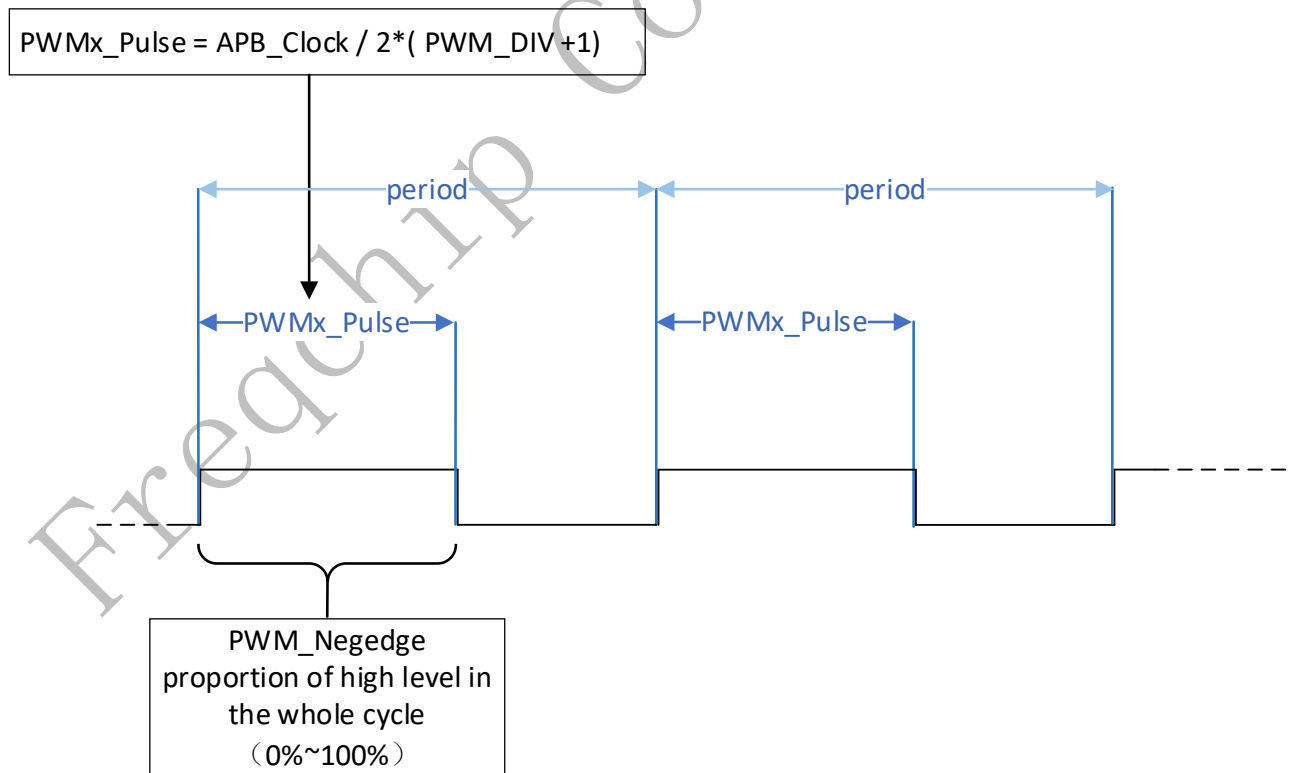


Figure 13-1 PWM_DAC timing diagram

Input Capture Function

The FR8000 series chip input captures the rising edge by default. When two rising edges are captured, the count value between the two rising edges will be automatically calculated. Count value clock is controlled by APB_Clock/[Capture Prescale](#) Prescale.

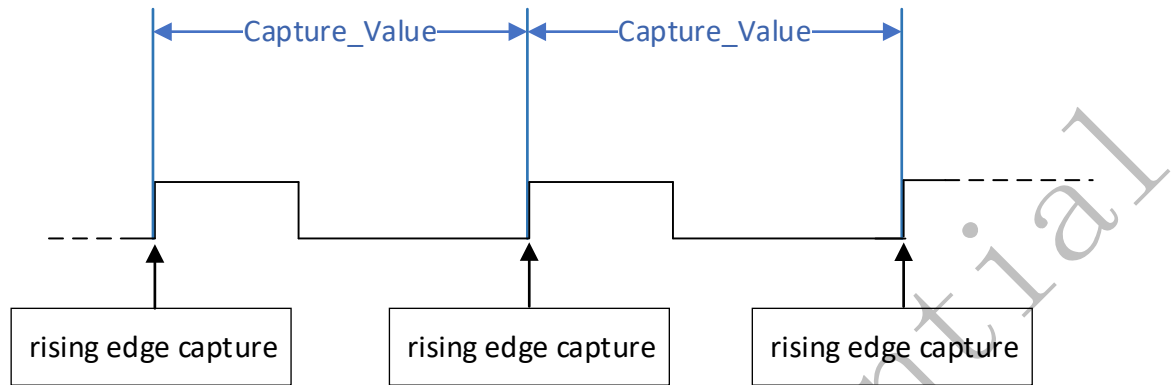


Figure 13-2 Input Capture timing diagram

14. PMU

14.1. Overview

PMU integrates power management, logic control and other functions, as described below.

- Power on detection
- Battery charging management
- Low voltage alarm
- Power module control in low power consumption mode
- SOC power supply module management: IO-LDO, DC-DC, DLDO, RFLDO, PKLDO
- Independent watchdog
- RTC
- Matrix keyboard scanning
- PWM
- IO detection wake-up

14.2. Access Control Interface

There are two modules in the PMU, Block0 and Block1, on which registers with different functions are allocated. MCU controls PMU functions by accessing registers on two modules.

For detailed register address and allocation, please refer to the following chapters.

14.3. PMU_IWDG

14.3.1. PMU_IWDG Function Description

IWDG (Independent Watchdog) uses internal low speed clock (LowPowerRC). Use a 24-bit counter to count down. When the count value counts from the configuration value of IWDG_LEN register to 0, an interrupt signal is generated. Continue to count down from the configuration value of IWDG_TOUT_LEN register. If the count reaches 0, a system reset will be generated.

Note: If the IWDG_TOUT_LEN register is not configured, only interrupts can be generated and system resets cannot be generated.

14.3.2. PMU_IWDG System block diagram

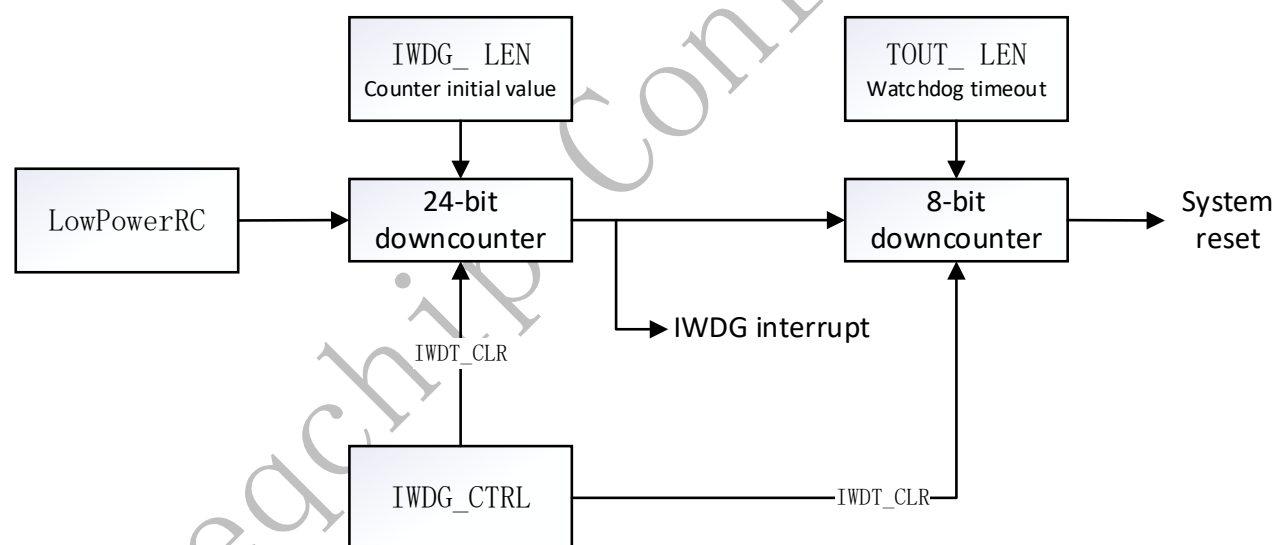


Figure 14-1 PMU_IWDG system diagram

14.3.3. PMU_IWDG Register Mapping

The IWDG Register is in the Block0 area of the PMU.

table 14-1 PMU_IWDG Register Mapping

Offset	Name	Description
0x26	IWDG_CTRL	Independent watchdog control register
0x27	IWDG_LEN_0	Independent watchdog counter initial value register (LSB[7:0])
0x28	IWDG_LEN_1	Independent watchdog count initial value register ([15:8])
0x29	IWDG_LEN_2	Independent watchdog count initial value register (MSB[23:6])
0x2A	IWDG_TOUT_LEN	Independent watchdog timeout register

14.3.4. IWDG_CTRL (Block0, 0x26)

table 14-2 IWDG_CTRL Register

Bit	Name	Access	Default	Description
7:3	Reserved	R/W	6'h0	Reserved
2	IWDT_IRQ_EN	R/W	1'b0	Independent watchdog interrupt enable
1	IWDT_CLR	R/W	1'b0	Clear count value, Count down from configuration value again
0	IWDT_EN	R/W	1'b0	Independent watchdog enable.

14.3.5. IWDG_LEN (Block0, 0x27, 0x28, 0x29)

table 14-3 IWDG_LEN Register

Bit	Name	Access	Default	Description
7:0	IWDG_LEN_0(0x29)	R/W	8'h00	Initial Count Value MSB[23:16]
7:0	IWDG_LEN_0(0x28)	R/W	8'h00	Initial Count Value [15:8]
7:0	IWDG_LEN_0(0x27)	R/W	8'h00	Initial Count Value LSB[7:0]

14.3.6. IWDG_TOUT_LEN (Block0, 0x2A)

table 14-4 IWDG_TOUT_LEN Register

Bit	Name	Access	Default	Description
7:0	TOUT_LEN	R/W	8'h00	The dog feeding timeout is configured. When the IWDG_LEN count is 0, an interrupt is generated, and the

Bit	Name	Access	Default	Description
				TOUT_LEN starts to count down. When the count is 0, the system resets. Note: If this value is 0, the system will not be reset, only interrupt will be generated

14.4. PMU_GPIO

14.4.1. PMU_GPIO Function Description

When using PMU to manage GPIO pins, can be configured through PMU series registers. Pull up/down configuration, input/output configuration, functions multiplex, IO pin change detection function.

14.4.2. PMU_GPIO Register Mapping

PMU_GPIO Registers are in **Block0** area of PMU

table 14-5 PMU_GPIO Register Mapping

Offset	Name	Description
0x43	PortA_SEL	PortA Control Selection
0x44	PortB_SEL	PortB Control Selection
0x45	PortC_SEL	PortC Control Selection
0x46	PortD_SEL	PortD Control Selection
0x47	PortE_SEL	PortE Control Selection
0x48	PortA_PE	PortA Pull up/down enable
0x49	PortB_PE	PortB Pull up/down enable
0x4A	PortC_PE	PortC Pull up/down enable
0x4B	PortD_PE	PortD Pull up/down enable
0x4C	PortE_PE	PortE Pull up/down enable
0x4D	PortA_PS	PortA Pull up/down enable
0x4E	PortB_PS	PortB Pull up/down enable
0x4F	PortC_PS	PortC Pull up/down enable
0x50	PortD_PS	PortD Pull up/down enable
0x51	PortE_PS	PortE Pull up/down enable
0x52	PortA_Monitor_EN	PortA Change detection enable
0x53	PortB_Monitor_EN	PortB Monitor enable
0x54	PortC_Monitor_EN	PortC Monitor enable

Offset	Name	Description
0x55	PortD_Monitor_EN	PortD Monitor enable
0x56	PortE_Monitor_EN	PortE Monitor enable
0x57	Analog_Monitor_EN	Analog Monitor enable
0x58	PortA_Reference_Status	PortA Reference Status
0x59	PortB_Reference_Status	PortB Reference Status
0x5A	PortC_Reference_Status	PortC Reference Status
0x5B	PortD_Reference_Status	PortD Reference Status
0x5C	PortE_Reference_Status	PortE Reference Status
0x5D	Analog_Reference_Status	Analog Reference Status
0x66	PortA_OEN	PortA Input Output Configuration
0x67	PortB_OEN	PortB Input Output Configuration
0x68	PortC_OEN	PortC Input Output Configuration
0x69	PortD_OEN	PortD Input Output Configuration
0x6A	PortE_OEN	PortE Input Output Configuration
0x6B	PortA_Data	PortA Input Output Configuration
0x6C	PortB_Data	PortB Output Data Register
0x6D	PortC_Data	PortC Output Data Register
0x6E	PortD_Data	PortD Output Data Register
0x6F	PortE_Data	PortE Output Data Register

14.4.3. PortA/B/C/D/E_SEL (Block0, 0x43, 0x44, 0x45, 0x46, 0x47)

table 14-6 PortA/B/C/D/E_SEL Register

Bit	Name	Access	Default	Description
7:0	PortA_SEL	R/W	8'hFF	PortA pin control selection: 0: Use PMU_ GPIO domain register control 1: Use system domain GPIO register to control Bit [0] corresponds to PortA0, 1 bit corresponds to 1 pin
7:0	PortB_SEL	R/W	8'hFF	Same as PortA
7:0	PortC_SEL	R/W	8'hFF	Same as PortA
7:0	PortD_SEL	R/W	8'hFF	Same as PortA
1:0	PortE_SEL	R/W	2'b11	Same as PortA

14.4.4. PortA/B/C/D/E_PE (Block0, 0x48, 0x49, 0x4A, 0x4B, 0x4C)

table 14-7 PortA/B/C/D/E_PE Register

Bit	Name	Access	Default	Description
7:0	PortA_PE	R/W	8'hFF	PortA Pull up/down enable: 0: Pull up/down Disable. 1: Pull up/down Enable.

Bit	Name	Access	Default	Description
				Bit [0] corresponds to PortA0, 1 bit corresponds to 1 pin
7:0	PortB_PE	R/W	8'hFF	Same as PortA
7:0	PortC_PE	R/W	8'hFF	Same as PortA
7:0	PortD_PE	R/W	8'hFF	Same as PortA
1:0	PortE_PE	R/W	2'b11	Same as PortA

14.4.5. PortA/B/C/D/E_PS (Block0, 0x4D, 0x4E, 0x4F, 0x50, 0x51)

table 14-8 PortA/B/C/D/E_PSRegister

Bit	Name	Access	Default	Description
7:0	PortA_PS	R/W	8'h00	PortA Pull up/down selection: 0: Pull down 1: Pull up Bit [0] corresponds to PortA0, 1 bit corresponds to 1 pin
7:0	PortB_PS	R/W	8'h00	Same as PortA
7:0	PortC_PS	R/W	8'h00	Same as PortA
7:0	PortD_PS	R/W	8'h00	Same as PortA
1:0	PortE_PS	R/W	2'b00	Same as PortA

14.4.6. PortA/B/C/D_Monitor_EN (Block0, 0x52, 0x53, 0x54, 0x55)

table 14-9 PortA/B/C/D Monitor EN Register

Bit	Name	Access	Default	Description
7:0	PortA_Monitor_EN	R/W	8'h00	PortA pin detection status change enable: 0: Enable 1: Disable Bit [0] corresponds to PortA0, 1 bit corresponds to 1 pin
7:0	PortB_Monitor_EN	R/W	8'h00	Same as PortA
7:0	PortC_Monitor_EN	R/W	8'h00	Same as PortA
7:0	PortD_Monitor_EN	R/W	8'h00	Same as PortA

14.4.7. PortE_Monitor_EN (Block0, 0x56)

table 14-10 PortE Monitor EN Register

Bit	Name	Access	Default	Description
7:6	Reserved	R/W	2'b00	Reserved

Bit	Name	Access	Default	Description
5	Reserved	R/W	1'b0	Reserved
4	Reserved	R/W	1'b0	Reserved
3	USB_DM Monitor_EN	R/W	1'b0	USB_DM Pin detection status change enable
2	USB_DP Monitor_EN	R/W	1'b0	USB_DP Pin detection status change enable
1:0	PortE_Monitor_EN	R/W	2'b00	PortE Pin detection status change enable: 0: Enable 1: Disable Bit [0] corresponds to PortA0, 1 bit corresponds to 1 pin

14.4.8. Analog_Monitor_EN (Block0, 0x57)

table 14-11 Analog_Monitor_EN Register

Bit	Name	Access	Default	Description
7:0	Reserved	R/W	1'b0	The functions of this part have been moved to Chapter 14.9.7

14.4.9. PortA/B/C/D_Reference_Status (Block0, 0x58, 0x59, 0x5A, 0x5B)

table 14-12 PortA/B/C/D Reference Status Register

Bit	Name	Access	Default	Description
7:0	PortA_Reference_Status	R/W	8'h00	PortA pin status change reference value. Write operation: write the reference status value. Example: write 0x01. And enable PA0 detection status change register , then when PortA0 is detected to change from high to low, an interrupt is triggered. Read operation: current state of the pin.
7:0	PortB_Reference_Status	R/W	8'h00	Same as PortA
7:0	PortC_Reference_Status	R/W	8'h00	Same as PortA
7:0	PortD_Reference_Status	R/W	8'h00	Same as PortA

14.4.10. PortE_Reference_Status (Block0, 0x5C)

table 14-13 PortE Reference Status Register

Bit	Name	Access	Default	Description
7:6	Reserved	R/W	2'b00	Reserved
5	Reserved	R/W	1'b0	Reserved
4	Reserved	R/W	1'b0	Reserved
3	USB_DM Reference_Status	R/W	1'b0	USB_DM pin status change reference value Write: Write the reference value. Read: Current state.
2	USB_DP Reference_Status	R/W	1'b0	USB_DP pin status change reference value Write: Write the reference value. Read: Current state.
1:0	PortE_Reference_Status	R/W	2'b00	PortE pin status change reference value. Write operation: write the reference status value. Example: write 0x01. And enable PE0 detection status change register , then when PortE0 is detected to change from high to low, an interrupt is triggered. Read operation: current state of the pin.

14.4.11. Analog_Reference_Status (Block0, 0x5D)

table 14-14 Analog Reference Status Register

Bit	Name	Access	Default	Description
7:4	Reserved	R/W	4'b0	Reserved
3	OTD Reference_Status	R/W	1'b0	OTD Reference Status Write: Write the reference value. Read: Current state. 0: The chip temperature has not risen to the detection threshold. 1: The chip temperature has risen to the detection threshold
2	LVD Reference_Status	R/W	1'b0	LVD Reference Status Write: Write the reference value. Read: Current state. 0: The VBAT voltage has not dropped to the detection threshold. 1: The VBAT voltage has dropped to the detection threshold
1	ACOK Reference_Status	R/W	1'b0	ACOK Reference Status Write: Write the reference value.

Bit	Name	Access	Default	Description
				Read: Current state. 0: The charging power supply is unplugged. 1: The charging power supply is plugged in.
0	BAT_FULL Reference_Status	R/W	1'b0	BAT_FULL Reference Status Write: Write the reference value. Read: Current state. 0: Not fully charged. 1: Fully charged.

14.4.12. PortA/B/C/D/E_OEN (Block0, 0x66, 0x67, 0x68, 0x69, 0x6A)

table 14-15 PortA/B/C/D/E_OEN Register

Bit	Name	Access	Default	Description
7:0	PortA_OEN	R/W	8'hFF	PortA Input/output selection: 0: Output 1: Input Bit[0] corresponds to PortA0, 1 bit corresponds to 1 pin
7:0	PortB_OEN	R/W	8'hFF	Same as PortA
7:0	PortC_OEN	R/W	8'hFF	Same as PortA
7:0	PortD_OEN	R/W	8'hFF	Same as PortA. Note: PD5 does not have independent input/output control. The input/output control of PD5 is determined by PD4.
1:0	PortE_OEN	R/W	2'b11	Same as PortA

14.4.13. PortA/B/C/D/E_Data(Block0, 0x6B, 0x6C, 0x6D, 0x6E, 0x6F)

table 14-16 PortA/B/C/D/E_Data Register

Bit	Name	Access	Default	Description
7:0	PortA_Data	R/W	8'h00	PortA output: 0: Low level 1: High level Bit[0] corresponds to PortA0, 1 bit corresponds to 1 pin
7:0	PortB_Data	R/W	8'h00	Same as PortA

Bit	Name	Access	Default	Description
7:0	PortC_Data	R/W	8'h00	Same as PortA
7:0	PortD_Data	R/W	8'h00	Same as PortA
1:0	PortE_Data	R/W	8'h00	Same as PortA

14.5. PMU_RTC

14.5.1. PMU_RTC Function Description

Use the internal low speed clock (LowPowerRC). Use 32-bit counter to count up. With alarm clock function, two alarm clock times can be configured.

Note: When using the alarm clock function, just enable the alarm clock A/B interrupt enable bit of the PMU interrupt control register. When the count value reaches the alarm clock setting value, the interrupt can be triggered.

14.5.2. PMU_RTC Register Mapping

The RTC register is located in the Block1 area of the PMU.

table 14-17 PMU_RTC Register Mapping

Offset	Name	Description
0x07	RTC_CTRL	RTC Control register
0x08	RTC_CNT	RTC Count value [7:0]
0x09	RTC_CNT	RTC Count value [15:8]
0x0A	RTC_CNT	RTC Count value [23:16]
0x0B	RTC_CNT	RTC Count value [31:24]
0x0C	ALARM_A	ALARM A[7:0]
0x0D	ALARM_A	ALARM A[15:8]
0x0E	ALARM_A	ALARM A[23:16]
0x0F	ALARM_A	ALARM A[31:24]
0x10	ALARM_B	ALARM B[7:0]
0x11	ALARM_B	ALARM B[15:8]

Offset	Name	Description
0x12	ALARM_B	ALARM B[23:16]
0x13	ALARM_B	ALARM B[31:24]

14.5.3. RTC_CTRL (Block1, 0x07)

table 14-18 RTC_CTRL Register

Bit	Name	Access	Default	Description
7:5	Reserved	R/W	3'b000	Reserved
4	RTC_EN	R/W	1'b0	RTC Enable. 0: RTC count Disable 1: RTC count Enable
3	Alarm_B_EN	R/W	1'b0	Alarm B Enable 0: Disable 1: Enable
2	Alarm_A_EN	R/W	1'b0	Alarm A Enable 0: Disable 1: Enable
1:0	Reserved	R/W	2'b00	Reserved

14.5.4. RTC_CNT (Block1, 0x08, 0x09, 0x0A, 0x0B)

table 14-19 RTC_CNT Register

Bit	Name	Access	Default	Description
7:0	RTC_CNT(0x0B)	R/W	8'h00	RTC Count value[31:24]
7:0	RTC_CNT(0x0A)	R/W	8'h00	RTC Count value [23:16]
7:0	RTC_CNT(0x09)	R/W	8'h00	RTC Count value [15:8]
7:0	RTC_CNT(0x08)	R/W	8'h00	RTC Count value [7:0] Write: Update count value Read: Current count value

14.5.5. ALARM_A (Block1, 0x0C, 0x0D, 0x0E, 0x0F)

table 14-20 ALARM_A Register

Bit	Name	Access	Default	Description
7:0	Alarm_A(0x0F)	R/W	8'h00	Alarm A setting value [31:24]
7:0	Alarm_A(0x0E)	R/W	8'h00	Alarm A setting value [23:16]
7:0	Alarm_A(0x0D)	R/W	8'h00	Alarm A setting value [15:8]
7:0	Alarm_A(0x0C)	R/W	8'h00	Alarm A setting value [7:0] Write: Update alarm value Reading: Current alarm value

14.5.6. ALARM_B (Block1, 0x10, 0x11, 0x12, 0x13)

table 14-21 ALARM_B Register

Bit	Name	Access	Default	Description
7:0	Alarm_B(0x13)	R/W	8'h00	AlarmB setting value[31:24]
7:0	Alarm_B(0x12)	R/W	8'h00	AlarmB setting value[23:16]
7:0	Alarm_B(0x11)	R/W	8'h00	AlarmB setting value[15:8]
7:0	Alarm_B(0x10)	R/W	8'h00	AlarmB setting value[7:0] Write: Update alarm value Reading: Current alarm value

14.6. PMU_KeyScan

14.6.1. PMU_KeyScan Fuction Description

The matrix keyboard uses an internal low-speed clock (LowPowerRC), which supports up to 8 rows and 20 columns. Support hardware filtering and anti shake, and support press and release interrupt response.

14.6.2. PMU_KeyScan Register Mapping

The KeyScan register is located in the Block1 area of the PMU.

table 14-22 PMU_KeyScan Register Mapping

Offset	Name	Description
0x18	KeyScan_CTL	Matrix keyboard scan control register
0x19	Scan_Interval	Scan Interval[7:0]
0x1A	Scan_Interval	Scan Interval [9:8]
0x1B	KeyScan_Config	KeyScan , filter configuration
0x1C	ROW_SEL	Row Selection
0x1D	COL_SEL	Column Selection [7:0]
0x1E	COL_SEL	Column Selection [15:8]
0x1F	COL_SEL	Column Selection [19:16]
0x20 ~ 0x33	KEY_VAL	Key value

14.6.3. KeyScan_CTL (Block1, 0x18)

table 14-23 KeyScan_CTL Register

Bit	Name	Access	Default	Description
7:5	Reserved	R/W	2'b00	Reserved
4	KEY_Status_CLR	R/W	1'b0	Key status Clear. When pressing or releasing occurs, set "1". If the interrupt bit is enabled, there will be a interrupt request. If the status bit is not cleared, frequent interrupts will be requested.
3	GHOST_KEY_EN	R/W	1'b0	GhostKey Enable When GhostKey is detected, the corresponding key interrupt will not be requested.
2	Scan_Lowpower_EN	R/W	1'b0	Key scan low power consumption mode. Stop column scanning when no key is used to save power consumption. When a key is detected to be pressed, scan the key value immediately.

Bit	Name	Access	Default	Description
1	Filter_EN	R/W	1'b0	Filter Enable. 0: Disable 1: Enable
0	KeyScan_EN	R/W	1'b0	Matrix key scan enable 0: Disable 1: Enable

14.6.4. Scan_Interval (Block1, 0x19, 0x1A)

table 14-24 Scan_Interval Register

Bit	Name	Access	Default	Description
1:0	Scan_Interval(0x1A)	R/W	2'b00	Matrix keyboard scan interval [9:8]
7:0	Scan_Interval(0x19)	R/W	8'h00	Matrix keyboard scan interval [7:0]

14.6.5. KeyScan_Config (Block1, 0x1B)

table 14-25 KeyScan_Config Register

Bit	Name	Access	Default	Description
7	Reserved	R/W	4'h0	Reserved
6:4	ROW_Select	R/W	3'b000	Row Configuration 001: PortA as row 010: PortB as row 011: PortC as row 100: PortD as row
3:0	Filter_Value	R/W	4'h0	Filter anti shake

14.6.6. ROW_SEL (Block1, 0x1C)

table 14-26 ROW_SEL Register

Bit	Name	Access	Default	Description
7:0	ROW_SEL	R/W	8'h00	Row selection. Up to 8 lines are supported. 1 bit represents 1 row. Example: ROW_Select=PortA. ROW_SEL=0x0F, select PA0~3 as row 0~3. ROW_SEL=0x05, select PA0 and PA2 as row 0 and row 2.

14.6.7. COL_SEL (Block1, 0x1D, 0x1E, 0x1F)

table 14-27 COL_SEL Register

Bit	Name	Access	Default	Description
3:0	COL_SEL (0x1F)	R/W	4'h0	Column Selection [19:16]。
7:0	COL_SEL (0x1E)	R/W	8'h00	Column Selection [15:8]。
7:0	COL_SEL (0x1D)	R/W	8'h00	Column Selection [7:0]。 Up to 20 columns. 1 bit represents 1 column. Example: COL_SEL(0x1D)= 0xFF, columns 0~7 are selected and 8 are enabled. See Appendix II for the pin corresponding to the column.

Note: Column pins cannot be selected repeatedly. For example, PA0 and PC4 can be used as KeyScan_COL[0], but only one can be selected at the same time.

14.6.8. KEY_VAL (Block1, 0x20 ~ 0x33)

table 14-28 KEY_VAL Register

Bit	Name	Access	Default	Description
7:0	KEY_VAL(0x20 ~ 0x33)	R/W	8'h00	Key value, 1 bit corresponds to 1 key.

14.7. PMU_PWM

14.7.1. PMU_PWM function description

The PMU domain provides 3 channel PWM and uses the internal low speed clock (LowPowerRC).

14.7.2. PMU_PWM IO allocation

PWMx	Pins
PWM0	PA0, PA4, PA3, PA7
	PB0, PB4, PB3, PB7
	PC0, PC4, PC3, PC7
	PD0, PD4, PD3, PD3
PWM1	PA1, PA5
	PB1, PB5
	PC1, PC5
	PD1, PD5
PWM2	PA2, PA6
	PB2, PB6
	PC2, PC6
	PD2, PD6

14.7.3. PMU_PWM Register Mapping

PMU_PWM register is in Block1 area of PMU.

table 14-29 PMU_PWM Register Mapping

Offset	Name	Description
0x34	PWM0_CTRL	PWM0 Control Register
0x35	PWM1_CTRL	PWM1 Control Register
0x36	PWM2_CTRL	PWM2 Control Register
0x37	Reserved	Reserved
0x38	PWM0_PRC	PWM0 Period Count[7:0]
0x39	PWM0_PRC	PWM0 Period Count[15:8]
0x3A	PWM0_HRC	PWM0 High Level Count [7:0]
0x3B	PWM0_HRC	PWM0 High Level Count [15:8]
0x3C	PWM1_PRC	PWM1 Period Count[7:0]
0x3D	PWM1_PRC	PWM1 Period Count[15:8]
0x3E	PWM1_HRC	PWM1 High Level Count[7:0]
0x3F	PWM1_HRC	PWM1 High Level Count[15:8]
0x40	PWM2_PRC	PWM2 Period Count [7:0]
0x41	PWM2_PRC	PWM2 Period Count [15:8]
0x42	PWM2_HRC	PWM2 High Level Count [7:0]
0x43	PWM2_HRC	PWM2 High Level Count [15:8]

14.7.4. PWM0 ~ 2_CTRL (Block1, 0x34, 0x35, 0x36)

table 14-30 PWM0~3_CTRL Register

Bit	Name	Access	Default	Description
7:5	Reserved	R/W	3'b000	Reserved
4	PWM_INV	R/W	1'b0	Inverted output enable. 0: Normal output 1: Inverted output
3	PWM_RST	R/W	1'b0	PWM counter reset. 0: Not reset 1: Reset
2	PWM_MODE	R/W	1'b0	Mode selection 0: Loop mode 1: Single mode
1	PWM_OUT_EN	R/W	1'b0	PWM Output Enable 0: Output Disable 1: Output Enable
0	PWM_EN	R/W	1'b0	PWM Enable

Bit	Name	Access	Default	Description
				0: Disable 1: Enable

Note: 3 channel PWM controller has the same content, PWM0~2_CTRL refers to the above.

14.7.5. PWM0_PRC (Block1, 0x38, 0x39)

14.7.6. PWM0_HRC (Block1, 0x3A, 0x3B)

14.7.7. PWM1_PRC (Block1, 0x3C, 0x3D)

14.7.8. PWM1_HRC (Block1, 0x3E, 0x3F)

14.7.9. PWM2_PRC (Block1, 0x40, 0x41)

14.7.10. PWM2_HRC (Block1, 0x42, 0x43)

table 14-31 PWM0~2_PRC Register

Bit	Name	Access	Default	Description
7:0	PWMx_PRC[15:8]	R/W	8'h00	PWMx Period counting time, upper eight bits
7:0	PWMx_PRC[7:0]	R/W	8'h00	PWMx Period counting time, lower eight bits

table 14-32 PWM0~2_HRC Register

Bit	Name	Access	Default	Description
7:0	PWMx_HRC[15:8]	R/W	8'h00	PWMx High level holding time, upper eight bits.
7:0	PWMx_HRC[7:0]	R/W	8'h00	PWMx high level holding time, lower eight bits.

Note: PWM Period = PWMx_PRC + 1. High level hold time = PWM Period - PWMx_HRC

Note: The contents of 3 channel PWM PRC and HRC are the same, refer to the above configuration.

14.8. PMU_IOMux

14.8.1. Function Description

When using PMU to control GPIO, users can configure pin multiplexing function of PMU, such as PWM output and matrix keyboard scanning.

14.8.2. PMU_IOMux Register Mapping

The PMU_IOMux register is in the **Block1** area of the PMU.

table 14-33 PMU_IOMux Register Mapping

Offset	Name	Description
0x6A	PortA_IOMux	PA0~PA3 Pin multiplexing function selection
0x6B	PortA_IOMux	PA4~PA7 Pin multiplexing function selection
0x6C	PortB_IOMux	PB0~PB3 Pin multiplexing function selection
0x6D	PortB_IOMux	PB4~PB7 Pin multiplexing function selection
0x6E	PortC_IOMux	PC0~PC3 Pin multiplexing function selection
0x6F	PortC_IOMux	PC4~PC7 Pin multiplexing function selection
0x70	PortD_IOMux	PD0~PD3 Pin multiplexing function selection
0x71	PortD_IOMux	PD4~PD7 Pin multiplexing function selection
0x72	PortE_IOMux	PE0~PE1 Pin multiplexing function selection

14.8.3. PortA_IOMux (Block1, 0x6A, 0x6B)

table 14-34 PortA IOMux Register

Bit	Name	Access	Default	Description
7:6	PA7_IOMux (0x6B)	R/W	2'b00	Multiplex configuration of PA7 in PMU
5:4	PA6_IOMux (0x6B)	R/W	2'b00	Multiplex configuration of PA6 in PMU
3:2	PA5_IOMux (0x6B)	R/W	2'b00	Multiplex configuration of PA5 in PMU
1:0	PA4_IOMux (0x6B)	R/W	2'b00	Multiplex configuration of PA4 in PMU
7:6	PA3_IOMux (0x6A)	R/W	2'b00	Multiplex configuration of PA3 in PMU
5:4	PA2_IOMux (0x6A)	R/W	2'b00	Multiplex configuration of PA2 in PMU
3:2	PA1_IOMux (0x6A)	R/W	2'b00	Multiplex configuration of PA1 in PMU

Bit	Name	Access	Default	Description
1:0	PA0_IOMux (0x6A)	R/W	2'b00	Multiplex configuration of PA0 in PMU

See [AppendixII](#) for relevant functions.

14.8.4. PortB_IOMux(Block1, 0x6C, 0x6D)

table 14-35 PortB IOMux Register

Bit	Name	Access	Default	Description
7:6	PB7_IOMux (0x6D)	R/W	2'b00	Multiplex configuration of PB7 in PMU
5:4	PB6_IOMux (0x6D)	R/W	2'b00	Multiplex configuration of PB6 in PMU
3:2	PB5_IOMux (0x6D)	R/W	2'b00	Multiplex configuration of PB5 in PMU
1:0	PB4_IOMux (0x6D)	R/W	2'b00	Multiplex configuration of PB4 in PMU
7:6	PB3_IOMux (0x6C)	R/W	2'b00	Multiplex configuration of PB3 in PMU
5:4	PB2_IOMux (0x6C)	R/W	2'b00	Multiplex configuration of PB2 in PMU
3:2	PB1_IOMux (0x6C)	R/W	2'b00	Multiplex configuration of PB1 in PMU
1:0	PB0_IOMux (0x6C)	R/W	2'b00	Multiplex configuration of PB0 in PMU

See [AppendixII](#) for relevant functions.

14.8.5. PortC_IOMux (Block1, 0x6E, 0x6F)

table 14-36 PortC IOMux Register

Bit	Name	Access	Default	Description
7:6	PC7_IOMux (0x6F)	R/W	2'b00	Multiplex configuration of PC7 in PMU
5:4	PC6_IOMux (0x6F)	R/W	2'b00	Multiplex configuration of PC6 in PMU
3:2	PC5_IOMux (0x6F)	R/W	2'b00	Multiplex configuration of PC5 in PMU
1:0	PC4_IOMux (0x6F)	R/W	2'b00	Multiplex configuration of PC4 in PMU
7:6	PC3_IOMux (0x6E)	R/W	2'b00	Multiplex configuration of PC3 in PMU
5:4	PC2_IOMux (0x6E)	R/W	2'b00	Multiplex configuration of PC2 in PMU
3:2	PC1_IOMux (0x6E)	R/W	2'b00	Multiplex configuration of PC1 in PMU
1:0	PC0_IOMux (0x6E)	R/W	2'b00	Multiplex configuration of PC0 in PMU

See [AppendixII](#) for relevant functions.

14.8.6. PortD_IOMux (Block1, 0x70, 0x71)

table 14-37 PortD IOMux Register

Bit	Name	Access	Default	Description
7:6	PD7_IOMux (0x71)	R/W	2'b00	Multiplex configuration of PD7 in PMU
5:4	PD6_IOMux (0x71)	R/W	2'b00	Multiplex configuration of PD6 in PMU
3:2	PD5_IOMux (0x71)	R/W	2'b00	Multiplex configuration of PD5 in PMU
1:0	PD4_IOMux (0x71)	R/W	2'b00	Multiplex configuration of PD4 in PMU
7:6	PD3_IOMux (0x70)	R/W	2'b00	Multiplex configuration of PD3 in PMU
5:4	PD2_IOMux (0x70)	R/W	2'b00	Multiplex configuration of PD2 in PMU
3:2	PD1_IOMux (0x70)	R/W	2'b00	Multiplex configuration of PD1 in PMU
1:0	PD0_IOMux (0x70)	R/W	2'b00	Multiplex configuration of PD0 in PMU

See [AppendixII](#) for relevant functions.

14.8.7. PortE_IOMux (Block1, 0x72)

table 14-38 PortE IOMux Register

Bit	Name	Access	Default	Description
7:4	Reserved	R/W	4'h0	Reserved
3:2	PE1_IOMux (0x72)	R/W	2'b00	Multiplex configuration of PE1 in PMU
1:0	PE0_IOMux (0x72)	R/W	2'b00	Multiplex configuration of PE0 in PMU

See [AppendixII](#) for relevant functions.

14.9. PMU_INT_CTRL

14.9.1. PMU_INT_CTRL Function Description

It includes PMU domain interrupt management, interrupt enable/disable, interrupt clearing, and interrupt query; BAT_Full, ACOK, LVD, OTD filtering configuration.

14.9.2. PMU_INT_CTRL Register Mapping

The PMU_INT_CTRL Registers are in the Block1 area of the PMU.

table 14-39 PMU_INT_CTRL Register Mapping

Offset	Name	Description
0x58	BAT_Full_Filter	BAT_Full Signal filtering configuration
0x59	ACOK_Filter	ACOK Signal filtering configuration
0x5A	LVD_Filter	LVD Signal filtering configuration
0x5B	OTD_Filter	OTD Signal filtering configuration
0x5E	INT_EN0	PMU Interrupt enable register 0
0x5F	INT_EN1	PMU Interrupt enable register 1
0x60	INT_CLR0	PMU Interrupt clear register 0
0x61	INT_CLR1	PMU Interrupt clear register 0
0x62	INT_STS0	PMU Interrupt status register 0
0x63	INT_STS1	PMU Interrupt status register 0
0x64	PortA_Monitor_Status	PortA pin status change detection
0x65	PortB_Monitor_Status	PortB pin status change detection
0x66	PortC_Monitor_Status	PortC pin status change detection
0x67	PortD_Monitor_Status	PortD pin status change detection
0x68	PortE_Monitor_Status	PortE pin status change detection
0x69	ANA_Monitor_Status	Analog status change detection

14.9.3. BAT_Full_Filter (Block1, 0x58)

table 14-40 BAT_Full_Filter Register

Bit	Name	Access	Default	Description
7:0	BAT_Full_Filter	R/W	8'h00	Battery full state detection filter filter clock use LowPowerRC Filter time = BAT_Full_Filter * 0xFF + 0x40 clock period.

14.9.4. ACOK_Filter (Block1, 0x59)

table 14-41 ACOK_Filter Register

Bit	Name	Access	Default	Description
7:0	ACOK_Filter	R/W	8'h00	Charging cable insertion state detection

Bit	Name	Access	Default	Description
				filter Filter clock uses LowPowerRC Filtering time = ACOK_Filter * 0xFF + 0x40 Clock cycle

14.9.5. LVD_Filter (Block1, 0x5A)

table 14-42 LVD_Filter Register

Bit	Name	Access	Default	Description
7:0	LVD_Filter	R/W	8'h00	Low voltage state detection filter. Filter clock uses LowPowerRC Filtering time = LVD_Filter * 0xFF + 0x40 Clock cycle

14.9.6. OTD_Filter (Block1, 0x5B)

table 14-43 OTD_Filter Register

Bit	Name	Access	Default	Description
7:0	OTD_Filter	R/W	8'h00	High temperature state detection filter Filter clock uses LowPowerRC Filtering time = OTD_Filter * 0xFF + 0x40 Clock cycle

14.9.7. Analog_Monitor_Level (Block1, 0x5C)

table 14-1 Analog_Monitor_Level Register

Bit	Name	Access	Default	Description
7:4	Reserved	R/W	4'b0000	Reserved
3	OTD_Level	R/W	1'b0	Over-temperature detection, interrupt trigger status selection. 0: The chip temperature exceeds the

Bit	Name	Access	Default	Description
				detection threshold. 1: The chip temperature does not exceed the detection threshold.
2	LVD_Level	R/W	1'b0	Low voltage detection, interrupt trigger status selection. 0: The VBAT voltage has dropped to the detection threshold. 1: The VBAT voltage has not dropped to the detection threshold
1	ACOK_Level	R/W	1'b0	Charging plug-in/pull-out, interrupt trigger status selection. 0: The charging power supply is plugged in. 1: The charging power supply is unplugged
0	BAT_FULL_Level	R/W	1'b0	Full charge, interrupt trigger status selection. 0: Trigger an interrupt when the battery is fully charged. 1: Trigger an interrupt when the battery is not fully charged

14.9.8. INT_EN0 (Block1, 0x5E)

table 14-44 INT_EN0 Register

Bit	Name	Access	Default	Description
7:5	Reserved	R/W	3'b000	Reserved
4	IWDG_INT_EN	R/W	1'b0	Independent watchdog interrupt enable
3	OTD_INT_EN	R/W	1'b0	OTD status, status change detection interrupt enable For details please refer to chapter Analog_Monitor_Level (Block1, 0x5C)

Bit	Name	Access	Default	Description
				and Analog Reference Status
2	LVD_INT_EN	R/W	1'b0	LVD status, status change detection interrupt enable For details please refer to chapter Analog Monitor Level (Block1, 0x5C) and chapter Analog Reference Status
1	ACOK_INT_EN	R/W	1'b0	ACOK status, status change detection interrupt enable For details please refer to chapter Analog Monitor Level (Block1, 0x5C) and Analog Reference Status
0	BAT_FULL_INT_EN	R/W	1'b0	BAT status, status change detection interrupt enable For details please refer to chapter Analog Monitor Level (Block1, 0x5C) and chapter Analog Reference Status

14.9.9. INT_EN1 (Block1, 0x5F)

table 14-45 INT_EN1 Register

Bit	Name	Access	Default	Description
7:6	Reserved	R/W	2'b0	Reserved
5	GPIO_Monitor_INT_EN	R/W	1'b0	GPIO pin state change detection interrupt enable
4	Reserved	R/W	1'b0	Reserved
3	KEY_SCAN_INT_EN	R/W	1'b0	Keyboard scan interrupt enable
2	AlarmB_INT_EN	R/W	1'b0	Alarm B interrupt enable
1	AlarmA_INT_EN	R/W	1'b0	Alarm A interrupt enable
0	Reserved	R/W	1'b0	Reserved

14.9.10. INT_CLR0(Block1, 0x60)

table 14-46 INT_CLR0 Register

Bit	Name	Access	Default	Description
7:5	Reserved	R/W	3'b000	Reserved
4	IWDG_INT_CLR	R/W	1'b0	Independent watchdog interrupt clear
3	OTD_INT_CLR	R/W	1'b0	OTD Interrupt Clear
2	LVD_INT_CLR	R/W	1'b0	LVD Interrupt Clear
1	ACOK_INT_CLR	R/W	1'b0	ACOK Interrupt Clear
0	BAT_FULL_INT_CLR	R/W	1'b0	BAT Interrupt Clear For all bits of this register, write '1' clear status first. Write '0' after 10us delay.

14.9.11. INT_CLR1 (Block1, 0x61)

table 14-47 INT_CLR1 Register

Bit	Name	Access	Default	Description
7:6	Reserved	R/W	2'b00	Reserved
5	GPIO_Monitor_INT_CLR	R/W	1'b0	GPIO pin state change detection interrupt clear
4	Reserved	R/W	1'b0	Reserved
3	KEY_SCAN_INT_CLR	R/W	1'b0	Keyboard scan interrupt clear
2	AlarmB_INT_CLR	R/W	1'b0	Alarm B interrupt clear
1	AlarmA_INT_CLR	R/W	1'b0	Alarm A interrupt clear For all bits of this register, write '1' clear status first. Write '0' after 10us delay.
0	Reserved	R/W	1'b0	Reserved

14.9.12. INT_STS0 (Block1, 0x62)

table 14-48 INT_STS0 Register

Bit	Name	Access	Default	Description
7:5	Reserved	R/W	2'b000	Reserved
4	IWDG_INT_STS	R	1'b0	Independent watchdog interrupt status
3	OTD_INT_STS	R	1'b0	OTD Interrupt status
2	LVD_INT_STS	R	1'b0	LVD Interrupt status
1	ACOK_INT_STS	R	1'b0	ACOK Interrupt status
0	BAT_FULL_INT_STS	R	1'b0	BAT Interrupt status

14.9.13. INT_STS1 (Block1, 0x63)

table 14-49 INT_STS1 Register

Bit	Name	Access	Default	Description
7:6	Reserved	R/W	2'b00	Reserved
5	GPIO_Monitor_INT_STS	R/W	1'b0	GPIO pin state change detection interrupt status
4	Reserved	R/W	1'b0	Reserved
3	KEY_SCAN_INT_STS	R/W	1'b0	Keyboard scan interrupt status
2	AlarmB_INT_STS	R/W	1'b0	Alarm B interrupt status
1	AlarmA_INT_STS	R/W	1'b0	Alarm A interrupt status
0	Reserved	R/W	1'b0	Reserved

14.9.14. PortA/B/C/D_Monitor_Status (Block1, 0x64, 0x65, 0x66, 0x67)

table 14-50 PortA/B/C/D_Monitor_Status Register

Bit	Name	Access	Default	Description
7:0	PortA_Monitor_Status	R/W	8'h00	PortA pin status detection: 0: No change 1: Change Bit[0] corresponds to PortA0, 1 bit corresponds to 1 pin
7:0	PortB_Monitor_Status	R/W	8'h00	Same as PortA

Bit	Name	Access	Default	Description
7:0	PortC_Monitor_Status	R/W	8'h00	Same as PortA
7:0	PortD_Monitor_Status	R/W	8'h00	Same as PortA

14.9.15. PortE_Monitor_Status (Block1, 0x68)

table 14-51 PortE_Monitor_Status Register

Bit	Name	Access	Default	Description
7:4	Reserved	R/W	4'b0000	Reserved
3	USB_DM Monitor_Status	R/W	1'b0	0: USB_DM pin status has not changed 1: USB_DM pin status has changed
2	USB_DP Monitor_Status	R/W	1'b0	0: USB_DP pin status has not changed 1: USB_DP pin status has changed
1:0	PortE_Monitor_Status	R/W	2'b00	PortE pin status detection: 0: pin status has not changed 1: pin status has changed Bit[0] corresponds to PortE0, 1 bit corresponds to 1 pin

15. LCD

15.1. Overview

The LCD interface conforms to the standard 8080, 6800 bus timing, which can be used to drive the LCD screen conforming to 8080 timing or 6800 timing, or other peripheral devices.

15.2. Feature

- Support 8080, 6800 bus timing
- Tx FIFO up to 128 bytes
- Support VSYNC and HSYNC detection
- Bus 8BIT and 16BIT are optional

15.3. LCD Block Diagram

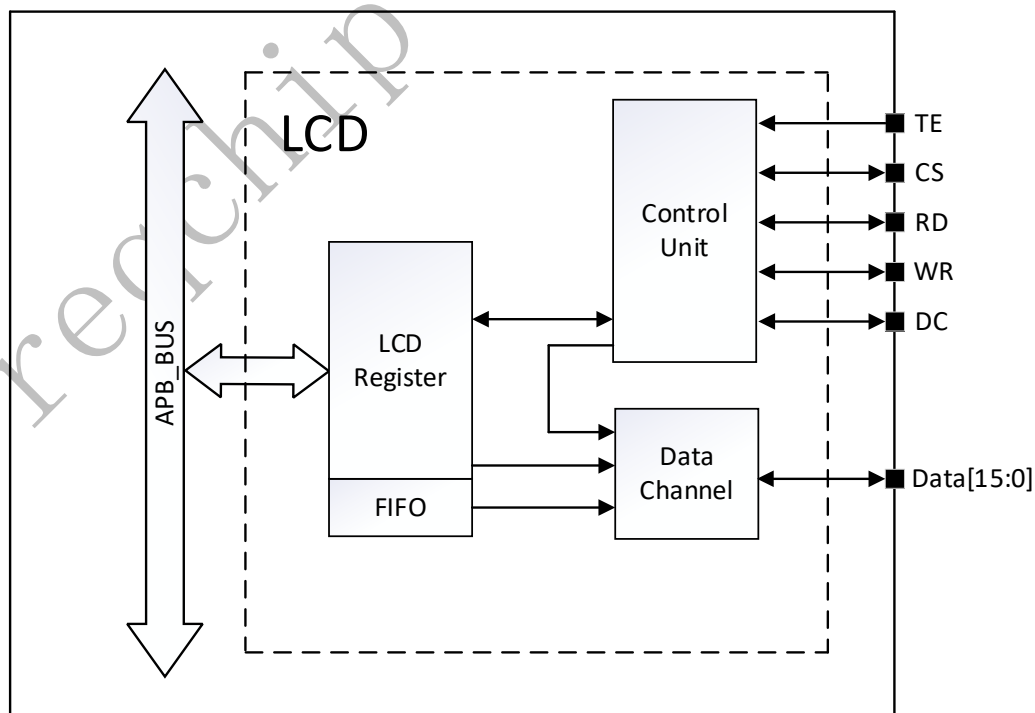


Figure 15-1 LCD Block Diagram

15.4. LCD Register

15.4.1. LCD Register Mapping

LCD Register Base Addr: 0x500D_0000

table 15-1 LCD Register Mapping

Offset	Name	Description
0x00	LCD_TXFF_PUSH	Tx FIFO Register
0x04	LCD_CFG	Configuration register
0x08	LCD_CFGWR_STATUS	Write Status Register
0x0C	LCD_CFGRD_STATUS	Read Status Register
0x10	LCD_APB_RD	Read Data Register
0x14	LCD_TXFF_AEMP_LV	Tx FIFO Remaining Data Depth
0x18	LCD_DAT_WR_LEN	Write Data Length Register
0x1C	LCD_DAT_WR_CFG	Data Configuration Register
0x20	LCD_DMA	DMA Configuration Register
0x24	LCD_CSX	CS Line Configuration Register
0x28	LCD_CRM	Read/write Clock Configuration Register
0x2C	LCD_TXFF_CLR	Tx FIFO Clearance Register
0x30	LCD_INTF_CFG	Interface Configuration Register
0x34	LCD_INT	Interrupt Enable, Query, Clearance Register
0x38	LCD_TE_VSYNC_CFG	Vertical Synchronization Detection Configuration Register
0x3C	LCD_TE_HSYNC_CFG	Horizontal Synchronization Detection Configuration Register

15.4.2. LCD_TXFF_PUSH (Offset 0x00)

table 15-2 LCD_TXFF_PUSH Register

Bit	Name	Access	Default	Description
31:0	LCD_TXFF_PUSH_DATA	W	32'h0	Tx FIFO

15.4.3. LCD_CFG (Offset 0x04)

table 15-3 LCD_CFG Register

Bit	Name	Access	Default	Description
31	LCD_CFG_WRT	R/W	1'b0	1: Write 0: Read
30	LCD_CFG_CDX	R/W	1'b0	Valid only when LCD_CFG_WR is 1 1: LCD_CFG_DATA is a parameter 0: LCD_CFG_DATA is a command
29:8	LCD_CFG_LEN	R/W	22'h00	Valid only when LCD_CFG_WR is 0, indicating the number of data to be read.
7:0	LCD_CFG_DATA	R/W	8'h00	The command or parameter to be sent.

15.4.4. LCD_CFGWR_STATUS (Offset 0x08)

table 15-4 LCD_CFGWR_STATUS Register

Bit	Name	Access	Default	Description
31:1	Reserved	R	31'h00	Reserved
0	LCD_CFGWR_STATUS	R	1'b0	After CFG WR operation (LCD_CFG_WR is 1, Write LCD_CFG register), this bit is 0. When the parallel bus completes the write operation, the status bit is 1.

15.4.5. LCD_CFGRD_STATUS (Offset 0x0C)

table 15-5 LCD_CFGRD_STATUS Register

Bit	Name	Access	Default	Description
31:1	Reserved	R	31'h00	Reserved
0	LCD_CFGRD_STATUS	R	1'b0	After CFG RD operation (LCD_CFG_WR is 0, Write LCD_CFG register), When the data is readable, the status bit is 1. After reading LCD_APB_RD, this bit is cleared

Bit	Name	Access	Default	Description
				to 0. When the data is readable again, the status bit is 1 again until all LCD_CFG_LEN data are read.

15.4.6. LCD_APB_RD (Offset 0x10)

table 15-6 LCD_APB_RD Register

Bit	Name	Access	Default	Description
31:16	Reserved	R	16'h00	Reserved
15:0	LCD_APB_RD_DATA	R	16'h00	When the data on the bus is stored in the register, the LCD_CFGRD_STATUS is 1, and the CPU can get the data through this address. After reading, LCD_CFGRD_STATUS clears to 0.

15.4.7. LCD_TXFF_AEMP_LV (Offset 0x14)

table 15-7 LCD_TXFF_AEMP_LV Register

Bit	Name	Access	Default	Description
31:5	Reserved	R	27'h00	Reserved
4:0	LCD_TXFF_AEMP_LV	R/W	5'h0	Tx FIFO empty threshold. When the data in TxFIFO is equal to or less than the empty threshold value, TxFIFO empty threshold status is set.

15.4.8. LCD_DAT_WR_LEN (Offset 0x18)

table 15-8 LCD_DAT_WR_LEN Register

Bit	Name	Access	Default	Description
31:24	Reserved	R	8'h00	Reserved
23:0	LCD_DAT_WR_LEN	R/W	24'h00	LCD DAT WR Data length

15.4.9. LCD_DAT_WR_CFG (Offset 0x1C)

table 15-9 LCD_DAT_WR_CFG Register

Offset	Name	Access	Default	Description
31:16	Reserved	R	0x0	Reserved
15:14	DATA_TRANS_SEQ_3	R/W	2'b11	In 8bit bus mode, the fourth transfer data selection, or In 16bit bus mode, MSB part of the second transfer data selection. 00: FIFO_DATA [7:0] 01: FIFO_DATA [15:8] 10: FIFO_DATA [23:16] 11: FIFO_DATA [31:24]
13:12	DATA_TRANS_SEQ_2	R/W	2'b10	In 8bit bus mode, the third transfer data selection, or In 16bit bus mode, LSB part of the second transfer data selection. 00: FIFO_DATA [7:0] 01: FIFO_DATA [15:8] 10: FIFO_DATA [23:16] 11: FIFO_DATA [31:24]
11:10	DATA_TRANS_SEQ_1	R/W	2'b01	In 8bit bus mode, the second transfer data selection, or In 16bit bus mode, MSB part of the first transfer data selection. 00: FIFO_DATA [7:0] 01: FIFO_DATA [15:8] 10: FIFO_DATA [23:16] 11: FIFO_DATA [31:24]
9:8	DATA_TRANS_SEQ_0	R/W	2'b00	In 8bit bus mode, the first transfer data selection, or In 16bit bus mode, LSB part of the first transfer data selection. 00: FIFO_DATA [7:0]

Offset	Name	Access	Default	Description
				01: FIFO_DATA [15:8] 10: FIFO_DATA [23:16] 11: FIFO_DATA [31:24]
7	DATA_BUS_16BIT	R/W	1'b0	1: 16 bit parallel data bus selected 0: 8 bit parallel data bus selected
6:4	WR_H_LEN	R/W	3'b1	During DAT WR operation, the number of high level holding wrclk
3	Reserved	R	1'b0	Reserved
2:0	WR_L_LEN	R/W	3'b1	During DAT WR operation, the number of low level holding wrclk.

15.4.10. LCD_DMA (Offset 0x20)

table 15-10 LCD_DMA Register

Bit	Name	Access	Default	Description
31:6	Reserved	R	0x0	Reserved
5	DMA_CONTROL	R/W	1'b0	1: enable 0: disable
4:0	DMA_TX_LEVEL	R/W	5'h10	When the number of data in TX FIFO is less than or equal to the threshold, a DMA Tx request is initiated.

15.4.11. LCD_CSX (Offset 0x24)

table 15-11 LCD_CSX Register

Bit	Name	Access	Default	Description
31:1	Reserved	R	0x0	Reserved
0	LCD_CSX	R/W	1'b1	CS bus signal control 1: Pull up the bus 0: Pull down the bus

15.4.12. LCD_CRM(Offset 0x28)

table 15-12 LCD_CRM Register

Bit	Name	Access	Default	Description
31:8	Reserved	R	0x0	Reserved
7:4	RDCLK_CFG	R/W	4'h0	Take m_Clk as the clock source, rdclk clock division is 3: 4 4: 6 5: 8 6: 16 7: 32 8: 64
3	Reserved	R	0x0	Reserved
2:0	WRCLK_CFG	R/W	3'b000	Take m_clk as the clock source, wrclk clock division is 0: 1 1: 2 2: 3 3: 4 4: 6 5: 8

15.4.13. LCD_TXFF_CLR (Offset 0x2C)

table 15-13 LCD_TXFF_CLR Register

Bit	Name	Access	Default	Description
31:2	Reserved	R	0x0	Reserved
1	LCD_TXFF_PUSH_CLR	R/W	1'b0	Reset and write FIFO pointer 1: reset push pointer 0: release
0	LCD_TXFF_POP_CLR	R/W	1'b0	Reset and read FIFO pointer 1: reset pop pointer

Bit	Name	Access	Default	Description
				0: release

15.4.14. LCD_INTF_CFG (Offset 0x30)

table 15-14 LCD_INTF_CFG Register

Bit	Name	Access	Default	Description
31:4	Reserved	R	0x0	Reserved
3	P68_RD	R/W	1'b0	In 6800 mode 1: Read (configured as 1 before CFG_RD and 0 after operation) 0: Write
2	P68_MODE	R/W	1'b0	8080 or 6800 selection: 1: 6800 0: 8080
1	CS_POLARITY	R/W	1'b0	CS polarity selection: 1: high active 0: low active
0	DC_POLARITY	R/W	1'b0	DC polarity selection: 1: high active 0: low active

15.4.15. LCD_INT (Offset 0x34)

table 15-15 LCD_INT Register

Bit	Name	Access	Default	Description
31:12	Reserved	R	0x0	Reserved
11	Hsync_int_clear	W	1'b0	Write '1' to clear Hsync status
10	Hsync	R	1'b0	Horizontal synchronization status
9	Hsync_int_enable	R/W	1'b0	Horizontal synchronization interrupt enable 0: Disable 1: Enable
8	Vsync_int_clear	W	1'b0	Write '1' to clear Vsync status

Bit	Name	Access	Default	Description
7	Vsync	R	1'b0	Vertical synchronization status
6	Vsync_int_enable	R/W	1'b0	Vertical synchronization status interrupt enable 0: Disable 1: Enable
5	txfifo_aempt	R	1'b1	Tx FIFO almost empty status
4	txfifo_aempt_int_enable	R/W	1'b0	Tx FIFO almost empty status interrupt enable 0: Disable 1: Enable
3	txfifo_empty	R	1'b1	Tx FIFO empty status
2	txfifo_empty_inten	R/W	1'b0	Tx FIFO empty status interrupt enable 0: Disable 1: Enable
1	txfifo_full	R	1'b0	Tx FIFO full status
0	txfifo_full_inten	R/W	1'b0	Tx FIFO full status interrupt enable 0: Disable 1: Enable

15.4.16. LCD_TE_VSYNC_CFG (Offset 0x38)

table 15-16 LCD_TE_VSYNC_CFG Register

Bit	Name	Access	Default	Description
31:24	Reserved	R	0x0	Reserved
23:0	TE_VSYNC_WIDTH_NUM	R/W	24'h1E	When the holding time of TE signal high level reaches TE_VSYNC_WIDTH_NUM pclk(cpu clock), Indicates that Vsync wide pulse is detected. If Vsync interrupt is enabled, the corresponding interrupt is set.

15.4.17. LCD_TE_HSYNC_CFG (Offset 0x3C)

table 15-17 LCD_TE_HSYNC_CFG Register

Bit	Name	Access	Default	Description
31:13	Reserved	R	0x0	Reserved
12	TE_HSYNC_POLARITY	R/W	1'b0	line number polarity 1: Rising edge line number + 1 0: Falling edge line number + 1
11:0	TE_HSYNC_LINE_NUM	R/W	12'h01	After Vsync wide pulse is detected, when the TE_HSYNC_LINE_NUM rising edge or falling edge appears, if te_hsync interrupt is enabled, the corresponding interrupt is set.

15.5. Process

15.5.1. Write command or parameter

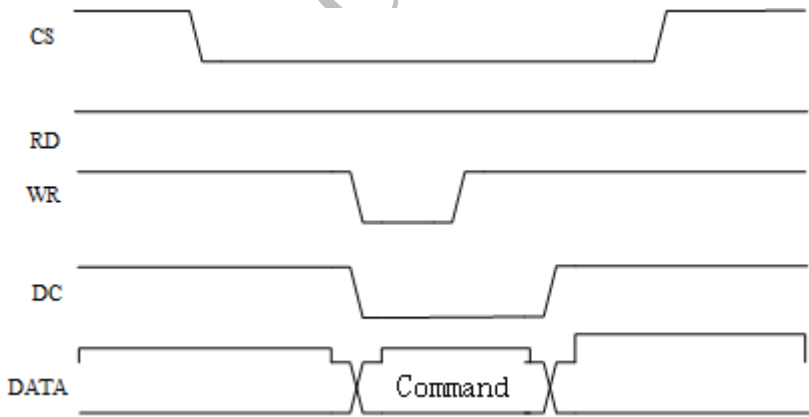


Figure 15-2 8080 Write Command Timing

1. First pull down the CSX signal.
2. Configure the LCD_CFG register. Because it is a write command and parameter operation, LCD_CFG_WR is set to 1. If CMD needs to be sent, LCD_CFG_CDX should be configured as 0. If parameter is to be sent, LCD_CFG_CDX should be configured as 1. Write command or parameter to LCD_CFG_DATA.
3. Read LCD_CFGWR_STATUS, ensure that the last transfer has been completed, and

repeat steps 2 and 3 for another transfer.

4. Finally pull up the CSX signal.

15.5.2. Write Data

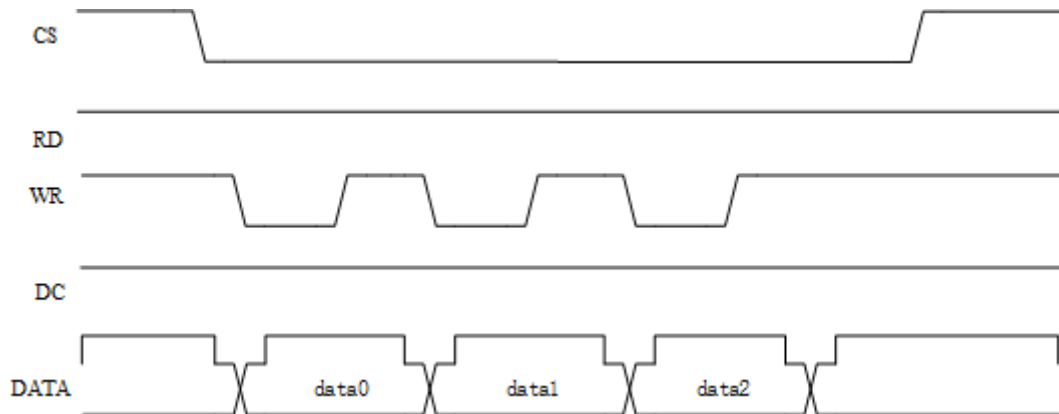


Figure 15-3 8080 Write Data Timing

1. First pull down the CSX signal.
2. Configure LCD_DAT_WR_LEN according to the number of data to be transfered.
3. Write data to FIFO.
4. Finally pull up the CSX signal.

15.5.3. Read Data

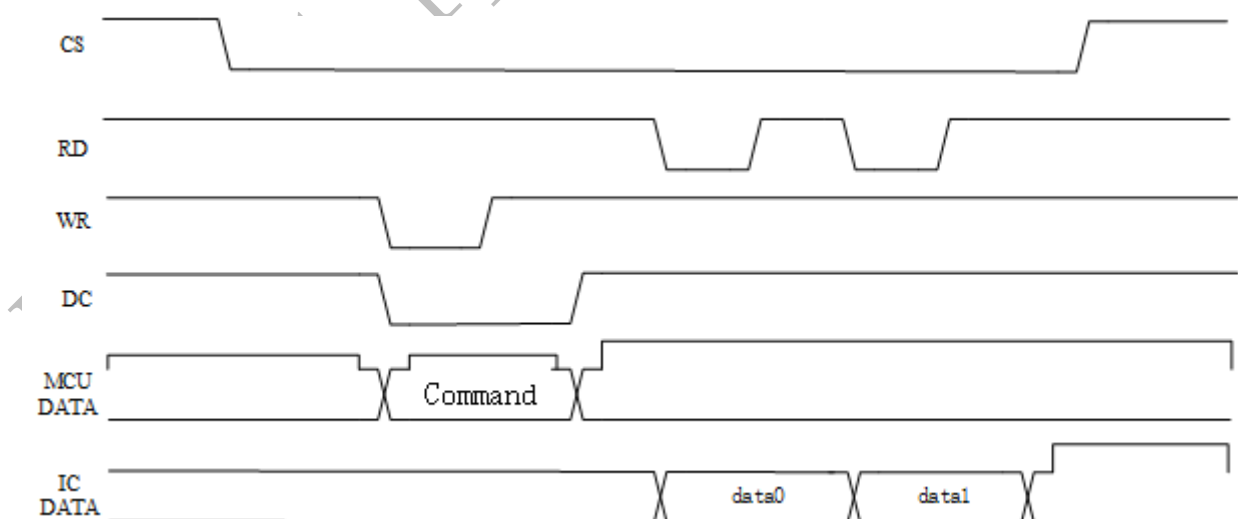


Figure 15-4 8080 Read Data Timing

1. First pull down the CSX signal.

2. Configure the LCD_CFG register. As it is a read operation, LCD_CFG_WR is set to 0. LCD_CFG_CDX does not need to be written. Set LCD_CFG_LEN to write the length of data to be read. Write the CMD to be read to LCD_CFG_DATA register (read data operation, send the CMD first, and then perform the read operation).
3. Read the LCD_CFGRD_STATUS register to get status, make sure the data is ready, read the LCD_APB_RD_DATA register to get the data, and repeat this step until all data which indicate by LCD_CFG_LEN register are read (after reading all data, the LCD will not be set again).
4. Finally pull up the CSX signal.

16. Timer

16.1. Overview

Two 32bit timers, which can count down, support loop counting, and interrupt triggering. Each timer is completely independent and does not share any resources with each other.

16.2. Feature

- 32bit count
- Support loop counting

16.3. System Block Diagram

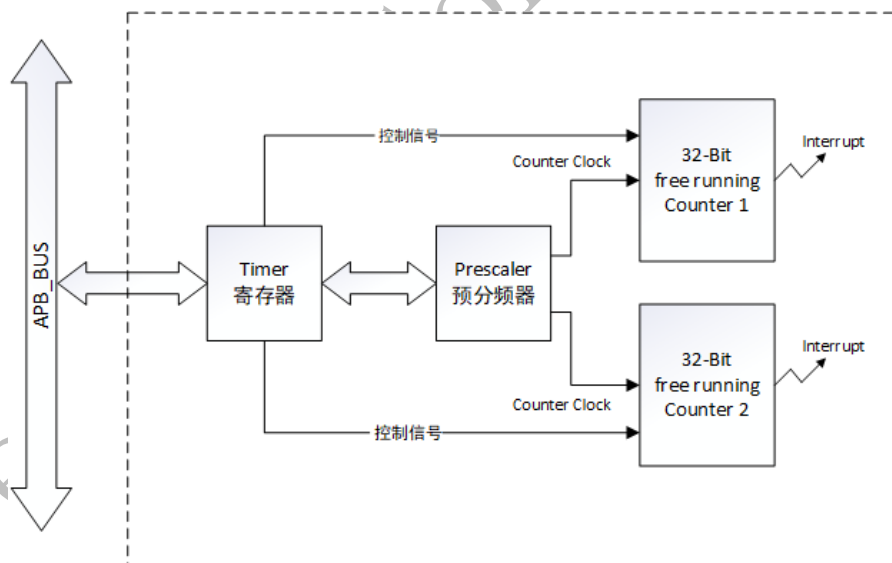


Figure 16-1 Timer Block Diagram

16.4. Timer Register

16.4.1. Timer Register Mapping

Timer0 Register Base Addr: 0x5001_0000

Timer0 Register Base Addr: 0x5001_0020

table 16-1 Timer Register Mapping

Offset	Name	Description
0x00	LOAD_VALUE	Count value load register
0x04	COUNT_VALUE	Current count value register
0x08	CONTROL	Control register
0x0C	INT_CLR	Interrupt control, query register

16.4.2. LOAD_VALUE (Offset 0x00)

table 16-2 LOAD_VALUE Register

Bit	Name	Access	Default	Description
31:0	LOAD	R/W	32'h00	Timer initial value

16.4.3. COUNT_VALUE (Offset 0x04)

table 16-3 COUNT_VALUE Register

Bit	Name	Access	Default	Description
31:0	CNT	R	32'hFFFFFFFF	Current count value

16.4.4. CONTROL (Offset 0x08)

table 16-4 CONTROL Register

Bit	Name	Access	Default	Description
31:8	Reserved	R/W	0x0	Reserved

Bit	Name	Access	Default	Description
7	CNT_EN	R/W	1'b0	Counter enable: 1: Start counting 0: Stop counting
6	CNT_MODE	R/W	1'b0	Counter mode selection: 0: single count 1: loop count
5:4	Reserved	R/W	2'b0	Reserved
3:2	PRESCALE	R/W	2'b0	Frequency division selection: 00: pclk 01: pclk/16 10: pclk/256
1:0	Reserved	R/W	2'b0	Reserved

16.4.5. INT_CLR (Offset 0x0C)

table 16-5 INT_CLR Register

Bit	Name	Access	Default	Description
31:16	Reserved	R/W	0x0	Reserved
15:0	INT_CLR	R/W	16'h00	Write '1' to clear the interrupt status.

16.5. Process

1. Write LOAD_VALUE register, configure timer initial value.
2. Write CONTROL register, configure timer frequency division, select loop count, enable count (timer will automatically enable timer interrupt).
3. The timer counts down from the initial value until the count is 0 to trigger interrupt.
4. Write INT_CLR register to clear interrupt.

17. ADC

17.1. Overview

10-bit ADC (Analog-to-digital converter). Up to 8 channels, and the A/D conversion of each channel can be performed in cycle mode.

17.2. Feature

- 10-bit resolution
- 8 independent sampling channels
- cycle conversion mode
- Data FIFO with 64 depth and 10 bit width
- Support battery voltage and core temperature detection

17.3. Block Diagram

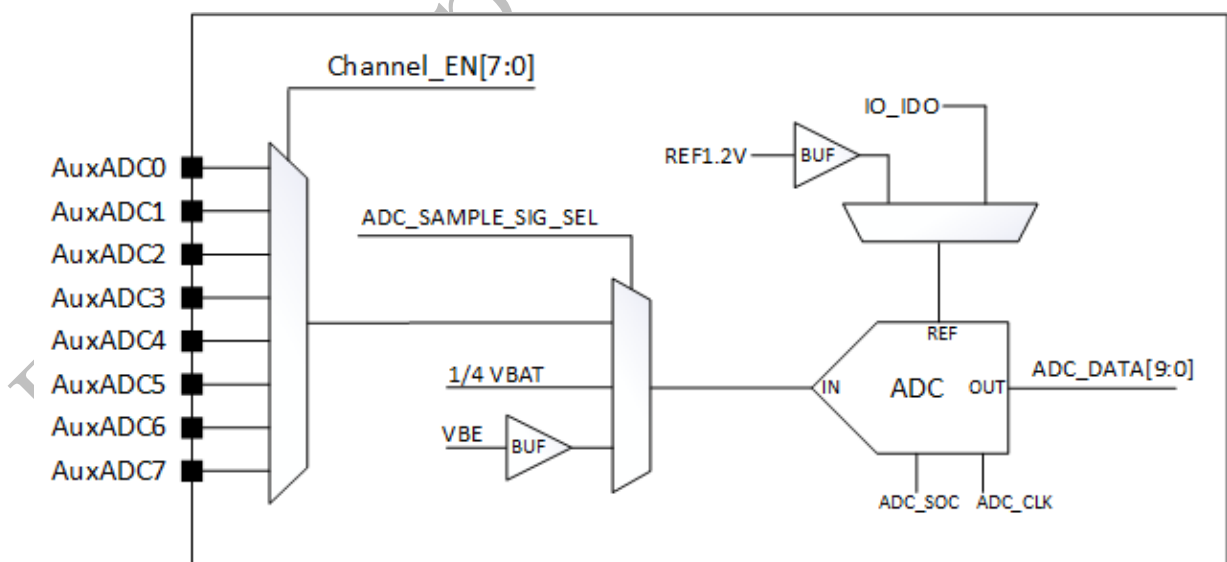


Figure 17-1 ADC Block Diagram

17.4. Function Description

For cycle conversion mode:

$$ADC_CLK = \frac{48MHz}{(CLK_DIV + 1) \times 2}$$

$$ADC_Channel_Period = ADC_CLK / 15$$

After enabling ADC sampling, the ADC controller starts from channel 0 and starts cycle conversion of enabled channels in Channel_EN. After each channel conversion, write the conversion value into the corresponding channel data register. If FIFO is enabled, the converted data will be stored in ADC_FIFO, the data in FIFO can be taken out by reading the [9: 0] bit of DATA register.

17.5. ADC Register

17.5.1. ADC Register Mapping

ADC Base Addr: 0x5009_0000

table 17-1 ADC Register Mapping

Offset	Name	Description
0x00	CONTROL	ADC Control Register
0x04	DATA0	Channel 0 to 7 data register
0x08	DATA1	
0x0C	DATA2	
0x10	DATA3	
0x14	DATA4	
0x18	DATA5	
0x1C	DATA6	
0x20	DATA7	
0x24	DATA	Data FIFO Register
0x28	FIFO_Status	FIFO Status Register

Offset	Name	Description
0x2C	INT	Interrupt status register
0x30	INTC	Interrupt status clear register
0x34	ADC_ANA_CTRL0	ADC analog control register 0
0x38	ADC_ANA_CTRL1	ADC analog control register 1
0x3C	ADC_ANA_CTRL2	ADC analog control register 2

17.5.2. CONTROL (Offset 0x00)

table 17-2 CONTROL Register

Bit	Name	Access	Default	Description
31	FIFO_CLR	R/W	1'b0	FIFO Clear. 0: Stop clearing 1: Clear FIFO Note: The software needs to write '1' to clear and then write '0' to stop.
30	ADC_TRIG_SW	R/W	1'b0	Conversion trigger conditions. 0: Software trigger 1: Disable Note: Software enable ADC conversion by setting ADC_EN as '1'.
29:24	DMA_Level	R/W	6'h30	DMA handling Level. Data in FIFO is equal to or greater than DMA_Level, generate a DMA request.
23	DMA_EN	R/W	1'b0	DMA Enable.
22	Reserved	R/W	0x0	Reserved
21:16	FIFO_Almost_Full_Level	R/W	6'h30	FIFO Almost full Level. When the data in FIFO is equal to or greater than FIFO_Almost_Full_Level, FIFO_ALMOST_FULL is set.
15:8	Channel_EN	R/W	8'h00	Channel Enable. 1 bit corresponds to 1 channel. Bit[8] ---- Channel0

Bit	Name	Access	Default	Description
				Bit[9] ---- Channel1 Bit[10] --- Channel2 Bit[11] --- Channel3 Bit[12] --- Channel4 Bit[13] --- Channel5 Bit[14] --- Channel6 Bit[15] --- Channel7
7	ADC_EN	R/W	1'b0	ADC conversion enable. 0: Disable 1: Enable
6:4	Reserved	R/W	0x0	Reserved
3	DATA_VALID	R	1'b0	Data valid signal. 0: Conversion not completed. 1: Internal signal conversion completed. Note: This bit is valid only when ADC_MODE is '1'. It indicates that the internal signal conversion is completed, and the conversion result value can be read from the DATA register.
2	ADC_MODE	R/W	1'b0	ADC mode selection. 0: External input signal cycle conversion mode. 1: Internal signal cycle conversion mode. Note: When set to '1', measure 1/4 VBAT and VEB.
1	FIFO_EN	R/W	1'b0	FIFO Enable. 0: Disable FIFO 1: Enable FIFO Note: After enable FIFO, the conversion result value will be saved in FIFO. Data in FIFO can be read out by reading DATA register.
0	ADC_CK_EN	R/W	1'b0	ADC sample clock enable

17.5.3. DATA0 ~ DATA7 (Offset 0x04 ~ 0x20)

table 17-3 DATA0~DATA7 Register

Bit	Name	Access	Default	Description
31:10	Reserved	R/W	0x0	Reserved
9:0	Channel0(Offset 0x04)	R/W	0x0	Channel 0 sampling conversion data.
9:0	Channel1(Offset 0x08)	R/W	0x0	Channel 1 sampling conversion data.
9:0	Channel2(Offset 0x0C)	R/W	0x0	Channel 2 sampling conversion data.
9:0	Channel3(Offset 0x10)	R/W	0x0	Channel 3 sampling conversion data.
9:0	Channel4(Offset 0x14)	R/W	0x0	Channel 4 sampling conversion data.
9:0	Channel5(Offset 0x18)	R/W	0x0	Channel 5 sampling conversion data.
9:0	Channel6(Offset 0x1C)	R/W	0x0	Channel 6 sampling conversion data.
9:0	Channel7(Offset 0x20)	R/W	0x0	Channel 7 sampling conversion data.

Note: When FIFO is not used, the data will be stored in the above registers.

17.5.4. DATA (Offset 0x24)

table 17-4 DATA Register

Bit	Name	Access	Default	Description
31:10	Reserved	R	0x0	Reserved
9:0	ADC_FIFO	R	0x0	When FIFO is enabled or internal signal is tested, the conversion result will be saved in FIFO, and the result can be read out from FIFO by reading this register.

Note: When using the FIFO, the data will be stored in the above Register.

17.5.5. FIFO_Status (Offset 0x28)

table 17-5 FIFO_Status Register

Bit	Name	Access	Default	Description
31:7	Reserved	R	0x0	Reserved
4	FIFO_UNDER	R	1'b0	FIFO underflow
3	FIFO_OVER	R	1'b0	FIFO overflow

Bit	Name	Access	Default	Description
2	FIFO_ALMOST_FULL	R	1'b0	FIFO almost full
1	FIFO_EMPTY	R	1'b1	FIFO empty
0	FIFO_FULL	R	1'b0	FIFO full

17.5.6. INT (Offset 0x2C)

table 17-6 INT Register

Bit	Name	Access	Default	Description
31:23	Reserved	R	0x0	Reserved
22	SW	R	1'b0	Corresponding interrupt status
21	ERR	R	1'b0	
20	FIFO_UNDER	R	1'b0	
19	FIFO_OVER	R	1'b0	
18	FIFO_ALMOST_FULL	R	1'b0	
17	FIFO_EMPTY	R	1'b0	
16	FIFO_FULL	R	1'b0	
15:7	Reserved	R	0x0	Reserved
6	SW_INT	R/W	1'b0	Software trigger conversion interrupt enable.
5	ERR_INT	R/W	1'b0	ADC error interrupt enable. If ADC Error is detected, ADC module should be reset.
4	FIFO_UNDER_INT	R/W	1'b0	FIFO under interrupt enable. Read DATA when FIFO is empty.
3	FIFO_OVER_INT	R/W	1'b0	FIFO overflow interrupt enable.
2	FIFO_ALMOST_FULL_INT	R/W	1'b0	FIFO almost full interrupt enable.
1	FIFO_EMPTY_INT	R/W	1'b0	FIFO empty interrupt enable.
0	FIFO_FULL_INT	R/W	1'b0	FIFO full interrupt enable. 0: Disable 1: Enable

17.5.7. INTC (Offset 0x30)

table 17-7 INTC Register

Bit	Name	Access	Default	Description
31:7	Reserved	R/W	0x0	Reserved
6	SW_CLR	W	1'b0	Write '1' to clear the corresponding status.
5	ERR_CLR	W	1'b0	
4	FIFO_UNDER_CLR	W	1'b0	
3	FIFO_OVER_CLR	W	1'b0	
2	FIFO_ALMOST_FULL_CLR	W	1'b0	
1	FIFO_EMPTY_CLR	W	1'b0	
0	FIFO_FULL_CLR	W	1'b0	

17.5.8. ADC_ANA_CTRL0 (Offset 0x34)

table 17-8 ADC_ANA_CTRL0 Register

Bit	Name	Access	Default	Description
31:16	Delay_For_Setup	R/W	16'h50	Time delay for setup The unit is 48MHz clock cycle.
15:8	CLK_DIV	R/W	8'h5	ADC sampling clock division. $48\text{MHz} / ((\text{CLK_DIV} + 1) * 2)$ 注: CLK_DIV >= 1。
7:5	ADC_SAMPLE_SIG_SEL	R/W	3'h0	Conversion source selection 0x3: IO(pin voltage) 0x6: VBE(temperature) 0x7: 1/4 VBAT(1/4 Battery voltage)
4	ADC_SACLK_CTL	R/W	1'b1	ADC internal clock control. Always set '1'.
3	ADC_CLK_EDGE	R/W	1'b0	Internal clock conversion edge selection. 0: conversion at rising edge 1: conversion at falling edge
2	ADC_ISON	R/W	1'b0	Isolation enable. 0: ADC isolation 1: ADC is not isolated. Note: ADC should be configured as non-

Bit	Name	Access	Default	Description
				isolated when it works normally and isolated when it is not in use.
1	ADC_RSTN	R/W	1'b0	Reset enable. 0: Reset 1: No reset
0	ADC_POWER	R/W	1'b1	ADC power control。 0: Power On 1: Power Down

17.5.9. ADC_ANA_CTRL1 (Offset 0x38)

table 17-9 ADC_ANA_CTRL1 Register

Bit	Name	Access	Default	Description
31:10	Reserved	R/W	0x0	Reserved
9	ADC_REF1.2V_BUF_EN	R/W	1'b0	1.2V reference voltage BUF enable. 0: Disable 1: Enable
8:5	Reserved	R/W	0x0	Reserved
4	ADC_VBAT_DIV_EN	R/W	1'b0	1/4VBAT division enable. 0: Disable 1: Enable
3	ADC_REFH_MODE	R/W	1'b0	ADC reference signal selection 0: IO_IDO 1: Internal 1.2V
2	ADC_BUF_IN_SEL	R/W	1'b0	ADC BUF input Selection。 0: Select VBE 1: invalid
1	ADC_BUF_EN	R/W	1'b0	ADC BUF enable. 0: Disable 1: Enable
0	Reserved	R/W	0x0	Reserved

17.5.10. ADC_ANA_CTRL2 (Offset 0x3C)

table 17-10 ADC_ANA_CTRL2 Register

Bit	Name	Access	Default	Description
31:24	Reserved	R/W	0x0	Reserved
23:16	SAMPLE_POINT	R/W	8'h8	Number of sampling points
15:7	Reserved	R/W	0x0	Reserved
6:2	CONV_TIME	R/W	5'hE	Conversion timeout
1	TOUT_MASK	R/W	1'b0	Timeout status mask. 0: Mask the ERR interrupt status. 1: Do not mask ERR interrupt status.
0	POLL_MODE	R/W	1'b0	Cyclic sampling. Always set '0'.

18. USB

18.1.1. Overview

USB peripherals realize the interface between USB2.0 full-speed bus and AHB bus. Provides communication connection with other USB compliant devices for the microcontroller.

18.1.2. Feature

- Complies with the USB2.0 specification for full-speed
- Configurable for up to 10 endpoints (5in + 5 out)
- Hardware CRC (cyclic redundancy check) generation/checking
- Hardware encoding/decoding, bit stuffing/unstuffing
- Hardware ACK/NACK
- All endpoints share 1024 byte FIFO, FIFO size of each endpoint can be configured, the total FIFO of all endpoints cannot exceed 1024 bytes

18.2. USB Block Diagram

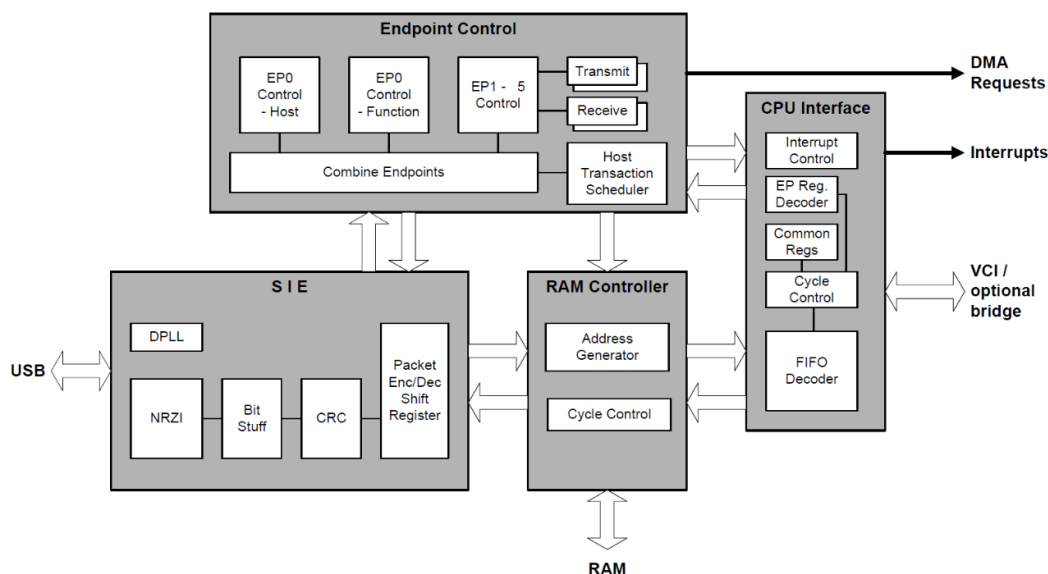


Figure 18-1 USB Block Diagram

18.3. USB Register

18.3.1. USB Register Mapping

USB Register map is split into three sections:

1. Common USB Registers (Offset 0x00~0x0F): These registers provide control and status for the complete core.
2. Endpoints Control/Status register (Offset 0x10~0x1F): These registers provide control and status for the endpoints.

Note: The content of endpoint 0 is slightly different from that of other endpoint registers; When peripheral mode (Device) or main mode (HOST) is selected, the register contents are different.

3. Endpoints FIFOs(0x20~0x34): This address range provides access to the endpoint FIFOs.

USB Base Addr: 0x2004_0000

table 18-1 USB Register Memory Mapping

Offset	Name	Description
1.USB Common Register		
0x00	Faddr	Function address register.
0x01	Power	Power management register.
0x02	IntrTx1	Interrupt register for Endpoint 0 plus Tx Endpoints 1 to 5.
0x03	IntrTx2	Reserved
0x04	IntrRx1	Interrupt register for Rx Endpoints 1 to 5.
0x05	IntrRx2	Reserved
0x06	IntrUSB	Interrupt register for common USB interrupts.
0x07	IntrTx1E	Interrupt enable register for IntrTx1.
0x08	IntrTx2E	Interrupt enable register for IntrTx2.
0x09	IntrRx1E	Interrupt enable register for IntrRx1.
0x0A	IntrRx2E	Interrupt enable register for IntrRx2.
0x0B	IntrUSBE	Interrupt enable register for IntrUSB.
0x0C	Frame1	Frame number bits 0 to 7.
0x0D	Frame2	Frame number bits 8 to 10.
0x0E	Index	Index register for selecting the endpoint status and control registers.
0x0F	DevCtl	USB device control register.
2.USB Endpoint registers, Device mode		
0x10	TxMaxP	Maximum packet size for peripheral IN endpoint. (Index register set to select Endpoints 1~5 only)
0x11	CSR0	Main Control Status register for Endpoint 0. (Index register set to select Endpoint 0)
	TxCSR1	Control Status register 1 for peripheral IN endpoint. (Index register set to select Endpoints 1~5)
0x12	CSR2	Subsidiary Control Status register for Endpoint 0, containing Flush FIFO Bit.
	TxCSR2	Control Status register 2 for peripheral IN endpoint. (Index register set to select Endpoint 1~5 only)
0x13	RxMaxP	Maximum packet size for peripheral OUT endpoint. (Index register set to select Endpoints 1~5 only)
0x14	RxCSR1	Control Status register 1 for peripheral OUT endpoint. (Index register set to select Endpoints 1~5 only)
0x15	RxCSR2	Control Status register 2 for peripheral OUT endpoint. (Index register set to select Endpoints 1~5 only)

Offset	Name	Description
0x16	Count0	Number of received bytes in Endpoint 0 FIFO. (Index register set to select Endpoint 0)
	RxCount1	Number of bytes in peripheral OUT endpoint FIFO (lower byte). (Index register set to select Endpoints 1~5)
0x17	RxCount2	Number of bytes in peripheral OUT endpoint FIFO (upper byte). (Index register set to select Endpoints 1~5)
0x18~0x1B	Reserve	Reserve
0x1C	TxFIFO1	IN endpoint FIFO configuration (Index register set to select Endpoints 1~5)
0x1B	TxFIFO2	IN endpoint FIFO configuration (Index register set to select Endpoints 1~5)
0x1D	RxFIFO1	OUT endpoint FIFO configuration (Index register set to select Endpoints 1~5)
0x1F	RxFIFO2	OUT endpoint FIFO configuration (Index register set to select Endpoints 1~5)
2.USB Indexed registers – Host mode		
0x10	TxMaxP	Maximum packet size for host OUT endpoint. (Index register set to select Endpoints 1~5 only)
0x11	CSR0	Control Status register for Endpoint 0.(Index register set to select Endpoint 0)
	TxCSR1	Control Status register 1 for host OUT endpoint. (Index register set to select Endpoints 1~5 only)
0x12	CSR2	Subsidiary Control Status register for Endpoint 0, containing FlushFIFO bit. (Index register set to select Endpoint 0)
	TxCSR2	Control Status register 2 for host OUT endpoint. (Index register set to select Endpoints 1~5 only)
0x13	RxMaxP	Maximum packet size for host IN endpoint. (Index register set to select Endpoints 1~5 only)
0x14	RxCSR1	Control Status register 1 for host IN endpoint. (Index register set to select Endpoints 1~5 only)
0x15	RxCSR2	Control Status register 2 for host IN endpoint. (Index register set to select Endpoints 1~5 only)
0x16	Count0	Number of received bytes in Endpoint 0 FIFO. (Index register set to select Endpoint 0)
	RxCount1	Number of bytes in host IN endpoint 0 FIFO (lower byte). (Index register set to select Endpoint 1~5)

Offset	Name	Description
0x17	RxCount2	Number of bytes in host IN endpoint 0 FIFO (upper byte). (Index register set to select Endpoint 1~5)
0x18	TxType	Sets the transaction protocol and peripheral endpoint number for the host OUT endpoint. (Index register set to select Endpoint 1~5 only)
0x19	NAKLimit0	Sets the NAK response timeout on Endpoint 0. (Index register set to select Endpoint 0)
	TxInterval	Sets the polling interval for an OUT interrupt endpoint, unit: ms. (Index register set to select Endpoint 1~5 only)
0x1A	RxType	Set the transaction protocol and peripheral endpoint number of the Rx endpoint. (Only endpoints 1 to 5 can be set)
0x1B	RxInterval	Set the polling interval for the Rx interrupt endpoint, unit: milliseconds. (Only endpoints 1 to 5 can be set)
0x1C	TxFIFO1	OUT endpoint FIFO configuration (Index register set to select Endpoints 1~5)
0x1B	TxFIFO2	OUT endpoint FIFO configuration (Index register set to select Endpoints 1~5)
0x1D	RxFIFO1	IN endpoint FIFO configuration (Index register set to select Endpoints 1~5)
0x1F	RxFIFO2	IN endpoint FIFO configuration (Index register set to select Endpoints 1~5)
3. FIFOs Register		
0x20	END_0_FIFO	FIFO for Endpoint 0(the same address for reading and writing)
0x24	END_1_FIFO	FIFO for Endpoint 1(the same address for reading and writing)
0x28	END_2_FIFO	FIFO for Endpoint 2(the same address for reading and writing)
0x2C	END_3_FIFO	FIFO for Endpoint 3(the same address for reading and writing)
0x30	END_4_FIFO	FIFO for Endpoint 4(the same address for reading and writing)
0x34	END_5_FIFO	FIFO for Endpoint 5(the same address for reading and writing)

18.3.2. USB Common Register (Offset 0x00~0x0F)

18.3.2.1. Faddr (Offset 0x00)

table 18-2 Faddr Register

Bit	Name	Access	Default	Description
7	Reserved	R	1'b0	Reserved
6:0	Func Addr	R/W	7'h0	In Peripheral mode (DevCtl.HostMode = 0), this register should be written with the address received through a SET_ADDRESS command, which will then be used for decoding the function address in subsequent token packets. In Host mode (DevCtl.HostMode = 1), this register should be set to the value sent in a SET_ADDRESS command during device enumeration as the address for the peripheral device.

18.3.2.2. Power (Offset 0x01)

table 18-3 Power Register

Bit	Name	Device Mode	Host Mode	Default	Description
7	ISO Update	R/W	R	1'b0	The USB controller waits for a SOF token packet before sending. If an IN token is received before a SOF token, a zero-length data packet will be sent. (Only valid during synchronous transfer in Device mode)
6	Vbus Val	R	R	1'b0	This bit indicates the state of the VbusVal input signal. 1: VBus above the VBus Valid threshold. 0: VBus below this threshold.
5	Vbus Sess	R	R	1'b0	This bit indicates the state of the VbusSess input signal.

Bit	Name	Device Mode	Host Mode	Default	Description
					1: VBus above the Session Valid threshold for a 'B' device 0: VBus below this threshold.
4	Vbus Lo	R	R	1'b0	This bit indicates the state of the VbusLo input signal. 1: VBus above the Session End threshold. 0: VBus below this threshold.
3	Reset	R	R/W	1'b0	Device mode: When there is a Reset signal on the bus, the bus Reset status can be read from this bit. Host mode: Writing '1' generates a Reset signal on the bus, and writing '0' restores it
2	Resume	R/W	R/W	1'b0	Generate Resume signaling when the device is in Suspend mode. In Peripheral mode, the CPU should clear this bit after 10ms (a maximum of 15ms), to end Resume signaling. In host mode, the CPU should clear this bit after 20ms.
1	Suspend Mode	R	W	1'b0	Device mode: Reading this bit indicates entering the suspend mode. Host mode: This bit is set by the CPU when entering the Suspend mode. When the CPU reads the interrupt register, or sets the Resume bit, or leaves the host mode, this bit will be cleared.
0	Enable Suspend	R/W	R	1'b0	Enable entering the Suspend mode when a Suspend signal is received on the bus. (Only valid in Device mode)

18.3.2.3. IntrTx1 (Offset 0x02)

table 18-1 IntrTx1 Register

Bit	Name	Access	Default	Description
7:6	Reserved	R	0x0	Reserved
5	EP5	R	1'b0	Tx Endpoint 5 interrupt.
4	EP4	R	1'b0	Tx Endpoint 4 interrupt.

Bit	Name	Access	Default	Description
3	EP3	R	1'b0	Tx Endpoint 3 interrupt.
2	EP2	R	1'b0	Tx Endpoint 2 interrupt.
1	EP1	R	1'b0	Tx Endpoint 1 interrupt.
0	EP0	R	1'b0	Endpoint 0 interrupt.

Note: Reading Register will clear all endpoint flags

18.3.2.4. IntrTx2 (Offset 0x03)

table 18-2 IntrTx2 Register

Bit	Name	Access	Default	Description
7:0	Reserved	R	8'h00	Reserved

18.3.2.5. IntrRx1 (Offset 0x04)

table 18-3 IntrRx1 Register

Bit	Name	Access	Default	Description
7:6	Reserved	R	0x0	Reserved
5	EP5	R	1'b0	Rx Endpoint 5 interrupt.
4	EP4	R	1'b0	Rx Endpoint 4 interrupt.
3	EP3	R	1'b0	Rx Endpoint 3 interrupt.
2	EP2	R	1'b0	Rx Endpoint 2 interrupt.
1	EP1	R	1'b0	Rx Endpoint 1 interrupt.
0	Reserved	R	1'b0	Reserved

Note: All active interrupts are cleared when this register is read.

18.3.2.6. IntrRx2 (Offset 0x05)

table 18-4 IntrRx2 Register

Bit	Name	Access	Default	Description
7:0	Reserved	R	0x0	Reserved

18.3.2.7. IntrUSB (Offset 0x06)

table 18-5 IntrUSB Register

Bit	Name	Access	Default	Description
7	VBus Error	R	1'b0	Set when VBus drops below the VBus Valid threshold during a session.
6	Sess Req	R	1'b0	Set when Session Request signaling has been detected.
5	Discon	R	1'b0	Set in Host mode when a device disconnect is detected. Set in Peripheral mode when a session ends.
4	Conn	R	1'b0	Set when a device connection is detected. (only valid in Host mode)
3	SOF	R	1'b0	Set when a new frame starts.
2	Reset	R	1'b0	Set in Peripheral mode when Reset signaling is detected on the bus.
1	Resume	R	1'b0	Set when Resume signaling is detected on the bus in Suspend mode.
0	Suspend	R	1'b0	Set when Suspend signaling is detected on the bus. (Only valid in Peripheral mode)

Note: All active interrupts will be cleared when this register is read.

18.3.2.8. IntrTx1E (Offset 0x07)

table 18-6 IntrTx1E Register

Bit	Name	Access	Default	Description
7:6	Reserved	R/W	0x0	Reserved
5	EP5	R/W	1'b1	Interrupt enable bit for endpoint 5
4	EP4	R/W	1'b1	Interrupt enable bit for endpoint 4
3	EP3	R/W	1'b1	Interrupt enable bit for endpoint 3
2	EP2	R/W	1'b1	Interrupt enable bit for endpoint 2
1	EP1	R/W	1'b1	Interrupt enable bit for endpoint 1
0	EP0	R/W	1'b1	Interrupt enable bit for endpoint 0

18.3.2.9. IntrTx2E (Offset 0x08)

table 18-7 IntrTx2E Register

Bit	Name	Access	Default	Description
7:0	Reserved	R/W	0x0	Reserved

18.3.2.10. IntrRx1E (Offset 0x09)

table 18-8 IntrRx1E Register

Bit	Name	Access	Default	Description
7:6	Reserved	R/W	0x0	Reserved
5	EP5	R/W	1'b1	Interrupt enable bit for endpoint 5
4	EP4	R/W	1'b1	Interrupt enable bit for endpoint 4
3	EP3	R/W	1'b1	Interrupt enable bit for endpoint 3
2	EP2	R/W	1'b1	Interrupt enable bit for endpoint 2
1	EP1	R/W	1'b1	Interrupt enable bit for endpoint 1
0	Reserved	R/W	0x0	Reserved

18.3.2.11. IntrRx2E (Offset 0x0A)

table 18-9 IntrRx2E Register

Bit	Name	Access	Default	Description
7:0	Reserved	R/W	0x0	Reserved

18.3.2.12. IntrUSBE (Offset 0x0B)

table 18-10 IntrUSBE Register

Bit	Name	Access	Default	Description
7	VBus Error	R/W	1'b0	Interrupt enable of VBus Error
6	Sess Req	R/W	1'b0	Interrupt enable of Sess Req

Bit	Name	Access	Default	Description
5	Discon	R/W	1'b0	Interrupt enable of Discon
4	Conn	R/W	1'b0	Interrupt enable of Conn
3	SOF	R/W	1'b0	Interrupt enable of SOF
2	Reset	R/W	1'b1	Interrupt enable of Reset
1	Resume	R/W	1'b1	Interrupt enable of Resume
0	Suspend	R/W	1'b0	Interrupt enable of Suspend

18.3.2.13. Frame1 (Offset 0x0C)

table 18-11 Frame1 Register

Bit	Name	Access	Default	Description
7:0	Lower 8 bits of Frame Number	R	8'h00	Frame1 is an 8-bit read-only register that in Peripheral mode holds the lower 8 bits of the last received frame number and in Host mode holds the lower 8 bits of the current frame number.

18.3.2.14. Frame2 (Offset 0x0D)

table 18-12 Frame2 Register

Bit	Name	Access	Default	Description
7:3	Reserved	R	5'h0	Reserved
2:0	Upper 3 bits of Frame Number	R	3'b000	Frame2 is a 3-bit read-only register that in Peripheral mode holds the upper 3 bits of the last received frame number and in Host mode holds the upper 3 bits of the current frame number.

18.3.2.15. Index (Offset 0x0E)

table 18-13 Index Register

Bit	Name	Access	Default	Description
7:4	Reserved	R	4'h0	Reserved
3:0	Selected Endpoint	R/W	4'h0	Index is a 4-bit register that determines which endpoint control/status registers are accessed at addresses. See details 端点选择

18.3.2.16. DevCtl (Offset 0x0F)

table 18-14 DevCtl Register

Bit	Name	Access	Default	Description
7	CID	R	1'b1	This bit indicates the state of the CID input signal to the USB. 0: Host mode 1: Device mode
6	FSDev	R	1'b0	This bit is set when a full-speed device has been detected being connected to the port. Only valid in Host mode.
5	LSDev	R	1'b0	This bit is set when a low-speed device has been detected being connected to the port. Only valid in Host mode.
4	PUCON	R	1'b0	This Read-only bit is set when a pull-up resistor is required on the USB D+ line for operation as a peripheral. The PUCON output will also have gone high.
3	PDCON	R	1'b1	This Read-only bit is set when a pull-down resistor is required on the USB D+ line (alongside a similar pull-down resistor on the USB D- line) for operation as a host. The PDCON output will also have gone high.
2	HostMode	R	1'b0	This Read-only bit is set when acting as a Host.

Bit	Name	Access	Default	Description
1	HostReq	R/W	1'b0	When set, USB will initiate the Host Negotiation when Suspend mode(no activity on the bus for 3 ms) is entered. It is cleared when Host Negotiation is completed.
0	Session	R/W	1'b0	When operating as HOST, this bit is set or cleared by software to start or end a session. When operating as device, this bit is set/cleared by the USB controller when a session starts/ends. No software intervention required. It can also be set by the software to initiate the Session Request Protocol, and cleared to execute a soft disconnect from the host (which then ends the session).

18.3.3. USB endpoint 0 control status Register (Offset 0x10~0x1F)

18.3.3.1. CSR0 (Offset 0x11)

IN Device Mode:

table 18-15 Device Mode CSR0 Register

Bit	Name	Access	Default	Description
7	ServicedSetupEnd	W	1'b0	write 1 to this bit to clear the SetupEnd bit. It is cleared automatically.
6	ServicedRxPktRdy	W	1'b0	write 1 to this bit to clear the RxPktRdy bit. It is cleared automatically.
5	SendStall	W	1'b0	write 1 to this bit to terminate the current transaction. The STALL handshake will be transmitted and then this bit will be cleared automatically.
4	SetupEnd	R	1'b0	This bit will be set when a control transaction ends before the DataEnd bit has been set. An interrupt will be generated and the FIFO flushed at this time. the bit is cleared by

Bit	Name	Access	Default	Description
				software writing a 1 to the ServicedSetupEnd bit.
3	DataEnd	W	1'b0	Set this bit in the following cases: 1. When setting TxPktRdy for the last data packet. 2. When clearing RxPktRdy after unloading the last data packet. 3. When setting TxPktRdy for a zero length data packet. It is cleared automatically.
2	SentStall	R	1'b0	This bit is set when a STALL handshake is transmitted. Write 0 to clear it.
1	TxPktRdy	R/W	1'b0	This bit is set after loading a data packet into the FIFO. It is cleared automatically when the data packet has been transmitted. An interrupt is generated when the bit is cleared.
0	RxPktRdy	R	1'b0	This bit is set when a data packet has been received. An interrupt is generated when this bit is set. The CPU clears this bit by setting the ServicedRxPktRdy bit.

IN Host Mode:

table 18-16 Host Mode CSR0 Register

Bit	Name	Access	Default	Description
7	NAK Timeout	R	1'b0	During a transmission, if the time during which the endpoint continuously receives NAK packets exceeds the configured timeout period (the timeout is configured in the NAKLimit0 register), this bit will be set. Endpoint 0 will stop working. Writing '0' will clear this bit, and endpoint 0 will continue to work.
6	StatusPkt	R/W	1'b0	Sets this bit at the same time as the TxPktRdy or ReqPkt bit is set, to perform a status stage transaction. Setting this bit ensures that the data toggle is set to 1 so that a DATA1 packet is used for the Status Stage

Bit	Name	Access	Default	Description
				transaction.
5	ReqPkt	R/W	1'b0	Sets this bit to request an IN transaction. It is cleared when RxPktRdy is set.
4	Error	R	1'b0	When three attempts are made to execute a transaction without receiving a response from the peripheral device, this bit will be set and an endpoint 0 interrupt will be generated. Endpoint 0 will stop working. Writing '0' will clear this bit, and endpoint 0 will continue to work.
3	SetupPkt	R/W	1'b0	The CPU sets this bit, at the same time as the TxPktRdy bit is set, to send a SETUP token instead of an OUT token for the transaction.
2	RxStall	R	1'b0	It is set when a STALL is received. Endpoint 0 stops working. Writing '0' clears this bit, and endpoint 0 continues to work.
1	TxPktRdy	R/W	1'b0	Sets this bit after loading a data packet into the FIFO. It is cleared automatically when the data packet has been transmitted. An interrupt is generated when the bit is cleared.
0	RxPktRdy	R	1'b0	This bit is set when a data packet has been received. An interrupt is generated when this bit is set. Write 0 to clear.

18.3.3.2. CSR2 (Offset 0x12)

table 18-17 CSR2 Register

Bit	Name	Access	Default	Description
7:6	Reserved	R	0x0	Reserved
0	FlushFIFO	W	1'b0	The CPU writes a 1 to this bit to flush the next packet to be transmitted/read from the Endpoint 0 FIFO. The FIFO pointer is reset and the TxPktRdy/RxPktRdy bit (above) is

Bit	Name	Access	Default	Description
				cleared. Note: FlushFIFO has no effect unless TxPktRdy/RxPktRdy is set.

18.3.3.3. Count0 (Offset 0x16)

table 18-18 Count0 Register

Bit	Name	Access	Default	Description
7	Reserved	R	0x0	Reserved
6:0	Endpoint 0 Count	R	7'h0	It indicates the number of received data bytes in the Endpoint 0 FIFO. The value returned is valid while RxPktRdy (CSR0.D0) is set.

18.3.3.4. NAKLimit0 (HostMode Only Offset 0x19)

table 18-19 NAKLimit0 Register

Bit	Name	Access	Default	Description
7:0	Endpoint 0 NAK Limit	R/W	8'h00	The number of frames (ranging from 2 to 255) in which endpoint 0 continuously receives NAK packets. If the number of continuously received NAK packets exceeds the set value, the NAK Timeout flag will be set. (The number of frames is calculated according to the SOF packets)

18.3.4. USB endpoints 1 to 5 control status Register (Offset 0x10~0x1F)

18.3.4.1. TxMaxP (Offset 0x10)

table 18-20 TxMaxP Register

Bit	Name	Access	Default	Description
7:0	Maximum Packet Size	R/W	8'h00	This register holds the maximum packet size for transactions in units of 8 bytes. If sets to 128, the maximum packet size to 1023. In setting this value, should note the constraints placed by the USB Specification on packet sizes. See details Endpoint FIFO configuration

18.3.4.2. TxCSR1(Offset 0x11)

IN Device Mode:

table 18-21 Device Mode TxCSR1 Register

Bit	Name	Access	Default	Description
7	Reserved	R	1'b0	Reserved
6	ClrDataTog	W	1'b0	Write 1 to this bit to reset the endpoint data toggle to 0.
5	SentStall	R	1'b0	This bit is set when a STALL handshake is transmitted. Write 0 to clear.
4	SendStall	R/W	1'b0	Write 1 to this bit to issue a STALL handshake. Clears this bit to terminate the stall condition. Note: This bit has no effect where the endpoint is being used for Isochronous transfers.
3	FlushFIFO	W	1'b0	Write 1 to this bit to flush the next packet to be transmitted from the endpoint FIFO. The FIFO pointer is reset and the TxPktRdy bit (below) is cleared.

Bit	Name	Access	Default	Description
				Note: FlushFIFO has no effect unless TxPktRdy is set.
2	UnderRun	R	1'b0	This bit is set if an IN token is received when the TxPktRdy bit not set. Write 0 to clear.
1	FIFONotEmpty	R	1'b0	This bit is set when there is at least 1 packet in the Tx FIFO. Write 0 to clear.
0	TxPktRdy	R/W	1'b0	This bit is set after loading a data packet into the FIFO. It is cleared automatically when a data packet has been transmitted. An interrupt is generated (if enabled) when the bit is cleared.

IN Host Mode:

table 18-22 Host Mode TxCSR1 Register

Bit	Name	Access	Default	Description
7	NAK Timeout	R	1'b0	This bit will be set when the Tx endpoint is halted following the receipt of NAK responses for longer than the time set as the NAK Limit by the TxInterval register. This bit should be cleared to allow the endpoint to continue. Note: Valid only for Bulk endpoints.
6	ClrDataTog	W	1'b0	Write 1 to this bit to reset the endpoint data toggle to 0.
5	RxStall	R	1'b0	This bit is set when a STALL handshake is received. The FIFO is flushed and the TxPktRdy bit is cleared. Write 0 to clear this bit.
4	Reserved	R	1'b0	Reserved
3	FlushFIFO	W	1'b0	Write 1 to this bit to flush the next packet to be transmitted from the endpoint FIFO. The FIFO pointer is reset and the TxPktRdy bit is cleared. Note: FlushFIFO has no effect unless TxPktRdy is set.
2	Error	R	1'b0	This bit is set when 3 attempts have been

Bit	Name	Access	Default	Description
				made to send a packet and no handshake packet has been received. Write 0 clear this bit. An interrupt is generated when the bit is set. Valid only when the endpoint is operating in Bulk or Interrupt mode.
1	FIFO NotEmpty	R	1'b0	This bit is set when there is at least 1 packet in the Tx FIFO. Write 0 clear.
0	TxPktRdy	R/W	1'b0	This bit is set after loading a data packet into the FIFO. It is cleared automatically when a data packet has been transmitted. An interrupt is generated (if enabled) when the bit is cleared.

18.3.4.3. TxCSR2 (Offset 0x12)

table 18-23 TxCSR2 Register

Bit	Name	Access	Default	Description
7	AutoSet	R/W	1'b0	If set this bit, TxPktRdy will be automatically set when data of the maximum packet size(value in TxMaxP) is loaded into the Tx FIFO. If a packet of less than the maximum packet size is loaded, then TxPktRdy will have to be set manually.
6	ISO	R/W	1'b0	set this bit enable the Tx endpoint for Isochronous transfers, and clears it to enable the Tx endpoint for Bulk or Interrupt transfers.
5	Mode	R/W	1'b1	Set this bit to enable the endpoint direction as Tx, and clears it to enable the endpoint direction as Rx. Note: This bit only has any effect where the same endpoint FIFO is used for both Tx and Rx transactions.

Bit	Name	Access	Default	Description
4	DMAEnable	R/W	1'b0	Set this bit to enable the DMA request for the Tx endpoint.
3	FrcDataTog	R/W	1'b0	Set this bit to force the endpoint data toggle to switch and the data packet to be cleared from the FIFO, regardless of whether an ACK was received.
2	DMAMode	R	1'b0	Set this bit to select DMA Mode 1 (in which a DMA request (but no interrupt) is automatically generated for packets of size TxMaxP bytes) and clears this bit to select DMA Mode 0 (in which a DMA request and an interrupt are generated for all packets).
1:0	Reserved	R	1'b0	Reserved

18.3.4.4. RxMaxP (Offset 0x13)

table 18-24 RxMaxP Register

Bit	Name	Access	Default	Description
7:0	Maximum Packet Size	R/W	8'h00	RxMaxP is an 8-bit register that holds the maximum packet size for transactions through the currently-selected Rx endpoint – in units of 8 bytes, except that a value of 128 sets the maximum packet size to 1023 (the maximum size for an Isochronous packet transferred in a full-speed transaction) rather than 1024. The maximum data packet should not exceed the USB specification requirements. Endpoint FIFO Configuration

18.3.4.5. RxCSR1 (Offset 0x14)

IN Device Mode:

table 18-25 Device Mode RxCSR1 Register

Bit	Name	Access	Default	Description
7	ClrDataTog	W	1'b0	Write a 1 to this bit to reset the endpoint data toggle to 0.
6	SentStall	R	1'b0	This bit is set when a STALL handshake is transmitted. The CPU should clear this bit.
5	SendStall	R/W	1'b0	The CPU writes a 1 to this bit to issue a STALL handshake. The CPU clears this bit to terminate the stall condition. Note: This bit has no effect where the endpoint is being used for Isochronous transfers.
4	FlushFIFO	W	1'b0	The CPU writes a 1 to this bit to flush the next packet to be read from the endpoint Rx FIFO. The FIFO pointer is reset and the RxPktRdy bit (below) is cleared. Note: FlushFIFO has no effect unless RxPktRdy is set.
3	DataError	R	1'b0	This bit is set when RxPktRdy is set if the data packet has a CRC or bit-stuff error. It is cleared when RxPktRdy is cleared. Note: This bit is only valid when the endpoint is operating in ISO mode.
2	OverRun	R	1'b0	This bit is set if an OUT packet cannot be loaded into the Rx FIFO. The CPU should clear this bit. Note: This bit is only valid when the endpoint is operating in ISO mode.
1	FIFOFull	R	1'b0	This bit is set when no more packets can be loaded into the Rx FIFO.
0	RxPktRdy	R	1'b0	This bit is set when a data packet has been received. The CPU should clear this bit when the packet has been unloaded from the Rx FIFO. An interrupt is generated when the bit is set.

IN Host Mode:

table 18-26 Host Mode RxCSR1 Register

Bit	Name	Access	Default	Description
7	ClrDataTog	W	1'b0	Write a 1 to this bit to reset the endpoint data toggle to 0.
6	RxStall	R	1'b0	When a STALL handshake is received, this bit is set and an interrupt is generated. The CPU should clear this bit.
5	ReqPkt	R/W	1'b0	The CPU writes a 1 to this bit to request an IN transaction. It is cleared when RxPktRdy is set.
4	FlushFIFO	W	1'b0	The CPU writes a 1 to this bit to flush the next packet to be read from the endpoint Rx FIFO. The FIFO pointer is reset and the RxPktRdy bit (below) is cleared. Note: FlushFIFO has no effect unless RxPktRdy is set.
3	DataError/NAK Timeout	R	1'b0	When operating in ISO mode, this bit is set when RxPktRdy is set if the data packet has a CRC or bit-stuff error and cleared when RxPktRdy is cleared. In Bulk mode, this bit will be set when the Rx endpoint is halted following the receipt of NAK responses for longer than the time set as the NAK Limit by the RxInterval register. The CPU should clear this bit to allow the endpoint to continue.
2	Error	R	1'b0	The USB sets this bit when 3 attempts have been made to receive a packet and no data packet has been received. The CPU should clear this bit. An interrupt is generated when the bit is set.
1	FIFOFull	R	1'b0	This bit is set when no more packets can be loaded into the Rx FIFO.
0	RxPktRdy	R	1'b0	This bit is set when a data packet has been received. The CPU should clear this bit when the packet

Bit	Name	Access	Default	Description
				has been unloaded from the Rx FIFO. An interrupt is generated when the bit is set.

18.3.4.6. RxCSR2 (Offset 0x15)

table 18-27 RxCSR2 Register

Bit	Name	Access	Default	Description
7	AutoClear	R/W	1'b0	If the CPU sets this bit then the RxPktRdy bit will be automatically cleared when a packet of RxMaxP bytes has been unloaded from the Rx FIFO. When packets of less than the maximum packet size are unloaded, RxPktRdy will have to be cleared manually.
6	Device Mode: IOS Host Mode: AutoReq	R/W	1'b0	In Peripheral mode, the CPU sets this bit to enable the Rx endpoint for Isochronous transfers, and clears it to enable the Rx endpoint for Bulk or Interrupt transfers. If the CPU sets this bit in Host mode, the ReqPkt bit will be automatically set when the RxPktRdy bit is cleared.
5	DMAEnable	R/W	1'b0	The CPU sets this bit to enable the DMA request for the Rx endpoint.
4	DMAMode	R/W	1'b0	Two modes of DMA operation are supported: DMA Mode 0 in which a DMA request is generated for all received packets, together with an interrupt (if enabled); DMA Mode 1 in which a DMA request (but no interrupt) is generated for Rx packets of size RxMaxP bytes and an interrupt (but no DMA request) is generated for Rx packets of any other size.
3:0	Reserved	R	0x0	Reserved

18.3.4.7. RxCount1 (Offset 0x16)

table 18-28 RxCount1 Register

Bit	Name	Access	Default	Description
7:0	Endpoint Rx Count lower 8 bits	R	8'h00	xCount1 is a 8-bit read-only register that holds the lower 8 bits of the number of received data bytes in the packet in the FIFO associated with the currently-selected Rx endpoint. The value returned is valid while RxPktRdy (RxCSR1.D0) is set.

18.3.4.8. RxCount2 (Offset 0x17)

table 18-29 RxCount2 Register

Bit	Name	Access	Default	Description
7:3	Reserved	R	0x0	Reserved
2:0	Endpoint Rx Count upper 3 bits	R	3'b000	RxCount2 is a 3-bit read-only register that holds the upper 3 bits of the number of received data bytes in the packet in the FIFO associated with the currently-selected Rx endpoint. The value returned is valid while RxPktRdy (RxCSR1.D0) is set.

18.3.4.9. TxType (HostMode Only. Offset 0x18)

table 18-30 TxType Register

Bit	Name	Access	Default	Description
7:6	Reserved	R	0x0	Reserved
5:4	Protocol	R/W	2'b00	protocol for the Tx endpoint: 00 : Illegal 01: Isochronous 10: Bulk 11: Interrupt

Bit	Name	Access	Default	Description
3:0	Target Endpoint Number	R/W	4'h0	The CPU should set this value to the endpoint number contained in the Tx endpoint descriptor.

18.3.4.10. TxInterval (HostMode Only Offset 0x19)

table 18-31 TxInterval Register

Bit	Name	Access	Default	Description
7:0	Tx Polling Interval/NAK Limit (n)	R/W	8'h00	for Interrupt and Isochronous transfers, defines the polling interval for the currently-selected Tx endpoint. (1~255ms); For Bulk endpoints, this register sets the number of frames (2~255). If continuously received NAK packets exceed the set value, the NAK Timeout flag will be set. (The number of frames is calculated according to the SOF packets)

18.3.4.11. RxType (HostMode Only Offset 0x1A)

table 18-32 RxType Register

Bit	Name	Access	Default	Description
7:6	Reserved	R	0x0	Reserved
5:4	Protocol	R/W	3'b00	required protocol for the Rx endpoint: 00 : Illegal 01: Isochronous 10: Bulk 11: Interrupt
3:0	Target Endpoint Number	R/W	4'h0	endpoint number contained in the Rx endpoint descriptor.

18.3.4.12. RxInterval (HostMode Only Offset 0x1B)

table 18-33 RxInterval Register

Bit	Name	Access	Default	Description
7:0	Rx Polling Interval/NAK Limit (n)	R/W	8'h00	Interrupt or Synchronous Transmission: Polling Interval (1 - 255 ms) Bulk Transmission: The number of frames of consecutive NAK packets received by the endpoint (2 - 255). If the number of consecutive NAK packets received exceeds the set value, the NAK Timeout flag will be set. (The number of frames is calculated based on the SOF packet)

18.3.4.13. TxFIFO1、TxFIFO2 (Offset 0x1C、0x1D)

table 18-34 TxFIFO1、TxFIFO2 Register

Bit	Name	Function										
Tx FIFO1		Start address of the endpoint FIFO in units of 8 Tx FIFO1 bytes, FIFO start address 0x000, end address 0x3FF.										
7:0	AD[7:0]											
Tx FIFO2												
3:0	AD[11:8]											
		<table><tr><th>AD[11 : 0]</th><th>Start Address</th></tr><tr><td>0x000</td><td>0x000</td></tr><tr><td>0x001</td><td>0x008</td></tr><tr><td>0x002</td><td>0x010</td></tr><tr><td>0x080</td><td>0x400</td></tr></table>	AD[11 : 0]	Start Address	0x000	0x000	0x001	0x008	0x002	0x010	0x080	0x400
AD[11 : 0]	Start Address											
0x000	0x000											
0x001	0x008											
0x002	0x010											
0x080	0x400											
4	DPB	Defines whether double-packet buffering supported. When '1', double-packet buffering is supported. (Not recommended). When '0', only single-packet buffering is supported. (single package buffering is used by default.)										
7:5	SZ[2:0]	Maximum packet size to be allowed for <table><tr><th>SZ[2 : 0]</th><th>Packet Size(byte)</th></tr><tr><td>3'b000</td><td>8</td></tr><tr><td>3'b001</td><td>16</td></tr><tr><td>3'b010</td><td>32</td></tr><tr><td>3'b011</td><td>64</td></tr></table>	SZ[2 : 0]	Packet Size(byte)	3'b000	8	3'b001	16	3'b010	32	3'b011	64
SZ[2 : 0]	Packet Size(byte)											
3'b000	8											
3'b001	16											
3'b010	32											
3'b011	64											

Bit	Name	Function		
			3'b100	128
			3'b101	256
			3'b110	512
			3'b111	1024
If DPB = 0, the FIFO will also be this size; if DPB = 1, the FIFO will be twice this size.				

18.3.4.14. RxFIFO1、RxFIFO2 (Offset 0x1E、0x1F)

table 18-35 RxFIFO1、RxFIFO2 Register

Bit	Name	Description																		
RxFIFO1		Start address of the endpoint FIFO in units of 8 TxFIFO1 bytes, FIFO start address 0x000, end address 0x3FF. <table><tr><td>AD[11 : 0]</td><td>Start Address</td></tr><tr><td>0x000</td><td>0x000</td></tr><tr><td>0x001</td><td>0x008</td></tr><tr><td>0x002</td><td>0x010</td></tr><tr><td>0x080</td><td>0x400</td></tr></table>	AD[11 : 0]	Start Address	0x000	0x000	0x001	0x008	0x002	0x010	0x080	0x400								
AD[11 : 0]	Start Address																			
0x000	0x000																			
0x001	0x008																			
0x002	0x010																			
0x080	0x400																			
7:0	AD[7:0]																			
RxFIFO2																				
3:0	AD[11:8]																			
4	DPB	Defines whether double-packet buffering supported. When '1', double-packet buffering is supported. (Not recommended). When '0', only single-packet buffering is supported. (single package buffering is used by default.)																		
7:5	SZ[2:0]	Maximum packet size to be allowed for <table><tr><td>SZ[2 : 0]</td><td>端点 FIFO 大小(byte)</td></tr><tr><td>3'b000</td><td>8</td></tr><tr><td>3'b001</td><td>16</td></tr><tr><td>3'b010</td><td>32</td></tr><tr><td>3'b011</td><td>64</td></tr><tr><td>3'b100</td><td>128</td></tr><tr><td>3'b101</td><td>256</td></tr><tr><td>3'b110</td><td>512</td></tr><tr><td>3'b111</td><td>1024</td></tr></table> If DPB = 0, the FIFO will also be this size; if DPB = 1, the FIFO will be twice this size.	SZ[2 : 0]	端点 FIFO 大小(byte)	3'b000	8	3'b001	16	3'b010	32	3'b011	64	3'b100	128	3'b101	256	3'b110	512	3'b111	1024
SZ[2 : 0]	端点 FIFO 大小(byte)																			
3'b000	8																			
3'b001	16																			
3'b010	32																			
3'b011	64																			
3'b100	128																			
3'b101	256																			
3'b110	512																			
3'b111	1024																			

18.3.5. USB Endpoint FIFO Register (Offset 0x20~0x34)

table 18-36 USB Endpoint FIFO Register

Addr	Name		Default	Description
0x20	END_0_FIFO	R/W	0x0	Endpoint 0 FIFO address, read and write access to this address
0x24	END_1_FIFO	R/W	0x0	Endpoint 1 FIFO address, read and write access to this address
0x28	END_2_FIFO	R/W	0x0	Endpoint 2 FIFO address, read and write access to this address
0x2C	END_3_FIFO	R/W	0x0	Endpoint 3 FIFO address, read and write access to this address
0x30	END_4_FIFO	R/W	0x0	Endpoint 4 FIFO address, read and write access to this address
0x34	END_5_FIFO	R/W	0x0	Endpoint 5 FIFO address, read and write access to this address

18.4. Process

18.4.1. Endpoint index

Each Tx endpoint and each Rx endpoint have their own set of control/status registers. Only one set of Tx control/status and one set of Rx control/status registers appear in the memory map at any one time. Before accessing an endpoint's control/status registers, the endpoint number should be written to the Index register to ensure that the correct control/status registers appear in the memory map.

18.4.2. Endpoint FIFO Configuration

USB provides 1024 bytes of FIFO. The FIFO for Endpoint 0 is required to be 64 bytes in size. Other endpoints need to be configured through TxFIFO1, TxFIFO2 or RxTxFIFO1, RxTxFIFO2.

The FIFO size of the endpoint needs to conform to the USB specification:

- **Control Transfer:** The maximum packet size for control endpoints can be only 8 bytes for low-speed devices; 8,16,32, or 64 bytes for full-speed devices; and only 64 bytes for high-speed devices.
- **Bulk Transfer:** This transfer type can be used by full-speed and high-speed devices, but not by low-speed devices. The maximum packet size for bulk endpoints can be 8,16,32, or 64 bytes for full-speed devices, and 512 bytes for high-speed devices.
- **Isochronous Transfer:** The maximum packet size for Isochronous endpoints is 1024 bytes for high-speed devices and 1023 bytes for full-speed devices, but not by low-speed devices.
- **Interrupt Transfer:** The maximum packet size for interrupt endpoints can be 8 bytes or less for low-speed devices, 64 bytes or less for full-speed devices; and 1024 bytes or less for high-speed devices.

Note: The maximum packet size set for any endpoint cannot exceed the FIFO size. You should also note that the TxMaxP register should not be written to while there is data in the FIFO as unexpected results may occur.

18.4.3. IN TRANSACTION HANDLING AS A PERIPHERAL

When operating in Peripheral mode, data for IN transactions is handled through TxFIFOs.

As each packet to be sent is loaded into the Tx FIFO, the TxPktRdy bit in TxCSR1 needs to be set. If the AutoSet bit in TxCSR2 is set, the TxPktRdy bit will be automatically set when a maximum-sized packet is loaded into the FIFO. For packet sizes less than the maximum, TxPktRdy will always have to be set manually.

When the TxPktRdy bit is set, either manually or automatically, the FIFONotEmpty bit in TxCSR1 is also set and the packet is ready to be sent.

When the packet has been successfully sent, both TxPktRdy and FIFONotEmpty will be cleared and the appropriate Tx endpoint interrupt generated (if enabled). The next packet can then be loaded into the FIFO.

18.4.4. OUT TRANSACTION HANDLING AS A PERIPHERAL

When operating in Peripheral mode, data for OUT transactions is handled through Rx FIFOs.

When a packet is received and placed in the Rx FIFO, the RxPktRdy bit (D0) and the FIFOFull bit (D1) in RxCSR1 are set and the appropriate Rx endpoint is generated (if enabled) to signal that a packet can now be unloaded from the FIFO.

After the packet has been unloaded, the RxPktRdy bit needs to be cleared in order to allow further packets to be received. If the AutoClear bit in RxCSR2 (D7) is set and a maximum-sized packet is unloaded from the FIFO, the RxPktRdy bit is cleared automatically. The FIFOFull bit is also cleared. For packet sizes less than the maximum, RxPktRdy will always have to be cleared manually.

18.4.5. Device Mode Additional Actions

- Zero length out data packets in control transfers

A zero-length OUT data packet is used to indicate the end of a Control transfer. In normal operation, such packets should only be received after the entire length of the device request has been transferred (i.e. after the CPU has set DataEnd). If, however, the host sends a zero-length OUT data packet before the entire length of device request has been transferred, this signals the premature end of the transfer. In this case, the USB controller will automatically flush any IN token loaded by CPU ready for the Data phase from the FIFO and set SetupEnd.

- Stall issued to control transfer

The USB controller core will automatically issue a STALL handshake to a Control transfer under the following conditions:

1. The host sends more data during an OUT Data phase of a Control transfer than was specified in the device request during the SETUP phase.
2. The host requests more data during an IN data phase of a Control transfer than was

specified in the device request during the SETUP phase.

3. The host sends more than MaxP data with an OUT data token.
4. The host sends the wrong PID for the OUT Status phase of a Control transfer.
5. The host sends more than a zero length data packet for the OUT Status phase.

18.4.6. Device Mode Suspend

If when the MUSBFDR is operating in Peripheral mode, no activity occurs on the USB for 3ms and the EnableSuspend bit in the Power register is set, the USB will enter Suspend mode. If the Suspend interrupt has been enabled, an interrupt will be generated at this time.

When Resume signaling is detected, the MUSBFDR will exit Suspend mode and re-enable the system clock. If the Resume interrupt is enabled, an interrupt will be generated.

The CPU can also force the USB to leave Suspend mode by setting the Resume bit in the Power register. When this bit is set, the MUSBFDR will exit Suspend mode and drive Resume signaling onto the bus. The CPU should clear this bit after 10 ms (a maximum of 15 ms) to end Resume signaling. Note: No Resume interrupt is generated in this case.

18.4.7. Device Mode Start of frame

When the MUSBFDR is operating in Device mode, it should receive a Start-Of-Frame packet from the host once every millisecond. When the SOF packet is received, the 11-bit frame number contained in the packet is written into the two registers, Frame1 and Frame2, and an output pulse lasting one USB bit period is generated on SOF_PULSE. A SOF interrupt is also generated (if enabled).

Once the USB has started to receive SOF packets, it expects to receive one every millisecond. If no SOF packet is received after 1.00358 ms, it is assumed that the packet has been lost and a SOF_PULSE and SOF interrupt are generated though the frame registers are not updated. The USB will continue to generate a SOF_PULSE and SOF interrupt every millisecond until it is able to resynchronize these pulses to received SOF packets when

these packets are successfully received again.

18.4.8. In transaction handling as a Host

When the USB is operating as a host, IN transactions are handled in a similar manner to the way in which OUT transactions are handled when the USB is operating as a Device except that the transaction needs first to be initiated by setting the ReqPkt bit in RxCSR1. This indicates to the transaction scheduler that there is an active transaction on this endpoint. The transaction scheduler then sends an IN token to the target function.

As in Device OUT transactions, when the packet is received and placed in the Rx FIFO, the RxPktRdy bit in RxCSR1 is set and the appropriate Rx endpoint interrupt is generated (if enabled) to signal that a packet can now be unloaded from the FIFO.

When the packet has been unloaded, RxPktRdy should be cleared. As in Device OUT transactions, the AutoClear bit in the RxCSR2 register can be used to have RxPktRdy automatically cleared when a maximum sized packet has been unloaded from the FIFO. There is also an AutoReq bit in RxCSR2 which causes the ReqPkt bit to be automatically set when the RxPktRdy bit is cleared. The AutoClear and AutoReq bits can be used with an external DMA controller to perform complete Bulk transfers without CPU intervention.

If the target function responds to the IN token with a NAK, the USB will keep retrying the transaction until any NAK Limit that has been set has been reached. If the target function responds with a STALL, however, the USB will not retry the transaction but will interrupt the CPU with the RxStall bit in the RxCSR1 register set. If the target function does not respond to the IN token within the required time, the USB will retry the transaction. If after three attempts the target function has still not responded, the USB will clear the ReqPkt bit and interrupt the CPU with the Error bit in RxCSR1 set.

18.4.9. Out Transaction handling as a Host

As in peripheral IN transactions, the TxPktRdy bit in the TxCSR1 register needs to be set as each packet is loaded into the TxFIFO and the AutoSet bit in TxCSR2 can be used to cause the TxPktRdy bit to be automatically set when a maximum sized packet has been loaded into the FIFO. Again, the AutoSet bit can be used with an external DMA controller to perform complete Bulk transfers without CPU intervention.

If the target function responds to the OUT token with a NAK, the USB will keep retrying the transaction until any NAK Limit that has been set has been reached. If the target function responds with a STALL, however, the USB will not retry the transaction but will interrupt the CPU with the RxStall bit in the TxCSR1 register set. If the target function does not respond to the OUT token within the required time, the USB will retry the transaction. If after three attempts the target function has still not responded, the USB will flush the FIFO and interrupt the CPU with the Error bit in TxCSR1 set.

18.4.10. Host Mode Transaction Scheduling

When operating as a host, the USB maintains a 1 ms frame counter. If the target function is a full-speed device, the USB will automatically send an SOF packet at the start of each frame. If the target function is a low-speed device, a 'K' state will be transmitted on the bus to act as a "keep-alive" to stop the low-speed device going into Suspend mode.

After the SOF packet has been transmitted, the USB will cycle through all the configured endpoints looking for active transactions. An active transaction is defined as an Rx endpoint for which the ReqPkt bit is set or a Tx endpoint for which the TxPktRdy bit is set.

An active Isochronous or Interrupt transaction will only be started if found on the first transaction scheduler cycle of a frame and if the interval counter for that endpoint has counted down to zero. This ensures that only one Interrupt/Isochronous transaction will occur per endpoint per n frames (where n is the interval set in the TxInterval/RxInterval

register for that endpoint). Note: The USB 2.0 specification allows Isochronous transactions to be scheduled at intervals of up to 65s. The mechanism described above caters for intervals of up to 255ms. Intervals greater than 255ms will need to be implemented in software.

An active Bulk transaction will be started immediately, provided there is sufficient time left in the frame to complete the transaction before the next SOF packet is due. If the transaction needs to be retried (e.g. because a NAK was received or the target function did not respond) then the transaction will not be retried until the transaction scheduler has checked all the other endpoints for active transactions first. This ensures that an endpoint that is sending a lot of NAKs does not block other transactions on the bus.

18.4.11. Host Mode Reset

If the Reset bit in the Power register is set while the USB is in Host mode, the USB will generate Reset signaling on the bus. After the CPU has cleared the bit, the USB will start its frame counter and transaction scheduler.

18.4.12. Host Mode Suspend

If the Suspend Mode bit in the Power register is set, the USB will complete the current transaction then stop the transaction scheduler and frame counter. No further transactions will be started and no SOF packets will be generated.

To exit Suspend mode, the CPU should set the Resume bit and clear the Suspend bit in the Power register. While the Resume bit is high, the USB will generate Resume signaling on the bus. After 20ms, the CPU should clear the Resume bit, at which point the frame counter and transaction scheduler will be started.

Contact Information

Website: www.freqchip.com

Sales Email: sales@freqchip.com

Phone: +86-21-5027-0080

Feedback:

Freqchip welcomes feedback on this product and this document. If you have comments or suggestions, please send an email to docs@freqchip.com.

19. Appendix I

table 19-1 PortA MUX

引脚 功能	PA0	PA1	PA2	PA3	PA4	PA5	PA6	PA7
0x0	GPIO_A0	GPIO_A1	GPIO_A2	GPIO_A3	GPIO_A4	GPIO_A5	GPIO_A6	GPIO_A7
0x1	I2C0_CLK	I2C0_DAT	I2C1_CLK	I2C1_DAT	I2C0_CLK	I2C0_DAT	I2C1_CLK	I2C1_DAT
0x2	SPI0_M_CLK	SPI0_M_CS	SPI0_M_IO0	SPI0_M_IO1	SPI0_M_IO2	SPI0_M_IO3	SPI0_M_CLK	SPI0_M_CS
0x3	SPI_S_CLK	SPI_S_CS	SPI_S_MOSI	SPI_S_MISO	SPI_S_CLK	SPI_S_CS	SPI_S_MOSI	SPI_S_MISO
0x4	UART0_Rx	UART0_Tx	UART0_RTS	UART0_CTS	UART0_Rx	UART0_Tx	UART0_RTS	UART0_CTS
0x5	USB_DP	USB_DM	UART1_Rx	UART1_Tx	USB_DP	USB_DM	UART1_Rx	UART1_Tx
0x6	PWM0	PWM1	PWM2	PWM3	PWM4	PWM5	PWM6	PWM7
0x7	PDM_CLK	PDM_DATA	PDM_CLK	PDM_DATA	PDM_CLK	PDM_DATA	PDM_CLK	PDM_DATA
0x8					CLK_OUT	RF_PA_TXEN	RF_PA_RXEN	CLK_OUT
0x9	IrDA_IN	IrDA_OUT				IrDA_IN	IrDA_OUT	
0xA	I2S_CK	I2S_WS	I2S_SD_OUT	I2S_SD_IN	I2S_CK	I2S_WS	I2S_SD_OUT	I2S_SD_IN
0xB								
0xC								
0xD	LCD_CS	LCD_DC	LCD_WR	LCD_RD	LCD_TE	LCD_TE	LCD_CS	LCD_DC
0xE								
0xF								

table 19-2 PortB MUX

引脚 功能	PB0	PB1	PB2	PB3	PB4	PB5	PB6	PB7
0x0	GPIO_B0	GPIO_B1	GPIO_B2	GPIO_B3	GPIO_B4	GPIO_B5	GPIO_B6	GPIO_B7
0x1	I2C0_CLK	I2C0_DAT	I2C1_CLK	I2C1_DAT	I2C0_CLK	I2C0_DAT	I2C1_CLK	I2C1_DAT
0x2	SPI0_M_CLK	SPI0_M_CS	SPI0_M_IO0	SPI0_M_IO1	SPI0_M_IO2	SPI0_M_IO3	SPI0_M_CLK	SPI0_M_CS
0x3	SPI_S_CLK	SPI_S_CS	SPI_S_MOSI	SPI_S_MISO	SPI_S_CLK	SPI_S_CS	SPI_S_MOSI	SPI_S_MISO
0x4	UART0_Rx	UART0_Tx	UART0_RTS	UART0_CTS	UART0_Rx	UART0_Tx	UART0_RTS	UART0_CTS
0x5	USB_DP	USB_DM	UART1_Rx	UART1_Tx	USB_DP	USB_DM	UART1_Rx	UART1_Tx
0x6	PWM0	PWM1	PWM2	PWM3	PWM4	PWM5	PWM6	PWM7
0x7	PDM_CLK	PDM_DATA	PDM_CLK	PDM_DATA	PDM_CLK	PDM_DATA	PDM_CLK	PDM_DATA
0x8					CLK_OUT	RF_PA_TXEN	RF_PA_RXEN	CLK_OUT
0x9	IrDA_IN	IrDA_OUT				IrDA_IN	IrDA_OUT	
0xA					I2S_CK	I2S_WS	I2S_SD_OUT	I2S_SD_IN
0xB								
0xC								
0xD	LCD_D0	LCD_D1	LCD_D2	LCD_D3	LCD_D4	LCD_D5	LCD_D6	LCD_D7
0xE								
0xF								

table 19-3 PortC MUX

Pin Func	PC0	PC1	PC2	PC3	PC4	PC5	PC6	PC7
0x0	GPIO_C0	GPIO_C1	GPIO_C2	GPIO_C3	GPIO_C4	GPIO_C5	GPIO_C6	GPIO_C7
0x1	I2C0_CLK	I2C0_DAT	I2C1_CLK	I2C1_DAT	I2C0_CLK	I2C0_DAT	I2C1_CLK	I2C1_DAT
0x2	SPI1_M_CLK	SPI1_M_CS	SPI1_M_IO0	SPI1_M_IO1	SPI1_M_IO2	SPI1_M_IO3	SPI1_M_CLK	SPI1_M_CS
0x3	SPI_S_CLK	SPI_S_CS	SPI_S_MOSI	SPI_S_MISO	SPI_S_CLK	SPI_S_CS	SPI_S_MOSI	SPI_S_MISO
0x4	UART0_Rx	UART0_Tx	UART0_RTS	UART0_CTS	UART0_Rx	UART0_Tx	UART0_RTS	UART0_CTS
0x5	LCD_CS	LCD_DC	UART1_Rx	UART1_Tx	LCD_WR	LCD_RD	UART1_Rx	UART1_Tx
0x6	PWM0	PWM1	PWM2	PWM3	PWM4	PWM5	PWM6	PWM7
0x7	PDM_CLK	PDM_DATA	PDM_CLK	PDM_DATA	PDM_CLK	PDM_DATA	PDM_CLK	PDM_DATA
0x8							SWCLK	SWDIO
0x9	IrDA_IN	IrDA_OUT				IrDA_IN	IrDA_OUT	
0xA	I2S_CK	I2S_WS	I2S_SD_OUT	I2S_SD_IN	I2S_CK	I2S_WS	I2S_SD_OUT	I2S_SD_IN
0xB								
0xC	QSPI0_IO3	QSPI0_CLK	QSPI0_CS	QSPI0_IO1	QSPI0_IO2	QSPI0_IO0		
0xD	LCD_D8	LCD_D9	LCD_D10	LCD_D11	LCD_D12	LCD_D13	LCD_D14	LCD_D15
0xE								
0xF								

table 19-4 PortD MUX

Pin Func	PD0	PD1	PD2	PD3	PD4	PD5	PD6	PD7
0x0	GPIO_D0	GPIO_D1	GPIO_D2	GPIO_D3	GPIO_D4	GPIO_D5	GPIO_D6	GPIO_D7
0x1	I2C0_CLK	I2C0_DAT	I2C1_CLK	I2C1_DAT	I2C0_CLK	I2C0_DAT	I2C1_CLK	I2C1_DAT
0x2	SPI1_M_CLK	SPI1_M_CS	SPI1_M_IO0	SPI1_M_IO1	SPI1_M_IO2	SPI1_M_IO3	SPI1_M_CLK	SPI1_M_CS
0x3	SPI_S_CLK	SPI_S_CS	SPI_S_MOSI	SPI_S_MISO	SPI_S_CLK	SPI_S_CS	SPI_S_MOSI	SPI_S_MISO
0x4	UART0_Rx	UART0_Tx	UART0_RTS	UART0_CTS	UART0_Rx	UART0_Tx	UART0_RTS	UART0_CTS
0x5			UART1_Rx	UART1_Tx			UART1_Rx	UART1_Tx
0x6	PWM0	PWM1	PWM2	PWM3	PWM4	PWM5	PWM6	PWM7
0x7	PDM_CLK	PDM_DATA	PDM_CLK	PDM_DATA	PDM_CLK	PDM_DATA	PDM_CLK	PDM_DATA
0x8	AuxADC7	AuxADC6	AuxADC5	AuxADC4	AuxADC3	AuxADC2	AuxADC1	AuxADC0
0x9	IrDA_IN	IrDA_OUT				IrDA_IN	IrDA_OUT	
0xA	I2S_CK	I2S_WS	I2S_SD_OUT	I2S_SD_IN	I2S_CK	I2S_WS	I2S_SD_OUT	I2S_SD_IN
0xB								
0xC								
0xD	LCD_WR	LCD_RD	LCD_TE	LCD_TE	LCD_CS	LCD_DC	LCD_WR	LCD_RD
0xE								
0xF								

table 19-5 PortE MUX

Pin Func	PE0	PE1						
0x0	GPIO_E0	GPIO_E1						
0x1	I2C0_CLK	I2C0_DAT						
0x2	SPI1_M_IO0	SPI1_M_IO1						
0x3	SPI_S_CLK	SPI_S_CS						
0x4	UART0_Rx	UART0_Tx						
0x5	UART1_Rx	UART1_Tx						
0x6	PWM0	PWM1						
0x7	PDM_CLK	PDM_DATA						
0x8								
0x9								
0xA								
0xB								
0xC								
0xD	LCD_TE	LCD_TE						
0xE								
0xF								

USB_PHY_DP	USB_DP	UART0_Rx	UART1_Rx
USB_PHY_DM	USB_DM	UART0_Tx	UART1_Tx

20. Appendix II

table 20-1 PMU PortA MUX

Pin \ Func	2'b00	2'b01	2'b10	2'b11
PA0	PMU_GPIO_A0	PMU_PWM0	KeyScan_COL[0]	
PA1	PMU_GPIO_A1	PMU_PWM1	KeyScan_COL[1]	
PA2	PMU_GPIO_A2	PMU_PWM2	KeyScan_COL[2]	
PA3	PMU_GPIO_A3	PMU_PWM3	KeyScan_COL[3]	
PA4	PMU_GPIO_A4	PMU_PWM0	KeyScan_COL[4]	
PA5	PMU_GPIO_A5	PMU_PWM1	KeyScan_COL[5]	
PA6	PMU_GPIO_A6	PMU_PWM2	KeyScan_COL[6]	
PA7	PMU_GPIO_A7	PMU_PWM3	KeyScan_COL[7]	

table 20-2 PMU PortB MUX

Pin \ Func	2'b00	2'b01	2'b10	2'b11
PB0	PMU_GPIO_B0	PMU_PWM0	KeyScan_COL[8]	
PB1	PMU_GPIO_B1	PMU_PWM1	KeyScan_COL[9]	
PB2	PMU_GPIO_B2	PMU_PWM2	KeyScan_COL[10]	
PB3	PMU_GPIO_B3	PMU_PWM3	KeyScan_COL[11]	
PB4	PMU_GPIO_B4	PMU_PWM0	KeyScan_COL[12]	
PB5	PMU_GPIO_B5	PMU_PWM1	KeyScan_COL[13]	
PB6	PMU_GPIO_B6	PMU_PWM2	KeyScan_COL[14]	
PB7	PMU_GPIO_B7	PMU_PWM3	KeyScan_COL[15]	

table 20-3 PMU PortC MUX

Pin \ Func	2'b00	2'b01	2'b10	2'b11
PC0	PMU_GPIO_C0	PMU_PWM0	KeyScan_COL[16]	
PC1	PMU_GPIO_C1	PMU_PWM1	KeyScan_COL[17]	
PC2	PMU_GPIO_C2	PMU_PWM2	KeyScan_COL[18]	
PC3	PMU_GPIO_C3	PMU_PWM3	KeyScan_COL[19]	
PC4	PMU_GPIO_C4	PMU_PWM0	KeyScan_COL[0]	
PC5	PMU_GPIO_C5	PMU_PWM1	KeyScan_COL[1]	
PC6	PMU_GPIO_C6	PMU_PWM2	KeyScan_COL[2]	
PC7	PMU_GPIO_C7	PMU_PWM3	KeyScan_COL[3]	

table 20-4 PMU PortD MUX

Pin \ Func	2'b00	2'b01	2'b10	2'b11
PD0	PMU_GPIO_D0	PMU_PWM0	KeyScan_COL[4]	
PD1	PMU_GPIO_D1	PMU_PWM1	KeyScan_COL[5]	
PD2	PMU_GPIO_D2	PMU_PWM2	KeyScan_COL[6]	
PD3	PMU_GPIO_D3	PMU_PWM3	KeyScan_COL[7]	
PD4	PMU_GPIO_D4	PMU_PWM0	KeyScan_COL[8]	
PD5	PMU_GPIO_D5	PMU_PWM1	KeyScan_COL[9]	
PD6	PMU_GPIO_D6	PMU_PWM2	KeyScan_COL[10]	
PD7	PMU_GPIO_D7	PMU_PWM3	KeyScan_COL[11]	

table 20-5 PMU PortE MUX

Pin \ Func	2'b00	2'b01	2'b10	2'b11
PE0	PMU_GPIO_E0	PMU_PWM0	KeyScan_COL[12]	
PE1	PMU_GPIO_E1	PMU_PWM1	KeyScan_COL[13]	